7 CHAPTER

Genetics-I
Classical & Population Genetics

CLASSICAL GENETICS

Mendel and His Work

Johann Mendel was the pioneer of classical geneticists. He was born in July 22, 1822 in Heinzendorf in Austrian. In 1866 his paper *"Experiments on plant hybridization"* published in volume 4 of the proceedings of the Natural Science Society.

Mendel's Selection of the Experimental Plant

For his hybridization experiments Mendel had certain consideration in his mind about the choice of a suitable material. Mendel's considerations about the material were as follows :

1. **Variation :** The organisms which are to be chosen for the genetical experiments, should have a number of detectable differences and at a time only single detectable character should be considered.
2. **Reproduction :** The chosen organisms should be sexually reproducing (*i.e.,* by fusion of male and female sex cells) because only then the offsprings will be able to receive different characters from both the male and female parents.
3. **Controlled mating :** The chosen organisms should be able to mate in controlled or well-planned conditions. Because in genetical experiments sometimes we have to rear genetically pure parents by methods of controlled mating. One should maintain careful records of the offsprings of many generation.
4. **Short life cycle :** The chosen organisms should have very short life cycles.
5. **Large number of offsprings :** The organisms which have been chosen for the genetical experiments should produce large number of offsprings after each successive mating because it will help in deducing the correct conclusions.
6. **Convenience in handling :** The experimental species should be of a type that can be raised and maintained coveniently and inexpensively in the laboratory.

Mendel's Material and Crossing Technique

Mendel found edible pea (*Pisum sativum*) a best material for his hybridization experiments. The pea plant has various contrasting characters among its different varieties such as stem may be *tall* or *dwarf*, cotyledons many be *green* or *yellow*, seeds may be *round* or *wrinkled*, seed coat may be *coloured* or *colourless*; the unripe pods may be *green* or *yellow*; the ripe pods may be *inflated* or *constricted* between the seeds, flowers may have *axial* or *terminal* positions and the colours of flowers may be *red* or *white*. Besides these contrasting characters, the pea plant is a very satisfactory material for the hybridization experiments due to its flower structure.

For each of seven pairs of characters, plants with one alternative trait were used as female and those with the other alternative as male. Reciprocal crosses were also made, *i.e.*. each of the crosses was made in two ways, depending on which phenotype is used as male or female. For example the following two crosses are reciprocal crosses :

Phenotype A (e.g., Tall) ♀ × phenotype B (*e.g.*, Dwarf) ♂

Phenotype B (e.g., Dwarf) ♀ × phenotype A (*e.g.*, Tall) ♂

The population obtained as a result of crossing plants showing contrasting characters is called **F_1 generation**. The progeny of F_1 plants was then obtained by self-fertilization and it forms the **F_2 generation**. Similarly, F_3, F_4, etc., generations can also be obtained.

Further, for getting the exact results in the breeding experiments, it was necessary for Mendel to rear genetically pure variety of pea plants for a single character. Mendel adopted self-fertilization technique for it. For instance, to get pure character for tallness, he self fertilized a tall pea plant for many generations till the resulted offsprings always produced only tall plants. Likewise, he got genetically pure variety for dwarf pea plants.

Mendel cross pollinated these two varieties of pea plants which were differing in a pair of contrasting characters, *viz.*, tallness and dwarfness of the stem. When he made observations on the offsprings of first generation he found only tall plants. He allowed the self pollination in the offsprings of first generation and made further observations on the offsprings of second generation. He was astonished to note both tall and dwarf offsprings in the second generation. This showed him that the character of dwarfness disappeared in first generation but again reappeared in second generation. Further, the tall and dwarf plants of second generation were always in the ratio of 3 : 1 (3 tall : 1 dwarf).

On the basis of the results of his experiments Mendel recognized the **phenomenon of dominance** and formulated following two laws :

1. Law of segregation ;
2. Law of independent assortment.

Phenomenon of Dominance

The cross between the pea plants differing in single pair of contrasting characters is known as **monohybrid cross**. Mendel described the phenomenon of dominance in following way : *in crossing between pure (homozygous) organisms for contrasting characters of a pair, only one character of the pair appears in the first filial generation.*

Mechanism of Dominance

Mendel carefully studied the results of his experiments and it became evident to him that there is a clear cut difference in between the actual visible character and that something which caused its production. Because the character cannot be present as such in sex cells (sperm and ova) which form the only link between a new individual and the parents. Therefore, there must be something which represents the characters and is responsible for their production. This something of Mendel is now called the **factor** or **gene**. The gene thus, can be considered as the unit of inheritance that is transmitted in a gamete and determines or controls the development of a character by interaction with the other genes, the cytoplasm and the environment.

Variation in Dominance Relation

Mendel reported full dominance and recessiveness for all the seven gene pairs (or allele pairs) he studied. He may have been selective in his choice of pea characters to study, because variations on the basic theme appear quite often in analysis. For instance, in some cases the phenotypes of heterozygotes are found to be different from either of the homozygotes :

1. In complete Dominance

Sometimes in a heterozygote dominant allele does not completely mask the phenotypic expression of the recessive allele and there occurs an intermediate phenotype in the heterozygote. This is called **in complete dominance**.

Examples. 1. When a red flowered pea plant (RR) is crossed with white flowered pea plant (rr) then the F_1 hybrid pea plants are found to have pink flowers. Is shows that gene for red colour could not completely dominate the gene fow **white** colour. In such a case, F_2 phenotypic ratio and genotypic ratio are the same, as follows :

F_2 phenotypic ratio = 1 Red : 2 Pink : 1 White

F_2 genotypic ratio = 1 RR : 2 Rr : 1 rr

2. Codominance

Sometimes both alleles of a gene in a heterozygote lack the dominant and recessive relationship, *i.e.*, each allele is capable of some degree of phenotypic expression. In a sense, codominance is no dominance at all, the heterozygote showing the phenotypes of both homozygotes. Hence, heterozygote genotype gives rise to a phenotype distinctly different from either of the homozygous genotypes.

Examples. 1. The coat colour of the Shorthorn breed of cattle represent a classical example of codominance. When a cattle of red coat (C^RC^R) is crossed with the cattle of white coat (C^WC^W), the F_1 heterozygote or hybrid is found to possess roan coat (C^RC^W). In roan coat the red and white hairs occur in definite patches but no hair has intermediate colour of red and white.

Law of Segregation

Mendel's first law–the law of segregation is also known as law of purity of gametes. The law *states that the hybrids or heterozygotes of F_1 generation have two contrasting characters or allelomorphs of dominant and recessive nature. These alleles though remain together for long time but do not contaminate or mix with each other and separate or segregate at the time of gametogenesis, so that each gamete receives only one allele of a character either dominant or recessive.*

Mechanism of Segregation

The mechanism of segregation, monohybrid cross between red and white flowered pea plants can be understood by assuming that the homozygous red flowered pea plant has the allele RR for redness and white flowered pea plant has the alleles rr for whiteness. The pea plant with RR alleles produces the gametes with single allele R and pea plant with rr alleles produces the gametes with the allele r. The gametes of both unite to form a hybrid or heterozygote having the alleles Rr both for redness and whiteness. Due to the phenomenon of incomplete dominance the allele R for red colour partially expresses itself in hybrids of F_1, while the allele r for white colour remains latent or recessive. Both the allele R and r remain together for long time but they do not effect each other. Neither they mix nor they contaminate each other. Because a gamete can contain only one chromosome of a homologous pair, therefore, each gamete can carry single allele R or r. At the time of gametogenesis two types of gametes are produced by F_1 hybrids in equal numbers . Half of the gametes carry the allele R and other half carry the allele r. These gametes during the process of fertilization can unite in three possible combinations, *viz.*, RR, Rr and rr, to produce three types of individuals in F_2 generation. Thus in F_2 75% individuals have coloured flowers and 25% white flowers. The appearance of white colour in F_2 generation indicates that in the hybrid the allele (r) for white colour remains along with allele (R) for red colour but does not mix with it or contaminated by it and it separates or segregates during gametogenesis.

Certain Other Examples of Law of Segregation

The law of segregation is universal in its application and it has been found to occur in both plants and animals. The examples which are cited in phenomenon of dominance can also be considered for the law of segregation. However, to understand the mechanism of segregation more clearly in animals it will be helpful to consider an original experiment of well known geneticists **T.H. Morgan** on *Drosophila*.

Morgan crossed a homozygous long-winged (wild type) *Drosophila* with a homozygous vestigial-winged (mutant) *Drosophila*. The F_1 heterozygotes or hybrids were found to be long winged. When the F_1 hybrids were allowed to mate among themselves, they produced long-winged and vestigial-winged hybrids in three and one ratio (3 : 1) in the F_2 generation.

Law of Independent Assortment

To formulate the law of dominance and law of segregation **Mendel** considered monohybrid crosses in which single pairs of contrasting characters were considered at a time. But he tried to find out how different characters would behave in relation to each other in their inheritance from generation to generation. For this purpose Mendel crossed two varieties of pea plants which were differing in two pairs of contrasting characters. Because such crosses yielded dihybrids and at a time two pairs of contrasting characters had been considered in them, therefore, these crosses were known as **dihybrid crosses**.

Mendel's Dihybrid Cross

In one of his hybridization experiment Mendel crossed a homozygous pea plant having **yellow round seeds** with the homozygous pea plant having **green wrinkled seeds**. The F_1 hybrids were found to have yellow round seeds. When the F_1 hybrids were allowed to cross among themselves they produced four types of seeds in the ratio of 9 : 3 : 3 : 1 given as follows :

1. Yellow Round – 9
2. Yellow Wrinkled – 3
3. Green Round – 3
4. Green Wrinkled – 1

Beside getting the ratio of 3 : 1 of the monohybrid crosses Mendel got the ratio of 9 : 3 : 3 : 1.

This irregularity in the ratio of F_2 offspring was explained by Mendel stating that *"when the parents differ from each other in two or more pairs of contrasting characters or factors then the inheritance of one pair of factors is independent to that of the other pair of factors"*. This is the Mendel's law of independent assortment.

Mechanism of Independent Assortment

The mechanism of independent assortment can be understood easily by assuming that the homozygous pea plant with yellow round seeds has the alleles YY and RR for the yellow colour and roundness of the seed, respectively.

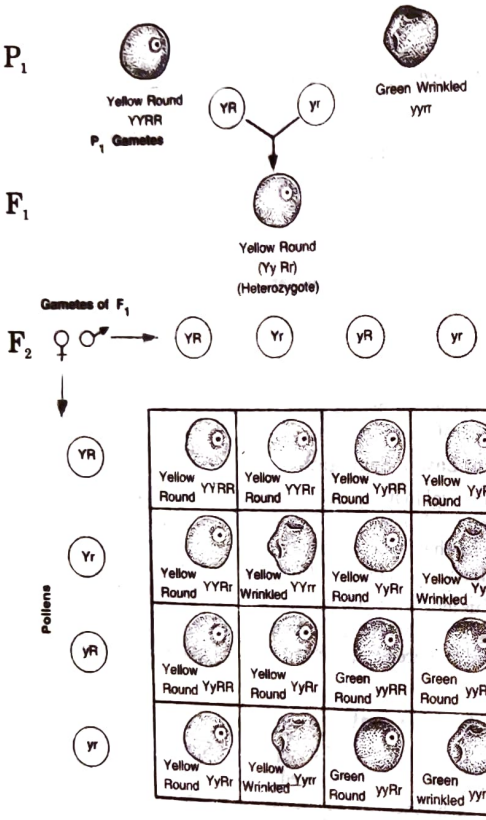


Fig. A dihybrid cross between a yellow round and a green wrinkled pea plant.

Similarly the homozygous pea plant with green wrinkled seeds contains the alleles yy and rr for the green colour and wrinkledness of seeds be located on r locus, but the gene is interrupted by a small DNA sequence (0.8. kilobases) called **transposon-like insertion**, due to which aberrant SBEl enzyme is produced. This defective enzyme leads to metabolic disturbances in the biosynthesis of starch, lipid and protein.

Back Cross and Test Cross

When F_1 individuals are crossed with one of the two parents from which they were derived, then such a cross is called **back cross**. In such back crosses, when F_1 is back crossed to the parent with dominant characters (phenotype), no recessive individuals are obtained in the progeny. On the other hand, when it is crossed with recessive parent, both phenotypes appear in the progeny. While both of these crosses are back crosses, only the cross with the recessive parents is known as **test cross**. It is called a test cross, because it is used to test whether an individual is homozygous (pure) or heterozygous (hybrid). For a monohybrid the test cross ratio remains 1 : 1, but, for a dihybrid test cross ratio becomes 1 : 1 : 1 : 1.

Example of Monohybrid Back Cross and Test Cross

In a monohybrid cross of homozygous tall (DD) and homozygous dwarf (dd) pea plants, when a F_1 heterozygous tall (Dd) plant is crossed either with its dominant parent to perform a back cross or with its recessive parent to perform a test cross following results are obtained :

P_1 :	Homozygous Tall DD	× ↓	Homozygous Dwarf dd
		Dd	
		Heterozygous Tall	
F_1 :			
A. Back cross :	F_1 Tall Dd	× ↓	P_1 Tall DD
Back cross progeny :	½DD Homozygous tall	: 	½Dd or all tall. Heterozygous tall
B. Test Cross :	F_1 Tall Dd	× ↓	P_1 Dwarf dd
Test cross progeny :	½Dd Heterozygous tall	: 	½dd or Test cross ratio = 1:1 Homozygous dwarf

Genetic Interaction and Lethal Genes

Non-allelic (Inter-Allelic) Genetic Interactions

In inter allelic genetic interactions, the independent (non-homologous) genes located on the same or on different chromosomes interact with one another for the expression of single phenotypic trait of an organism.

Originally a gene or locus which suppressed or masked the action of a gene at another locus was termed **epistatic gene**. The gene or locus which was suppressed by a epistatic gene was called **hypostatic gene**.

Difference Between Dominance and Epistasis

The phenomenon of dominance involves intra-allelic gene suppression, or the masking effect which one allele has upon the expression of another allele at the same locus, while the phenomenon of epistasis involves inter-allelic gene suppression or the masking effect wihch one gene locus has upon the expression of another. The classical phenotypic ratio of 9 : 3 : 3 : 1 observed in the progeny of dihybrid parents becomes modified by epistasis into ratios which are various combinations of the 9 : 3 : 3 : 1 groupings.

Non-epistatic Inter-allelic Genetic Interactions

In certain cases, two pairs of genes determine a same phenotype but assorted independently, produce new phenotypes by mutual non-epistatic interactions and the F_2 phenotypic ratio 9 : 3 : 3 : 1 remains unaltered. Two pairs of genes which interact to affect size and shape of comb but are independently transmitted, exist in chicken.

Kinds of Epistatic Interaction

When in dihybrid crosses, the epistatic interactions occur between two genes, less than four phenotypes appear in F_2. Such bigenic (two gene) epistatic interactions may be of following six types :

1. **Dominant Epistasis (12 : 3 : 1)**

   When out of two genes, the dominant allele (*e.g.,* A) of one gene masked the activity of alleles of another gene (*e.g.,* B) and expressed itself phenotypically, then A gene locus is said to be epistatic to the B gene locus. Because, the dominant allele A can express itself only in the presence of either B or b allele, therefore, such type of epistasis is termed as **dominant epistasis.** The alleles of hypostatic locus or gene B will be able to express themselves

phenotypically only when gene locus A may contain two recessive alleles (aa). Thus, the genotype AA BB or Aa Bb and AA bb or Aa bb produce the same phenotype whereas the genotype aa BB or aa Bb and aa bb produce two additional phenotypes. The dominant epistasis modify the classical ratio of 9 : 3 : 3 : 1 into 12 : 3 : 1 ratio.

Example

Dominant epistatis in dogs. Among dogs, the colours of coats depend upon the action of two genes. One gene locus has a dominant epistatic inhibitor allele (I) of coat colour pigment. The allele I prevents the expression of colour allele at another independently assorting, hypostatic gene locus (B or b) and produces white coat colour. The alleles of hypostatic gene locus (BB, Bb, or bb) express only when two recessive alleles ii occur on the epistatic locus, *i.e.*, ii BB or ii Bb produces black and ii bb produces brown individuals. When two such white coat colour dogs are crossed, in F_1 the white, black and brown coat colours appear in 12 : 3 : 1 ratio.

2. Recessive Epistasis (9 : 3 : 4)

Sometimes the recessive alleles of one gene locus (aa) mask the action (phenotypic expression) of alleles of another gene locus (BB, Bb of bb alleles). This type of epistasis is called **recessive epistasis**. The alleles of B locus express themselves only when epistatic locus has dominant alleles (*e.g.*, AA or Aa). Due to recessive epistasis the phenotypic ratio 9 : 3 : 3 : 1 becomes modified into 9 : 3 : 4 ratio.

Example

Recessive epistasis in mice. In mice various types of epistatic genetic interactions have been reported. The most interesting case is of recessive epistasis in coat colours. The common house mouse occurs in a number of coat colours, *i.e.*, agouti, black and albino. The agouti colour pattern is commonly occurred one (wild type) and is characterized by colour banded hairs in which the part nearest the skin is gray, then a yellow band and finally the distal part is either black or brown. The albino mouse lacks totally in pigments and has white hairs and pink eyes.

When a homozygous black (CC aa) is crossed with a homozygous albino (cc AA) in F_1 all agouti (Cc Aa) offsprings appear. When the F_1 agouti are crossed among themselves in F_2 agouti, black and albino offsprings appear in the ratio of 9 : 3 : 4.

Supplementary genes. In the cross between black (CCaa) and albino (ccAA) mice, one thing becomes apparent that two independent pairs of genes (*i.e.*, C–c and A–a) have interacted in the production of the phenotypic trait (*i.e.*, coat colour) in such a way that one dominant (C) produces its effect whether or not the second (A) is present, but the second (A) gene can produce its effect only in the presence of the first. These genes (*i.e.*, C and A) have been termed as **supplementary genes**.

3. Duplicate Genes with Cumulative Effect (9 : 6 : 1)

Certain phenotypic traits (*e.g.*, coat colouration) depend on the dominant alleles of two gene loci. When the dominant condition (homozygous or heterozygous) at either locus (but not both) produces the same phenotype, the F_2 ratio becomes 9 : 6 : 1.

Example

Cumulative effect in coat colour of pigs. In the Duroc-jersey breed of pigs, coat colour is influenced by two pairs of genes that interact in peculiar manner. Sandy coat colour results from a dominant gene S, and the homozygous recessive (ss) is white in colour. Sandy coat colour may also result from a non-allelic dominant gene R; its homozygous recessive (rr) is also white. When a sandy pig (SS rr) is crossed with a second sandy pig (ss RR), the F_1 offsprings were found with red coloured coats. Such interactions are said to be the result of **mutually supplementary genes**. When F_1 red coated pigs cross bred among themselves they produce red, sandy and white coats in the ratio of 9 : 6 : 1

4. Duplicate Recessive Genes (or Complimentary genes) (9 : 7)

If both gene loci have homozygous recessive alleles and both of them produce identical phenotypes, the F_2 ratio 9 : 3 : 3 : 1 would become 9 : 7. In such case, the genotypes aa BB, aa Bb, AA bb, Aa bb, and aa bb produce one phenotype. Both dominant alleles when present together complement each other and **are called complementary genes** and produce a different phenotype. A case of such complimentary inheritance, resulting from the combined action of complimentary genes is known in sweet peas.

Example

When a pure line variety of white flowered sweet pea (*Lathyrus odoratus*) was crossed with another pure line variety of white flowered sweet pea, in F_1 purple or red flowered plants were produced. The F_1 plants when self-pollinated or crossed among themselves, produced the F_2 generation with the phenotypic ratio 9 coloured and 7 white flowered plants.

5. Duplicate Dominant Genes (15 : 1)

If the dominant alleles of both gene loci produce the same phenotype without cumulative effect the 9 : 3 : 3 : 1 ratio is modified into a 15 : 1 ratio.

Example

The seed capsules of Shepherd's purse (genus *Capsella*) occur in two different shapes, *i.e.*, triangular and top-shaped. When a plant with triangular seed capsule is crossed with one having top-shaped capsule in F_1 only triangular, character appears. The F_1 offsprings by self crossing produce the F_2 generation with the triangular and top-shaped seed capsules in the ratio of 15 : 1

6. Dominant and Recessive Interactions (13 : 3)

Sometimes, the dominant alleles of one gene locus (A) in homozygous (AA) and heterozygous (Aa) condition and the homozygous recessive alleles (bb) of another gene locus (B) produce the same phenotype, the F_2 phenotypic ratio becomes 13 : 3 instead of 9 : 3 : 3 : 1. In such case, the genotype AA BB, AA Bb, Aa BB, Aa Bb, AA bb and Aa bb produce same phenotype and the genotype aa BB, aa Bb and aa bb produce another but same phenotype.

Example

In Leghorn type of fowl the white colour of feather is caused by the dominant genotype CC II, similarly the white colour of feathers of **Plymouth Rock** breed is caused by the recessive genotype cc ii when both white varieties of fowl are crossed in F1 white coloured hybrids appear. The F_1 hybrids in F_2 produce the white and coloured offsprings in the ratio of 13 : 3.

Atavism or Reversion

While we were discussing the cross between two white flowered sweet pea plants we observed that F_1 hybrid had purple colour unlike their immediate parents but like their remote ancestors. The appearance of such offsprings which resemble with their remote ancestors are called throwbacks, atavisms or reversions.

Lethal Genes

Lethal genes are mutant genes and result in the death of the individual which carries them. Death of the individual occurs either in the prenatal or postnatal period prior to sexual maturity. A **fully** (completely) **dominant lethal allele** kills both in homozygous and heterozygous states. Individuals with a dominant lethal allele die before they give rise to their progeny. Therefore, the mutant dominant lethal is removed from the population in the same generation in which it arose. **Recessive lethal genes** kill only when they are in a homozygous state and they may be of two kinds :

1. one which has no obvious phenotypic effect in heterozygous and
2. one which exhibits a distinctive phenotype when heterozygous.

The completely lethal genes usually cause death of the zygote, later in the embryonic development or even after birth or hatching. Such lethal genes which handicap but do not destroy their possessor are called **subvital**, **sublethal** or **semilethal** genes. The lethal alleles modify the 3 : 1 phenotypic ratio into 2 : 1.

In the chicken an incompletely dominant gene (cp) in heterozygous condition (cp/+) cause "creeper" condition. The creeper birds have much shortened and deformed legs and wings giving them a squatty appearance and creeping gait. A cross of two creeper birds yields viable offsprings in the ratio of 2 creepers : 1 normal. The homozygous creepers having such a gross deformities that they die during incubation.

F_1 :	cp/+	X	cp/+		
	Creeper	↓	Creeper		
	(Heterozygous)			:	1+/+
F_2 :	1 cp/cp	:	2cp/+		Normal
	Creeper		Creeper		(Homozygous)
	(Homozygous; dies)		(Heterozygous)		

Penetrance

The ability of a given gene or gene combination to be expressed phenotypically to any degree is called penetrance. It is of following two kinds :

1. **Complete Penetrance**

   Most dominant and recessive genes in homozygous conditions and many completely dominant genes even in heterozygous conditions give their complete phenotypic expressions. Such genes are called to have **complete penetrance.**

   Examples of Complete Penetrance

   1. In pea the alleles (RR) for red flowers and the alleles (rr) for white flowers have complete penetrance in homozygous conditions.
   2. In *Drosophila* the recessive alleles for vestigial wings in homozygous conditions have complete penetrance.
   3. In guinea pigs the dominant allele 'B' for black coat has complete penetrance both in homozygous and heterozygous conditions.

2. **Incomplete Penetrance**

   Some genes in homozygous as well as in heterozygous conditions fail to provide complete phenotypic expression of them. Such genes are called to have **incomplete penetrance.**

   Examples of Incomplete Penetrance

   (*i*) Polydactyly in man is thought to be produced by a dominant gene P. The normal condition with five digits on each limb is produced by the recessive genotype (pp). Some heterozygous individuals (Pp) are not polydactylus and therefore, has a penetrance of less than 70%

Expressivity

A trait though penetrant may be quite variable in its phenotypic expressions. The degree of effect produced by a penetrant genotype is called **expressivity**.

Example of expressivity. In man the polydactylous condition may be penetrant in the left hand (6 fingers) are not in the right (5 fingers); or it may be penetrant in the feet and not in the hands.

Pleiotropism

Most genes have their multiple effects and are called **pleiotropic genes**. The phenomenon of multiple effect (multiple phenotypic expressions) of a single gene is called pleiotropism.

Examples of Pleiotropism

1. In *Drosophila* the recessive gene for vestigial wings cause vestigial wings in homozygous conditions. However, careful observations show that other traits as well are affected —
   (*i*) the tiny wing- like balancer behind the wings;
   (*ii*) certain bristles;
   (*iii*) the structure of the reproductive organs;
   (*iv*) egg production is lowered, and,
   (*v*) longevity is reduced.

Linkage

The hereditary units or genes which determine the characters of an individual are carrier in the chromosomes and an individual usually has many genes for the determination of various different characters. If the genes are situated in the same chromosome and are fairly close to each other, they tend to be inherited together. This type of coexistence of two or more genes in the same chromosome is known as linkage

Morgan's Views on Linkage

Morgan stated that the pairs of genes of homozygous parent tend to enter in the same gametes and to remain together, whereas same genes from heterozygous parent tend to enter in the different gametes and remain apart from each other. He further stated that the tendency of linked genes remaining together in original

combinations is due to their location in the same chromosome. According to him, the degree or strength of linkage depends upon the distance between the linked genes in the chromosome. Morgan's concept about the linkage developed the theory of linear arrangement of genes in the chromosomes which helped the cytogeneticists in the construction of genetic of linkage maps of chromosomes.

Chromosome Theory of Linkage

Morgan along with **Castle** formulated the chromosome theory of linkage which is as follows:

1. The genes which show the phenomenon of linkage are situated in the same chromosomes and these linked genes usually remain bounded by the chromosomal material so that they cannot be separated during the process of inheritance.
2. The distance between the linked genes determines the strength of linkage. The closely located genes show strong linkage than the widely located genes which show the weak linkage.
3. The genes are arranged in linear fashion in the chromosomes.

Kinds of Linkage

T.H. Morgan and his co-workers by their investigation on the *Drosophila* and other organisms have found two types of linkage, *viz.*, complete linkage and incomplete linkage.

1. **Complete Linkage**

   The complete linkage is the phenomenon in which parental combinations of characters appear together for two or more generations in a continuous and regular fashion. In this type of linkage genes are closely associated and tend to transmit together.

   Complete linkage in male Drosophila

   T.H. Morgan mated gray bodied and vestigial winged (b^+vg/b^+vg) fruit flies with flies having black bodies and normal wings (bvg^+/bvg^+). F_1 progeny had gray bodies and normal long wings (b^+vg/bvg^+), indicating thereby that these characters are dominant. When F_1 males (b^+vg/bvg^+), were backcrossed (*i.e.*, test crossed) to double recessive female (bvg/bvg or black vestigial), only two types of progeny (one with gray bodies and vestigial wings, b+vg/bvg and the other with black bodies and normal wings, to bvg^+/bvg instead of four types of phenotypes were obtained.

Parents:	Gray, Vestigial b^+vg/b^+vg	×	Black, Long bvg^+/bvg^+
Gametes:	(b^+vg)		(bvg^+)
F_1:		All Gray, Long b^+vg/bvg^+	
Test cross:	F_1 male Gray, Long b^+vg/bvg^+	×	Female Black, Vestigial bvg/bvg
Gametes:	(b^+vg) (bvg^+)		(bvg)

(only two types of gametes due to complete linkage)

2. **Incomplete Linkage**

   The linked genes do not always stay together because homologous non-sister chromatids may exchange segments of varying length with one another during meiotic prophase. This sort of exchange of chromosomal segments in between homologous chromosomes is known as **crossing over**. The linked genes which are widely located in chromosomes and have chances of separation by crossing over are called **incompletely linked genes** and the phenomenon of their inheritance is called **incomplete linkage.**

   Incomplete Linkage in maize. In *Zea mays* (maize) a case of incomplete linkage between the alleles for colour and shape of the seed has been observed by **Hutchison.** When a maize plant with seeds having coloured and full endosperm (CS/CS) is crossed with another plant having recessive alleles for colourless, shrunken seeds (cs/cs), the F_1 heterozygotes are found with the phenotype of coloured full and genotype of CS/cs. When F_1 hybrid is test crossed with double recessive parent (cs/cs) four classes of descendants are obtained.

Parents :	Coloured full CS/CS	×	Colourless Shrunken cs/cs	
F_1 :	Coloured full (CS/cs)			
Test cross : F_1 Coloured Full × Colourless Shrunken				
Test cross results :	CS/cs Coloured : Full	cs/cs Coloured : shrunken	Colourless : Full	Colourless Shrunken
	CS/cs 48%	Cs/cs 2%	cS/cs 2%	cs/cs 48%

The test cross results are clearly showing that parental combination of alleles (*e.g.*, CS/cs and cs/cs) are those expected from complete linkage and appear in 96% cases, the other two are new combinations (*e.g.*, Cs/cs and cS/cs) and appear in 4% cases. Thus, in 4% cases crossing over has occurred between linked genes.

Linkage Groups

All the linked genes of a chromosome form a linkage group. Because, all the genes of a chromosome have their identical genes (alleomorphs) on the homologous chromosome, therefore linkage groups of a homologous pair of chromosome is considered as one. The number of linkage group of a species thus, corresponds with haploid chromosome number of that species.

Example.

1. *Drosophila* has 4 pairs of chromosomes and 4 linkage groups.
2. Man has 23 pairs of chromosomes and 23 linkage groups.
3. Corn (*Zea mays*) has 10 pairs of chromosomes and 10 linkage groups.

Example.

1. Female human beings : = 22 pairs of autosomes or non-sex chromosomes + 1 pairs of homomorphic X chromosomes
   = 22 autosomal linkage groups + 1 X chromosomal linkage group
   = 23 linkage groups.
2. Male human beings : = 22 pairs of autosomes + 2 heteromorphic sex chromosomes, *i.e.*, 1 X chromosome + 1 Y chromosome
   = 22 autosomal linkage group + 1 X chromosomal linkage group + 1 Y chromosomal linkage group
   = 24 linkage groups.

Significance of Linkage

The phenomenon of linkage has one of the great significance for the living organisms that it reduces the possibility of variability in gametes unless crossing over occurs.

Genetic and Cytological Mapping of Chromosomes

A diagrammatic, graphical representation of relative distances between linked genes of a chromosome is called linkage or genetic map. Because, such a **linkage** or **genetic map** is the outcome of crossing over studies, is also called **cross over map**.

A. Construction of a Linkage Map or Genetic Mapping

The method of construction maps of different chromosomes is called **genetic mapping**. The genetic mapping includes following processes:

1. Determination of Linkage Groups

Before starting the genetic mapping of the chromosomes of a species, one has to know the different linkage groups of a species.

2. Determination of Map Distance

Geneticists use an arbitary unit to measure the **map unit**, to describe distances between linked genes. A map unit is equal to 1 per cent of crossovers (recombinants); that is, it represents the linear distance along the chromosome for which recombination frequency of 1 **percent** is observed. These distances can also be expressed in **morgan units**; one morgan unit represents 100 **percent** crossing over. Thus, 1 **percent** crossing over can also be expressed as **1centimorgan (1cM)**, 10 **percent** crossing-over as **1 decimorgan** and so on. The Morgan unit is named in honour of **T.H. Morgan;** however, most geneticists prefer map units.

Examples

1. If a F_1 hybrid having the genotype Ab/aB produces 8% of cross over gametes AB and ab, then the distance between A and B is estimated to be 16 map units or centimorgan.
2. If the map distance between the gene loci B and C is 12 centimorgan, then 12% of gametes of genotype BC/bc should be crossover types, *i.e.*, 6% bC.

Because, each chiasmata produces 50% crossover products, 50 percent crossing over is equivalent to 50 map units or centimorgans. If the mean number of chiasmata is known for a chromosome pair, the total length of the map for that linkage group may be predicted :

Total length = mean number of chiasmata × 50

Two Point Test Cross

The percentage of crossing over between two linked genes is calculated by test crosses in which a F_1 dihybrid is crossed with a double recessive parent. Since such crosses involve crossing over at two points, they are called **two point test crosses**. For example, a dihybrid having the genotype Ac/ac is test crossed with a double recessive parent (ac/ac), then among F_2 test cross hybrids we may get 37% dominant genes at both gene loci (AC/ac), 37% recessive genes at both gene loci (ac/ac), 13% dominant gene at first gene locus and recessive gene at the second gene locus (Ac/ac), and 13% recessive gene at first gene locus and dominant gene at second gene locus (aC/ac). The last two groups (*i.e.*, 13% Ac/ac and 13% aC/ac) were produced by crossover gametes (13+13) from the dihybrid parent. Thus, 26% of all gametes (13+13) were of cross over types and the distance between the loci A and C is estimated to be 26 centimorgans.

Three Point Test Cross

A three point test cross or trihybrid test cross (involving three genes) gives us information regarding relative distances between these genes, and also shows us the linear order in which these genes should be present on a chromosome. Such a three point test cross may be carried out if three points or gene loci on a chromosome pair can be identified by marker genes. If, in addition to genes A and C indicated above, a third marker gene B is located in fairly close proximity in the same linkage group, all three markers may be used together in conducting a more precise analysis of the map distance and the relative position of the three points.

Suppose that we testcross trihybrid individuals of genotype ABC/abc and find in the progeny the following :

36% ABC/abc	9%Abc/abc	4%ABc/abc	1%AbC/abc
36% abc/abc	9%aBC/abc	4% abC/abc	1%aBc/abc
72%Parental type:	18%Single crossovers : between A and B. (region I)	8% Single crossovers : between B and C. (region II)	2%Double crossover

To find the distance A–B we must count all crossovers (both singles and doubles) that occurred in region I = 18% + 2% or = 20% or 20 map units between the loci A and B. To find the distance B-C we must again count all crossovers (both singles and doubles) that occurred in region II = 8%+2% = 10% or 10 map units between the loci B and C. The A–C distance is therefore, 30 map units when double crossovers are detected in a three point linkage experiment and 26 map units when double crossovers are undetected in the two-point linkage experiment above.

Without the middle marker (B), double crossovers would appear as parental types and hence we underestimate the true map distance (crossover percentage). In this case the 2% double crossovers would appear with the 72% parental types, making a total of 74% parental types and 26% recombinant types. Therefore, for any three linked genes whose distances are known, the amount of detectable crossovers between the two outer

markers A and C when the middle marker B is missing is; (A–B crossover percentage) plus (B–C crossovers percentage) minus (2 × double crossover percentage).

3. Determination of Gene Order

After determining the relative distances between the genes of a linkage group, it becomes easy to place genes in their proper linear order.

Now, if we suppose that the distance between the genes A – B = 12, B – C = 7, A – C = 5, we can determine the order of genes correctly in the following manner :

Case I. Let us assume that gene A is in the middle (*e.g.*, B – A – C) :

B	12	A	A	5	C

B	7	C

In this case because, the distances between B – C are not equitable, gene A cannot be in the middle.

Case II. Let us assume that gene B is in the middle (*e.g.*, A – B – C) :

A	12	B	B	7	C

B	5	C

In this case, the distance between A – C are not equitable, therefore, gene B cannot be in the middle.

Case III. Let us assume that gene C is in the middle (*e.g.*, A – C – B).

A	5	C	A	5	C

A	12	B

In this case, the distance between A – B are equitable, therefore, gene C must be in the middle.

Thus, the relative distance and ordering of genes in a linkage group are determined in separate segments by two point test crosses or three point crosses, as the case may be.

4. Combining Map Segments

Finally, the different segments of maps of a complete chromosome are combined to form a complete genetic map of 100 centimorgans long for a chromosome.

Example. For example, suppose we have to combine following three map segments.

I | A | 8 | b | 10 | c |

II | c | 10 | b | 22 | d |

III | A | 30 | e | 2 | d |

We can superimpose each of these segments by aligning the genes shared in common.

I | | | A | 8 | b | 10 | c |

II | | c | 10 | b | 22 | d |

III | A | 30 | e | 2 | d |

Then finally we may combine the three segments into one map :

The a to d distance = (d to b) – (a to b) = 22 – 8 = 14

The a to e distance = (a to d) – (d to e) = 14 – 2 = 12

d	2	e	12	a	8	b	10	c

Interference & Coincidence

In most higher organisms it has been found that one chiasma formation reduces the probability of another chiasma formation in an immediately adjacent region of the chromosome, probably because of physical inability of the chromatids to bend back upon themselves within certain minimum distances. The tendency of one crossover to interfere with the other crossover is called **interference**. Thus, the proximity of one crossover to another decreases the probability of another which is very close by. The centromere has a similar interference effect; frequency of crossing over is also reduced near the ends of the chromosome arms. The net result of this interference is the observation of fewer double crossover types than would be expected according to map distances. The strength of interference varies in different segments of the chromosome and is usually expressed in terms of a **coefficient of coincidence**, or the ratio between the observed and the expected double crossovers.

$$\text{coefficient of coincidence} = \frac{\%\text{observed double crossovers}}{\%\text{expected double crossovers}}$$

The coincidence is the complement of interference, so :

Coincidence + Interference = 1.0

When interference is complete (1.0), no double crossovers will be observed and coincidence becomes zero. When, interference decreases, coincidence increases. Coincidence values ordinarily vary between 0 and I. Coincidence is generally quite small for short map distance. There is no interference across centromere.

Example

For explaining interference and coincidence, we can consider the results of one of the experiment of **Hutchison** (1922). He reported the map distances for three genes, c (colourless aleurone), Sh (shrunken grains), and wx (waxy endosperm) of corn and observed following crossing over frequencies between these genes :

Crossing over frequencies between genes c, Sh and wx of corn.

Regions	Ganes	Percentage crossovers	Map-distances (in map units)
I	c–Sh	3.4	3.4 + 0.1 3.5
II	Sh–wx	18.3	18.3 + 0.1 = 18.4
Double crossover	c–Sh–wx	0.1	

If crossing over in region I and II were independent, we should predict 0.035 × 0.184 = 0.6 percent double crossovers, where as only 0.1 per cent was observed

So, coincidence $= \frac{0.1}{0.6} = 0.167$

Linkage Maps of Different Organisms

By adopting the above mentioned techniques, geneticists have constructed the linkage or genetic maps of various organisms, such as, viruses, bacteria, fungi, tomato, barley, wheat, rice, sorghum, morning glory, garden pea, maize, *Drosophila*, chickens, mice, man, etc. The first linkage map has been constructed for two chromosomes of *Drosophila* by **Strutevant** in 1911. The linkage maps of other chromosomes of *Drosophila* have been constructed by **C.B. Bridges**. The linkage or genetic mapping in maize has been done by **Mc Clintock** under the leadership of R.A. Emerson.

Syntenic Genes

If two or more specific human gene products and a given human chromosome are both present in the same hybrid cells, then those genes are located in the same chromosome: that is, they are **syntenic**. The term **synteny** refers to genes that are located on the same chromosome, whether or not they show recombination; **linkage**.

B. Chromosome, Physical Or Cytological Mapping

It has been found that linkage map distances between genes are not necessarily proportional to physical linear measurements. Special cytological techniques have been used to determine the physical locations of a gene in a chromosome. Localization has been accomplished by identifying a gene locus with relation to some

visible landmark such as a chromomere or cross band. In this process of **chromosome, physical or cytological mapping**, the polytene chromosomes of salivary glands of *Drosophila* have been found very useful. **T.H. Painter** was the first geneticist who used polytene chromosomes of *Drosophila* in the cytological verification of genetic data. He related the bands on the giant **chromosomes** to genes, but he was more interested in the morphology of the chromosomes and applications concerning speciation than in the association of chromosome sections with particular genes.

Cytological Mapping of Chromosomes of Drosophila

About 5000 single cross bands have been noted on the four pairs of salivary gland chromosomes in *D. melanogaster*. This number is considered a minimum approximation of the number of genes in that animal. Some genes have been associated with individual bands. Bridges system of designating parts of chromosomes with numbers, subdivision with letters, and numbering bands within subdivision had made it possible for investigators to discuss precise locations. In this system, fairly uniform divisions are numbered in order throughout each entire chromosome set from O at the beginning of the X chromosome to 102 at the end of chromosome 4. Subdivisions within the areas are identified with letters from A to F, and bands within subdivisions are numbered from left to right. For example, the gene (w) for white eyes is in bands 3C2. In linkage units this gene is located at 1.5 in the X chromosome.

Differences Between Genetic and Chromosome Maps

The frequency of crossing over usually varies in different segments of the chromosome but is a highly predictable even between any two genes loci. Therefore, the actual physical distances between linked genes bears no direct relationship to the map distance calculated on the basis of crossover percentages. However, the linear order of linked genes is identical in both maps.

Uses of Genetic Maps

The genetic maps have following uses :

(*i*) The chromosome maps display the exact location, arrangement and combination of genes in a linkage group or chromosomes.

(*ii*) They are useful in predicting results of dihybrid crosses.

Sex Linked Inheritance

The genes that are carried by sex chromosomes through generations are called *sex-linked genes*, and the inheritance determined by the genes is termed *sex-linkage* or *sex-linked inheritance*. The phenomenon of sex-linkage is examplified by the following :

A. X-Linked Genes :

Most of the sex-linked genes are located on the X-sex chromosomes. e.g; *Drosophila, man poultry and other similar unisexual organisms*. In the above given examples in the male individuals, the pair of sex chromosomes is heteromorphic. i.e; one-X-chromosomes and one-Y-chromosomes (XY), while it is homomorphic (XX) in female. However, this situation is reversed in birds where the females have heteromorphic pair of sex chromosomes (*i.e;* ZW) and males have homomorphic pair (ZZ).

In males the single X-linked dominant or recessive gene expresses the trait phenotypically, while in females two-X-linked genes are necessary for the determination or expression of a single phenotypic trait or character related to sex. The recessive X-linked genes have characteristic *criss cross inheritance* (*i.e;* when male, for example, transmits his X-linked recessive genes to his grandson in F_2 generation through his daughter of F_1 generation.)

1. Sex-Linkage in Drosophila :

In *Drosophila melanogaster* over 100 sex-linked characters have been identified. They affect every part of the body *i.e.;* eye colour, body colour, wing size, wing venation, eye shape and distribution of abdominal markings, etc.

The inheritance of X-linked genes in *Drosophila* (Eye colour) takes place by following methods :

I. White eyed female × Red eyed male :

T.H. Morgan (1910) found the first X-linked gene in *Drosophila*. This was the recessive white eye mutant gene. When a white eyed female (*w w*; recessive) is crossed to a red eyed male (W, dominant), all the females are red eyed and all the males are white eyed in the F_1 generation. When these females and males of F_1 generation are intercrossed, the female population will consist of 50% red eyed and 50% white eyed in the F_2 generation. Similarly the males will be 50% red eyed and 50% white eyed in this generation (Fig.)

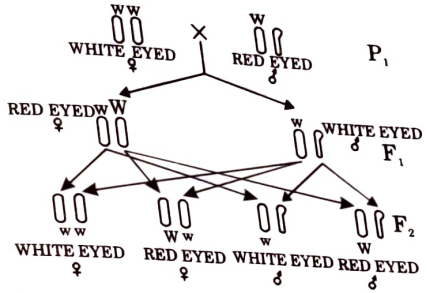


Fig. Cross between white eyed ♀ and red eyed ♂ in Drosophila.

II. Red-eyed female × White eyed male.

Likewise in other experiment, when a red eyed female (WW) is crossed with a white eyed male (w), the males and the females would be red eyed in the F_1 generation. When males and females of F_1 generation are inter crossed, the females would be exclusively red eyed, however, in the males 50% would be red eyed and 50% white eyed in the F_2 generation. (Fig.)

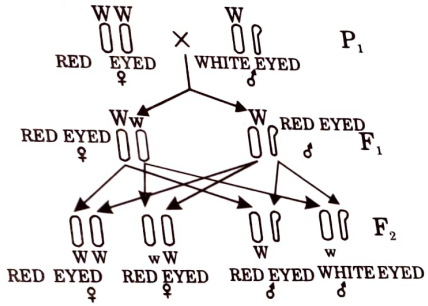


Fig. Cross between red eyed ♀ and white eyed ♂ in Drosophila.

Thus we see in the above reciprocal experiments that the character (located on the X-chromosome) is always transferred from the mother to the son and never from father to the son.

2. Sex-Linkage in Man :

In Man about 56 sex-linked genes have been identified. The most common X-linked genes of man are such as colour or red green blindness, night blindness, myopia (short sightedness), hemophilia, juvenile glaucoma (hardening of eyeball), anhidrotic ectoderma (non-functional sweat glands) and white forelock etc. These diseases are dependent upon the X-linked recessive genes. So sex-linked characters in man will also follow the same pattern as in *Drosophila*.

I. Colour Blindness in Man :

Some men are not able to distinguish red and green colours in the spectrum. They are said to be red and green colour blind. The colour blindness gene is located on the X-chromosome. The gene for colour blindness is recessive to its normal allele and denoted by 'c'. The inheritance colour blindness can be studied by following two types of marriages :

(i) Colour Blind Female × Normal Visioned Male :

When a colour blind female (X^cX^c) marries a normal visioned male (X^+Y), all males would be colour blind and females will be normal visioned in the F_1 generation. It is because males receive on X-linked recessive gene for colour blindness from colour blind mother. However, the male does not carry the gene for colour blindness, denotes '+', the normal allele of 'c' in his chromosome 'X'. The 'Y' chromosome is inert *i.e.*; having on allele for this gene. So his genotype is X + Y or X^+Y. The females, therefore, receive a X^+ -linked dominant gene for normal vision from father and on X^c-linked recessive gene for colour blindness from the mother. Such females are referred to as *carriers* for colourb lindness.

If the carrier females marry with colour blind men, they produce a colour blind homozygous female, a normal visioned heterozygous female, a normal visioned hemizygous male and a hemizygous colour blind male in the F_2 generation (Fig.)

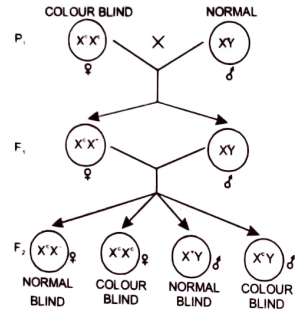


Fig. Cross between a red-green colour blind female and a normal visioned male in man.

(ii) Normal Visioned Female × Colour Blind Male :

Like-wise, in another experiment, when a normal visioned female (X^+X^+) marries with a colour blind male (X^cY), all the males would be normal with X^+Y and all the females would be carriers with genotype X^+X^c, in the F_1 generation. Thus, in the F_2 generation, two normal visioned females, one normal visioned male and one colour blind male would be borne (Fig.)

Thus the gene for colour blindness from a male passes on to his grand males (sons) through his females (daughters). This inheritance of human colour blindness is of **crisscross** type.

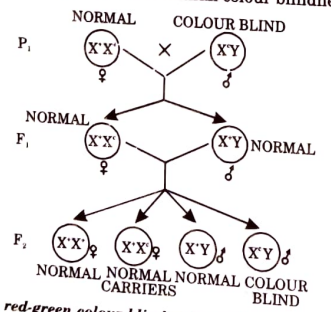


Fig. A cross between a red-green colour-blind male and a normal visioned female in man

II. Hemophilia in Man :

It is also a sex lined recessive trait. In this disease, the blood coagulation is inhibited. Therefore, in the event of an injury, the patient bleeds continuously till the last drop of the blood is drained out. **John Cotto (1803)** reported this hereditary disease first of all in man. This disease was most common in royal families of Russia and Queen Victoria of England. Queen Victoria was a carrier for this hemophilic trait and among her four sons Leopold was hemophilic.

III. Night Blindness in Man :

Some people suffer from night blindness disease. They feel difficulty in seeing in dim light. This trait is also a sex-linked recessive one.

B. Inheritance of Y-linked or Holandric genes :

Such Y-linked genes inherit directly from male to male only. In man, the Y -linked gene (*ichthyosis hystrix graris, hyportrichosis i.e,;* excessive development of hairs on pinna or ear, etc.) are inherited directly from father to son.

C. Inheritance of XY-linked Genes :

The genes located in homologous sections of 'X' and 'Y' chromosomes are termed as XY-linked genes. These genes inherit like the autosomal genes. *e.g.;* In *Drosophila melanogaster*, an XY-linked gene *i.e.*; bobbed gene for bobbed bristles located in the X-chromosome or its allele may be located in the Y-chromosome (recessive). This bobbed allele is originated due to mutation of nucleolar organizer region.

In human, several diseases like total colourblindness, a skin pigmentation and cancerous disease (**Xeroderma pigmentosum), Epidermolysis bullosa, Retinitis pigmentosa** (degeneration of retina due to deposition of pigment in the eye) and nephritis (a kidney disease) are XY-linked.

Sex Determination

The Chromosomal Mechanism of Sex determination :

Mclung (1901) reported in some insects that their males contained an odd number of chromosomes and after spermatogenesis these "accessory" chromosomes were present in half of the spermatozoa but were absent in the other half. He suggested that sex of these individuals was associated with the presence or absence of the "accessory chromosome".

Sex determination in Dioecious Organisms

The dioecious animals have distinct male and female individuals. The chief mode of sex determination in most ogranisms is by chromosomes. The principle of chromosomal mechanism of sex determination is much the same.

A. Male Heterogamety :

Males form two types of gametes. One gamete possesses 'X' chromosome and other lacks it. In some cases, male may possess a 'Y' chromosome. Due to this reason the males are called heterogametic. Females form only one type of gametes which contains 'X' chromosome.

In diploid organisms, there is a special chromosome (X) or chromosome pair (XX or XY). These are concerned with determination of sex. This duplet is known as "Sex chromosome" pairs, while rest of the chromosome pairs are alike in both the sexes and are called autosomes.

Types : Male heterogamety is of two types :

I. 'XX-XO' type :

1. Male contains autosomes and one 'X' chromosome. They are designated as XO.
2. Female contains autosomes and two 'X' chromosomes. They are designated as XX.
3. Production of sperms are of two types. One contains 'X' chromosome and the other lacking 'X' chromosome or possessing 'O'.
4. At the time of fertilization, if the sperm with a 'X' chromosome fuses with the egg (X), the result is female. Since the XX number is restored as one X has come from the sperm and other from the egg.

5. If the sperm without X chromosome happens to fertilize an egg (X), the result is male because in this zygote there is only one X chromosome which was present in the egg (Fig.)

Since the number of sperms with AX is equal to those with AO. The sex ratio is always expected to be 1:1. Examples are grasshoppers (lower animal) and in plants like *Dioscorea sp.*

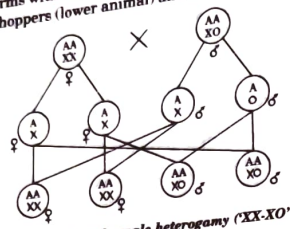


Fig. A cross showing the male heterogamy ('XX-XO' type)

II. 'XX-XY' type :

1. Male possesses AA and XY while female has AA and XX.
2. The male gametes produced are of two types. One containing a X and the other containing a Y chromosome.
3. When male gamete (X) fertilizes an egg (X). On the contrary, if the 'Y' containing male gamete fertilizes an egg (X), the result is a male (Fig).

Examples are human being and *Drosophila* and in plant slike coccinia *Hydrilla*, *Smilax*, *Cannabis*, *Elodea* and *Melandrium* etc.

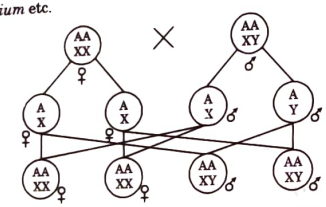


Fig. A cross showing the male heterogamy ('XX-XY' type).

B. Female Heterogamety :

Female produce two types of eggs. One egg contains an 'X' and other either lacking it (X) or containing 'Y' chromosome. Thus the male is AAXX and female AAXO or AAXY. For the sake of convenience and to avoid confusion the standard practice is that in these organisms 'X' chromosome is denoted to as 'Z' and 'Y' as 'W'. Thus, the female heterogamety is of two types :-

I. 'ZO-ZZ' type :

1. Female has 'ZO' and male has 'ZZ'.
2. Females produce two types of eggs, one possesses a 'Z' and the other lacking a 'Z'.

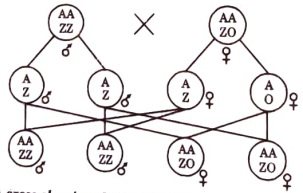


Fig. A cross showing the famale heterogamy ('ZO-ZZ' type).

3. If one female gamete possessing 'Z' is fertilized, a male (ZZ) is formed, while 'Z' lacking gametes are fertilized, the individual would be female (ZO) (Fig.)

The moth *Fumea* is an example.

Among Bryophytes, it has been found that the haploid female contains an X chromosome while the male contains at Y chromosome. The zygote and the sporophyte have the XY pair which on reduction division forms X and Y spores in equal numbers resulting in a equal number of male and female gametophytic plants.

II. 'ZW-ZZ' type :

1. Female has ZW type while male has ZZ type.
2. Two types of eggs are produced by females, one containing a Z and the other having a W chromosome.
3. The eggs (Z-chromosomes) on fertilization produce males (ZZ) and W-containing eggs produce females (ZW) (Fig.)

This is 'Avian-moth type' being widely found among birds and futterflies. Some fishes and a few plants like *Fragaria* or strawberry show this type determination of sex.

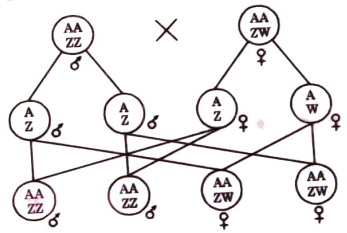


Fig. A cross showing the female heterogamy ('ZW-ZZ' type).

Modified Chromosomal methods

In *Drosophila*, Bridges observed certain female flies which has 3 sets of outosomes and 3 'X' chromosomes (triploid). The cross between triploid females and diploid males gave rise to a progeny which contained very interesting types of individuals.

The triploid female produces four kinds of eggs:

(*a*) Egg-containing 2 autosome sets and 2 'X's (AAXX).

(*b*) Egg-containing one autosome set and one 'X' (AX).

(*c*) Egg-containing two autosome set and one 'X' (AAX). and

(*d*) Egg-having one autosome set and two 'Xs' (AXX).

The diploid male on the other hand produces only two types of sperms :

(*a*) One containing 'AX', and (b) the other having 'AY'.

The fertilization of the four kinds of eggs with two kinds of sperms would in the production of the following eight kinds of zygotes. These zygotes develop into distinct types of flies (Fig.)

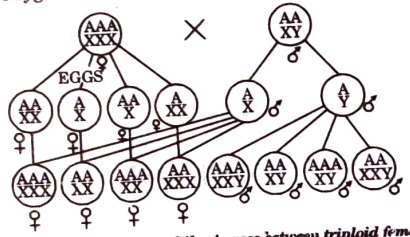


Fig. Modified chromosomal methods in Drosophila. A cross between triploid females and diploid males.

Results : These are as follows :

1. Individuals are triploid females *i.e.*; having 3 As and 3Xs (AAA XXX).
2. Flies are normal diploid (AAXX).
3. Flies are neither males nor females but intersexes (3As and 2 Xs) *i.e;* AAAXX
4. Individuals are super females with (2As and 3 X).
5. Flies are also inter sexes with 3A and 2X.
6. Individuals are normal males with 2A, 1X and 1Y.
7. Flies are also superfemales with 3A, IX and I Y.
8. Some flies are normal diploid females with 2A, 2X and 1 Y.

Due to the above observations. Bridges pointed out that in *Drosophila* the femaleness is not simply determined by the presence of two 'X' and maleness by the presence of one 'X' and one 'Y' because cross gave rise to certain flies which had two 'X', but these were not females. So these were flies which had a Y-chromosome, were still females. He, therefore, concluded that the sex in this insect is determined by the interaction of genes on he autosome with those on the X-chromosomes. According to him, the autosomes carry genes for maleness and 'Xs' for femaleness.

Quantitative Inheritance

Quantitative traits

Most classic genetic traits are discontinuous, different genotypes produce quite different phenotypes with no overlap (e.g. brown or blue eyes).

Quantitative traits such as height vary on a continuous distribution. All values of height within a range are possible. They are controlled by the cumulative effects of many genetic loci, and also by the effects of differences in the environments experienced by individuals. Quantitative traits are also called multifactorial, polygenic or multilocus.

There is also a newly recognized category which varies discontinuously, but is controlled quantitatively (continuously). A combination of genetic and environmental factors cause some individuals to cross a threshold from one phenotypic state to another. Examples are diabetes and cancer where genotype affects risk, but a combination of genotype and environment push some individuals over the threshold from health to disease.

Measuring continuous variation

Continuously variable characters are described by their mean and variance, usually assuming a normal distribution (x_i = individual height measurements of n individuals).

$$\text{mean} = \bar{x} = \frac{\sum(x_i)}{n}$$

$$\text{variance } s^2 = V = \frac{\sum(x_i - \bar{x})^2}{(n-1)}$$

Components of variation

The total phenotypic variance is equal to the sum of the variance due to all causes, namely the environment (E), gene–environment interactions (GE), additive genetic effect (GA), dominant genetic effects (GD), and epistatic interactions (GI) between different genes (loci) respectively as in the formula:

$$V_{total} = V_{Ph} = V_E + V_{GE} + V_{GA} + V_{GD} + V_{GI}$$

This is often simplified by putting all the genetic components together :

$$V_{ph} = V_E + V_G$$

Broad sense heritability

Broad sense heritability (H_B^2, also written H_B, H^2 or h_B^2) is the proportion of total phenotypic variance in a population sample that is due to genetic effects.

$$H_B^2 = \frac{V_G}{V_{ph}} = \frac{V_G}{V_E + V_G}$$

Narrow sense heritability

Narrow sense heritability (H_N^2, also h^2 or H_N) is predictive. It is the component of variation caused by alleles or genes with additive effects. It is passed from parent to offspring, each parent contributing 0.5 of the offspring's character:

$$H_N^2 = \frac{V_A}{V_{PL}}$$

Measuring heritability

(*i*) Controlled breeding. Environmental effects can be measured by comparing inbred (homozygous) lines in crop plants or monozygotic twins in humans. The extra variance between unrelated individuals in the same environment must be the broad sense heritability. Controlled crosses can be used to produce genetically uniform F_1 and variable F_2 and backcross generations. By making the correct comparisons, the environmental and additive, dominant and epistatic genetic contributions to the range of phenotypes can be calculated. The F_2 generation contains all possible genotypes, and can be used to calculate broad sense heritability.

$$H_B^2 = \frac{(V_{phF2} - V_E)}{V_{phF2}}$$

Similar calculations can be performed in humans by suitable comparisons between people with different degrees of relatedness, and adoptive relatives.

(*ii*) Realized heritability. Realized heritability is the response of a population to selection, and can be calculated after a selective breeding program. It is the ratio of change in character to selection differential:

$$H = \frac{M_o - M_T}{M_P - M_T}$$

where M_T is original mean, M_o is the mean in the next (offspring) generation after selection, and M_p is the mean of the individuals selected to be parents.

(*iii*) Offspring-parent regression. A graph of offspring values (y) against the average (mid-point value) of their two parents (x) for a particular trait (characteristic) will have a slope (b) equal to the narrow sense heritability of that trait. Put simply, the correlation between a characteristic in parents and offspring reveals how much the offspring inherit that characteristic from their parents. In humans they also share their family environment, which confuses the measurement. The formula for a straight line graph is:

$y = a + bx$ where a is a constant and b = slope

so, $H_N^2 = b$

Limitations of heritability

Heritability changes as conditions change because it is the ratio of genetic to total variance. High environmental variance causes lower heritability than low environmental variance, although genetic effects stay constant. Heritability only means something in the population and conditions where it is measured because numerical values cannot be compared meaningfully between different populations or conditions.

Quantitative trait loci

Quantitative trait loci are locatable genetic markers (typically variable number tandem repeats) which are closely linked (physically nearby in the DNA) to the genes affecting interesting character traits. The correlation between the markers and the trait is used to find the genetic locations of the genes controlling the trait.

Human studies

Monozygotic twins are genetically identical, so all their differences should be due to their different environments. Dizygotic twins are only 50% genetically identical, and can be used as a comparison. Twins adopted separately or together at birth indicate the environmental difference between families. All the components of genetic and environmental variation in a trait can be calculated by examining the correlations between monozygotic (identical) and dizygotic twins and sibs to each other and to natural and adoptive parents and sibs. Concordant events affect both twins of a pair (discordant events affect only one twin) and can be used to examine heritability of qualitative traits.

Human quantitative traits

Many interesting human traits are quantitative, the most intensely studied being intelligence and personality. Genetic loci that contribute to variation in these traits have been identified. The advantage of knowing the comparative effects of the environment and genetics (nature and nurture) is that resources can be directed where they are most beneficial, however there is a risk of the information being distorted and employed politically for racial or class-biased motives.

Extranuclear Inheritance and Maternal Effect

Mitochondrial and chloroplast genomes

Eukaryote cells contain more than one genome. Most genetic studies concentrate on the nuclear genome, but the mitochondria and chloroplasts carried in the cytoplasm also contain their own DNA. It is widely accepted that these organelles represent the ancestors of prokaryote organisms that were 'captured' by primitive eukaryotes and have since lived in a symbiotic relationship. The organelle genomes differ greatly from the nuclear genome in that they resemble their prokaryote ancestors. The mitochondrial and chloroplast genomes of several species have been characterized and completely sequenced. In most plant species chloroplast genomes are circular, and range from 50 to 200 kb in size. Many copies of the genome can be found in any one chloroplast. Mitochondrial genomes are also circular but vary greatly in size between different species. Human mitochondria contain 16.6 kb of DNA, whereas the yeast mitochondrial genome is approximately five times larger. Plants have mitochondrial genomes of about 1 Mb.

In animals, mitochondrial DNA mutates at a higher rate than nuclear DNA. Despite the high mutation rate almost all mitochondria in any one individual are identical (**homoplasmy**), because only small numbers are transferred to the next generation. This allows mitochondrial DNA to be used to monitor divergence in animal populations. It is worth noting that plant mitochondria behave very differently and evolve slowly. Both mitochondria and chloroplasts carry their own ribosomal RNA and transfer RNA genes, and their ribosomes are similar in size to prokaryotic ribosomes. Some of the proteins found in these organelles are synthesized on their own ribosomes from mRNA transcribed from the organelle genome, but most proteins are coded by nuclear genes and are imported from the cytoplasm. This dependence on host genes gives an indication of the large amount of co-evolution that has taken place between the organelle and the eukaryote cell.

Maternal inheritance

Although there are some exceptions, notably gymnosperms, conifers, most plant pollen does not transfer chloroplasts or mitochondria to the zygote. In animals, sperm do not contribute mitochondria. For this reason any phenotypes coded by organelle genomes show a pattern of inheritance in which a phenotype can only be inherited from the mother; this is known as **maternal inheritance**. There are numerous examples of this.

Variegated plants, showing white patches or stripes on the leaves, are often due to the presence of a proportion of chloroplasts that have a mutation which results in loss of the ability to form chlorophyll. At each cell division the chloroplasts segregate randomly between the daughter cells, and occasionally a daughter cell will be formed that contains only mutant chloroplasts. All of its descendants will lack chlorophyll, producing a white stripe in grasses or a white patch in broad-leaved plants. The frequency of such an event will depend on the number of chloroplasts per cell and the ratio of mutant to normal chloroplasts.

Cytoplasmic inheritance is also observed in fungi. Mutations in the mitochondrial DNA of two species have been associated with reduced growth rates. In the Baker's yeast *S. cerevisiae* these are referred to as 'petite' mutants. Two forms of mitochondrial petite mutants are known. Neutral petites are recessive and when normal yeast are crossed with petite yeast the resulting ascospores have normal growth rates. This is because each cell contains many mitochondria, and although the hybrid contains a mixture of mutant and normal mitochondria, it is highly improbable that at any cell division the distribution of mitochondria between daughter cells would result in a cell containing only mutant mitochondria. Other petite mutants, suppressor petites, are not recessive and when crossed with normal yeast produce normal and petite progeny in non-Mendelian ratios. Petite mutants are incapable of supporting aerobic respiration. The precise defect differs between mutants, but frequently this involves major deletions of mitochondrial DNA. Suppressor petites appear to be able to induce changes in the DNA of normal mitochondria in the same yeast cell.

Mutations in mitochondrial DNA can lead to antibiotic resistance in animal cells grown in culture, and certain inherited human disorders show a pattern of maternal inheritance and can be traced to mutations in mitochondrial DNA. An example of one such disease is Leber's hereditary optic atrophy. Here, as in other examples, nuclear genes interact with the expression of defective mitochondrial genotypes; there is growing evidence that an accumulation of damage to mitochondrial DNA may play a part in the aging process.

Cytoplasmic male sterility (CMS)

CMS, first observed by M. Rhoades, is an example of cytoplasmic inheritance in plants. In this condition the plant does not produce active pollen but the female reproductive process and fertility are normal. This type of sterility is not controlled by nuclear genes but is transmitted through the egg cytoplasm from generation to generation. Cytoplasmic male sterility in maize results from the expression of an abnormal protein in the mitochondria of the anthers. Although the mutation is present in all the mitochondria of the plant, only the anthers show a phenotype. It is not yet understood how the abnormal proteins disrupt mitochondrial function. However, these result suggest that efficient mitochondria function is particularly important during pollen development.

Endosymbiont

In eukaryotes, a variety of cytoplasmically transmitted traits result from the presence of bacteria living in the cytoplasm of certain cells. One of the best known is the killer phenomenon found in certain strains of the protozoan *Paramecium aurelia*. Killer strains of paramecium release toxin to the surrounding medium called *paramicin* that is lethal to many other strains of protzozoan called sensitive strain. The killer phenotype requires the presence of cytoplasmic bacteria referred to as **kappa particles**. The kappa particles are the cells of *Caedobacter taeniospiralis* which produce the killer substance. Kappa is not the only infective agent known in *Paramecium*. Another agent is the **mu particles**. Killer cells containing mu particles, termed mate killers, do not release a toxin into the environment, but instead kill their mates during conjugation.

Maternal effects

Some confusion can occur between the maternal inheritance of genes encoded in organelle genomes and situations where the phenotype of progeny is affected by its mother, to the extent where its own genotype is masked. This can occur to a greater or lesser degree in many organisms because of maternal contribution of protein or RNA to the embryo. An extreme example of this is displayed by the pattern of shell coiling in the water snail *Limnae*. In this species a single gene with two alleles determines whether the snail shell follows a right-handed (dextral) or left-handed (sinistral) coil. There are two alleles, D for dextral, and *d* for sinistral coiling, dextral being dominant over sinistral. The water snail is a hermaphrodite and can either self-fertilize or mate with other snails. When a *D/D* dextral snail is mated with a *d/d* sinistral snail the progeny all have the same shell coiling pattern as the mother. If these F1 individuals are allowed to self-fertilize, the progeny are all dextral. A further round of self-fertilization produces an F3 generation with a ratio of 3 dextral: 1 sinistral. Note that the predicted Mendelian ratios appear one generation later than would be expected in the case of conventional Mendelian genetics. Although this has some similarities with cytoplasmic maternal inheritance it is clear that the phenotype does not consistently follow that of the female parent.

Why is the true ratio delayed for a generation? The development of the snail is predetermined by the mother. Its influence directs the manner in which the shell coils. Hence the phenotype of each snail reflects the genotype of its mother (or single parent in a self-fertilization) rather than its own genotype. This is the best known example of maternal effect, but other examples are found in insects where the phenotype of the young of the next generation resembles the mother, but this is lost with age. In the case of the snail the coiling of the shell is not reversible.

Cytogenetics

A chromosome is an organized structure of DNA and protein that is found in the nucleus of a cell. The study of the structure, function and abnormalities of chromosome is called *cytogenetics*, a discipline that combines cytology with genetics.

Human karyotype

The number, sizes, and shapes of the metaphase chromosomes constitute the **karyotype** or karyogram which is distinctive for each species. Karyotype consisting of photograph or diagram of all the metaphasic chromosomes arranged in homologous pairs according to decreasing length, thickness, position of centromere, shapes etc is described as **idiogram.**

Table : Symbol used in describing a karyotype

Symbol	Meaning
p	Short arm
q	Long arm
13p	Short arm of chromosome 13
13q	Long arm of chromosome 13
del	Deletion
del(2)	Deletion in chromosome 2
dup	Duplication
dup(1)	Duplication in chromosome 1
inv	Inversion
inv(4)	Inversion in chromosome 4
t	Translocation
t(2;5)	Reciprocal translocation between a chromosome 2 and a chromosome 5
ter	Telomere
cen	Centromere
+ or -	Indicate gain or loss of part of chromosome
2q-	Deletion of the long arm of chromosome 2

Tjio and *Levan* (1956) of Sweden found that human cells have 23 pairs or 46 chromosomes. Of the 23 pairs of human chromosome, 22 are perfectly matched in both males and females and are called autosomes. The remaining pair, the sex chromosomes, consists of two similar chromosomes in females and two dissimilar chromosomes in males. In human, females are designated XX and males XY.

Denver system

According to 'Denver system' of classification, the 22 pairs of human chromosomes are placed in seven groups as

	Group	Position of centromere	Idiogram number
I	(A)	Metacentric or submetacentric	1,2,3
II	(B)	Submetacentric	4,5
III	(C)	Submetacentric	6,7,8,9,10,11,12 and X
IV	(D)	Acrocentric	13, 14, and 15
V	(E)	Metacentric or submetacentric	16, 17 and 18
VI	(F)	Metacentric	19 and 20
VII	(G)	Metacentric	21, 22 and Y

Chromosome banding

Chromosome banding is a cytological procedure of differential staining of mitotic chromosome along the longitudinal axis. The differential staining reactions reflect the heterogeneity and complexity of the chromosome along its length. The molecular mechanisms involved in producing the various banding patterns are not precisely defined.

Features of commonly used banding technique are described in the table.

Table : Chromosome banding techniques

Technique	Procedure	Banding Pattern
G – banding	Mild proteolysis followed by staining with Giemsa	Dark bands are AT-rich Pale bands are GC-rich
R – banding	Heat denature followed by staining with Giemsa	Dark bands are GC-rich Pale bands are AT-rich
Q – banding	Stain with quinacrine	Dark bands are AT-rich Pale bands are GC-rich
C – banding	Denature with barium hydroxide and then stain with Giemsa	Dark bands contain constitutive heterochromatin

The bands and the regions to which they belong are identified by numbers, with the centromere serving as the point of reference for the numbering scheme. In designating a particular band, four items are required : the chromosome number, the arm symbol, the region number and the band number within that region.

For example,

13q14 Region 1, band 4 of the long arm of chromosome 13

13q14.2 Subband 2 of 13q14

Chromosomal Aberrations (Ploidy)

Each species has a tixed chromosomal complement called its **karyotype**. The fixed number of chromosomes may, however, be disturbed due to irregularities in the processes of gamete formation (gametogenesis) and fertilization. Any change in the number of chromosomes is called a **chromosomal anomaly** or **aberration**. These changes can be classified into two general categories as follows:

(1) **Aneuploidy** (*aneu* = **uneven**): Sometimes, the homologous chromosomes of one or more pairs fail to separate from each other in the anaphase of the reduction division (first maturation division) during gametogenesis. This is called **nondisjunction,** of homologous chromosomes. It results into the formation of gametes possessing chromosomes in greater or lesser number than the normal haploid number of the genome. When such gametes participate in fertilization, the resultant offsprings contain aberrant chromosomal complements. Since each chromosome carries genes of a number of hereditary traits, offsprings with aberrant chromosomal complements exhibit important phenotypic variations from normal individuals in respect of several characteristics. Hence, such phenotypes are called, **syndromes.**

(2) **Euploidy** (*eu* = **even**): The change in chromosomal number in euploidy always amounts to multiples of complete haploid sets, such as triploid (3X), tetraploid (4X), pentaploid (5X), and so on. That is why, it is also called **polyploidy**. It is hardly found in animals, but often found in plants. It results either from failure of reduction division during gametogenesis, or from incomplete mitosis during second maturation division of gemetogenesis, or due to fertilization of an ovum by two or more sperms.

Some Important Human Syndromes

(A) Autosomal Syndrome

Down's syndrome : One in 'about 700 children, on an average, is a Down's syndrome. Twenty-first pair of autosomes in this syndrome has an additional chromosome. *i.e.* there are three homologous chromosomes (**trisomic condition**), instead of two at this number, and the karyotype is indicated, not by 2X, but by 2X +1 (chromosome number 21) = 47. Thus, a Down's syndrome illustrates **autosomal aneuploidy**. The phenotype of this syndrome is a short-statured and feeble- minded child who does not usually survive the first year after birth, but may have a life expectancy of 20 years with proper medical care. Its head is small and spherical, skin warty, tongue massive, furrowed and protruding, digits stubby, nose flattened, mouth always opened, eyes. slanting, hand and fingers short and broad, and eyelids folded as in the Mongols. This syndrome is, therefore, also called **'Mongoloid Idiocy'**. The genitalia are normal. yet the male syndromes are sterile.

Contrary to Down's sindromes, children who are **monosomic** for this number of chromosomes, *i.e.* who inherit a single chromosome at twentyfirst position, exhibit characteristics opposite to those of Down's syndromes.

Similarly, trisomics for autosomes of 18th and 13th numbers and monosomics for autosomes number five are also on record.

(B) Sex Syndromes And Other Genetic Sex Abnormalities

Individuals displaying **aberrant sexual differentiation** due to genetic abnormalities have been observed in certain unisexual plant and animal species, including man. Genetic sex abnormalities result from nondisjunction of sex chromosomes either during gametogenesis or during zygotic cleavages in embryonic development. Gametogenetic nondisjunction of sex chromosomes results into individuals which are monosomic or trisomic for sex chromosomes. Such persons are **sex syndromes**. Important sex syndromes are—

1. **Turner's syndromes** : These syndromes inherit a single X chromosome instead of two sex chromosomes, *i.e.*, these are monosomics for sex chromosomes (2X-1 or 2A+XO). On an average, one in about 5,000 persons is a Turner's syndrome. Its body is short-statured, chest is flat and broad, neck is short and webbed, gonads absent or rudimentary, and accessory genital organs and external genitalia are also poorly developed. Hence, these syndromes are sterile females (eunuchs). These result either from fertilization of an ovum by a sperm carrying no sex chromosome due to nondisjunction, or from fertilization of an ovum carrying no sex chromosome by an X-carrying sperm.
2. **Klinefelter's syndromes :** These syndromes are trisomics for sex chromosomes (2X+1) = 47. Commonly, they inherit 2A+XXY chromosomes. Obviously, they are genetically males due to presence of a Y chromosome. Their bodies are quite like normal males but, due to an extra dose of X-born genes, the testes remain underdeveloped and fail to produce sperms. Hence, these syndromes are also sterile. Often breasts develop (**gynaecomastia**) and there may be some mental retardation. On an average, one in 500 men is a Klinefelter's syndrome.

   Klinefelter's syndromes, inherting XXXY, XXXXY, XXYY and XXXYY sex chromosomes are also on record. Similarly, persons inheriting XYY sex chromosomes (supermales),are also known. These are tall, less intelligent, very aggressive, prone to Violence and criminality, and suffer from persistent erection. Superfemales with three, four, or five X chromosomes are also known. Some of these may be normal and fertile, but usually these are infertile and mentally retarded.
3. **Gyanandromorphs** : Zygotic nondisjunction of sex chromosomes in unisexuals results into **true hermaphrodite** individuals or **sex mosaics**. Rarely, a sex mosaic may have **harmaphrodite** gonads, called **ovotestes**, with male germinal cells in testicular and female germinal cells in ovarian parts of the gonads. Usually, sex mosaics are **gynandromorphs**, having separate male and female parts or tissues in the same body. These are commonly seen in fruit flies, bees and months. If nondisjunction of sex chromosomes occurs, in the very first cleavage of xygote, a bilateral gynandromorph, with one half of body being male and the other half being female, results. If non-disjunction occurs in later cleavages, the resulting gynandromorph possesses irregularly distributed male and female tissues.

POPULATION GENETICS

Hardy-Weinberg Principle

Populations genetics is the study of alleles of genes in populations (often called demes), and the forces which maintain or change the frequencies of particular alleles and genotypes in populations. The Hardy-Weinberg equilibrium is a means of calculating expected genotype frequencies from the allele frequencies determined in the same population (and vice versa) assuming random matting, equal reproductive success, no mutations and no effects of selection or migration affecting particular genotypes. If a population does not fit Hardy-Weinberg predictions then that is evidence of some real effect (e.g. natural selection) operating to disturb it.

The genetic equilibrium is defined as: 'The relative frequencies of various kinds of genes in a large and randomly mating sexual panmictic population tend to remain constant from generation to generation in the absence of mutation, selection and gene flow.'

Salient features of Hardy-wein berg's principle :

(1) Random mating.

(2) Large population size

(3) Biparental mode of reproduction

(4) Homogenous age structure

(5) Absence of evolutionary forces-mutation migration, selection, genetic drift.

Significance of Hardy-Weinberg's Principle : The Hardy-Weinberg's law is important primarily because it describes the situation in which there is genetic equilibrium and no evolution. Thus :

1. It provides a theoretical baseline for measuring evolutionary change.
2. The equilibrium tends to conserve gains which have been made in the past and also to avoid rapid changes.
3. Equilibrium maintains heterozygosity in the populations.
4. Equilibrium prevent's evolutionary progress.

Measuring Allele Frequency : Allele frequency is found by adding up the number of copies of each allele in a population and expressing it as a frequency. A population of N diploid individuals has 2N alleles. Each As heterozygote has one A allele, each AA homozygote has two A alleles, so:

$$\frac{(n_{Aa} + 2n_{AA})}{2N}$$

Frequency of allele A is where n is the number of individuals with the respective genotype. The frequency of A is usually, called p, the frequency of other (non A) alleles is denoted by q. as a check, p + 1 = 1, but this only checks your arithmetic, not whether p and q are correct.

Equilibrium Genotype Frequencies: Equilibrium genotypes frequencies will be reached in zygotes after one round of random mating. The expected genotype frequencies are predicted by the Hardy-Weinberg equation :

$$p^2 + 2pq + q^2 = 1$$

If there are two alleles, A and a, with frequencies p and q respectively, then the expected frequency of AA homozygotes is p^2, of Aa heterozygotes is 2pq, and of aa homozygotes is q^2. As a check of your arithmetic $p^2 + 2pq + q^2 = 1$. When the frequency q of an allele is low, the frequency of occurrence of homozygotes, q^2, is very low (e.g. recessive genetic diseases). Uncommon alleles (small value of q) are usually present in heterozygotes, so cannot be identified if they are recessive and are not expressed, so selection cannot act against them.

Distorting : A range of natural phenomena can distort genotype frequencies from those predicted by the Hardy-Weinberg equation.

(*i*) Selection reduces the fertility or survival of certain genotypes.

(*ii*) Migration replaces local individuals with immigrants from a populations where the allele frequencies, and hence expected genotype frequencies, are different, If some genotypes are more likely to migrate than others this will also change their frequency locally.

(*iii*) Assortative mating of either similar or dissimilar genotypes produces an excess of homozygotes or heterozygotes respectively.

(*iv*) Sub population exits; these are locally mating groups in larger, possibly continuous populations. This may be common in humans due to ethnic or class groupings, and also in organisms with low mobility and always increases the frequency of homozygotes. In an extreme case this becomes inbreeding.

(*v*) Mutations will produce new alleles, but the rate is too low to be noticeable practically, and selection will exactly counteract this if the mutants are deleterious.

(*vi*) Drift (chance) will also cause small deviations from the predicted frequencies, especially, in small samples, but these deviations are expected and allowed for in statistical tests.

Testing the fit: A chi-square (X^2) test is used to compare observed genotype frequencies to expected frequencies estimated assuming Hardy-Weinberg equilibrium:

$$x^2 \text{ sum} \left[\frac{(\text{observed} - \text{expected})^2}{\text{expected}} \right]$$

For n allele, there are $(n^2 - n)/2$ degrees of freedom because once the frequency (p) of one allele (of two alleles is known, the frequency of the other is fixed (p + q = 1) and so are the expected genotype frequencies. Small values of X^2 indicate a good fit to expectation; large values of X^2 indicate a large deviation.

Estimating Allele Frequency : If heterozygotes cannot be identified, the frequency of recessive alleles can still be estimated from the frequency of homozygotes by assuming Hardy-Weinberg equilibrium.

The frequency of homozygous recessives is q^2.

So The frequency of the recessive allele is q, and

The frequency of the dominant allele is p = (1 – q)

There is a large error associated with this method. It is difficult to measure q^2 accurately because:

(*i*) it is often very small so a very large population must be sampled; and

(*ii*) Hardy-Weinberg equilibrium may not apply, usually because of populations structure and local inbreeding subpopulations.

Population and Evolution

Evolution can be defined as changes in gene frequencies in the gene pool of populations and a population is considered to be the primary unit in evolution. During evolution, it is the gene pool of the populations that changes or evolves and not the individual. Diversity in the population is maintained by mutation, recombination and reassortment of genes or chromosomes. Each population or its gene pool contains a certain collection of genotypes. In a population, when the adaptive peak is moving back and forth over the seasons, it indicates that individuals with one genotype have a higher fitness at one time, but in altered seasonal variations, the greater fitness shifts to another genotype.

Populations has reached an adaptive peak, further evolution will depend upon the origin of new selective environment which will lead to the formation of new adaptive peaks.

Genes in Populations

Introduction: The total genetics stock of populations is its gene pool. Hardy-Weinberg equilibrium is a means of calculating expected genotype frequencies from allele frequencies determined in the same population (and vice versa) assuming random mating. Equal reproductive success, no mutations and no effects of selection or migration affecting particular genotypes.

$$\frac{(n_{Aa} + 2n_{AA})}{2N}$$

Frequency of allele is A :

Where n is the number of individuals with the respective genotype.

Uncommon alleles (small value of q) are usually present in heterozygotes, so cannot be identified if they are recessive and are not expressed, so selection cannot act againt them.

Distorting effects :

i. Selection reduces the fertility or survival of certain genotypes.

ii. Migration replaces local individuals with immigrants from a population where the allele frequencies, and hence expected genotype frequencies, are different.

iii. Assortative mating of either similar or dissimilar genotypes produces an excess of homozygotes or heterorozygotes respectively.

iv. Subpopulations exist. These are locally mating groups in a large, possibly continuous population. This may be common in humans due to ethnic or class groupings, and also in organisms with low mobility, and always increases the frequency of homozygotes.

v. Mutations will produce new alleles.

vi. Drift (chance) will also cause small deviations from the predicted frequencies.

There are $(n^2 - n)/2$ degrees of freedom because once the frequency (p) of one allele (of two allete) is known, the frequency of the other is fixed (p + q = 1) and so are the expected gonotype frequencies. Small values of X^2 indicate a good fit to expectation, large values of indicate X^2 a large deviation.

It is often very small so Hardy-Weinberg equilibrium may not apply, usually because of population structure and local inbreeding subpopulations.

POPULATIONS PROBLEM

Hardy-Weinberg states:

P = the frequency of the dominant allele (so the % of A are in the population)

Q = the frequency of the recessive allele (so the % of a in a population)

So p + q = 1

Using the equations $p^2 + 2pq + q^2 = 1$

P^2 = frequency of *AA* (homozygous.dominant) 2pq = frequency of *Aa* (deterozygous)

q^2 = frequency of *aa* (homozygous recessive) So, a population genetics problem would look like this.

In a certain population of 1000 fruit flies, 640 have red eyes while the remainders have sepia eyes. The sepia eye trait is recessive to red eyes. How many individuals would you expect to be homozygous for red eye color?

Answer :

1. q^2 for this population is 360/1000 = 0.36
2. $q = \sqrt{0.36} = 0.6$
3. p = 1–q = 1 – 0.6 = 0.4
4. The homozygous dominant frequency = $p^2 = (0.4)(0.4) = 0.16$.
5. Therefore, we can expect 16% of 1000, or 160 individuals to be homozygous dominant

EXERCISE – I

MCQ TYPE QUESTIONS

1. The term 'allelomorphic' implies.
(*a*) Any two characters.
(*b*) A pair of contrasting characters.
(*c*) Sex linked characters.
(*d*) A pair of non-contrasting characters.

2. When a true breeding pea plant that has yellow seeds is pollinated by a plant with green seeds all the F1 plants have yellow seeds. This means that the allele for yellow is:
(*a*) Heterozygous (*b*) Dominant
(*c*) Recessive (*d*) Lethal

3. Which if the following is heterozygous for two pairs of alleles
(*a*) TTRR (*b*) TrRR
(*c*) ttrr (*d*) TrtR

4. The cross used to ascertain whether the plant is homozygous or heterozygous is
(*a*) Monohybrid Cross (*b*) Dihybrid Cross
(*c*) Test cross (*d*) Back cross

5. Mendel crossed a pure white -flowered recessive pea plant with a dominant pure red - flower plant. The first generation on hybrids from the cross should show
(*a*) 50% white -flower and 50%red flower plants
(*b*) All red -flower plants
(*c*) 75% red flowered and 25% white flowered plants
(*d*) All white flowered plants

6. If a cross is made between AA and aa, the nature of F1 progeny will be
(*a*) Genotypically AA, phenotypically a
(*b*) Genotypically Aa phenotypically a
(*c*) Genotypically Aa, phenotypicaly A
(*d*) Genotypically aa, phenotypicaly A

7. A cross in which an organism showing a dominant phenotype is crossed with recessive parent in order to know its genotype is called
(*a*) Monohybrid cross (*b*) Back cross
(*c*) Test cross (*d*) Dihybrid cross

8. The F2 dihybrid ratio 9:3:4 is explained on the basis of
(*a*) Epistatic gene
(*b*) Supplementary gene
(*c*) Allelic interaction
(*d*) Complementary gene interaction

9. A human male produces sperms with the genotypes AB, Ab, aB and ab pertaining to two diallelic characters in equal proportions. What is the corresponding genotype of the person?
(*a*) AaBb (*b*) AaBB
(*c*) AABb (*d*) AABB

10. In *Mirabilis* a hybrid for red(RR) and white(rr) flower produces pink(Rr) flower. A plant with pink flower is crossed with white flower. The expected phenotypic ratio is
(a) Red:Pink:White(1:2:1)
(b) Pink:White(1:1)
(c) Red:Pink(1:1)
(d) Red:White(3:1)

11. In shorthorn cattle genes for red(r1) and white(r2) coat colour occur. Crosses between red(r1r2) and white(r2r2) produced (r1r2) roan. This is an example of
(a) Complementary genes
(b) Epistasis
(c) Co-dominance
(d) Incomplete dominance

12. The gene interaction when one gene masks th effect
(a) Complementary gene action
(b) Supplementary gene action
(c) Duplicate gene action
(d) Epistasis

13. After crossing two plants, the progenies are found to be male sterile. The phenomenon is found to be maternally inherited and is due to some genes which reside in
(a) Nucleus (b) Chloroplast
(c) Mitochondria (d) Cytoplasm

14. Which of the following is associated with multiple phenotypes?
(a) Epistasis
(b) Pleiotropy
(c) Polygenic inheritance
(d) Mutation

15. The evidence that crossing over occurs at four stranded stage and not at two stranded stage of the chromosomes, comes from
(a) 2:2:2:2 arrange of ascospores in *Neurospora*
(b) 4:4 arrangement of ascospores in *Neurospora*
(c) Studies of meiosis in maize
(d) Studies on linkage maps of chromosomes in *Drosophila*

16. Coupling and repulsion are the two faces of
(a) Crossing over
(b) Linakge
(c) Chiasmata
(d) Mutation

17. The map distance between genes A and B is 3 units, between B and C 10 units and between C and A 7 units. The order of the genes in a linkage map constructed on the above data would perhaps be
(a) A,B,C (b) A,C,B
(c) B,C,A (d) B,A,C

18. Which one of the following pairs is correctly matched

(a) Morgan	Discovered the process of linkage
(b) Linus Pauling	Isolated DNA for the first time
(c) Francis Crick	Discovered the phenomenon of tranformation
(d) H.Khorana	Discovered about triplet sequence

19. Two genes R and Y are located very close on the chromosomal linkage map of maize plant. When RRYY and rryy genotypes are hybridized the F_2 segregation will show
(a) Higher number of the recombinant types
(b) segregation in the expected 9:3:3:1 ratio
(c) segregation in 3:1 ratio
(d) Higher number of parental types

20. Linkage decreases the frequency of
(a) Hybrid (b) Dominant allele
(c) Recessive allele (d) Both (a) and (b)

21. Crossing over in diploid organism is responsible for
(a) Dominance of genes
(b) Linkage between genes
(c) Segregation of alleles
(d) Recombination of linked allele

22. What will be the number of linkage groups in maize if it has 10 pairs of chromosomes
(a) 5 (b) 10
(c) zero (d) 20

23. Chiasma shows the sites of
(a) Spindle formation (b) Synapsis
(c) Crossing over (d) None of these

24. ABO blood grouping is controlled by gene I which has three alleles and show co-dominance. There are six genotypes. How many phenotypes in all are possible?
(a) Six
(b) Three
(c) Four
(d) Five

25. Inheritance of ABO blood group system is an example of
(a) Multiple allelism
(b) Partial dominance
(c) Epistasis
(d) Dominance

26. Antisera used to detect Rh blood group
(a) Anti A (b) Anti B
(c) Anti C (d) Anti D

27. Mating among close relations is referred
(a) Permanent marriage
(b) Line breeding
(c) Inbreeding
(d) Cross breeding

28. For a child having blood group B, if father has blood group A, what may be the blood group of mother
(a) O or A (b) O
(c) B or AB (d) A

29. Parents of blood group O and AB cannot have a child of group AB because
(a) Gene O is dominant over gene A
(b) Gene O is dominant over gene B
(c) Gene A or B is absent in one of the parents
(d) Gene A and B are absent in one of the parents

30. A person with antigens A and B and no antibodies belongs to blood group or In which blood group antibodies are absent
(a) A (b) B
(c) AB (d) O

31. Universal donors have no antigens in RBC and have both a and b antibodies. They belong to blood group
(a) A (b) B
(c) AB (d) O

32. Blood bank of the body or reservoir where the blood is stored and can be mobilized is
(a) Heart (b) Liver
(c) Bone marrow (d) Spleen

33. The probability of having a child with blood group O to parents with blood groups A and B is
(a) 4 out of 4 (b) 3 out of 4
(c) 2 out of 4 (d) 1 out of 4

34. Which of the following substances, if introduced into the blood stream, would cause coagulation of blood at the site of its introduction
(a) Fibrinogen (b) Prothrombin
(c) Heparin (d) Thromboplastin

35. If an albino man marries a normal woman and 50 offspring's are albino and 50 are normal, the woman is
(a) Heterozygous normal
(b) Homozygous normal
(c) Heterozygous carrier
(d) None of these

36. To be evolutionary successful, a mutation must be
(a) Germplasm DNA (b) Somatoplasm DNA
(c) Cytoplasm (d) RNA

37. The monosomic conditions in human beings depicted as XO is referred to as
(a) Criminal syndrome
(b) Down's syndrome
(c) Klinefelter's syndrome
(d) Turner's syndrome

38. The number of chromosomes in Down's syndrome is
(a) 23rd pair with one less = 45
(b) 21st pair with one more = 47
(c) 17th pair with one more = 47
(d) One extra sex chromosome = 47

39. The most striking example of point mutation is found in a disease called
(a) Night blindness (b) Thalassemia
(c) Down's syndrome (d) Sickle-cell anemia

40. The frequency of a mutant gene in a population is expected to increase, if the gene is
(a) Recessive (b) Dominant
(c) Sex linked (d) Favourably selected

41. Height is
(a) Somatogenic variation
(b) Discontinuous variation
(c) Continuous variation
(d) Blastogenic

42. Discontinuous variations are
(a) Mutations
(b) Acquired characters
(c) Essential features
(d) Nonessential features

43. If a diploid cell is treated with colchicine, then it becomes
(a) Tetraploid (b) Diploid
(c) Triploid (d) Monoploid

44. The genotype of boy having sexual characters of a girl is
(a) XXX (b) XXY
(c) XO (d) XYY

45. The sex determination pattern in honeybee is called

(a) Female haploidy (b) Haplodiploidy

(c) Gametic diploidy (d) Gametogony

46. Meta-females have

(a) XX (b) XO

(c) XXXX (d) XXXXXX

47. Chromosomal abnormality of an unborn baby (while in mother's womb) can be found out by a technique called

(a) Amniocentesis (b) CAT scanning

(c) Ultrasound (d) Tissue culture

48. Based on Lyon's hypothesis, what will be the number of Barr bodies found in a human female suffering from Down's syndrome?

(a) 0 (b) 1

(c) 2 (d) 3

49. Gynandromorphs develop in Drosophila when the two cells in the two-celled pro embryo will have one of the following chromosomal sets

(a) 2A + XX in one cell and 2A + X in the other

(b) 2A + X in both the cells

(c) 2A + XXX in both the cells

(d) All of the above

50. A medical technician while observing a human blood smear under the microscope notes the presence of Barr body close to the nuclear membrane in the WBC. This indicates that person under investigation is

(a) Colour blind (b) Hemophilic

(c) Normal female (d) Normal male

51. Random genetic drift in a population probably results from

(a) Large population size

(b) Highly genetically variable individuals

(c) Interbreeding within small isolated population

(d) Constant low mutation rate

52. Which one of the following conditions correctly describes the manner of determining the sex in the given example?

(a) Homozygous sex chromosome(XX) produce male in Drosophila

(b) Homozygous sex chromosome(ZZ) determine female sex in birds

(c) XO type of sex chromosomes determine male sex in grasshopper

(d) XO condition in humans as found in Turner Syndrome, determines female sex.

53. A colour blind man has colour blind sister but a normal brother than phenotype of its parents is

(a) Father colour blind and mother normal

(b) Father normal and mother colour blind

(c) Father and mother both are colour blind

(d) Father and mother both are normal

54. In which of the following colour blindness is inherited

(a) In males

(b) In females

(c) Both males and females

(d) none of the above

55. The most common type of haemophilia results from the congential absence of

(a) Factor II (b) Factor V

(c) Factor VIII (d) Factor XI

56. Fraternal twins are produced when

(a) A fertilized egg divided into two

(b) An egg is fertilized by two sperms

(c) A divided egg has two set of chromosomes

(d) Two eggs are fertilized simultaneously.

57. Identical twins are-

(a) Heterozygous (b) Homozygous

(c) Monozygotic (d) Dizygotic

58. What is the probability in families of six for

A. 1 boy and 5 girls B. 3 boy and 3 girls

C. all six girls

(a) $\frac{6}{64}, \frac{20}{64}, \frac{3}{64}$ (b) $\frac{6}{64}, \frac{30}{64}, \frac{60}{64}$

(c) $\frac{6}{64}, \frac{20}{64}, \frac{1}{64}$ (d) $\frac{6}{64}, \frac{6}{64}, \frac{6}{64}$

59. How do multiple alleles originate?

(a) Phenocopies provide one approach to the study of physiological genetics

(b) Mutations are inherited; phenocopies are not inherited

(c) By independent changes at various sites within an ancestral gene

(d) All of these

60. For order of dominance of alleles in rabbits is C^+ - colored, C^{ch} - chinchilla, C^h - himalayan; C - albino. What phenotypes and ratios would be expected from the following crosses

a) $C^+C \times CC$ b) $C^+C^h \times C^+C$

c) $C^+C^h \times C^+C^{ch}$ d) $C^{ch}C \times CC$

e) $C^+C^h \times C^+C$ f) $C^hC \times CC$

(a) a) all colored: b) 3 colored: 1 albino c) 3 colored: 1 chinchilla

d) 1 chinchilla : 1 albino e) 3 colored: 1 himalayan f) 1 himalayan : 1 albino

(b) a) 3 colored : 1 albino b) all colored c) 3 colored : 1 chinchilla

d) 1 chinchilla: 1 albino e) 3 colored : 1 himalayan f) 1 himalayan : 1 albino

(c) a) all colored b) 3 colored:1 albino c) 3 colored: 1 chinchilla

d) 3 colored : 1 himalayan e) 1 chinchilla: 1 albino f) 1 himalayan : 1 albino

(d) None of these

61. A homozygous Rh +ve man (RR) married an Rh - ve (rr) women. Their 1st child was normal and their 2nd child has the hemolytic disease of new born. What genetic explanation might be offered

(a) All children would be heterozygous and Rh +ve

(b) All children would be homozygous and Rh +ve

(c) All children would be heterozygous and Rh –ve

(d) All children would be homozygous and Rh –ve

62. A homozygous Rh +ve man (RR) married an Rh –ve (rr) women. Their 1st child was normal and their 2nd child has the hemolytic disease of new born. What prediction might be made concerning future children by this couple?

(a) All future children would be expected to be not affected

(b) All future children would be expected to be affected

(c) 50% future children expected to be affected

(d) Only 1 child would be expected out of 100

63. Tendency of certain loci or alleles to be inherited together.

(a) Crossing over (b) Recombination

(c) Genetic linkage (d) All of these

64. Gametes that contain recombinant combinations of genetic markers located on the same chromosome are produced by a phenomenon called

(a) Linkage (b) Crossing over

(c) Transduction (d) All of these

65. When will the crossing over occur?

(a) In the post replication stage

(b) In the replication stage

(c) In the pre-replication stage

(d) None of these

66. A fully heterozyous gray-bodied (b+), normal-winged (vg+) female – F1 frut fly crossed with black-bodied (b), vestigial winged (vg) male gave the following results; gray, normal 126; gray, vestigial 24, black, normal 26 and black vestigial 124. Does this indicate linkage?

(a) Yes (b) No

(c) 1 in 1000 (d) None of these

67. A student has 2 dominant trais dependent on single gene cataract, which he inherited from his mother; and polydactyl, he inherited from his father. His wife has neither trait. If the gene for these 2 traits are closely linked, would the student's be more apt to have

(a) Either cataract on polydactyly

(b) Cataract and polydactyly

(c) Neither trait

(d) Only cataract

68. Use of F′ factors in genetic analysis

(a) Producing wild-type of prototrophic bacteria

(b) Genetic mapping

(c) Complementation tests in bacteria

(d) Both (b) and (c)

69. How are F′ factors formed?

(a) F′ factors are formed by normal excision of F factors from Hfr chromosomes

(b) F′ factors are formed by abnormal excision of F factors from Hfr Chromosomes

(c) F′ factors are formed when Hfr cells in F+ become integrated

(d) None of these

70. By what mechanism does sexduction occur?

(a) By the conjugative transfer of an F′ factors from the recipient cell to the donor cell

(b) By the conjugative transfer of the F′ factors from a donor cell to a recipient F-cell

(c) By the transformation of Hfr cell to donor cell

(d) All of these

71. What is the difference between phage and prophage

(a) Prophage is a phage chromosome that has become integrated into the host chromosome

(b) The prophage is dormant in the sense that the phage genes involved in lytic development are repressed

(c) Both (a) and (b)

(d) The prophage is dormant in the sense that the phage genes involved in Lysogenic development are repressed.

72. The length of IS an element is
(a) 8-80 nucleotide pairs
(b) 80-780 nucleotide pairs
(c) 800-1400 nucleotide pairs
(d) 1400-80,000 nucleotide pairs

73. What roler(s) do its elements play in recombination
(a) Mediate recombination between homologous DNA molecules
(b) Illegltimate recombination between homologous DNA sequences
(c) Both (a) and (b)
(d) Mediate recombination between non-homologous DNA molecules

74. What does the term co-transduction mean?
(a) Simultaneous transduction of 2 similar genetic marker to a single recipient cell
(b) Simultaneous transduction of only 1 genetic marker to single recipient cell
(c) Simultaneous transduction of 2 different genetic marker to a single recipient cell
(d) All of these

75. Following which recombination process is sensitive to DNase
(a) Transformation (b) Transduction
(c) Conjugation (d) Both (b) and (c)

76. Which of the following recombination process required cell contact?
(a) Transformation (b) Transduction
(c) Conjugation (d) All of these

77. Transformation was first discovered in
(a) E.coli
(b) Virus
(c) Diplococcus pneumoniae
(d) Streptococcus pneumoniae

78. Which step is specific for homologous DNA in transformation?
(a) Integration (b) DNA recombination
(c) Both (a) and (b) (d) Conversion

79. Bacteriophage λ usually mediates transduction of
(a) The gal genes of E.coli
(b) The bio genes of E.coli
(c) Both (a) and (b)
(d) The pal genes of E.coli

80. Which phages always multiply and lyse the host cell after infection
(a) Virulent phages (b) Temperate phages
(c) Prophages (d) Both (b) and (c)

81. Abortive transductants are used to carry out which tests?
(a) Widal tests
(b) Precipitation tests
(c) Complementation tests
(d) Blotting tests

82. If the progeny particles produced from such an infected cell are analyzed in
(a) 3H density gradient
(b) CsCl density gradient
(c) $CaCl_2$ density gradlent
(d) None of these

83. If markers a+ and b+ are co-transduced
(a) Markers b+ and c+ are cotransduced
(b) Markers a+ and c+ are not cotransduced
(c) Both (a) and (b)
(d) Markers a+ and c+ are cotransduced

84. F factors are responsible for
(a) DNA transfer during conjugation
(b) DNA molecules carrying genes for resistance to various antibiotics
(c) Code proteins
(d) All of these

85. A cell carrying the F factor in the integrated state is called
(a) Hfr (b) F′
(c) F^+ (d) F^-

86. Match the following

ISC elements	Nucleotide-pairs in
1. IS1	P. 768
2. IS2	Q. 1327
3. IS3	R. 1300
4. IS4	S. 1426

(a) 1-P; 2-R; 3-S; 4-Q (b) 1-P; 2-Q; 3-R; 4-S
(c) 1-S; 2-P; 3-Q; 4-R (d) 1-R; 2-S; 3-P; 4-Q

87. The E.coli K12 chromosome contains how many copies of IS1 and IS2?
(a) 8-IS1, 4-IS2 (b) 4-IS1, 8-IS2
(c) 8-IS1, 5-IS2 (d) 4-IS1, 4-IS2

88. The conjugative transfer of donor chromosomal genes carried by F′ factors to recipient cells is called
(a) Transformation
(b) Transduction
(c) Sexduction
(d) Both (b) & (c)

89. Transposable elements were discovered by BrMc clintock through an analysis of genetic instability in
(a) Rice (b) Tomato
(c) Maize (d) all of these

90. The length of inverted terminal repeats
(a) 4-8 nucleotide pairs
(b) 9-40 nucleotide pairs
(c) 41-423 nucleotide pairs
(d) 1-2 nucleotide pairs

91. Which IS elements are in direct orientation with respect to each other?
(a) Tn9 (b) Tn5
(c) Tn10 (d) None of these

92. Nucleotide pain in length of the CpDNAs of marchantia and tobacco are
(a) 121, 1024 and 155, 844
(b) 155, 844 and 121, 024
(c) 121 and 155
(d) 121, 844 and 844, 024

93. When killers are subjected to low temperatures, their killing capacityis
(a) Increased suddenly (b) Increased gradually
(c) Decreased gradually (d) Non change

94. Kappa bacteria possess 'R' body and are called
(a) Killers (b) Brights
(c) Resistant (d) RNA

95. Maternal effect in snail shell coiling is determined by
(a) Genotype of mother (b) Phenotype of mother
(c) Both (a) and (b) (d) None of these

96. In a pedidree; males are represented by
(a) □ (b) ○
(c) ● (d) ■

97. All statements are correct in case of pUC118 and PUC0119 except
(a) Small, supercolled, covalently closed circular DNA
(b) Carry the amp gene as a selectable marker
(c) High copy number up to 500 copies/bacterium
(d) Low copy number up to 20 copies/bacterium

98. The polycloning regions are present in pUC118 and pUC119 in
(a) Opposite orientations
(b) Same orientations
(c) Both (a) and (b)
(d) No orientations

99. Which segment encodes the amino terminal portion B-galactosidase?
(a) Amp gene (b) Lac 'z' gene
(c) Lac 'y' gene (d) Both (b) & (c)

100. What are linkers?
(a) Short, double stranded DNA's
(b) Long, double stranded DNA's
(c) Short, single stranded DNA's
(d) Long, double stranded RNA's

101. The mechanism of structural aberration consists of the breaking and rejoining of chromosomes, resulting in
(a) Deficiencies, duplications
(b) Inversions and translocations
(c) Both (a) and (b)
(d) Duplications and inversion

102. In inversions
(a) Parts of chromosomes become attached turn through 180° and reversed order
(b) Parts of chromosomes become detached, turn through 180° and reversed order
(c) Parts of chromosomes become detached, turn through 180° and reversed order
(d) Both (b) and (c)

103. Cross over within the loops of pericentric inversions result in
(a) Duplications (b) Deficiencies
(c) Both (a) and (b). (d) Deletions

104. A cross between two Bd files. Since the gene Bd is homozygous lethal, ratio is
(a) $\frac{2}{3}$ Beaded, $\frac{1}{3}$ wild type
(b) $\frac{1}{4}$ wild type, $\frac{2}{3}$ Beaded
(c) $\frac{1}{4}$ Beaded, $\frac{2}{3}$ wild type
(d) $\frac{1}{2}$ Beaded, $\frac{1}{2}$ wild type

105. The centric region are stained by
(a) G-banding (b) C-Banding
(c) Q-banding (d) None of these

106. The chromosome deficiency in cri-du-chat.
(a) Short arm of chromosome 5
(b) Short arm of chromosome 11
(c) Long arm of chromosome 3
(d) Short arm of chromosome 3

107. Burkitt's lymphoma was related to

(*a*) Translocation of chromosome 18

(*b*) Deletion of chromosome 14

(*c*) Translocation of chromosome 14

(*d*) Deficiency of chromosome 18

NUMERICAL TYPE QUESTIONS

1. Ratio of progeny when a red colored heterozygote is crossed with a white colored plant in which red color is dominant over white color.

2. How many types of genotypes are formed in F2 progeny obtained from self-pollination of a dihybrid F1?

3. When a tall plant with round seeds (TTRR) crossed with a dwarf plant with wrinkle seeds (ttrr). In F1 generation consists of tall plants with round seeds. What would be the proportion of dwarf plant with wrinkle seeds in F1 generation?

4. How many different kinds of gametes may be produced by an organism with genotype RrYy

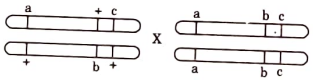
5. Which of the following is genotypic ratio of Mendel's monohybrid cross

6. A dihybrid for qualitative trait is crossed with homozygous recessive individual of its type , the phenotypic ratio is

7. If a man who is colour blind marries a woman who is pure normal for colour vision, the chances of their sons having colour blindness is

8. If a senes of 4 alleles is known to exist in a given in a diploid species, how many would be present in a chromosome pair

9. In the following test cross, genes a and b are 20 map units apart and genes b and c are 10 map units apart.



If the coefficient of coincidence is 0.5, how many triple homozygous individuals would you expect to find among 1000 genes from the cross?

10. The Frequency of an allele in large randomly population is 0.2. The frequency of carriers is

11. The incidence of recessive albinism is 0.004 human populations. What is the frequency of recessive allele (Assume random mating)?

12. A locus has 3 alleles A_1, A_2 and A_3, with frequency 0.6, 0.3 and 0.1. Assuming random mating, what is the frequency of all the heterozygote in the population

13. Human beings carrying the dominant allele T can taste the substance phenyl thiocarramide. In a randomly mating population in which the frequency of this allels is 0.4. What is the probability that a particular taster is homozygous?

(*a*) 0.025 (*b*) 0.25

(*c*) 0.75 (*d*) 0.5

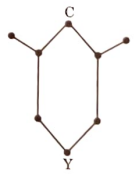
14. Assuming that Q and R in the pedigree below aroseifrom random mating; what is the Inbreeding coefficient of X?

15. The locus is segregating 2 alleles, A and a, with frequencies of 0.6 and 0.4, respectively. After a long history of random mating, the population goes through one cycle in which all the mating are between full sibs. What is the expected frequency of heterozygote among the offspring of these full sib mating?

16. A, B and C are inbred strains of mice, assumed to be completely homozygous. A is mated to B and B to C. Then the A × B hybrids are mated to C and the offspring are mated to the B × C hybrids. What is the inbreeding coefficient of the offspring of this last mating?

17. In mice there are several different loci that segregate recessive alleles causing albinism. A cross between albinos from 2 highly inbred strains produces normally pigmented F_1 animals. If these F1 mice are sib-mated, what fraction of their progeny will be albinos?

18. In pedigree below, what is the inbreeding coefficient of the common ancestor, C, if the inbreeding coefficient of Y is $\frac{1}{16}$?



19. The frequency of newborn infants homozyous for a recessive lethal allele is about 1 in 25000. What is the expected frequency of this allele in the population (Assume random rating)

20. Assume that pattern baldness in humans is controlled by an allele, B, of an autosomal gene.

Further assume that this allele is dominant in males and recessive in females that is sex influence. If a bald man and non bald women in a randomly mating population in which 51% of the men are bald have a child, what is the probability that the child will eventually become bald?

EXERCISE - II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. Match the type of chromosomal inheritance **(Column-I)** with the corresponding genetic disease or trait **(Column-II).**

Column-I

P. Autosomal recessive inheritance

Q. Autosomal dominant inheritance

R. X-linked inheritance

S. Y-linked inheritance

Column-II

1. Huntington disease
2. Hairy ears
3. Cystic fibrosis
4. Hemophilia

Codes:

	P	Q	R	S
(*a*)	1	4	3	2
(*b*)	4	3	2	1
(*c*)	3	1	4	2
(*d*)	4	2	3	1

2014

2. In a relatively large but finite and closed population of sexually reproducing diploid organisms, the frequency of homozygous genotype PP changes from 0.40 to 0.50 and that of pp changes from 0.40 to 0.41 in a span of 10 generations. Which of the following is the most likely cause for the above change in frequency of the PP genotype?

(*a*) Non-random mating

(*b*) Random genetic drift

(*c*) Selection

(*d*) Combination of non-random mating and random genetic drift

3. In a genetic cross between the genotypes WWXX and wwxx, the following phenotypic distributions were observed among the F_2 progeny: WX, 562; wx, 158; Wx, 38; and wX, 42. Likewise, a cross between XXYY and xxyy yielded the following results: XY, 675; xy, 175; Xy, 72; and xY, 78. Similarly, a cross between WWYY and wwyy yielded: WY, 292; wy, 88; Wy, 12; and wY, 8. In all the genotypes, capital letters denote the dominant allele. Assume that the F_1 progeny were self-fertilized in all three crosses. Also, double cross-over does not occur in this species. Which of the following is correct?

(*a*) Relative position: W-X-Y Distances: W-X = 5 map units, X-Y = 17 map units

(*b*) Relative position: X-Y-W Distances: X-Y = 15 map units, Y-W = 11 map units

(*c*) Relative position: Y-W-X Distances: Y-W = 5 map units, W-X = 11 map units

(*d*) Relative position: X-W-Y Distances: X-W = 5 map units, W-Y = 10 map units

4. A dioecious plant has XX sexual genotype for female and XY for male. After double fertilization, what would be the genotype of the embryos and endosperms?

(*a*) 100% ovules will have XXX endosperm and XX embryo

(*b*) 100% ovules will have XXY endosperm and XY embryo

(*c*) 50% ovules will have XYY endosperm and XY embryo, while other 50% will have XXY endosperm and YY embryo

2013

5. Of the two diploid species, species I has 36 chromosomes and species II has 28 chromosomes. How many chromosomes would be found in an allotriploid individual?

(*a*) 42 or 54 (*b*) 46 or 50

(*c*) 74 or 86 (*d*) 84 or 108

6. Hypophosphatemia is manifested by an X-linked dominant allele. What proportion of the offsprings from a normal male and an affected heterozygous female will manifest the disease?

(*a*) ½ sons and ½ daughters

(*b*) all daughters and no sons

(*c*) all sons and no daughters

(*d*) ¼ daughters and ¼ sons

2012

7. Match the terms in Group I with the ploidy in Group II.

Group I	Group II
P. Disome	1. 2n + 1
Q. Monosome	2. 2n –1
R. Nullisome	3. n –1
S. Trisome	4. n + 1

Codes :

	P	Q	R	S
(*a*)	4	2	3	1
(*b*)	4	3	1	2
(*c*)	2	3	4	1
(*d*)	1	4	3	2

8. A disease is inherited by a child with a probability of $\frac{1}{4}$. In a family with two children, the probability that exactly one sibling is affected by this disease is

(*a*) $\frac{1}{4}$

(*b*) $\frac{3}{8}$

(*c*) $\frac{7}{16}$

(*d*) $\frac{9}{16}$

2011

9. Diploid *Drosophila* has eight chromosomes. Which one of the following terms should NOT be used to describe *Drosophila* with sixteen numbers of chromosomes?

(*a*) Polyploid

(*b*) Aneuploid

(*c*) Euploid

(*d*) Tetraploid

Common Data for Questions 10 and 11:

Red-green colour blindness is inherited as a recessive X-linked trait.

10. What will be the probability of having the colour-blind son to a woman with phenotypically normal parents and a colour-blind brother, and married to a normal man? (Assume that she has no previous children)

(*a*) 100 % (*b*) 50 %

(*c*) 25 % (*d*) 12.5 %

11. What will be the probability of having the colour-blind daughter to a phenotypically normal woman, who already had one colour-blind son, and is married to a colour-blind man?

(*a*) 75 % (*b*) 50 %

(*c*) 25 % (*d*) 15 %

Statement for Linked Answer Questions 12:

The abdomen length (in millimeters) was measured in 15 male fruit flies, and the following data were obtained:

1.9, 2.4, 2.1, 2.0, 2.2, 2.4, 1.7, 1.8, 2.0, 2.0, 2.3, 2.1, 1.6, 2.3 and 2.2.

12. The value of standard deviation (SD) will be

(*a*) 0.061 (*b*) 0.25

(*c*) 0.61 (*d*) 0.85

2010

13. In transgenics, alterations in the sequence of uncleotide in genes are due to

P. Substitution

Q. Deletion

R. Insertion

S. Rearrangement

(*a*) P and Q

(*b*) P, Q and R

(*c*) Q and R

(*d*) R and S

14. True breeding **Drosophila** flies with curved wings and dark bodies were mated with true breeding short wings and tan body **Drosophila**. The F1 progeny was observed to be with curved wings and tan body. The F1 progeny was again allowed to breed and produced flies of the following phenotype, 45 curved wings tan body, 15 short wings tan body, 16 curved wings dark body and, 6 short wings dark body.

The mode of inheritance is

(*a*) Typical Mendelian with curved wings and tan body being dominant

(*b*) Typical non-Mendelian with curved wings and tan body not following any pattern

(*c*) Mendelian with suppression of phenotypes

(*d*) Mendelian with single crossover

NUMERICAL TYPE QUESTIONS

2016

1. A crossing was performed between the genotypes ***DdEeFfgg*** and ***ddEeFfGg***. Assuming that the allelic pairs of all genes assort independently, the proportion of progeny having the genotype ***ddeeffgg*** is expected to be _____%.

2015

2. A heterozygous tall plant (*Tt*) was crossed with a homozygous dwarf plant (*tt*). The resultant seeds were collected. If five seeds are chosen at random, then the probability (in %) that exactly two of these seeds will yield dwarf plants is _____.

2014

3. Consider a population of 10,000 individuals, of which 2500 are homozygotes (PP) and 3000 are heterozygotes (Pp) genotype. The frequency of allele p in the population is _____.

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (*b*)	**2.** (*b*)	**3.** (*d*)	**4.** (*c*)	**5.** (*b*)	**6.** (*c*)	**7.** (*c*)	**8.** (*b*)	**9.** (*a*)	**10.** (*b*)
11. (*c*)	**12.** (*d*)	**13.** (*d*)	**14.** (*b*)	**15.** (*a*)	**16.** (*b*)	**17.** (*d*)	**18.** (*a*)	**19.** (*d*)	**20.** (*a*)
21. (*d*)	**22.** (*b*)	**23.** (*c*)	**24.** (*c*)	**25.** (*a*)	**26.** (*d*)	**27.** (*c*)	**28.** (*c*)	**29.** (*d*)	**30.** (*c*)
31. (*d*)	**32.** (*d*)	**33.** (*d*)	**34.** (*d*)	**35.** (*c*)	**36.** (*a*)	**37.** (*d*)	**38.** (*b*)	**39.** (*d*)	**40.** (*d*)
41. (*c*)	**42.** (*a*)	**43.** (*a*)	**44.** (*b*)	**45.** (*b*)	**46.** (*c*)	**47.** (*a*)	**48.** (*b*)	**49.** (*a*)	**50.** (*c*)
51. (*c*)	**52.** (*c*)	**53.** (*a*)	**54.** (*c*)	**55.** (*c*)	**56.** (*d*)	**57.** (*c*)	**58.** (*c*)	**59.** (*c*)	**60.** (*a*)
61. (*a*)	**62.** (*b*)	**63.** (*c*)	**64.** (*b*)	**65.** (*a*)	**66.** (*a*)	**67.** (*a*)	**68.** (*d*)	**69.** (*b*)	**70.** (*b*)
71. (*d*)	**72.** (*c*)	**73.** (*c*)	**74.** (*c*)	**75.** (*a*)	**76.** (*c*)	**77.** (*c*)	**78.** (*c*)	**79.** (*c*)	**80.** (*a*)
81. (*c*)	**82.** (*b*)	**83.** (*c*)	**84.** (*a*)	**85.** (*a*)	**86.** (*b*)	**87.** (*c*)	**88.** (*c*)	**89.** (*c*)	**90.** (*b*)
91. (*a*)	**92.** (*a*)	**93.** (*c*)	**94.** (*b*)	**95.** (*a*)	**96.** (*a*)	**97.** (*c*)	**98.** (*a*)	**99.** (*b*)	**100.** (*a*)
101. (*c*)	**102.** (*c*)	**103.** (*c*)	**104.** (*a*)	**105.** (*b*)	**106.** (*a*)	**107.** (*c*)			

Numerical Type Questions

1. 1:1	**2.** 4	**3.** 0	**4.** 4	**5.** 1:2:1	**6.** 1:1:1:1
7. 0%	**8.** 2	**9.** 5	**10.** 0.32	**11.** 0.002	**12.** 0.54
13. 0.25	**14.** $\frac{3}{8}$	**15.** 0.36	**16.** $\frac{3}{8}$	**17.** $\frac{7}{16}$	**18.** $\frac{1}{16}$
19. 0.012	**20.** 0.4				

EXERCISE – II

MCQ Type questions

1. (*c*)	**2.** (*c*)	**3.** (*c*)	**4.** (*d*)	**5.** (*b*)	**6.** (*a*)	**7.** (*a*)	**8.** (*b*)	**9.** (*b*)	**10.** (*b*)
11. (*b*)	**12.** (*b*)	**13.** (*b*)	**14.** (*a*)						

Numerical Type Questions

1. (1.3 : 1.8) **2.** (31%) **3.** (0.6 to 0.6)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. Individuals having a pair of contrasting characters are known as allelomorphs.
2. Dominant factor: it is a pair of allele which expresses itself in both homozygous and heterozygous conditions.
3. TrtR heterozygous for Tt and Rr alleles
4. Test cross is the cross to know whether an individual is homozygous or heterozygous for dominant character. The individual is crossed with recessive parent.
5. All the red flowered plants, according to Mendel's law of dominance

   RR * rr ⟶ Rr (red)

   (Red) (WHITE)
6. Genotypically Aa, phenotypically A

   AA *aa – P ⟶ Aa-F1
7. Test cross include cross of F1 of the recessive parents i.e., (Tt*tt)
8. The independent pairs of genes which interact to produce a new trait together, but each dominant gene alone produces its own trait are called **Supplementary Genes**.
9. By the rule of 2^n where n = number of heterozygous pairs of genes.

   Thus, 2^2 = 4 type of combinations are formed.

   AaBb = AB, Ab, aB, ab
10. *Mirabilis jalapa* shows incomplete dominance.

    RR X rr -> Rr

    Red White Pink

    Rr X rr -> Rr, Rr, rr, rr

    Pink White 2 Pink, 2 White
11. here, both types of traits are transmitting in a single individual. It means both are dominant at a time or co-dominant; e.g., Biood groups of human beings and roan colour in cattle.
12. In **epistasis**, an allele of one gene suppresses or masks the phenotypic expression of a gene at another locus. The gene which is masked is called Hypostatic gene.
13. **Cytoplasmic** inheritance is due to extra-nuclear genes present in mitochondria and chloroplasts. These genes do not follow Mendelian inheritance.
14. Responsibility of a single gene for more than one phenotypic effect, often seemingly unrelated is known as **pleiotropy**.
15. As Neurospora produce 4 types of ascospores two of one parental type, two of other parental type , two of one recombinant and another two of second recombinant type.
16. The tendancy of parental combinations to remain together is called linakge.
17. B-^{3}--A---^{7}----C The distance between B and C is 10 that is sum of distance between B and A and distance between A and C
18. The process of linkage on the basis of his breeding experiments in fruitfly Drosophila melanogaster was discovered by Morgan .
19. When two gnese are located very closely on chromosome they are called as "linkage genes" and they inherit together therefore two types of gametes will be formed in maximum numbers. Thus higher number of the parental type off springs will be seen in F_2 generation.
20. Because distance between the linked genes is the major factor which controls the frequency of crossing over
21. During crossing over non-sister homologous chromosome exchange genes
22. As n=10 so linkage groups = 10
23. Morgan said that chiasmata leads to crossing over by breakage and reunion of homologous chromosomes.
24. A, B, AB, O are the four possible number of phenotypes.
25. ABO blood groups are controlled by a series of three multiple alleles. Different combinations of these alleles produce four blood types A, B, AB, and O.
26. Antisera D (also called D Antigen) is used for testing to the blood group of Rh, because Antisera D containing antibody against protein D.
27. The process of mating among closely related individuals is known as breeding.
28. Alleles for blood group B (of child) are absent in father (blood group A), therefore they should come from mother and they are present in persons having blood group B or AB.
29. Heterozygous IA/IB individual are blood group AB. Both the A antigen (Product of the IA alleles) and the B antigen (Product of the IB alleles) are produced by them.

30. A person with blood group AB possess both the antigen A and B but their blood plasma does not possess any of the antibodies.
31. O Individuals produce neither A or B antigen, but have both type of antibodies. So their blood can be transfused into any recipient.
32. Spleen acts as a great reservoir of blood. It is observed that the spleen may release about 150 ml of blood to circulation.
33. Since the blood groups of parents are A and B, therefore their genotypes are AO and BO so the possible genotypes of their children are AO,BO,AB and OO their blood groups are A,B,AB and O.
34. Stage 1 of blood clotting is concerned with the formation of thromboplastin released from damaged tissue or platelets. Thromboplastin helps in the formation of the enzyme thrombokinase.
35. Albinism is an autosomal recessive trait. The genotype of such mother would be Aa i.e. one recessive gene a responsible for the absence of melanin and one dominant gene responsible for presence of melanin would be present. So the mother will be heterozygotic carrier.
36. Because germplasm DNA is inherited from one generation to another.
37. Individuals with Turner's syndrome have one X chromosome due to disjunction of sex chromosome in their parent.
38. The individuals with this syndrome are trisomic for chromosome 21. Chromosomal complement is 46+ additional chromosome 21 = 47.
39. Sickle cell Anemia is a hereditary disorder of autosomal nature which is caused by mutation of the gene controlling B-chain of hemoglobin.
40. Natural selection will preserve the adaptive mutation.
41. Continuous variations are small and graded these includes variation in colour, shape, size, weight and structure of body parts, height. These variations may be Somatogenic or Blastogenic.
42. Mutation causes discontinuous variations.
44. XXY genotype is found in Klinefelters syndrome. In which individuals are male, but they can show some female secondary sexual characters and are usually sterile.
45. Haplodiploidy is a type of sex determination in which the male is haploid while female is diploid .it occurs in some insects like honey bees, ants and wasps.
46. Super or meta-females have 47(44 + XXX), 48(44 + XXXX) or 49(44 + XXXXX) chromosomes.
47. Amniocentesis is the most widely used method for prenatal detection of many genetic disorders.
48. Down's syndrome is a chromosomal abnormality in humans and is associated with an extra chromosome 21(chromosome formula is 45 + XX = 47); while one of the two X-chromosomes becomes genetically inert and heteropycnotic and forms the Barr body. Hence human female suffering from Down's syndrome contains (2X - 1 = 1) one Barr body.
49. Gynandromorphs are formed due to misdivision of chromosomes and start as female with 2A + 2X chromosomes. One of the X-chromosomes is lost during the division of cell with the result that one of the daughter cells possesses 2A + 2X chromosomes and the other 2A + X.
50. Barr body is characteristic feature of female.
51. Random genetic drift in a population probably result from interbreeding within the small isolated population.
52. X/A ratio determines sex in Drosophila. In 'XO type' the 'o' determines maleness.
58. Father colour blind and mother normal.
54. Colour blindn ess is X linked disease influencing man in heterozygous condition and woman in homozygous condition.
55. Haemophilia A is a recessive sex linked X Chromosome desorder. It is the most common desorder (clotting factor VIII deficiency). Haemophilia covers blood plasma clotting factors levels of the cogulation factors needed for normal clotting process.
56. Fraternal twins are formed when a woman ovulates and releases more than one egg, and both eggs are fertilized by different sperm. Fraternal twins can be the same or different genders because sperm can be male or female and gender is based on which sperm fertilized the egg. Imagine two circles on their own from step one rather than one splitting. Each child would have its own DNA and genetic composition because they were never one single egg.
57. Identical twins are produced when one fertilized egg divides into two blastomeres and both give rise to young ones.
61. All children would be heterozygous (Rr) and Rb the. The mother bcecame immunized during the 1st pregnancy and the next child was affected.
63. Genetic linkage is the tendency of certain loci or alleles to be inherited together. Genetic loci that are physically close to one another on the same chromosome tend to stay together during meiosis, and are thus genetically linked.
64. Recombination's of genes on the same chromosame is accomplished by crossing over, a process by which parts of homologous dromospmes are interchange

65. Crossing over occurs at pachytene after the synapsis of the homologous chromosomes has occurred in prophase I of meiosis. Since chromosome replication occurs during interphase, merotic crossing over occurs in the post replication tetrad stage

67. Either cataract/polydactyly. The genes would be in repulsion. In the separation at meiosis, each gamete would get one/the other. A crossover would be required to produce a gamete with both or neither.

68. F' factors are useful for genetic mapping and particularly for cis & trans or complementation tests in bacteria

71. A prophase is a phage chromosome that has become intergrated into the host chromosome. The prophase is dormant in the series that the phage genes involved in lytic development (replication/maturation) are repressed. The prophage replicates during lost chromosome replication just as any other segment of the host chromosome.

72-73.

IS elements (or insertion sequences) are short (800 - 1400 nucleotide - pairs) DNA sequences that are transposable, that is capable of moving from one position in a chromosome to another position or from one chromosome to another chromoscme. Is elements appear to mediate recombination between nonhomologous DNA molecules/sequences so called illeoitimate recombination.

74. Cotransduction refers to the simultaneous transduction of 2 different genetic markers to a single recipient, well since bacteriophage particles can package only $\frac{1}{100}$ to $\frac{1}{50}$ of the total bacterial chromosome, only markers that are relatively closely linked can be cotransduced.

75-76.

CRITERION		
Recombinant process	Cell contact required	Sensitive to DNASE
Transformation	No	Yes
Transduction	No	No
Conjugation	Yes	No

77. Transformation was first discovered in pathogenic strains of diplococus pneumoniae by Griffith in 1928.

78. The 1st steps in transformation binding up take and degradation of the one strand of the double stranded DNA are not specifie for homologous DNA. However, the integration, or DNA recombination step, is specific for homologous DNA. That is not to say that the integration of segments of heterologous (foreign) DNA never occurs. If it does occur, however it does not so at frequencies very much lower than the frequencies observed using homologous DNA

79. Bacteriophage λ, the best known specialized transducing phage, for example, usually mediates transduction of only the gal and biogenes of E.coli.

80. Generalized transduction

Bacteriophages have been classified into 2 types on the basis of their interactions with the bacterial cell. Virulent phages always multiply and lyse the cell after infection.

81. The genes located on the transduced chromosome fragments may be expressed, even if they are not integrated cells carrying nonintegrated transducing fragment are called abortive transductants. They are partially diploid and can be used to carry one complementation tests. The complementation test provides the operational definition of the gene.

82. If the progeny particles produced from such an infected cell are analyzed in CsCl density gradient, any transducing particles present should band at a heavier than normal position

83. Genetic markers can be ordered by cotransduction patterns. If 1 markers a^+ and c^+ are cotransduced, (2) markers b^+ and c^+ but (3) markers a^+ and c^+ are cot cotransduced.

84. Cells carrying an F-factor (for fertility factor, also called, "sex factor" and "F" plasmld), cells carrying an F factor (donor cells) from conjugation tubes and initiate DNA transfer after making contact with cells not carrying an F factor, called F cells (recipient cells)

86. The 1st 4 is elements to be extensively characterized and sequenced - IS1, IS2, IS3, IS4 are 768, 1327, 1300 and 1426 nucleotide pairs in length, respectively.

87. The E.coli K12 chromosome apparently contains 8 copies of IS1 and 5 copies of IS2, pus one or more copies of IS3 and IS4. The E.coli K12 F factor contains one copy of IS2 and 2 copies of IS3.

89. Transposable elements was discovered by B.Mc-clintock through an analysis of genetic instability in maize

90. IS element are compactly organized. Typically there is a single coding sequence with short, identical or nearly identical sequences at both ends. These terminal sequences are always in inverted orientation with respect to each other. So they are called inverted terminal repeats. Their length rannes fro 9-40 nucleotide pairs.

91. Composite transposons are created when 2 IS elements insert near each other. The sequence between them can then be transposed by the joint action of the flacking elements

In Tn9, the flanking IS element are in direct orientation with respect to each other

92. The complete nucleotide pair sequences of the CpDNAs of the liverwort marchantia polymorpha and tobacco (nicotiana tobacum) have been deformined. The cpDNAs of the marrchantia and tobacco are 121, 024, and 155, 844 nucleofid pairs in length.

93. Paramecia from strains capable of producing the toxic substance are called killers. When killers are subjected to low temperatures, their killing capacity gradually disappears. The toxic effect also decreases after repeated cell divisions.

94. The kappa bacteria posses a refractile protein containing 'R' body and are called "brights" becoz they are infected with a virus that controls the synthesis of viral protein.

95. One of the earlieest and best known example of a material effect is that for the direction of coilling in shells of the snail lymnaea peregra. Since strains of this species have dextral shells, which coil to the right. Other have sinistral shells, which coil to the left. This characteristic is determined by the genotype of the mother (not her phenotype) rather than by the genes of the developing snail. Allele st for right handed coiling is dominant over allele s for colling to the left.

97. The important factures of the pUC 118 and pUC 119 cloning vectors

1. Small supercoiled covalently closed circular DNA
2. Carry the ampr gene as a selection marker
3. High copy number, up to 500 copies/bacterium
4. Carry a poly cloning uregion containing a veriety of restriction enzyme cleavage sites

98. The polycloning regions are present in PUC 118 and PUC 119 in opposite orientation thus if PUC 118 packages one strand of a cloned gene, PUC 119 will package the complimentary strand.

99. The E-coli z gene encondes B-galactosidase, the enzyme that cleaves lactose into glucose and galactose

100. Linkers are short, double - stranded DNAs that contain restriction enzyme cleavage sites, they are synthesized from mononucleotides by simple protocols of organic chemistry be any desired sequence can be synthesized

101. 4 different kinds of structural changes

1. Deficiencies (parts of chromosomes lost/ deleted)
2. Duplications (parts added/duplicated)
3. Inversions (sections detached and reunited in reverse order).
4. Translocations (parts of chromosomes detected and joined to non homologous chromosomes

102. Inversions occurs when parts of chromosomes become detached, turn through 180° and are reinserted in such a way that the genes are in reversed order.

103. Crossover within the loops of pericentric inversions results in duplications and deficiencies. The fate of these cells varies in animals and plants

104. $Bd \parallel Bd^+ \times Bd \parallel Bd^+$

$Bd \parallel Bd^+$	$Bd^+ \parallel Bd$	$Bd^+ \parallel Bd^+$
Deaded	Beaded	Wild type

Summary – $\frac{2}{3}$ Beaded, $\frac{1}{3}$ wild type

105. The centric region and other heterochromatic areas are stained by c-banding.

106. IQs of Cri du Chat studied are in the range of 20 – 40. The chromosome deficiency is in the short arm of chromosome 5 and is designated 5p – A kary otype (46 × 5p) of a newborn.

107. Burkitt's lymphoma was related to a translocation of chromosome $14[t(89-314q^+)(q^{24}; q^{32})]$

NUMERICAL TYPE QUESTIONS

1. The cross of heterozygous dominant with its recessive parent is called attest cross and it gives a ratio of 1:1 in monohybrid condition and 1:1:1:1 in dihybrid condition.

2. Number of gene pair (n) = 2.

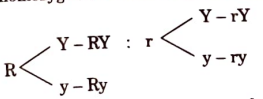
Number of F2 genotype = 3 × 3 × 3 = 9.

Number of kinds of gametes = 2 × 2 = 4.

3. The population of dwarf plant with wrinkled seeds will be 0 in F1 generation.

4. RrYy 2*2 = 4 gamete

In heterozygous condition 2 gametes and in homozygous condition 1 gamete is produced



5. 1:2:1 one non homozygous dominant, two heterozygous dominant and one homozygous recessive.

6. A dihybrid test cross give a 1:1:1:1 ratio indicating that two pairs of factors are segregation and assorting independently

   Tt Rr* ttrr

7. All the progeny will be normal, and there is o% chances of their sons having colour blindness.

9. $\frac{(0.5)(0.02)(1000)}{2}$

10. $(2)(0.2)(0.8) = 0.32$

11. $\sqrt{0.0004} = 0.02$

12. Compute the probability of not being homozygous: $1 - 0.36 - 0.09 - 0.01 = 0.54$

13. The frequency of homozygous tasters is $(0.4)^2 = 0.16$ and the frequency of heterozygous tastes is $(2)(0.4)(0.6) = 0.48$ thus, $0.16/(0.16 + 0.48) = 0.25$

14. There are 6 intreeding loops; assuming that and R come from random mating

$$Fx = (4)\left(\frac{1}{2}\right)^5 + 2\left(\frac{1}{2}\right)^3$$

$$= \frac{3}{8}$$

15. F = 0.25 and the allele frequencies are 0.6 and 0.4 therefore $(2)(0.6)(0.4)(1 - 0.25) = 0.36$

16. Note that for inbred strain F = 1; thus;

$$\left(\frac{1}{2}\right)^3(1+1) + \left(\frac{1}{2}\right)^4(1+1) = \frac{3}{8}$$

17. Assuming 2 unlinked loci are involved

$$\frac{1}{4} + \frac{1}{4} - \frac{1}{16} = \frac{7}{16}$$

18. $F_c = 1$ to satisfy the equation $\left(\frac{1}{2}\right)^5(1 + F_c) = \frac{1}{16}$

19. $\sim 2\sqrt{0.0004} = 0.012$

20. ♂ BB | BB Bald ♂ ♀ BB Bald ♀

    B'B' nondald | BB' / B'B' non bald

$$p = \frac{\text{Total population}}{\text{frequency B allele}}$$

q = frequency B' allele

$p^2 + 2pq = 0.51$

$q^2 = 0.49$

$q = 0.7, p = 3$

Mating subpopulation

$$\text{bald men} \begin{cases} BB: \frac{p^2}{p^2 + 2pq} = \frac{0.09}{0.51} = 0.176 \\ BB': \frac{2pq}{p^2 + 2pq} = \frac{0.42}{0.51} = 0.824 \end{cases}$$

$$\text{nonbald women} \begin{cases} BB': \frac{2pq}{2pq + q^2} = \frac{0.42}{0.91} = 0.462 \\ B'B': \frac{q^2}{2pq + q^2} = \frac{0.49}{0.91} = 0.538 \end{cases}$$

$$P \text{ male} = \frac{1.76 + 0.824}{2} = 0.588$$

$$q \text{ male} = \frac{0.824}{2} = 0.412$$

$$q \text{ female} = \frac{0.462}{2} t = 0.538 = 0.769$$

$$\text{probability bald child} = p_f p_m + \frac{p_f q_m}{2} + \frac{f p_m q_r}{2}$$

$$= 0.1358 + 0.0476 + 0.2261 \equiv 0.41$$

EXERCISE – II

MCQ TYPE QUESTIONS

1. The common Chromosomal inheritances including Autosomal Recessive inheritance, Autosomal dominant inheritance, X linked inheritance, Y linked inheritance are associated with Cystic Fibrosis, Huntington Disease, Hemophilia, Hairy ears.

2. The genotype of PP changes from 0.4 to 0.5 while in case of pp; it changes from 0.4 to 0.41 in a closed population of sexually reproducing diploids. So, it is clearly suggestive of the fact that the genotype PP is naturally selected and adapted by the environment and so its genotype has a higher frequency.

3. In the various genetic crosses, it can be seen that W and Y are more close to each other than X and Y. At the same time, X and W are close to each other while Y is farthest from X. So, the relative positioning of the genes are Y-W-X and y-w-x.

   Now, Relative distances of genes with respect to each other are as follows:

   Distance of W gene from Y, Y - W = 5 units and W from X, W - X = 11 units

4. Dioecious plants have male (staminate) flowers on one plant, and female (pistillate) flowers on another plant. Some of these plants are polygamo-dioecious, with some male flowers on female plants and some female flowers on male plants. A dioecious plant having XX femal genotype and XY mal genotype undergoes double fertilization. So, on double fertilization, we have:

   50% ovules : XXX endosperm and XX embryo

   And rest 50% ovules : XXY endosperm and XY embryo

5. Given, Species I has 36 (2n = 36) chromosomes and species II has 28 (2n = 28) chromosomes. Now, species I and II are crossed to obtain an allotriploid individual. That means it has a total chromosome of 3n. When the two diploid species are crossed, the chromosome set of one species would not undergo reduction division for the purpose of obtaining the allotriploid hybrid. That means the 3n would be contributed by 2n and n. So, the 2n can be from species I or II and so the n can be from species II or I respectively. Hence, the required chromosome number would be:

   Species I (2n) + Species II (n) = 3n

   $\Rightarrow$ 36 + 14 = 50

   Or Species I (n) + Species II (2n) = 3n

   $\Rightarrow$ 18 + 28 = 46

6. Given, Hypophosphatemia is caused due to X-linked dominant allele.

   Now, when a normal male (XY) and a diseased female (X^HX) are crossed together, the F1 generation will have offspring: XX, X^HX, XY and X^HY. So from this it can be seen that half of the male offspring and half of the female offspring bear the diseased allele.

7. Disome is a chromosome set having members paired (as in a normal somatic cell, n+1). Monosome is a chromosome having no homologue, especially an unpaired X chromosome (2n–1, because only one copy of some specific chromosome is present instead of the usual two found in its diploid progenitor. especially an unpaired X chromosome). Nullisome is a genetic condition involving the lack of one of the normal chromosomal pairs for a species (2n–2). A trisomy is a type of polysomy in which there are three instances of a particular chromosome, instead of the normal two (2n+1).

8. *Given*: $P(E) = \frac{1}{4}$,

$$P(\bar{E}) = \frac{3}{4}$$

$$P(x = 1) = 2\,C_1 \times \left(\frac{1}{4}\right) \times \left(\frac{3}{4}\right) = \frac{3}{8}$$

9. Aneuploidy is an error in cell division that results in the daughter cells having the wrong number of chromosomes. It is the second major category of chromosomes mutation in which chromosome number is abnormal. An aneuploid is an individual organism whose chromosome number differs from the wild type by part of a chromosome set. Generally, the aneuploid chromosome set differs from wild type by one or small number of chromosomes. Aneuploids have chromosome number either greater or smaller than of wild type. Aneuploid nomenclature is based on the number of copies of the specific chromosome in the aneuploid state. So, for drosophila with 16 chromosomes, it is not aneuploid in condition.

10. The probability of having the colorblind son to woman with phenotypically normal parent and a colorblind brother, and married to a normal man than may be the woman is a carrier of blind trait or else not.

    Then we can say mother can be X^RX a carrier or XX normal and have a cross with a normal man.

	X	Y
X^R	X^RX	X^RY
X	XX	XY

OR

	X	Y
X	XX	XY
X	XX	XY

    Therefore, there is 1 chance of having a colorblind son out of 4 possible progenies.

    Hence, probability of one child to be colorblind = ¼ = 0.25 = 25% .

    Again 2 sons could be born, out of which 1 could be colorblind;

    Probability for colorblind son = ½ = 50%

11. Let say woman can be a carrier or normal and married to a colorblind man.

    So, woman can be either X^RX or XX and man is X^RY

    Therefore, $X^RX * X^RY$ and $XX * X^RY$

	X^R	X
X^R	X^RX^R	X^RX
Y	X^RY	XY

	X	X
X^R	X^RX	X^RX
Y	XY	XY

    Therefore, out of 4 children, there is only one chance of having a colorblind daughter.

    The probability for 1 out 4 children = ¼ = 0.25 = 25% ;

    Now there are 2 probable daughters.

    Probability for daughter to be colorblind = ½ = 50%

12. We know, Standard deviation, S.D. = √(Variance)
$= \sqrt{\sigma^2}$

Now, Variance = 0.061 (from previous question)

Hence, Standard deviation = √(0.061)

= 0.25 (approx.)

13. Transgenics is based on insertion of foreign genes from a different species, or deletion of existing nucleotides from the gene sequence in normal host cells. Also substitution of bases by transition(Pu by Pu/ Py by Py) or transversion(Pu replaced by Py of vice-versa). Usually, the process of random gene rearrangement is not included in transgenic experiments, because it is a natural process not yielding desired transgenic traits. Thus, in actual practice, insertion, deletion and/ or substitution is used for yielding transgenics.

14. Let us denote the dominant curved wings allele by C, short wings by c, tan body by D, dark body by d. Assumption is made that curved wings and tan body are recessive traits and that they follow Mendelian genetics.

Now, F_1 progeny CcDd (Since, F_1 progeny was observed to be with curved wings and tan body)

Cross breed→ CcDd X CcDd

F_2 progeny → CCDD/CcDd/CCDd/CcDD (45) : ccDD/ccDd (15) : Ccdd/CCdd (16) : ccdd (6)

9 : 3 : 3.2 : 1.2

which approximately becomes 9: 3: 3: 1, which is the phenotypic ratio of F_2 generation according to Mendelian genetics

NUMERICAL TYPE QUESTIONS

1. Crossing of ***DdEeFfgg*** & ***ddEeFfGg***. was performed hence.

Dd × dd : Dd dd Dd dd i.e. dd = 0.5

Ee × Ee : EE Ee eE ee i.e. ee = 0.25

Ff × Ff : Ff fF fF ff i.e. ff = 0.25

gg × Gg : gG gG gg gg i.e. gg = 0.50

∴ ddeeffgg = (0.5) (0.25) (0.25) (0.5)

= 0.0156 = 0.0156 × 100% =1.56%

2.

	T	t
t	Tt	tt
t	Tt	tt

The probability of dwarf seeds from resultant seeds

= 0.5 = 50%

Then, probability of selection of 2 seeds which yield dwarf plant from selected 5 seeds

By applying Bionomial theorem

$$P = \frac{n!}{s! \times t!} a^s b^t$$

n = The total number of events

s = The number of times outcome a occurs

t = The number of times out come b occurs

Therefore $n = s + t$.

$$P = \frac{5!}{3! \times 2!}\left[\left(\frac{1}{2}\right)^3 \times \left(\frac{1}{2}\right)^2\right]$$

$$= \frac{5 \times 4}{2}\left(\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}\right)$$

$$P = \frac{20}{64}$$

$$\%P = \frac{20}{64} \times 100 = 31.25$$

3. We have, according to Hardy-Weinberg's law of population dynamics and equilibrium,

$P^2 + 2PQ + Q^2 = 1$

Where, P is the allelic genotype or the frequency of 'P' and Q is the allelic frequency of 'p'.

Now, Total population = 10000

Therefore, $P^2 = 2500/10000 = 0.25$

and $2PQ = 3000/10000 = 0.3$

Therefore, $Q^2 = 1 - P^2 - 2PQ$

$= 1 - 0.25 - 0.3 = 0.45$

So, Q = 0.671 which is the frequency of the allele *p*

8 CHAPTER

Genetics-II
Molecular Genetics

Genome

The genome is the entire collection of genes and all other functional and non-functional DNA sequences in an organism in a haploid set of chromosomes. It includes genes, regulatory genes and nonfunctional nucleotide sequences. Therefore, genome comprises of :

- The **structural genes** are DNA segments that code for some specific RNAs or proteins. They encode for mRNAs, tRNAs, snRNAs and scRNAs.
- The **functional sequences** or **regulatory** genes occur as regulatory elements or as initiation sites, promoter sites, operator sites and enhancer sites, etc.
- The **nonfunctional sequences** include introns and repetitive sequences. They are needed for coding, regulation and replication of DNA. Their number is found to be much more than the functional sequences.

The C-value Paradox

C-value is the total amount of DNA present in the haploid genome of an organism; it is a characteristic value for each living species and is usually represented in base pairs (bp). It ranges from merely $<10^6$ for a mycoplasma to 10^{11} for some plants and amphibians. In general, there is an increase in the C value with an increase in the complexity of organisms. For example, the unicellular eukaryotes like yeast have only five times as much DNA as the bacterium *E coli*, while the first truly multicellular organisms like the nematode *C. elegans* show more than 100-times of *E. coli* DNA content. The lowest C-value ($6.6. \times 10^5$ bp) for a living organism is found, in fact, in a eukaryote, the unicellular alga *Pyronomas salina*.

Nucleic Acid Reassociation Kinetics

Studies of DNA reassociation kinetics have provided information on the existence of repetitive DNA sequences as well as on the number of copies for genome of such sequences. These studies have permitted the classification of repetitive sequences into highly repetitive ($> 10^5$ copies per genome) and moderately repetitive (2 to $<10^5$ copies/genome) DNA. In such a study, a sample of DNA is generally sheared (broken) into fragments of small size (say of 400-500 bp). It is then subjected to high temperature or high pH, which leads to the separation of complementary strands producing single-stranded DNA; this is known as ***denaturation***. When a denatured DNA sample is allowed to cool (in case of heat denaturation) or its pH is lowered (in case of alkali denaturation), the complementary single strands reassociate through complementary base pairing to produce double-stranded DNA; this phenomenon is called ***renaturation***.

If RNA is added into a sample of denatured DNA, RNA strands will associate with their complementary DNA strands producing DNA : RNA hybrid molecules; this is referred to as ***DNA. RNA hybridization*** If, however, two samples of denatured DNA are mixed, DNA : DNA hybrid molecules will be formed such a process is known as ***DNA hybridization***. Both these phenomena taken together constitute the technique of ***nucleic acid hybridization***.

Renaturation rate in a denatured sample of DNA is mainly a function of DNA concentration and time allowed for renaturation; these two parameters are expressed as *Cot* (initial concentration × time) which is estimated as moles per litre × seconds as follows.

$$\frac{C}{C_0} = \frac{1}{1 + k \cdot Cot}$$

where, C_0 and C are the concentrations in moles of nucleotides per litre at the initial time t_0 and after reassociation reaction has progressed for time t, respectively, and k is a reassociation rate constant. Thus the parameter controlling reassociatioin reaction is the product of initial concentration of single-stranded DNA

(C_o) and the time of incubation for reassociation (t); this is simply described as the *Cot*. The rate of reassociation reaction is governed by the equation given below.

$$\frac{dC}{dt} = -kC^2$$

The rate of renaturation of a given DNA segment is proportional to the number of copies of that segment present in a unit DNA solution; the more the number of copies, the faster would be its renaturation.

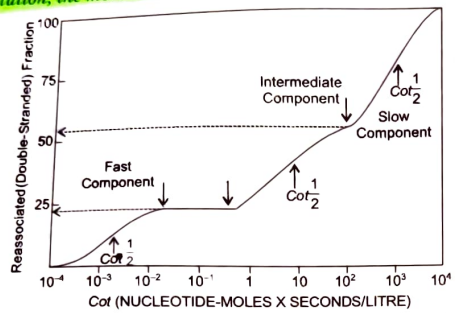


Fig. The renaturation pattern of an eukayotic DNA sheared into fragments of about 400 nucleotides. The earliest renaturation occurs in the highly repetitive DNA, followed by moderately repetitive DNA and, atlast, unique sequences.

A useful parameter is obtained as $Cot\ \frac{1}{2}$, i.e., the value required for half reassociation. $Cot\ \frac{1}{2}$ is the product of the concentration and time (in seconds) required for reassociation to proceed halfway and is expressed either in nucleotide-moles × seconds/litre as

$$\frac{C}{C_0} = \frac{1}{2} = \frac{1}{1 + k \cdot Cot\frac{1}{2}}$$

or in units of per litre-nucleotide moles per second as

$$Cot\ \frac{1}{2} = \frac{1}{k}$$

A greater value of *Cot* implies a slower reaction. The value of $Cot\ \frac{1}{2}$ is directly related to the amount of DNA in the genome, i.e., the greater the amount of DNA in a genome, the larger will be the value of $Cot\ \frac{1}{2}$. This is because the same absolute concentration of DNA measured in moles of nucleotides per litre (the C_o) will contain a larger number of copies complementary to a given DNA sequence in case of smaller genome size than it would in the case of a larger genome size.

The $Cot\ \frac{1}{2}$ of a DNA sample indicates the total length (in bp) of different sequences present in the DNA, which is usually referred to as ***complexity of the genome***. The renaturation of DNA of a genome or a part of a genome gives a $Cot\ \frac{1}{2}$ value proportionate to its complexity. The complexity of a genome can be determined by comparing its $Cot\ \frac{1}{2}$ with that of a DNA of known complexity; usually, *E. coli* DNA is used for this purpose. The *E. coli* genome has 4.2×10^6 bp and is assumed to contain only unique DNA. The complexity of a genome is estimated from the following formula.

$$\frac{Cot\ \frac{1}{2} \text{ of DNA of a genome}}{Cot\ \frac{1}{2} \text{ of } E.\ coli \text{ genome}} = \frac{\text{Complexity of the genome}}{4.2 \times 10^6 \text{ bp i.e., size of } E.\ coli \text{ genome}}$$

Transposable Elements

Transposable (Tn) elements are DNA sequences that move from one location in a chromosome to another within the same chromosome or into another non-homologous chromosome. Transposable elements are of two main types: (1) those that propagate themselves by transposition as DNA copies, and (2) those that mobilize themselves not by transposition but by making DNA copies of their RNA transcripts. Insertion sequences and composite transposons belong to the first group of transposons, while retroposons belong to the second category.

Insertion Sequences (IS)

Insertion Sequences are short DNA sequences of about 800-1400 bp and have the ability to transpose. They also promote recombination between nonhomologous chromosomes (***illegitimate recombination***) and are considered to be involved in the integration of episomes into bacterial chromosomes. The first four *IS* elements discovered in *E. coli* were designated as *IS*1 (768 pb; 8 copies/genome), *IS*2 (1327 pb; 5 copies/genome), *IS*3 (1300 bp; one or more copies/genome), and *IS*4 (1428 bp; one or more copies/genome) *IS* elements usually have 9-24 pb inverse-repeats (the same sequence when read in the opposite directions) at their two ends; the repeated sequences at the two ends of an *IS* element are closely related, rather than identical *IS* elements are the simplest forms of transposons and are so called because they were discovered as spontaneous insertions within bacterial operons that prevented transcription or translation of the concerned gene into which insertion had occurred.

Composite Transposons

Composite transposons (*Tn*) are more complex transposable elements; they are usually more than 2,000 bp long, carry one or more genes unrelated to their transposability (e.g., genes for antibiotic resistance), and have *IS*-like elements repeated at their ends. The *IS* elements at the two ends of a *Tn* element may be either in the same orientation (direct terminal repeats), e.g. *IS*1 at the ends of *Tn*9 of *E coli*, or in the opposite orientation (**inverse terminal repeats**), e.g., *IS*10 in *Tn*10 of *E. coli*. The terminal *IS* elements form the two (left and right) arms of the Tn elements, while the resistance and other genes are present in their central region.

Transposable Elements of Eukaryotes

Eukaryotes are known to have a variety of transposable elements that originally discovered in maize and described as ***controlling elements***. Insertion of a controlling element in a chromosome may affect the expression of adjacent gene. These elements may also produce deletions, duplications, inversions and translocations at the sites in which they are present. Chromosome breakage is the common consequence of the presence of some elements. In maize, the activities of controlling elements are regulated during development; they transpose and promote genetic rearrangements at characteristic time and frequencies during plant development.

Eukaryotic transposable elements fall into the following two general categories : (1) transposons and (2) retroposons. ***Transposons*** are comparable to the bacterial transposons, and have no life outside the genome, e.g., the controlling elements of maize. Alongwith the bacterial transposons and *IS* elements, they form a group of elements often regarded as ***selfish DNA*** since they are chiefly concerned with their own propagation; the *IS* elements appear to have no other role whatsoever. ***Retroposons*** are either DNA copies of RNA viruses called ***retroviruses*** (and hence yield infective virus particles on transcription) or derivatives from such copies and may even have lost their ability to transpose. These elements also generate short direct repeats in target DNA during their insertion.

P Factors in Drosophila

In *Drosophila melanogaster*, hybrids derived from certain strains show a series of defects, including mutations, chromosomal aberrations, distorted segregation and sterility. For example, when *P* (paternal contributing) type of males are mated to M (maternal contributing) type females, the F_1 progeny are sterile, while those from the reciprocal cross ($P♀ \times M♂$) are normal fertile. The *P* type flies have a large number (30-50 copies) of *P* factors (transposable elements), which become active in the cytoplasm of M flies. The active *P* factors transpose readily creating chromosome breaks and new mutations. These events are responsible for the abnormalities, including mutations, chromosomal aberrations, distorted segregation and sterility in the F_1; the phenomenon is known as ***hybrid dysgenesis***.

Retroposons

The retroviruses seem to have given rise to transposons, which are closely related to them; such elements are called ***retroposons***. Yeast *Ty* (transposon yeast) elements are an example of such transposons. The *Ty* elements are considerably divergent; there are two major classes, viz, *Ty*1 and *Ty* 917. They transpose at a frequency of $- 10^{-7} - 10^{-8}$ through and RNA intermediate, which is reverse-transcribed to yield a DNA duplex that becomes inserted into the yeast genome generating a 5 bp direct repeat in yeast DNA. Their transposition is controlled by genes within the *Ty* elements; they form noninfective virus-like particles, which accumulate within such cells in which transposition has been induced. Only some of the yeast *Ty* elements are active, while most of them have lost the ability to transpose.

Each *Ty* element is 6.3 kb long, and has 330 bp direct repeats at its ends; the repeats are called ***delta (δ) repeats***. All *Ty* elements share δ repeats, a long region at the left end another region in the centre and a short region adjacent to the right δ repeat.

Gene

Genes are unit of inheritance. These are located in the chromosomes in linear fashion and are carried from parent to offspring by gametes. These control the expression of characters. **Mendel** was first to visualise a gene as a unit of inheritance and called it factor. The word gene was introduced by **Johannsen** in 1909.

Modern Definition of Gene

According to modern concept, a gene is a segment of DNA with specific nucleotide sequence which has coded information for a specific character. The specific sequence of nitrogenous bases in a gene provides information for the sequence of amino acids in the polypeptide chain during synthesis of a particular protein. A gene is now described as a unit of function (**cistron**), a unit of mutation (**muton**) and a unit of regulation (**operon**).

Types of Genes

Based on the functional activity, the genes are classified as follows:

1. **House-keeping Genes or Constitutive Genes**

   These genes are functional in all types of body cells of multicellular organisms and all the time. These are needed for basic cellular activities. Such genes are called **house-keeping genes** and their function is called as **house-keeping** or **constitutive activity**. For example, genes associated with encoding the enzymes of glycolysis are active in all cell types and all the time throughout life. Their expression is not regulated. Such enzymes are called **constitutive enzymes**.

2. **Luxury Genes or Nonconstitutive Genes**

   These genes remain inactive for most of the time in the life span of an individual and are expressed in certain cells or at certain time only when their products are needed.

   These are also called **specialist genes** or **luxury genes**. For example, some genes are functional in kidney cells, some in liver cells and some in intestine or stomach but are present in all the body cells.

3. **Structure Genes**

   These genes code for substances which are needed for the morphological or functional traits of the cell and are called **cistrons**. These occur in split form, segmented into **introns** and **exons** in eukaryotes but in prokaryotes these are continuous. The structure genes are concerned with the synthesis of a polypeptide chain or a number of polypeptide chains and also with the synthesis of different types of RNA required during protein synthesis.

4. **Control Genes**

   Some genes produce proteins that exercise a positive or negative control on the functioning of structure genes. These are of the following types :

   1. **Regulator Genes :** These code for repressor proteins which regulate transcription of **cistrons**.
   2. **Operator Genes :** These act as switch and turn on or off the transcription of a structural gene as and when required by the cell.
   3. **Promoter Genes:** These genes provide sites for the binding of ***RNA polymerase*** enzyme for the transcription of RNAs by structure genes.
   4. **Terminator Genes :** These DNA segments lie in the end of messages and stop or terminate transcription activity of structure genes.

30+20+50+10+30+10+20+30+48

5. **Interrupted Genes or Discontinuous Genes or Split Genes**

   In prokaryotes (eubacteria) the genes and proteins are collinear, i.e. the sequence of nucleotides in the gene corresponds exactly with the sequence of amino acids in the protein. Each gene is a continuous stretch of DNA whose length is related to the size of protein it synthesises. The colinearity of gene was identified in the gene that codes for ***tryptophan synthetase*** in ***E. coli***.

   (i) **Split Genes :** In eukaryotes, the message coded in the genes is not continuous. It is found to be split into several distinct coding units separated by regions of noncoding DNA. The coding units containing biological information are called **exons** and the intervening noncoding DNA segments are called **introns**. Such genes are called **discontinuous genes** or **interrupted genes** or **split genes**. The concept of **interrupted genes** was introduced in 1977.

   (ii) **Introns :** These are present in the genes of eukaryotes, viruses and archaebacteria. The transcription of an interrupted gene produces the primary transcript RNA. It is a faithful copy of the interrupted gene and is only the precursor. The functional RNA is formed by the removal of introns and the rejoining of its exons. This process is called **RNA splicing**.

6. **Alternative Genes**

   These are formed when exons from different discontinuous genes get connected forming several new combinations. The alternative forms thus formed produce proteins in which one part is common while the other part is different. This concept was proposed by **Gilbert** and is known as **Gilbert hypothesis**.

7. **Overlapping Genes**

   In overlapping genes, the same nucleotide sequence codes for more than two different proteins by employing two different reading frames. It means the same specific nucleotide sequence is shared in the transcription of two different mRNAs for the synthesis of two non-homologous proteins.

8. **Transposons (Jumping Genes)**

   Nobel Prize winner, **Barbara McClintock,** described the occurrence of DNA segments in maize, which can move from one position to another in the genome of a cell.

   In bacteria, these were first detected in *E. coli* and are called **insertion** or ***IS* elements** or **transposable elements**. The process of insertion of transposon between two base pairs in a recipient DNA molecule is called **transposition**.

9. **Gene Families and Pseudogenes**

   Prokaryotic genes are mostly represented only once in the genome. But many eukaryotic genes are present in multiple copies. These multiple copies of genes are called **gene families.** Members of many gene families are clustered in the same region of DNA, while some gene families are dispersed to different chromosomes.

   Example : The α and β sub-units of human haemoglobin are encoded by gene families. Different members of these families are expressed in embryonic, foetal and adult tissues.

10. **Smart Genes**

    Smart genes are artificially constructed genetic molecules composed of one or more genes that code for functional proteins or enzymes. Smart genes are expressed only under specific conditions such as presence of some particular substrate molecules. Thus a smart gene has a conditional expression. Because of this property, the constructed smart genes form ideal therapeutic vehicle for gene therapy.

Organisation of Chromatin

The chromatin is formed of about 60 percent **proteins**, 35 percent **DNA** and 5 percent **RNA**. The DNA molecule of individual human chromosomes ranges from 1.7 to 8.5 cm in length when uncoiled. It contains $50 \times 10^6 - 250 \times 10^6$ nucleotide pairs. The information stored in DNA is organised, replicated and read with the help of a variety of **DNA-binding proteins**. These fall into two categories:

- Structural proteins – **Histones**
- Functional proteins or control proteins – **Non-histones**

Structural proteins or **packaging proteins** are non-specifically binding proteins. These are bound to DNA along most of its length and help to package it without preventing the access of other DNA-binding proteins. They are basic proteins and are called **histones**. The **non-histones** or **functional proteins** are associated with gene regulation and other functions of chromatin.

Histones

Histones are the main structural proteins found in eukaryotic cells. These are low molecular weight proteins with high proportion of positively charged amino acids (basic amino acids, the **arginine** and **lysine**). The positive charge helps histones to bind to DNA and play a crucial role in the packaging of long DNA molecules.

Type of Histones

There are five different types of histones that fall into two categories:

1. **Nucleosomal Histones :** The nucleosomal proteins are eight molecules of four types of small proteins. They are responsible for coiling of DNA into nucleosomes. They are represented by two molecules of each H2A, H2B, H3 and H4 proteins. Each of them is formed of about 102–135 amino acids. These form the inner core of nucleosome. Nucleosomal histones exhibit great similarity because they have not changed during evolution. These are present in equimolar amounts, two of each type being present after every 200 base pairs, thus forming a **histone octamer** or **core particle**.
2. **H1-histones :** These are large (of about 200 amino acids) proteins and are tissue specific. These are present one molecule per 200 base pairs and are loosely associated with DNA. Only one H1 molecule is present with each nucleosome. H1 histones are responsible for packing of nucleosomes into 30 nm nucleoprotein fibre.

Functions of Histones

Histones in eukaryotic chromosomes serve two functions :

- These serve as structural elements and help in coiling and packing of long DNA molecules.
- These cover or repress specific segments of DNA so that these segments are unable to transcribe. Their transcription is possible only when histones are removed in response to certain molecular signals.

Nucleosome Concept or Oudet Concept of Chromatin Structure

Nucleosomes are the fundamental packing units of chromatin and give chromatin the 'beads **on a string**' appearance. Each nucleosome is disc-shaped about 11nm in diameter. It consists of a **core particle** and a smaller **spacer** or **linker** DNA.

1. **Core Particle :** The core particle consists of the octamer of histones having two copies each of **H2A**, **H2B**, **H3** and **H4**. It is about 11nm in diameter and 6nm in height. A strand of DNA having 146 base pairs is tightly wrapped around this core forming two circles (73 nucleotides per turn).
2. **Spacer DNA or Linker DNA :** It is a small segment of DNA having just four base pairs. One unit of histone H1 is associated with it. There is considerable variation in the length of spacer region in different species varying from one base pair to about 80 base pairs in sea-urchin sperm.

   On an average, nucleosomes repeat at an interval of about 200 nucleotide pairs. Thus each human cell with 6×10^9 DNA nucleotide pairs contains 3×10^7 nucleosomes.

Nucleosome Packing or Formation of Chromatin Fibre

Nucleosomes are packed upon one another to form **a chromatin fibre** or **nucleoprotein fibre** with a diameter of about 30nm. It appears to have a beaded appearance. The beads are **nucleosomes**. These are connected by linker DNA.

The 11 nm nucleoprotein fibre represents the first level of organisation of chromatin and is seen in interphase nucleus. The thick chromatin fibre of 30nm arises during early prophase of cell division by spiral coiling of thin 11nm chromatin fibre. The 30nm fibre has a solenoid type of ultrastructure and has 6 to 7 nucleosomes per turn.

This type of packing requires one molecule of histone H1 per nucleosome. H1 gives polarity to chromatin fibre. The 30nm fibre is called a **solenoid**. This solenoid is further folded and packed into the **supersolenoid** of 300nm diameter. In mitotic chromosomes, the supersolenoid is further condensed to produce the final shape and dimension of metaphase or anaphase chromosome. In metaphase chromosome, the two daughter DNA molecules are separately folded to produce two sister chromatids; held together by a single centromere. Each chromatid is about 700nm thick.

Chromosomes

Different Regions Recognised in Chromosomes

1. **Primary Constriction and Centromere :** A part of the chromosome is marked by a constriction. It is comparatively narrow than the remaining chromosome and is known as *primary constriction*. Its position is constant for a given chromosome and forms a feature of identification. The primary constriction divides the chromosome into two *arms*. It shows a faintly positive Feulgen reaction, indicating presence of DNA of repetitive type. This DNA is called *centromeric heterochromatin*.

   Centromere or kinetochore lies in the region of primary constriction. The microtubules of the chromosomal spindle fibres get attached to the centromere during metaphase and anaphase. Therefore, centromere is associated with the chromosomal movement during cell division.

A. Types of Chromosomes Based on the Number of Centromeres

Usually, there are two centromeres one on each chromatid but there may be more or none.

Depending upon the number of centromeres, the chromosomes may be:

- **Monocentric** with only one centromere.
- **Dicentric** with two centromeres.
- **Polycentric** with more than two centromeres.
- **Acentric** without a centromere. Such chromosomes represent freshly broken segments of chromosomes which do not survive for long and are digested into their constituent nucleotides.
- **Diffused** or **nonlocated** are with indistinct centromere which remains diffused throughout the length of chromosome. The microtubules of spindle fibres are attached to the chromosome arms-at-many points. The diffused centromeres are found in insects, some algae and some groups of plants, (e.g. *Luzula*).

B. Types of Chromosomes Based on the Position of Centromere

Based on the location of centromere the chromosomes are categorised as follows:

- **Acrocentric :** These are rod-shaped chromosomes with centromere occupying a terminal position. They have only one arm, the other is absent.
- **Telocentric :** These are rod-shaped chromosomes having subterminal centromere. One arm is very long and the other arm is very short.
- **Submetacentric :** These are S-shaped chromosomes with centromere slightly away from the mid-point so that the two arms are unequal.
- **Metacentric :** These are V-shaped chromosomes in which centromere lies almost in the middle of chromosome so that the two arms are almost equal.

Ultrastructure of Centromere

Under electron microscope the centromere appears as a plate-like or cup-like disc, plastered upon the primary constriction. It is about 0.20 – 0.25μ in diameter and is formed of some non-chromatin material. In cross section it usually appears to consist of:

- **An electron dense layer :** It is 30 to 40nm thick with convex outer surface. Microtubules of spindle fibres get attached to it and penetrate it reaching the chromatin fibres.
- **An inner less dense layer :** It is 15 to 30nm thick which lies between the electron dense layer and the underlying chromatin fibres.
- **An outer fibrillar material :** It is on the convex face of centromere and forms a kind of **electron dense corona**.

Functions of Centromere

The centromere is considered to perform following two functions:

- It provides attachment to the microtubules of chromosomal spindle fibres and helps in chromosomal movement during cell division.
- The centromere serves as a nucleation centre for the polymerisation of tubulin, the protein used in the formation of microtubules. It might help in the formation of microtubules. Thus it might help in the formation of spindle fibres during prometaphase and metaphase.

2. Secondary Constriction or Nucleolar Organiser

Sometimes one or both the arms of a chromosome are marked by a constriction other than the primary constriction. During interphase this area is associated with the nucleolus and is found to participate in the formation of nucleolus. It is, therefore, known as **nucleolar organiser region** or the **secondary constriction**. Its location on the chromosome is marked with a lightly stained constricted region.

3. Nucleolar Organiser Region (NOR)

In certain chromosomes, the secondary constriction is intimately associated with the nucleolus during interphase. It contains genes coding for 18S and 28S ribosomal RNA and is responsible for the formation of nucleolus. It is, therefore, known as **nucleolar organiser region** (**NOR**). Its location in the chromosome is marked by a lightly stained constricted area and is constant for a given chromosome. In man, the nucleolar organisers are located in the secondary constriction of chromosomes 13, 14, 15, 20 and 22.

4. Tertiary Constriction

The tertiary constrictions are present in nearly all the chromosomes. Their significance is not known. However, these help to distinguish one chromosome from the other.

5. Telomeres

The tips of the chromosomes are rounded and sealed and are called **telomeres**. These provide stability to the chromosomes and protect their individuality, because the telomeric ends do not form any permanent association with other parts of the homologous or nonhomologous chromosomes, whereas the broken ends may join.

The terminal part of a chromosome beyond secondary constriction is called **satellite**. It is attached to the main body of chromosome by a delicate chromatin filament. The satellite may appear as a rounded or elongated knob. It has a constant shape and size for a particular chromosome. The chromosome with satellite is known as **sat-chromosome**.

6. Chromatids

At metaphase stage a chromosome consists of two chromatids joined at the common centromere. In the beginning of anaphase when centromere divides, the two chromatids acquire independent centromere and each one changes into a chromosome.

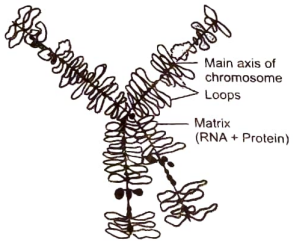


Fig. Structure of a lampbrush chromosome

Special Types of Chromosomes

Lampbrush Chromosomes

History : Lampbrush chromosomes were first observed by **Flemming** (1882) in amphibian oocyte. A detailed study was made by **J. Ruckert** (1892) in the oocytes of sharks.

Occurrence: Lampbrush chromosomes are found in the oocytes of insects, sharks, amphibians, reptiles and birds which produce large and yolky eggs. These have also been found in plants and some invertebrates like *Sagitta*, *Sepia* and *Echinaster*.

Size : Lampbrush chromosomes are large enough to be seen under light microscope. They may be as long as 1,000μ or more and about 20μ in width. In salamander oocyte, they may attain a length of about 5,900μ.

Structure : A lampbrush chromosome (in diplotene stage) consists of two homologous chromosomes which are in contact only at certain points, called **chiasmata**. Each chromosome of the pair is formed of two **chromatids** which lie parallel and form the **chromosomal axis** or the **main axis**. The axis is differentiated into alternate regions of high density loops which are lightly coloured, and arise on both sides of the chromosomal axis. The chromosomal axis, the chromomeres and the loop axis all are formed of DNA.

The chromomeres are found in pairs, one chromomere on each chromatid. These are about 0.25 to 2.0 μm in diameter and are spaced about 2μm from centre to centre along chromatid axis. These probably represent heterochromatic regions where axial filament remains tightly coiled.

The lateral loops arise from the chromomeres either in two or in multiples of two. These extend on either side of the chromosomal axis about 550μm and are about 30 – 50Å (3–5 nm) in diameter. Each loop consists of an axial fibre formed of DNA. It is surrounded with the matrix composed of RNA and proteins. This gives fuzzy appearance to lateral loops.

Polytene Chromosomes

Occurrence

Polytene chromosomes are special types of interphase chromosomes that are large and visible with the naked eye. These giant chromosomes are found in the cells of salivary glands of *Drosophila* and *Chironomus* larvae and in the cells of fat bodies of larval stages of certain dipterans. They were first observed in the salivary glands of *Chironomus* and hence are called **salivary gland chromosomes**.

Nature : Polytene chromosomes are **multistranded**, being composed of a large number of chromonemata.

Kodani (1942) : Believes that the paired homologoues consist of four chromonemata (similar to somatic chromosomes), which have become structures swollen by the lateral enlargement of chromotin material and form bands. They are darkly coloured and are separated by light coloured interbands.

1. Bands, Interbands and Genes

A polytene chromosome presents a distinct pattern of transverse banding. It consists of dark coloured **bands** alternating with the light coloured **interbands**.

The bands are Feulgen positive, rich in DNA and, therefore, darkly stained. They are regarded to be chromomeres where DNA fibre is supercoiled. Adjacent chromatid fibres are aligned by their chromomeres forming dark bands. The bands are specific for a particular chromosome and banding pattern is almost similar in both the homologous chromosomes of a pair. Therefore, during meiosis the pairing between homologous chromosomes is band to band.

Initially, it was believed that each band represented a single gene. Now it is estimated that an average band contains about 30,000 base pairs or 20 times the amount of DNA required to code for an average **protein**. It means each band is **polygenic.**

2. Chromosome Puffs and Gene Expression (Balbiani Rings)

The structure of giant chromosomes is modified in certain regions due to enlargement and extension of bands into lateral loops. Such enlargements or puffs are known as **Balbiani rings** or **chromosomal puffs**.

The puffs form large rings around the chromosomes and are presumed to be formed by the unwinding or uncoiling of chromonemata, which are otherwise closely folded or coiled in the bands. These then project out into a series of loops. These loops increase the thickness or diameter of the chromosome and provide fuzzy appearance.

The formation of puffs is controlled by specific genes and takes place at definite time. They are concerned with the synthesis of RNA because besides DNA and protein, puffs contain large amount of RNA also. The puffs are chiefly associated with the metabolic activities of the chromosomes because of high concentration of RNA.

The puffs have been observed not only in the giant chromosomes of salivary glands of *Drosophila* but also in different chromosomes in different tissues of dipterans.

Supernumerary Chromosomes

In certain plants and animals one or more additional chromosomes were observed in addition to the normal number. Such accessory chromosomes are very small and generally inert, being composed of heterochromatin. Normally, their presence in the nucleus does not affect the phenotype but if they are too many they reduce the fertility and vigour.

Physiology of Chromosomes

Chromosomes are known as hereditary vehicles. They are formed of strands of DNA molecules which contain information for the development of different characteristics and performance of various metabolic activities of the cells. The coordination of various functions is brought about through the formation of enzymes which are complex protein molecules. The information for the synthesis of these protein -molecules is contained in the DNA molecules in the sequence of nitrogen bases. The sequence of three nitrogen bases that codes for one molecule of a particular amino acid constitutes a *triplet codon*. DNA of a particular cistron transcribes its **mRNA**. It carries genetic code similar to its DNA. It moves out of the nucleus and gets attached to the ribosome. The tRNA molecules pick specific activated amino acids and recognise the codons on the mRNA by their specific anticodons. The amino acids are thus linked together forming a polypeptide chain, which is finally changed into functional protein either singly or by the association of several polypeptide chains.

Molecular Structure of DNA

DNA is a long chain **polymer** was understood in late 1930s. But only in 1953, **D.S. Watson** and **F.H.C. Crick** presented a working model of DNA. This model illustrates not only its chemical structure but also the mechanism by which it duplicates itself.

Nucleosides : A nitrogenous base with a molecule of deoxyribose (without phosphate group) is known as deoxyribonucleoside or nucleoside. The nitrogenous base is attached to first carbon atom C–1′ of deoxyribose through N–glycosidic bond.

Nucleotides are the Monomers of DNA

A nucleotide is formed of one molecule of deoxyribose, one molecule of phosphoric acid and one of the four nitrogenous bases. Thus a nucleotide is a **nucleoside monophosphate** (dNMP).

Nucleotides are Covalently Linked to form Polynucleotide Chain

DNA is a macromolecule formed by the linking of several thousands of nucleotides. These are called **monomers** or **building blocks** of DNA. In a nucleotide, the phosphate (phosphoric acid) molecule is attached to fifth carbon atom (C–5′) of the deoxyribose molecule. The phosphate molecule of a nucleotide is joined to the third carbon atom of the deoxyribose of the next nucleotide by **covalent bond**. Such 5′ – 3′ linkages between adjacent nucleotides are called **phosphodiester bonds.**

Biological Significance of Watson and Crick's Double Helical Model of DNA

The biological significance of Watson and Crick's double helical model of DNA is as follows :

1. **Information Storage :** Genetic material is expected to act as a repository of information. Its structure must be related to the intimate features of gene-product, i.e. the protein.
2. **Information Transfer :** The process of transferring message encoded in DNA and ultimate synthesis of polypeptide chain can be broadly separated into two steps: Transcription and Translation.
3. **Self Replication :** Each strand of double helix of DNA can construct its complementary strand by acting as a template, because the sequence of nitrogenous bases in one strand could easily determine the laying down of bases of the second or complementary strand.
4. **Variations :** Genetic material should be capable of undergoing change in its composition. Its structure should be able to undergo mutations so as to produce diversity in the protein-making information.
5. **DNA Repair :** DNA molecule is relatively fragile and is easily damaged simply by bending or shear forces or by local change in pH. Its base sequence changes by chemical agents from environment and also by X-ray and ultraviolet radiations. Still it preserves its integrity after several millions of replications. Actually, the double helical structure of DNA presents the most stable structure. It is most suited for the long term preservation of the coded information. Because the base sequence in the two polynucleotide chains is complementary, the coded information in a DNA molecule are preserved in duplicate. If anyhow, the sequence of bases on one chain is disturbed, the correct sequence of the other chain may help in correcting the errors that occur in the complementary strand and sends the correct message.

Denaturation-renaturation

The double-stranded structure of DNA can be melted in solution by increasing temperature or decreasing salt concentration or by using certain denaturation reagents. The phenomenon of separation of two strands of DNA molecule by heating is known as **denaturation** or **melting**. Since the two strands of DNA molecule are held together by weak hydrogen bonds, these get separated easily.

The strands of a DNA molecule separate over a temperature range. The midpoint of this range is called **melting temperature** and is represented by T_m. It is influenced by base composition. DNA rich in G-C pairs melts at a higher temperature.

Denaturation of DNA may be caused by some chemicals which either break hydrogen bonds or weaken hydrophobic interactions. These are called **denaturants**.

When heated DNA solution is slowly cooled, the two strands pair again forming double helix. This is known as **renaturation** or **reannealing**. This property of DNA was discovered by **Julius Marmur** and coworkers at Harvard University in 1960. This has helped in synthesising hybrid DNA. The technique of hybrid DNA synthesis is utilised in establishing genetic similarity between various taxonomic groups. More length of hybrid DNA denotes taxonomic nearness.

Heteroduplex DNA

Hybrid DNA formed as a result of recombination is called **heteroduplex DNA**. It contains mismatched base pairs or heterologous base sequences. Heteroduplex DNA is also called **heterozygous DNA**.

Single-Stranded DNA

Sinsheimer (1959) isolated unusual form of DNA from a small bacteriophage ϕ 174 which attacks *E. coli*. This DNA contains 5,500 nucleotides and is formed of a single strand. It does not have helical configuration. Its molar ratio of bases does not show the equivalence of **A** and **T** or **G** and **C**. Subsequent evidences indicated that the DNA of bacteriophage ϕ 174 is circular and consists of a single strand.

The process of replication of single-stranded DNA is also very interesting. On entering *E. coli*, its *DNA polymerase* synthesises a 'minus strand'. This is complementary to the plus parental strand and the two together from a **'Plus-minus' DNA helix**. This is known as **replicative form (RF)**.

In Vitro Synthesis of DNA

In 1967, **A. Kornberg** and his coworkers were able to synthesise DNA molecules outside the cell in a test tube. They isolated enzyme ***DNA polymerase*** from *Escherichia coli* bacteria that could be used for *in vitro* synthesis of DNA. The reaction mixture in the test tube contained the ***polymerase enzyme***, four different types of **deoxyribonucleotides** in the form of triphosphates (i.e. deoxyadenosine triphosphate, deoxythymidine triphosphate, deoxycytidine triphosphate and deoxyguanosine triphosphate), magnesium ions and some previously formed DNA. The amount of DNA synthesised was about 20 or more times greater than the amount of initial DNA, The enzymes hook together the freely available nucleotide units into a DNA strand. The deoxyribonucleotide triphosphates are added to the hydroxyl group at number 3′ carbon atom of deoxyribose sugar at one end of the chain.

The nitrogenous base of the newly added triphosphate is then spontaneously bounded to the complementary base of the primer strand through hydrogen bond. A phosphodiester bond is then formed between 3′–OH group of the newly added nucleotide and the 5′– PO_4 group of the next arriving nucleotide. This shows that the new and complementary nucleotides are added to the 5′ → 3′ strand from 5′ end and move towards the 3′ end.

Supercoiling of DNA in Viruses and Prokaryotes

Supercoiling of DNA was discovered by **Vinograd** and coworkers in 1963. The DNA molecule in viruses and bacteria is twisted upon itself. The supercoiled DNA molecules sediment more rapidly and are more compact. The coiling of DNA could be of following two types:

1. **Negative Supercoiling:** When DNA molecule is coiled or twisted in a direction opposite to the direction of twisting of duplex, the two polynucleotide strands tend to unwind. This type of supercoiling that results in unwinding is called negative supercoiling. Negative supercoiling is found in circular DNAs found in mitochondria, viral and bacterial chromosomes.
2. **Positive Supercoiling:** When DNA molecule is overwound it presents positive supercoiling. This is found in eukaryotic chromosomes. It makes eukaryotic DNA to be compacted into a limited space in the nucleus. Enzymes ***topoisomerases*** uncoil or unwind supercoiled DNA in both prokaryotes and eukaryotes. Supercoiling plays key role in allowing DNA to be compacted so as to fit inside the limited space.

Repeated DNA Sequences in Eukaryotic DNA

Roy Britten and **David Kohne** found that mammalian genome has numerous DNA fragments that have repeated nucleotide sequences. These are separated into three classes:

1. Highly repeated DNA sequences
2. Moderately repeated DNA sequences
3. Non-repeated DNA sequences

1.Highly Repeated DNA Sequences

In vertebrate genome about 10 per cent of total DNA is formed of **repeated fractions**. These fractions are formed of a few hundred nucleotides in which a given sequence of nucleotides is repeated over again and again without interruption. These show tandem arrangement. The highly repeated sequences fall into three categories : **satellite DNAs**, **minisatellite DNAs** and **microsatellite DNAs**.

***(i)* Satellite DNAs :** These are short nucleotide sequences having 5 to 1,000 base pairs. These sequences are clusteres of tandem repeats. They are repeated over again and again forming regions of inert DNA fragments with up to 100 million base pairs. These are called **satellite DNAs** because DNA fragments with these sequences can be separated into distinct satellite bands during density gradient centrifugation.

Satellite DNA is inert and is permanently coiled to form heterochromatin. The satellite DNA around the centromere of chromosomes is presumed to have some structural role providing attachment to the spindle fibres during cell division.

***(ii)* Minisatellite DNAs or VNTRs :** Minisatellites are DNA segments consisting of short tandem repeats and have only fewer copies. They are also called **variable number of tandem repeats (VNTRs)** because the number of repeats in a given minisatellite varies greatly among individuals.

Typically each VNTR is formed of 5 to 50 tandem repeats and each repetitive segment of about 15bp or more. For example, one human VNTR has 64bp. Some hyper-variable VNTRs have as many as 1,000 repeats. Minisatellite sequences are considerably shorter than satellite sequences.

Minisatellites are found scattered over the genome, but are concentrated close to the telomeres. They are non-coding DNA segments. However, because of highly variable nature, VNTRs are used for the identification of individuals by **DNA fingerprinting**.

***(iii)* Microsatellite DNAs :** These are shortest nucleotide sequences formed of 2–5 base pairs and are repeated about 100 times or less. These are scattered throughout DNA length. At least 30,000 different microsatellites are known in human genome. For example, in human beings a gene FMRI under normal conditions has 5 to 60 copies of a triplet CCG. If their repeat number rises above 200, the individual is mentally retarded. This condition is called fragile X syndrome because the microsatellite is present in the fragile site on X-chromosome.

2.Moderately Repetitive DNA Sequences

This fraction of DNA includes sequences that are repeated in the genome from a few times to several hundred thousand times. It may form 20 to 80 per cent of the total DNA. The moderately repeated sequences are of coding type and noncoding type.

***(i)* Moderately Repetitive Noncoding DNA Sequences:** These form bulk of moderately repeated DNA sequences. These do not code for any product. These are found scattered throughout the genome and do not form clusters as in the case of highly repetitive sequences. These are mostly separated into two classes, namely **short interspersed elements (SINEs)** with less than 500 base pairs and **long interspersed DNA elements (LINEs)** with more than 1,000 base pairs.

***(ii)* Moderately Repetitive Coding DNA Sequences:** These fragments of DNA include multiple copies of those genes that code for transfer RNAs, ribosomal RNAs and chromosomal proteins, the **histones**. This DNA constitutes about 25 per cent of human DNA. Each type of these repeated coding sequences are present in tandem array. Since tRNAs and rRNAs are needed in large quantities, multiple copies of their genes are found in the genome. For example, prokaryotic cells have about half a dozen copies of genes for rRNA and tRNA, whereas in eukaryotic cells, hundreds or thousands of copies of these genes are present.

3.Non-repeated or Non-repetitive DNA Sequences

Extension : Noncoding DNA

The moderately repetitive noncoding DNA is more abundant than the moderately repetitive coding DNA in eukaryotic cells. It is also called 'junk DNA'. It is represented in two forms :

- **Short Interspersed Elements (SINEs)** are highly repetitive DNA segments, consisting of thousands to millions of copies of noncoding sequences. They form about 10% of human genome. The best known example of SINE is the 300 bp long **Alu element**.
- **Long Interspersed Elements (LINEs)** are larger and more numerous than SINEs. Mammalian genome has about 20,000–50,000 copies of the LINE–I family. They are considererd to be derived from retrovirus-like ancestors. LINEs contain sequence rearrangements that disrupt the coding sequences rendering them nonfunctional.

These are single copy DNA sequences which include functional or coding sequences. It means these sequences represent genes that code for virtually all enzymatic proteins associated with inheritance. These are always localised to a particular site on a particular chromosome. These are also called **unique sequences**.

Selfish DNA and Junk DNA

Transposable segments of DNA, that are solely concerned with their own survival and replication and are of no use to the host cell in any way, are referred to as '**selfish DNA**'. It accumulates in the genome of slow growing multicellular organisms. On the other hand **junk DNA** is the defective selfish DNA that can no longer move from its location and can not express itself or its gene. It consists of repetitive noncoding sequences, including some introns.

Origin of Selfish DNA and Junk DNA

The eukaryotic cells can have foreign DNA from two different sources :

1. Viral DNA that gets inserted into the chromosome of eukaryotic host cell.
2. Transposable elements of mobile DNA.

The accumulation of selfish DNA depends on replication and reinsertion of selfish DNA. In rapidly dividing, single-celled organisms, selfish DNA tends to be eliminated, whereas in slowly dividing multicellular organisms it gets accumulated. Though originally useless, some defunct selfish DNA sequences may take up the role of maintaining chromosome structure.

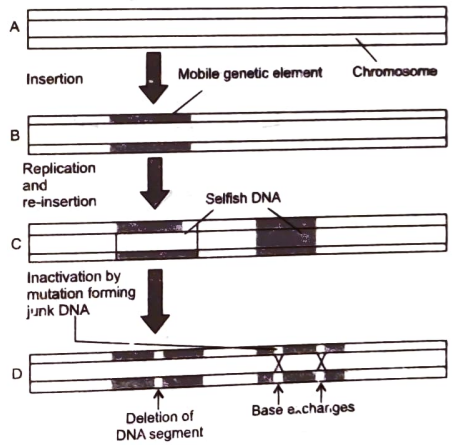


Fig. Origin of selfish DNA and junk DNA

The origin of repetitive DNA or junk DNA involves following processes:

- The original sequences of either viral DNA or transposable elements get inserted into the chromosome of eukaryotic cell.
- Both types of elements replicate and the duplicate copies get inserted at more locations in the chromosomes of infected cell.
- Many of these sequences mutate producing family of related sequences, most of which become nonfunctional because of mutations.

Palindromes and Inverted Repeats

Palindromes or palindromic sequences are sequences of nitrogenous bases that read the same backward and forward. Since DNA is double-stranded, two types of palindromes are theoretically possible.

1. **Mirror-like palindromes** that are similar to palindromic words and that read the same forward and backward on the same strand.
2. **Inverted repeat type of palindromes** are those DNA sequences that are the same when read in 5′ → 3′ direction in both the strands of a DNA, i.e. the sequence reads the same forward on one strand and backward on the complementary strand.

Inverted repeats are more common and of major biological significance. They act as recognition sites on DNA for the binding of a variety of proteins. Regulatory proteins, restriction enzymes and also the modification enzymes recognise inverted repeats.

Mitochondrial DNA (mtDNA)

Mitochondrial DNA (mtDNA) is a double-stranded circular DNA. It occurs in two to six copies per mitochondrion and is attached to the inner mitochondrial membrane and lies in the mitochondrial matrix. The total amount of mtDNA is much less than 1 per cent of the total cellular DNA of a eukaryotic somatic cell. In size a mtDNA resembles the DNA of a small bacteriophage. It has a molecular weight of only about 10 million.

mtDNA differs in base composition from nuclear DNA of the same cell. mtDNA is responsible for the synthesis of mitochondrial proteins. Therefore, it carries out synthesis of mRNA, tRNA and rRNA. However, the transcription and translation activities of mtDNA are under the control of nuclear DNA.

Mitochondrial DNA is presumed to have originated from some free-living bacteria-like ancestors. Its size gradually reduces from lower to higher organisms.

Chloroplast DNA (ctDNA)

Like mitochondria, chloroplasts also possess double-stranded circular DNA. It is represented by several copies that lie in the matrix. The functioning of ctDNA is controlled by nuclear DNA.

Catalytic DNA (Deoxyribozymes)

Single-stranded DNA molecules are found to greatly accelerate transformations in a substrate, a variety of chemical reactions occuring in the cells and reactions during DNA cloning. DNA are found to catalyse cleaving of RNA molecules using cofactors such as histidine.

The single-stranded DNA molecules, acting as deoxyribozymes form intricate structures by forming binding pockets for small molecules. This enhances their catalytic activity.

Significance of Cataltytic DNA

The study of catalytic DNA can enhance our understanding of enzymatic catalysis and biopolymer function in general and also in the development of useful biotech tools.

DNA Replication

Replication is an essential property of genetic material. DNA replication is the process whereby DNA makes exact copies of itself. Replication occurs at the rate of between 50 nucleotide per second (in mammals) and 500 nucleotides per second (in bacteria).

Watson and **Crick** suggested a very simple mechanism of DNA replication on the basis of its double helical structure.

Each strand of the double helix DNA serves as a template or model on which its complementary strand is synthesised. This method of DNA replication is described as semi-**conservative**, because each daughter DNA molecule is a hybrid conserving one parental polynucleotide strand and the other one newly synthesised strand.

Basic Rules for DNA Replication

The basic rules that are followed strictly during DNA replication can be summarised as follows:

1. There is specificity of base pairing during DNA replication, i.e., adenine pairs with thymine and guanine with cytosine.
2. Nucleotide monomers are added one by one to the 3′ end of the growing strand by the enzyme *DNA polymerase.*
3. The sequence of nitrogenous bases in each daughter strand being formed is complementary to the base sequence in the template strand.
4. C-3′ carbon of deoxyribose present on the 3′ end of the new polynucleotide chain of DNA has OH group and is free to bind to another nucleotide. C-5′ carbon of deoxyribose on the 5′ end of polynucleotide chain has a phosphate. Therefore, the new polynucleotide chain is always synthesised in 5′ → 3′ direction.

Basic Requirements for DNA Replication

The cells ready to undergo DNA replication are required to possess following precursors, primers, enzymes and proteins :

1. **Precursor Nucleotide Molecules :** Cells need all the four deoxyribonucleotide 5′-triphosphates (dNTPs) namely dATP, dGTP, dCTP and dTTP.
2. **Template DNA :** A single strand of DNA is needed which determines the sequence of nucleotides during polymerisation of new strand.
3. **RNA Primer :** It is needed for the initiation of polymerisation of new polynucleotide strand.
4. **Enzymes :** Enzymes required for DNA replication are **DNA polymerases, polynucleotide ligases and helicases**, etc.
5. **Proteins :** A number of proteins are associated with DNA replication. They help in identifying origin of replication, recognising initiation point, unzipping of nucleotide strands, stabilising the unzipped single DNA strands and recoiling of new or daughter DNA molecules.

DNA Replication is Semi-Conservative

There can be three possible modes of DNA replication :

1. **Semi-conservative Replication :** In semi-conservative mode of replication, each replicated DNA molecule consists of one old and one new strand. During this replication, the double helix of DNA unzips, its two polynucleotide strands separate and new complementary strand is synthesised on each of the two parental template strands. This mechanism ensures the faithful copying of genetic information at each cell division.
2. **Conservative Replication :** In this method of replication, the two newly synthesised polynucleotide chains or strands are brought together. These join to form a new DNA molecule, while the parental strands reassociate to form the parent DNA molecule. It means original DNA helix retains itself while the new one is formed as its copy.
3. **Disruptive Replication :** In this case parental strands are dispersed in the two daughter DNA helices so that each strand is formed of some parts of parental DNA and some parts of new DNA. This would involve cleavage of parental strands during replication.

Synthesis of New DNA Strands Always Proceeds in 5′ → 3′ Direction

During DNA replication, the polymerisation of new strands on both the parental strands (3′ → 5′ strand and 5′ → 3′ strand) proceeds only in 5′ → 3′ direction. This is because of the following two reasons :

1. DNA polymerase enzyme can carry out polymerisation in 5′ → 3′ direction only.
2. When DNA is synthesised in 5′ → 3′ direction, the energy required for polymerisation is derived from the hydrolysis of triphosphate group of the free dNTP when it is being added to 3′ hydroxyl group (3′ OH) of the growing polynucleotide chain. If the nucleotides are added in the 3′ → 5′ direction, the energy of polymerisation would be derived from the hydrolysis of 5′ -triphosphate group of the terminal nucleotide already added to the growing strand of DNA. This would eliminate the possibility of proof reading, because removal of a mismatched terminal nucleotide would also remove the 5′ -triphosphate group needed for polymerisation and further chain elongation.

The Two Strands of DNA are Assembled in Different Ways

(Semi-discontinuous Replication of DNA)

Anti-parallel strands of DNA exhibit continuous and discontinuous DNA synthesis. This is called semi-discontinuous replication. It their replication on 3′ → 5′ strand and on 5′ → 3′ strand occurs differently and includes following two procedures :

1. Leading Strand Synthesis or Continuous Strand Synthesis

Continuous replication occurs in 5′ → 3′ direction using 3′ → 5′ strand of parental DNA as template. DNA strand synthesised in 5′ → 3′ direction is called leading strand.

2. Lagging Strand Synthesis

Lagging strand is synthesised on 5′ → 3′ strand of template DNA in short segments of 1,000 to 2,000 nucleotides in 5′ → 3′ direction. These segments are called Okazaki fragments. This was described by **Reiji Okazaki** and colleagues. Synthesis of each Okazaki fragment begins with RNA primer, synthesised by primase and is carried out by polymerase III as in leading strand. Thus, synthesis of lagging strand of DNA is discontinuous and needs a series of RNA primers, one for each Okazaki fragment.

Molecular Mechanism of DNA Replication in Prokaryotes

In E. coli the entire process of DNA replication involves following four steps. Each step is governed by specific enzymes. Thus, more than a dozen enzymes and proteins are involved in initiation and completion of DNA replication.

I.Activation of Deoxyribonucleotides

The deoxyribonucleotide monophosphates (AMP, GMP, CMP, TMP) occur in the nucleoplasm. These are activated into triphosphates (ATP, GTP, CTP and TTP) by utilising ATP. This process is known as **phosphorylation** and is catalysed by the enzyme phosphorylase.

II.Initiation

DNA replication is initiated at a definite sequence of nucleotides, called initiation point or **origin of replication** (*oriC*). This consists of a specific sequence of 100 to 200 or more base pairs. In prokaryotic chromosomes, there is only one origin of replication per chromosome. Each point of origin along with the part of DNA to be replicated from it forms one **replicon**.

1. **Recognition of Initiation Point:** Specific initiator protein, **DnaA**, recognises the initiation point on DNA. A single complex of about 30 molecules of DnaA protein binds to the four repeats of 9 base pairs sequence in one of the strands of the duplex and initiates the separation of intertwined polynucleotide chains at all the three mers of 13 base pairs.
2. **Unwinding or Opening of DNA Double Helix:** Initiator protein, **DnaA**, after binding to the origin of replication, denatures DNA in the region of three repeats of 13 base pairs sequence. This process requires ATP and the bacterial histone like protein (HU). About 45bp are melted at the origin to open up the two strands. This initiates the formation of **pre-priming complex**, which is formed by five proteins, namely DnaA, DnaB DnaC, SSB and gyrase or topoisomerase. Firstly, the two hexamers of *helicase* enzyme along with **DnaB**, bind into each strand of unwounded DNA close to DnaA. Each DnaB is loaded onto DNA strand with the help of six DnaC proteins. Two molecules of protein DnaB unwind DNA bidirectionally and create two potential replication forks. A topoisomerase enzyme, called *DNA gyrase*, promotes separation of DNA strands by inducing negative supercoiling in advance of helicase.

   Strand separation is maintained by several molecules of single strand binding proteins (SSB) by preventing the renaturation of separated strands and stabilises them so that these can serve as templates.

III.Elongation (Synthesis of New Strands of DNA)

The synthesis of new strands on the separated strands of template DNA involves following six steps:

1. RNA Priming or Formation of RNA Primer

The *primase* enzyme (also called *DNA directed RNA polymerase*) binds to the first priming sequences on the leading strand and synthesises a short **primer RNA** segment consisting of 10–60 nucleotides only at the origin of replication complementary to the separated strand of DNA.

Synthesis of RNA primer is essential because *DNA polymerase* can not initiate synthesis of new DNA strand. It can only polymerise growth of DNA chain.

2. Assembling of Complementary Strands

Deoxyribonucleoside triphosphates pair with the appropriate nitrogenous bases of template DNA strands according to base pairing rule.

3. Conversion of Deoxyribonucleoside Triphosphates to Monophosphates

The deoxyribonucleoside triphosphate molecules on pairing with the nitrogenous bases of template strand se', free pyrophosphate (p ~ p) molecules and change to deoxyribonucleotides. The enzyme *pyrophosphatase* hydrolyses the pyrophosphates into inorganic phosphate groups (Pi) and releases energy.

4. Formation of New DNA Chains on RNA Primers (Polymerisation of DNA)

The energy so released is utilised in joining the adjacent nucleotides to form the polynucleotide chain. The process is catalysed by the enzyme *DNA polymerase* along with Mn^{++} and Mg^{++} ions.

The new strand of DNA is formed in the 5′ → 3′ direction on template DNA by the additon of deoxyribonucleotides to the 3′ end of primer RNA. The addition is carried out by the enzyme *DNA polymerase* III (poly III – copol III) in the presence of ATP. Once the synthesis of DNA strand is initiated, it proceeds continuously, keeping pace with the unwinding of DNA at the replication fork.

Therefore, the new strand on 3′ → 5′ template parent strand grows towards the replication fork in 5′ → 3′ direction by the addition of nucleotides to the growing 3′ end. It means the 5′ → 3′ strand grows as one piece and is called **leading strand**. Its synthesis is described as **leading strand synthesis**. In contrast, the new strand on 5′ → 3′ template strand elongates away from the replication fork. It is synthesised discontinuously as a series of short segments of nucleotides, called **Okazaki fragments** which are joined by the enzyme **DNA ligase**. Each Okazaki fragment is formed of 100 – 200 nucleotides in eukaryotes and 1,000 – 2,000 nucleotides in prokaryotes and needs its own RNA primer for its polymerisation. This 3′ → 5′ new strand synthesised discontinuously on 5′ → 3′ template parent strand is called lagging strand.

Since, the synthesis of leading strand is continuous and that of lagging strand discontinuous, the overall replication of DNA is said to be **semidiscontinuous**.

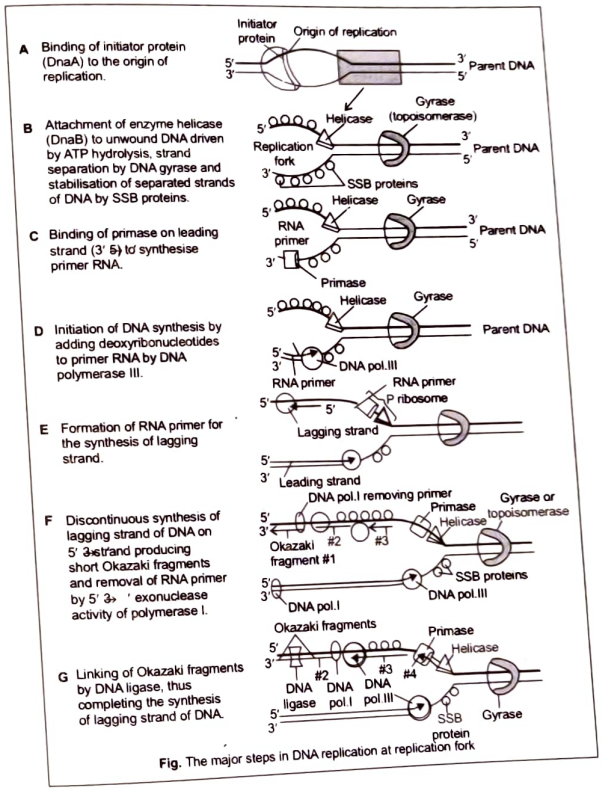


Fig. The major steps in DNA replication at replication fork

5. Excision of RNA Primers

Once a small segment of an Okazaki fragment has been formed, the nucleotides of RNA primer are removed from the 5′ end one by one by the action of 5′ → 3′ exonuclease activity of *DNA polymerase I* and are replaced with DNA nucleotides.

6. Joining of Okazaki Fragments

The gaps left between Okazaki fragments formed by the removal of RNA primer are filled with complementary deoxyribonucleotide residues by **DNA polymerase I**. Finally, the adjacent 5′ and 3′ ends of Okazaki fragments are joined by DNA ligase.

IV. Termination

With the completion of replication in the circular chromosome of E. coli, the two replication forks meet at a terminus region containing multiple copies of a 20-base pairs sequence called **Ter sites** (for terminus). The Ter sequences have a 23bp consensus sequence that binds to **tus** protein. They are arranged on the chromosome to create a sort of trap for replication fork and act as binding sites for protein **Tus** (terminal utilisation substance). The **tus-ter** complex arrests replication fork from only one direction. Finally, few hundred base pairs of DNA between tus-ter complex are replicated completing two topologically interlinked circular chromosomes.

Extension : Termination sites (Ter-sites)

DNA replication terminates at **Ter-sites** located in the termination region. *TerC*, *TerB* and *TerF* prevent clockwise movement of replication forks and *TerA*, *TerD* and *TerE* prevent counter clockwise movement.

V. Helix Formation

The two polynucleotide chains of daughter DNA molecules formed as a result of DNA replication become spirally coiled to assume double helical structure.

Mechanism of DNA Replication in Eukaryotes

The essential features of DNA replication are the same in both eukaryotes and prokaryotes except for the following differences :

1. There are many origins of replication in each eukaryotic chromosome. Thus, replication of eukaryotic DNA occurs simultaneously at several points. Therefore, a number of replication forks can be seen along the length of one replicating DNA molecule.
2. There are large number of replicons in a eukaryotic chromosome. Each replicon has about 50,000 to 3,00,000 nucleotide pairs. All replicons do not replicate simultaneously. Clusters of 20-80 replicons are activated together.
3. Each origin of replication consists of about 150 base pairs. These autonomously replicating sequences are called **replicators**.
4. Initiation of replication in all eukaryotes requires a multisubunit protein, the **origin recognition complex** (ORC)
5. **Replication bubbles** are formed at the origins due to separation of two strands of parent DNA at origins of replication.
6. In eukaryotic cells, *DNA polymerase-α* is a multisubunit enzyme with similar structure in all eukaryotic cells. it functions only in the initiation of a new strand. Two small proteins associated with it help in the synthesis of short primers containing RNA and a short segment of DNA for okazaki fragments of the lagging strand. This short segment of DNA is 4 nucleotide long and is called *iDNA* or ***initiator DNA***. ***DNA polymerase*** α lacks proof reading 3′ → 5′ exonuclease activity. These primers are extended by ***DNA polymerase*** δ, which is also a multisubunit enzyme. Along with specific protein, it functions analogous to the β-subunit of *polymeraseIII* of *E. coli*. Therefore, *DNA polymerase δ* has 3′ → 5′ proof reading exonuclease activity and also carries out synthesis of both lagging and leading strands.

   DNA polymerase ε functions during DNA repair, acts analogous to *polymerase*I at the replication fork and also removes primers of Okazaki fragments on lagging strand.
7. Termination of replication on linear chromosomes of eukaryotes involves synthesis of special structures called *telomeres* at the end of chromosomes with the help of enzyme *telomerase*.

Enzymology of DNA Replication

Studies on enzymology of DNA replication were first carried out by **Arthur Kornberg** and colleagues in 1957. They isolated an enzyme from lysates of *E. coli* that was able to direct DNA synthesis in *vitro*. This enzyme is now called *DNA polymerase* I. For this discoveries he was awarded by Nobel prize.

About 20 or more different proteins and enzymes are required during DNA replication. They collectively form a **DNA replicase system** or **replisome**.

The enzymes associated with DNA replication are :

- DNA polymerase enzymes
- Polynucleotide ligase
- Exonuclease

DNA Polymerase Enzymes in Prokaryotes

Enzymes that add successive nucleotides to a growing DNA strand are called **DNA polymerases**. These have three sites for attachment.

1. **Template site** for the attachment to the template DNA.
2. **5′ – triphosphate** site for attachment of deoxyribonucleoside 5′ –triphosphate, and
3. **Primer terminus site** for the attachment of the 3′ – OH end of the RNA primer. DNA polymerases are associated with the addition of deoxyribonucleoside triphosphate nucleotides to primer RNA in the direction from 5′ $-PO_4$ end to the 3′ –OH end.

There are three *DNA polymerase* enzymes that participate in the process of DNA replication and play distinct roles in the replication and repair of DNA. These enzymes are :

1. *DNA polymerase I* (Pol. I)
2. *DNA polymerase II* (Pol. II)
3. *DNA polymerase III* (Pol. III)

All the DNA polymerases show following fundamental properties :

1. All polymerases synthesise DNA in 5′ → 3′ direction adding a dNTP to the 3′ –hydroxyl group of the growing chain.
2. DNA polymerases can simply add a new dNTP to a preformed primer strand that is hydrogen bonded to template.
3. They can not initiate DNA synthesis *de novo* by catalysing polymerisation of free dNTPs.

1. **DNA Polymerase I :** It was the first enzyme discovered that could castalyse DNA synthesis. It was isolated from *E.coli* by **Arthur Kornberg** and his colleagues in 1957. It is roughly spherical with a diameter of about 6.5nm. It has a molecular weight of 1,90,000 D and is formed of a single polypeptide chain of about 1,000 amino acid residues. It possesses a sulphydroxyl group, single interchain disulphide and one zinc molecule at the active site.

   A *DNA polymerase I* molecule, in reality, is a complex structure formed of:

   (*a*) *DNA polymerase 3′→ 5′ exonuclease*

   (*b*) *5′ → 3′ exonuclease*

   (*c*) *3′ → 5′ proof reading exonuclease.*

   There are five specific binding sites on the spherical molecule of *DNA polymerase I*

   - *Template site* for binding the template DNA.
   - *Primer site* for binding primer strand of RNA.
   - *Primer terminus site* for 3′ –hydroxyl terminus of primer.
   - The *ε-subunit* proof-reads 3′ → 5′ for mistakes.
   - The *θ subunit* of *polymerase III* is associated with α and ε subunits to form the core polymerase enzyme.
   - The *subunit γ* of *polymerase III* is found in mitochondria. It carries out replication of mitochondrial DNA only.

Enzymes γ, δ, χ, ψ form **clamp-loading complex**. A single clamp-loading complex consists of six subunits of five types $\gamma^2\ \delta\ \delta'\ \gamma\ \psi$. The two core polymerases are linked together by a dimer of τ(tau) subunits. This dimeric polymerase then associates with a single clamp-loading complex, completing *DNA polymerase III*. *Polymerase α, β* and *ε* are most active in the dividing cells and are, therefore, associated with DNA replication.

DNA Polymerases in Eukaryotes

In eukaryotes, DNA polymerases have been isolated from yeasts, rat liver cells and human tumour cells. They have the following five types of DNA polymerases:

(*i*) **DNA polymerase α : *Alpha polymerase*** is a large, high molecular weight protein. It is found in both nucleus and cytoplasm and is, therefore, known as *cytoplasmic polymerase*. It is associated with the replication of mitochondrial DNA.

(*ii*) **DNA polymerase β :** The *beta polymerase* is found only in the nucleus and is called *nuclear polymerase* or *small polymerase*. It occurs only in vertebrate cells.

(*iii*) **DNA polymerase γ :** The gamma polymerase is encoded in the nucleus of mammalian cells.

Table : Subunits of *DNA polymerase III of E. coil*

Subunits	No of subunits per holoenzyme	Molecular wt. of subunit	Function of subunit
α	2	1,32,000	Polymerisation activity. It synthesises short primers containing RNA and DNA for Okazaki segments on lagging strand. – α
ε	2	27,000	3′ → 5′ exonuclease activity,
θ	2	10,000	carries out proof reading.–ε, θ
τ	2	71,000	Associated with stable template binding, core enzyme dimerisation. – τ
γ	2	52,000	Clamp-loading complex that loads β sub-units on lagging strand at each Okazaki fragment. γ, δ, δ′, χ, ψ
δ	1	35,000	
δ′	1	33,000	
χ	1	15,000	
ψ	1	12,000	
β	4	37,000	DNA clamp that slides over DNA, required for optimal processivity. – β

(*iv*) **DNA polymerase δ :** The *delta polymerase* enzyme is found in mammalian cells. Its action is assisted by proliferating cell nuclear antigen (PCNA). This enzyme, along with *polymerase* α, is associated with DNA replication.

(*v*) **DNA polymerase ε :** The *epsilon polymerase* enzyme is PCNA-protein independent enzyme. It occurs in mammalian Hela cells and in budding yeast cells. It was previously known as *DNA polymerase δII*.

2.Polynucleotide Ligase or DNA ligase Enzyme

DNA ligase can join the ends of two fragments of DNA strand by catalysing the synthesis of a phosphodiester bond between a 3′ – hydroxyl group at the end of one chain and 5′ – phosphate group at the end of other chain. *DNA ligase,* obtained from *E. coli,* is formed of a single polypeptide chain having molecular weight about 74,000 to 77,000. About 200 to 400 copies of this enzyme are present in each *E. coli* cell.

DNA ligase performs the following functions :

- Joins the Okazaki segments of lagging strand of DNA in the process of DNA replication.
- Repairs single-strand nicks in duplex DNA.
- Links the ends of linear DNA duplexes to yield circles.
- Joins segments of DNA in the process of recombination, which occurs during meiosis, transduction or in genetic transformation.

Proteins Associated with DNA Replication

At least nine different proteins participate in the initiation of DNA replication.

1. **Initiator Proteins :** These proteins recognise origin of replication in the two strands of DNA. DnaA protein recognises sequence of nucleotides in the origin of replication site (four repeats of nine nucleotides) and opens the duplex at specific site.

2. **DNA Unwinding Proteins or Helicases :** The unwinding proteins or **rep proteins** were discovered by **B. Alberts** and his colleagues in phage T_4 bacteriophage. A number of them are found both in prokaryotic and eukaryotic cells. Their molecular weight varies from 10,000 to 75,000. They possess specific binding sites for short segments of about 9 nucleotide residues. One particular protein called **DnaA** identifies origin of replication and initiates unwinding of the helix and separation of the two strands of DNA by breaking hydrogen bonding between base pairs of antiparallel strands of DNA. A number of molecules of this unwinding protein bind to each of the unit of binding site in advance of the formation of replication fork. Other unwinding proteins like **DnaB, DnaC**, etc., help in opening up and destabilising the DNA helix. These unwinding proteins are called **helicases**, because they open and destabilise DNA helix.

   These enzymes use energy from ATP to unwind DNA in advance of replication fork. In *E. coli* two different helicases are found. One helicase attaches to the lagging strand template and moves in 5′ → 3′ direction and the other attaches to the leading strand template and moves in 3′ → 5′ direction.

3. **Accessory Proteins**

   Two types of proteins, namely the **sliding clamp proteins** and **brace proteins** are required for **DNA replication**. These are also known as **clamp-loading proteins**. These load the *polymerase III* in *E. coli* and *polymerase* δ in eukaryotes onto the primer RNA and maintain a stable association with the template.

   1. The **brace proteins** are called **γ-complex** in prokaryotes and **replication factor C (RFC)** in eukaryotes. They recognise binding site at the junction of primer and template and bind to it.
   2. The **sliding clamp proteins** are β proteins in *E. coli* and proliferating cell nuclear antigens (PCNA) in eukaryotes. These proteins bind adjacent to the brace proteins and form a ring around the template DNA.

4. **Untwisting Proteins or Swivelases or Topoisomerases**

   DNA strands separation creates topological stress in helical DNA. This stress is relieved by *swivelases* or *topoisomerases*. These enzymes cause reversible breakage and rejoining of DNA strands. There are two types of topoisomerases :

   1. ***Type I topoisomerases*** break just one strand of DNA.
   2. ***Type II topoisomerases*** introduce simultaneous break in both the strands. Earlier they were named gyrases.

5. **Single Stranded DNA Binding Proteins (SSBP)**

   The separated strands of DNA helix are stabilised and protected from the hydrolytic cleavage of phosphodiester bonds by **DNA-binding proteins**. These are also called **single strand binding proteins (SSBPs)**.

6. **Primases**

   Primases are specific kind of DNA dependent RNA polymerases. They function only during DNA replication. They synthesise a small piece of RNA, called **RNA primer**, which serves as a primer on which deoxyribonucleotides are added and a new polynucleotide strand complementary to template strand of DNA is synthesised. It means primases can initiate formation of polynucleotide strand which DNA polymerases can not.

Replication Models

Following models of DNA replications have been identified in prokaryotes and eukaryotes:

1. **Theta (θ) Replication of DNA**

   This model of DNA replication was studied in the circular DNA of bacteria *E. coli*. Replication begins with the unwinding of DNA double helix at initiation point, also called **origin of replication**. An **initiator protein** binds at this point. Using energy from ATP, the initiator protein slightly unwinds the double helix at the initiation point, separating the two strands. This produces two forks, called **replication forks** or **Y-junctions**.

   In this replication, the circular DNA molecule does not break. It maintains its circular form throughout the replication process. However, in the replication of circular DNA an intermediate **theta structure** is formed, similar in shape to Greek letter thaet (θ). Therefore, DNA replication of circular chromosome in prokaryotes is called **θ replication**.

2. **Rolling Circle Model**

   Rolling circle model of DNA replication occurs in F⁺ or Hfr cells of bacteria, in F-plasmids and in some bacteriophages. To start with, a nick or break is formed in one of the strands by a site-specific endonuclease. This opens up one of the two strands of DNA helix. The open strand has a free 3′ – OH end and a free 5′ – PO_4 end. The complementary closed strand of circular DNA serves as a template and ***DNA polymerase*** adds nucleotides to the free 3′ end of the open strand. There is no need for a primer, because the original break produces a primer configuration (3′ – OH). As nucleotides are added to 3′ end of the open strand, the unbroken circular template strand rolls and the 5′ – PO_4 end of open strand is displaced as **tail**. Since the molecular shape created in this replication resembles Greek letter sigma (σ), the rolling circle replication has been called **σ-mode of DNA replication**.

3. **D-Loop Model**

   This type of replication of DNA is found in circular DNA molecules of chloroplasts and mitochondria. In them, origin of replication is located at different points on each of the two parental template strands. Replication begins on one strand at its origin of replication, displacing the other. The displaced strand forms a **displacement loop** or **D-loop**.

4. **Inhibitors of DNA Replication**

   A variety of substances inhibit DNA replication. These act at various levels during DNA replication.

   Some inhibitors interfere with nucleotide biosynthesis. Substances like **methotrexate** and **fluorodeoxyuridylate** interfere with nucleotide biosynthesis. **Methotrexate** is used for the treatment of cancers. It inhibits actions of dihydrofolate reductase causing accumulation of dihydrofolate in the cell.

5. **Fluorouracial** (5FU) also inhibits the activity of ***thymidylate synthetase*** causing depletion of dTMP and dTTP to the cells and consequently DNA replication. Therefore, 5FU is also used as a powerful anticancer drug.

DNA Repair

Necessity of DNA Repair

DNA molecules are quite fragile and are constantly subjected to radiations and intracellular physical and chemical stresses. Oxygen, UV light, X-rays, alkylating agents and free radicals all can cause damage and introduce changes in nucleotide sequences. The commonly occuring changes in DNA can be due to formation of pyrimidine dimers, alkylation or deamination of nitrogenous bases. Both spontaneous and induced damages in DNA, if left uncorrectred, may result in mutations that disrupt the function of essential enzymes or structural proteins and pose a significant risk to the existence of the cell.

Mistakes in DNA

Types of Mistakes

The various mistakes that are often introduced in the nucleotide sequences in DNA molecules can be classified into following categories:

- Single strand breaks in DNA
- Double strand breaks in DNA
- Alternation of bases such as intrastrand dimer formed by two adjacent pyrimidines (thymine dimer by UV radiation).

Table : Types and causes of DNA mistakes

Mistakes	Causes
1. Insertion of incorrect base in one strand (i.e. replacement of purine by pyrimidine or vice-versa)	Replication error Spontaneous deamination of pyrimidine C to U Spontaneous deamination of purine A to hypoxanthine.
2. Alteration of bases	Introduced by : Ionising radiations and UV rays. They lead to the formation of thymine dimers.
3. Loss of bases	Introduced by alkylating agents that cause depurination by breaking N-glycosidic bond.
4. Single strand breaks	Caused due to the breakage of phosphodiester bonds by chemicals (peroxides, sulphydryl containing compounds) by ionising radiation and by free radicals in water.
5. Double strand breaks	Are single strand breaks introduced in separate strands of DNA at the same or at different levels.
6. Cross-linking of two chains of the helix	Are antibody-mediated covalent linkages between a base of one strand and an opposite base in complementary DNA strand.

Agents Causing Mistakes and Damage

The various agents causing damage to DNA or introducing mistakes in nucleotides sequence can be placed under following categories:

1. **Radiation**

   Ionising radiation, for example, X-rays, naturally occurring cosmic rays, electron beams, γ-rays and radiation beams of various kinds used in radiotherapy are associated with breaks in polynucleotide strands of DNA.

   The strand breaks are caused by the action of secondary electrons produced by a β-particle or an X-ray photon or by the production of free radicals in water that can attack the phosphodiester bonds.

2. **Chemicals**

   Strong chemicals like alkylating agents such as methyl ethanesulphonate (**MES**), ethyl ethanesulphonate (**EES**) and nitrosoguanidine, etc., react primarily with guanine by adding an alkyl group to N-7 of purine ring. This weakens the N-glycosidic bond and causes **depurination**.

   Aflutoxin, benzopyrene, nitrosamines, nitrosourea, psoralen and cisplatin are some of the chemical agents or drugs which cause damage to DNA and introduce gene mutations.

Biochemical Mechanism of DNA Repair

Cells possess a variety of mechanisms to repair the damages in DNA structure, caused by various mutagenic agents. These are :

1. **Repair of Damage Caused by Incorrect base Insertion (Mismatch Repair)**

   During DNA replication, both the enzymes ***polymerase*** I and ***polymerase III*** occasionally incorporate wrong or incorrect base in the daughter strand under synthesis. A wrong base fails to form a hydrogen bond with the template base in the parental or template strand. Such errors are usually corrected by the editing function of these enzymes.

   The errors missed by the editing function of these enzymes are corrected by a second correcting system, called **mismatch repair**. In mismatch repair, a pair of non hydrogen bonded base is recognised as incorrect and a polynucleotide segment with unmatched pair is excised, thereby removing the unmatched pair. The gap is filled by ***polymerase I***.

The mismatch repair system requires a complex of three proteins, MutS, MutH and MutL. The protein MutS recognises and binds to the mismatched base. **MutH** binds to DNA at the hemimethylated GATC sites and cleaves the unmethylated (newly systhesised) strand. **MutL** binds to MutS and MutH and helps in final repair steps.

2. Repair of Thymidine Dimers

This can be discussed under the following heads:

Formation of Pyrimidine Dimers

UV radiations produce intrastrand photodimers between two adjacent pyrimidine bases of the same strand. The dimers result from the formation of **cyclobutyl ring** by the joining of 5th and 6th carbon atoms of two pyrimidine rings. Such cyclobutyl pyrimidine dimers are most common photodimers and thymidine dimers are commonest of all.

Effect of Dimers : The presence of thymine dimer of any other pyrimidine dimer causes distortion of DNA helix because the two thymines are pulled closer. This produces a '**kink**' in the polynucleotide strand in which dimer is formed. The kink causes inhibition of the advancement of replication fork and thus interferes with DNA replication.

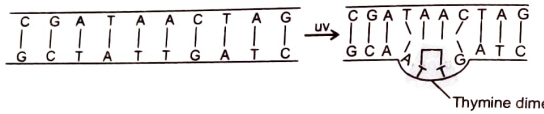


Fig. Distortion of DNA helix and formation of 'Kink' caused by thymine dimer

3. Mechanism of Dimer Repair : There are four major pathways for the repair of damage in DNA caused by thymidine dimer. These are separated into two categories :

1. Photo-reactivation or (light induced repair)
2. Light independent repair
   - Excision repair
   - Recombinational repair
   - SOS repair

(i) **Photoreactiviation or Light Induced Repair :** Photoreactivation is light induced enzymatic cleavage of pyrimidine dimer (thymidine dimer). It is catalysed by enzymes called ***photolyases*** or **PR enzymes.** They absorb radiant energy of wavelengths 300 to 600 nm and split the cyclobutyl ring between two pyrimidine molecules of the dimer. As a result, the cyclobutyl thymidine dimers are converted to pyrimidine monomers.

(ii) **Excision Repair or Dark Repair :** The excision repair mechanism is a multistep enzymatic process that can remove and replace individual nitrogenous bases (**base excision repair**) and also the longer nucleotide chains (**nucleotide repair**). This repair mechanism does not require sunlight. So it is also called **dark repair**.

Base excision repair includes enzymatic removal of chemically modified base (such as uracil, 5-methylcytosine or 3-methyladenine), followed by removal of the nucleotide. The gap thus created, is filled by a complementary base pair. It is carried out in the following steps with the aid of several enzymes.

(a) The enzyme ***DNA glycosylase*** removes the chemically modified base, causing apurination or apyrimidation.

(b) The enzyme ***endonuclease*** cleaves the phosphodiester bond in the sugar phosphate backbone at the apurinic or apyrimidinic site, forming a nick.

(c) Enzyme ***deoxyribose phosphodiesterase*** removes the deoxyribose phosphate molecule.

(d) The repair polymerase (***DNA polymerase β***) adds the single nucleotide to the 3′ end of the nicked strand at the gap.

(e) Enzyme ***DNA-ligase*** seals the nick by joining the two parts of the polynucleotide strand.

- **Nucleotide excision repair** operates to remove pyrimidine dimer (or thymidine dimer) and fills the gap by complementary nucleotides. It requires the entire DNA replication machinery. The enzyme ***repair-endonuclease*** recognises the distortion in the polynucleotide strand produced by a thymidine dimer. It cleaves the sugar-phosphate backbone of the DNA strand on both the sides of the kink. This exposes 3′ –OH group at the 5′ incision site. The oligonucleotide consisting of about 30 nucleotides is removed by ***DNA helicase***. The resulting gap is repaired by ***DNA polymerase δ*** **or** ***ε*** and is closed by ***DNA ligase***.

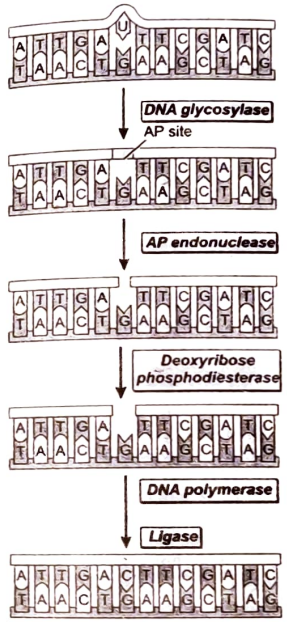


Fig. Diagrammatic representation of base excision repair when methylated cytosine, i.e. uracil enters DNA strand during replication

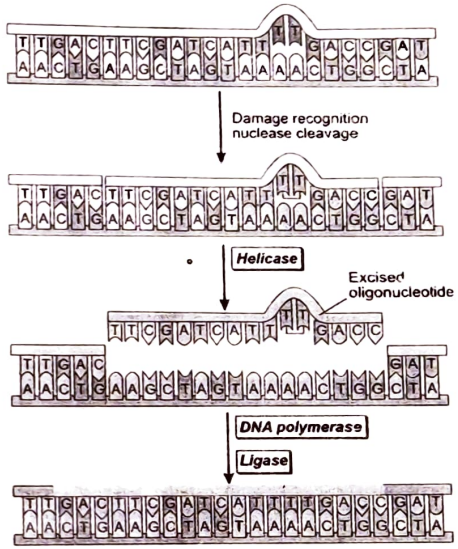


Fig. Nucleotide excision repair of a pyrimidine dimer utilising DNA replication machinery

(iii) **Recombinational Repair :** In recombinational repair, the undamaged strand of parental DNA is copied during replication to produce a normal daughter molecule. During DNA replication, ***polymerase III*** is stalled at the site of thymine dimer. The enzyme passes over the dimer block and initiates chain growth beyond the block. Therefore, the daughter strands have large gaps, one for each thymidine dimer. Complete normal double-stranded DNA molecules are formed from such fragmented and kinked DNA molecules by sister-strand exchange mechanism.

In single strand exchange, the gap produced in the daughter strand is filled by recombination with the undamaged parental strand. Due to this exchange, a gap is formed in the previously intact parental strand. This is filled by the action of DNA polymerase and ligase enzymes. The pyrimidine dimer is later on removed by excision repair to produce two intact normal double stranded DNA molecules.

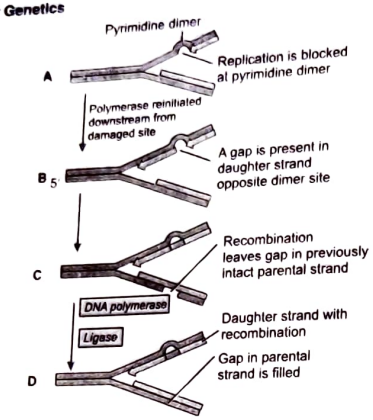


Fig. Post-replication repair by recombinations. **A.** Replicating DNA molecule with a pyrimidine dimer; **B.** DNA polymerase bypasses the dimer and starts postdimer synthesis of new strand; **C.** The resultant gap in the newly synthesised DNA strand is repaired by recombination with undamaged parental strand; **D.** Enzymes DNA polymerase and DNA ligase fill the resultant gap in the previously intact parental strand

(iv) **SOS Repair or Error prone Repair :** SOS repair is **a bypass repair system** that allows polymerisation to proceed across a dimer despite the distortion of the helix. During replication, the DNA polymerase complex bypasses the damaged site in the template strand and continues DNA synthesis. It inserts adenine nucleotides at the gap regardless of original DNA sequence. Thus, this repair system is potentially mutagenic.

In SOS repair, *RecA* protein binds tightly to the dimer strand. When ***DNA polymerase III*** encounters the dimer, *RecA* interacts with the ε unit of *polymerase III* which is responsible for DNA editing. As a result, ε unit fails to proof read and edit the insertion of mismatch base and allows replication to proceed overlooking the insertion of wrong base. In *E. coli,* proteins UmuC and UmuD bind to *DNA polymerase* and alter the stringency of nucleotide incorporation. Adenines are inserted into the newly synthesised strand at the lesion site irrespective of base sequence in the parental template strand. If the damaged site is thymidine dimer, the newly synthesised DNA is not mutated. But if the damaged site has cytosine dimer, the newly synthesised strand is mutated.

SOS repair system operates only under potentially lethal conditions caused by extensive UV irradiation. Therefore, SOS repair system works on the principle that survival with mutation is better than no survival at all. This repair is also called **error prone repair** or **translation replication**.

Extension : Role of genes umuC and umuD in SOS repair

The proteins produced by *umuC* and *umuD* gene family are necessary for dimer bypass. They perform following functions.

- They help in tight binding of *RecA* protein at the small distortion caused by pyrimidine dimer.
- They help in binding of *DNA polymerase* III to distorted region.
- They help *polymerase III* to bypass the damaged site in template DNA and move forward in the direction of advancing replication fork.

Genes homologous to umuC and umuD are found in almost all animals and plants.

Extensive Damage Repair by Homologous Recombination

Homologous recombination repair mechanism is postreplicative repair process. When DNA damage includes a longer stretch of nucleotides, an alternative pathway is found to operate. It is called **homologous recombination**. ***DNA polymerase*** stops replication on reaching the damaged site in the template strand. DNA replication is resumed as the next priming site on the template strand in the form of Okazaki fragment with a new RNA primer.

The gap left in the daughter strand or newly synthesised strand may be of several hundred nucleotides. This gap is then repaired by homologous DNAs recombination with the complementary parental strand using specific recombination enzymes. Enzymes ***DNA polymerases*** and ***DNA ligases*** repair the resulting gap in the complementary parental strand.

Ribonucleic Acid (RNA)

Ribonucleic acid (**RNA**) is a single-stranded nucleic acid found in all living cells.

RNA is found chiefly in the cytoplasm but also in the nucleolus. Inside the cytoplasm it occurs freely as well as in the ribosomes. RNA can also be detected from mitochondria, chloroplasts and associated with the eukaryotic chromosomes. In some plant and animal viruses, RNA acts as a hereditary material.

Structure

The structure of RNA can be easily understood by comparing it with the structure of DNA.

1. More commonly RNA is a single-stranded structure consisting of an unbranched polynucleotide chain, but it is often folded back on itself forming helices.
2. RNA like DNA is formed of several hundreds or thousands of nucleotides arranged in a linear sequence and connected together by 3′ – 5′ phosphodiester bonds.
3. The sugar found in nucleotides of RNA is **ribose**, whereas it is deoxyribose in DNA. The nucleotides of RNA are called **ribonucleotides**.
4. The four nitrogenous bases found in RNA are adenine, cytosine, guanine, and uracil. Therefore, in RNA thymine of DNA is substituted by uracil.
5. The base composition of RNA does not agree to the A :: U = G ⁝⁝ C = 1.
6. Intramolecular pairing between the nucleotides of the single strand of RNA provides stability to RNA.
7. DNA is the hereditary material, whereas RNAs are of different types for performing different functions during protein synthesis. However, in most plant viruses and some animal viruses RNA also acts as hereditary or genetic material.

The Genetic RNA

In most plant viruses, some animal viruses and in many bacteriophages, DNA is not found and RNA acts as hereditary material. This RNA may be single-stranded or double-stranded. When it is double-stranded it follows the same rules of base pairing as in the case of DNA.

Replication of Genetic RNA (Viruses follow Reverse Transcription)

1. **In Retroviruses :** Viruses having RNA as the hereditary material possess an enzyme called ***reverse transcriptase***. When a virus particle enters the host cell, its RNA is released in the host cytoplasm. Here, with the help of ***reverse transcriptase*** a strand of complementary DNA is synthesised using RNA as template. This single DNA strand synthesises its complementary antiparallel strand completing the fomation of double-stranded DNA. This viral DNA then joins the DNA of host cell becoming its integral part. The integrated DNA copy of host DNA is called **provirus** and this process of DNA transcription is called **reverse transcription**. Utilising synthetic machinery of host cell, this DNA produces several copies of RNA and protein molecules to be used in the fomation of viral protein coat.

   RNA → ssDNA → dsDNA → Viral mRNA → Viral proteins
2. **In RNA-RNA Viruses :** These viruses synthesis new genomic RNA on RNA template with the help of an enzyme called ***replicase***. These new RNA molecules then synthesise mRNAs which guide synthesis of viral proteins using host's ribosomes, tRNA, amino acids and required enzymes.

   RNA → RNA → mRNA → Polypeptide

Non-Genetic RNA

In all other organisms, where DNA is the hereditary material all types of RNAs are non-genetic. The non-genetic RNAs are synthesised from DNA template. In general, three types of RNAs have been distinguished:

1. Messenger RNA or nuclear RNA (mRNA)
2. Ribosomal RNA (rRNA)
3. Transfer RNA (tRNA)

1.Messenger RNA or Nuclear RNA

Messenger RNA carries genetic information from chromosomal DNA to the cytoplasm, where it acts as a template for protein synthesis. It is complementary to DNA and carries the copy of the same base sequence as found in that part of DNA from which it is copied, except that thymine is substituted by uracil. It constitutes about 5 per cent of the total RNA of the cell.

Since the '**genetic blueprint**' is contained in the nucleus, and the '**work benches**' or '**sites**' of protein synthesis (i.e. ribosomes) are present in the cytoplasm, these RNA molecules serve to transport genetic messages from DNA to ribosomes. The term **messenger RNA** was used by **Francois Jacob** and **Jacques Monod** (1961). In the cytoplasm, mRNA molecules become attached to ribosomes and act as **template** for the polymerisation of amino acids into polypeptide chains. The sequence of amino acids in each polypeptide chain is determined by the sequence of codons in its mRNA molecule.

Because of great variation in the size of mRNA molecules, mRNA is also called **heterogeneous nuclear RNA** or **hnRNA**.

Heterogeneity of mRNA

The molecules of mRNA are heterogeneous because these occur in different sizes having different molecular weight. The heterogeneity depends upon two main factors.

1. The size and number of cistrons, and
2. The size of the protein molecule for which it carries codons.

Based on the number of cistrons, two types of mRNA have been recognised.

- **Monocistronic mRNA** molecule contains the codons of a single cistron, which codes for one complete molecule of protein. In eukaryotes, mRNA carries information only for one polypeptide chain. It is **monocistronic (monogenic)** because it is transcribed from a single cistron (gene) and has a single initiation codon and a single terminator codon.
- **Polycistronic mRNA** molecule contains the codons for more than one cistron which may lie close together. This type of mRNA synthesises more than one polypeptide chains. Therefore, a polycistronic mRNA molecule must possess a series of start and stop codons. If a mRNA molecule encodes three proteins the minimal requirement would be the sequence.

Start codon, protein 1, stop codon (1st sequence) start codon, protein 2, stop codon, (2nd sequence)

start codon, protein 3, stop codon. (3rd sequence)

Life Span of mRNA

Life span of mRNA in prokaryons (prokaryotes) is very short. In bacteria, it is about 2 minutes. But in the eukaryons, mRNA is metabolically more stable and can function for a number of hours and even for days.

Role of Messenger RNA (mRNA)

Genetic information for protein synthesis is encoded in the DNA of a gene. It is stored in the nucleotide sequences in DNA present in the nucleus. Protein synthesis occurs in the cytoplasm. The sequence of nucleotides in DNA determines the sequence of amino acids in the polypeptide chain. **Messenger RNA** (mRNA) acts as an intermediate in the flow of information from DNA to protein.

2.Ribosomal RNA(rRNA)

Occurrence : The ribosomal RNA (rRNA) constitutes bulk for the cellular RNA (over 80 percent of the weight of total RNA of the cell). It occurs in ribosomes which are nucleoprotein molecules.

Biogenesis of Ribosomal RNA : Cells contain millions of ribosomes, and each ribosome contains several molecules of rRNA. The DNA sequences encoding rRNA are normally repeated hundreds of times showing **tandem arrangement**. This DNA is called **rDNA**. It is clustered in one or more **nucleoli**. Thus, nucleolus is a ribosome-producing organelle. The part of chromosome containing rDNA is called **nucleolar organiser**. In bacteria about 10–20 cistrons are associated with rRNA synthesis, but in eukaryotes 210–2,100 tightly clustered rDNA cistrons are involved in rRNA synthesis. The process of rRNA synthesis involves following steps.

Types of rRNA : Inside the ribosome of eukaryotic cells, rRNA occurs in the form of particles of three different dimensions. These are designated 28S, 18S and 5.8S. The 28S and 5.8S molecules occur in large subunit (60S subunit) of ribosome, whereas 18S molecule is present in the small subunit (50S subunit) of ribosome. The 28S molecules of rRNA have a molecular weight $1.5 - 1.8 \times 10^6$ daltons, whereas 18S particles are only 0.7×10^6 daltons. In prokaryotic cells there are only 23S and 16S rRNAs with a molecular weight 1.2×10^6 and 0.6×10^6 daltons respectively.

Base Composition of rRNA : rRNA differs in base content from tRNA and mRNA. It is relatively rich in guanine and cytosine. The base components in rRNA of *E. coli* have a molar ratio of adenine 21: uracil 19: guanine 37: cytosine 23.

Function

The function of rRNAs is not known but recent evidences suggest that one of the subunit of rRNA serves to release mRNA from DNA. Largest molecule of rRNA (23S in prokaryotes and 28S in eukaryotes) is a ***ribozyme*** that catalyses the synthesis of peptide bonds between the amino acids. It functions as ***peptidyl transferase.***

3.Transfer RNA (tRNA):

The transfer RNA is a family of about 60 small-sized ribonucleic acids which can recognise the codons on mRNA and exhibit high affinity for 21 activated amino acids, combine with them and carry them to the site of protein synthesis. tRNA molecules have been variously termed as *soluble RNA* or *supernatant RNA* or *adaptor RNA*. tRNA is about 10-15 per cent of total weight of RNA of the cell.

Characteristics of tRNA

tRNA molecules have following characteristics:

- tRNA molecules are smallest, containing 75 to 93 nucleotides.
- Each tRNA molecule has a **3′-OH terminus** and a **5′-monophosphate terminus** (not 5′- triphosphate as in other RNA molecules).
- Their sedimentation coefficient is **4S** and molecular weight 24,000 to 31,000 daltons.
- Its polynucleotide chain undergoes secondary and tertiary foldings because of internal complementary base pairing. As a result, a tRNA molecule acquires a precise L-shaped **three-dimensional configuration**, resembling a twisted clover leaf. In 1965, **R.W. Holley** proposed a **Clover leaf model** of tRNA molecule. For this he got Nobel Prize with **Khorana** and **Nirenberg** in 1968.
- Some of the bases of the two arms are coiled over one another.
- The 3′ end of the polynucleotide chain of tRNA ends in **CCA** base sequence. This represents site for the attachment of activated amino acid. The 5′ end of the chain terminates with guanine base.
- The band in chain of each tRNA molecule contains a definite sequence of three nitrogenous bases which constitute the **anticodon**. It recognises the codon on mRNA.

Clover-Leaf Model of tRNA

According to Clover leaf model four different regions or special sites can be recognised in the molecule of tRNA. These are :

1. **Amino Acid Arm or Acceptor Arm (AA-arm) or Acceptor Stem :** It is double helical and stem-like. It possesses both 5′ and 3′ ends of the molecule. It has seven base pairs and the unpaired 3′ terminal always has a base triplet CCA with – OH at the tip. The – COOH group of a specific amino acid joins with the – OH group of adenosine base of CCA in the presence of ATP, forming **amino acyl tRNA**. It is common to all the tRNA molecules. Therefore, 3′ end of acceptor arm is called **carrier end**.
2. **Anticodon Arm :** It is a loop-like structure consisting of 7 to 11 unpaired nucleotides. It lies opposite to the AA arm. Three of the 11 nitrogenous bases of these ribonucleotides are complementary to one of the triplet codon of mRNA molecule. Since the base triplet on mRNA chain is called **codon**, its complementary base triplet on tRNA molecule is termed **anticodon**. Anticodon reads its appropriate codon on mRNA and temporarily binds to it. Therefore, the terminal end of this arm is called **recognition end**.
3. **DHU Loop (Dihydrouridine Loop) :** It is also a loop-like arm. It has enzyme recognition site that binds to specific amino acid activating enzyme which catalyses the union of a specific amino acid to tRNA molecule.

4. **Tψ/C loop :** It is also a loop-like or hairpin arm of the tRNA with a site for attachment to a ribosome. This site is common to all the molecules of tRNA. It is formed of five base pairs and a terminal loop with seven nitrogenous bases.
5. **Variable Arm :** It contains 4 to 21 nucleotides.

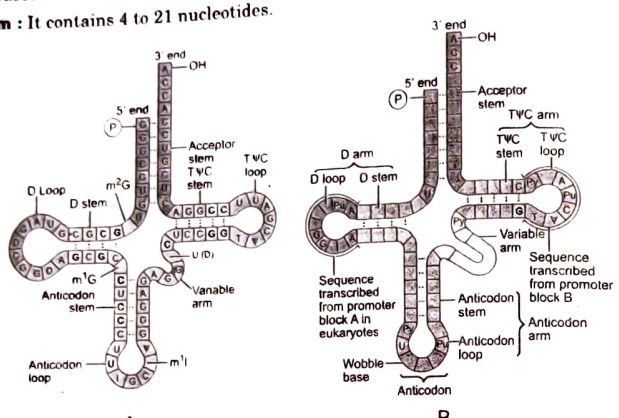


Fig. A Clover leaf model of a eukaryotic alanine tRNA molecule from yeast as determined by Holley.

Function : The transfer RNA plays a key role in protein synthesis. It picks up a specific amino acid (activated) from the cytoplasm, carries it to the site of protein synthesis and attaches itself to ribosome in accordance with the sequence specified by mRNA. Finally, it transmits its amino acid to the polypeptide chain which is being synthesised. Cells need at least 32 tRNAs to recognise amino acid codons, but some cells use more than 32 tRNA molecules.

Other types of RNAs

In addition to these three types of RNAs, following types of RNAs are also found:

1. **Small Nuclear RNAs (snRNAs) :** These are small RNA molecules formed of 100 to 215 nucleotides. These are found in the nucleus of eukaryotic cells. These are important for DNA replication and processing of long, newly transcribed heterogeneous nuclear RNA molecules.
2. **Enzymatic RNAs or Ribozymes : Thomas Cech** and **Sidney Altman** (1983 and 1985) independently discovered that certain RNAs act as biological catalysts. These are called **ribozymes**. These are associated with the maturation of **hnRNAs** (heterogeneous RNA). Even 23S and 28S ribosomal RNAs of ribosomes are found to catalyse the formation of peptide bonds during synthesis of polypeptide chains in prokaryotic and eukaryotic cells.

Transcription and Processing of RNA

The process of the formation of all the three families of RNA, i.e. mRNA, tRNA and rRNA from DNA templates is called **transcription**. It involves rewriting the genetic message coded in DNA into RNA molecule. The transcribed mRNA moves out of the nucleus to the ribosomes in the cytoplasm and directs protein synthesis.

Materials Required

RNA transcription requires :

1. The enzyme ***RNA polymerase*** or ***DNA directed RNA polymerase.***
2. DNA template – the transcription unit.
3. All the four types of ribonucleoside triphosphates (ATP, CTP, GTP and UTP).
4. Divalent metal ions Mg^{2+} or Mn^{2+} as cofactors.

No primer is needed for RNA synthesis.

RNA Polymerase in Prokaryotes

Bacterial ***RNA polymerase*** is a complex **holoenzyme** that catalyses transcription of RNA. In prokaryotes, single ***RNA polymerase*** (**RNAP**) enzyme carries out synthesis of all the three types of RNAs. But in eukaryotes three distinct *RNA polymerase* enzymes, namely ***RNA polymerase I***, ***RNA polymerase II*** and ***RNA polymerase III*** catalyse synthesis of rRNA, mRNA and tRNA respectively. In addition, mitochondria and chloroplasts have their own RNA polymerases, which resemble bacterial enzyme.

In prokaryotes, single RNA polymerase catalyses synthesis of all the three types of RNAs. In *E. coli*, ***RNA polymerase*** enzyme is a large protein with a molecular weight of 4,80,000. It is a holoenzyme and consists of **core enzyme** and **sigma subunit**.

1. **Core Enzyme**

   Core enzyme is formed of four tightly associated protein chains namely ββ′ αα with molecular weights of 1,60,000, 1,50,000, 90,000, and 4,000 respectively. It binds non-specifically to DNA template and migrates downstream till its sigma factor recognises the promoter region.

   (*i*) **The α Subunits:** The core enzyme has two molecules of α-polypeptide. This is coded by gene ***rpoA***. The α subunits are needed for the assembly of core enzyme and probaby help in the recognition of promoter.

   (*ii*) **The β Subunit:** Only one copy of β subunit is present in one molecule of core enzyme of RNA polymerase. It is encoded by gene ***rpoB***. The β subunit binds with the incoming nucleotide to be added to RNA chain and helps in the formation of the first phosphodiester bond.

   (*ii*) **The β′ Subunit:** It is encoded by ***rpoC*** gene and binds with the template strand or antisense strand of DNA. Therefore, both β and β′ are catalytic subunits of core enzyme.

2. **Sigma Subunit (σ)**

   Sigma subunit is formed of a single polypeptide chain, loosely attached to the core enzyme. It is encoded by the gene ***rpoD***. It recognises the start signal on DNA molecule and directs the core enzyme of ***RNA polymerase*** to bind to the promoter region upstream of initiation codon. The σ subunit recognises two special sequences of bases in the promoter region of the coding strand (i.e. nontemplate strand) of DNA. These are – **10 sequence** and – **35 sequence**.

Functions of RNA Polymerase

1. ***RNA polymerase*** unwinds about 15 bases of DNA around the initiation site to form an open promoter – DNA complex and provides single-strand of DNA to act as template for transcription.
2. ***RNA polymerase*** basically catalyses the formation of phosphodiester bonds between successive nucleotides of a polynucleotide chain during synthesis of both DNA and RNA. With sigma factor, it contributes to the synthesis of RNA because sigma factor recognises promoter site.

Transcription Unit

The transcription unit is a stretch of nucleotides in DNA that is transcribed into a single functional RNA molecule. It is called an **operon** in prokaryotes and a **gene** in eukaryotes. A typical transcription unit has following essential parts :

1. **Consensus Sequence in – 35 Region**

   The consensus sequence in – 35 region occurs as a variation of **5′... TTGACA...3′**. It lies about 35 bases upstream from the initiation site. Because of its location, this consensus sequence is called – 35 sequence.

2. **Upstream Promoter Element**

   It is another AT-rich recognition element in the promoter region. It is present only in the promoters of certain highly expressed genes and is located between – 40 and – 60 base sequences. This upstream promoter sequence binds to the α subunit of *RNA polymerase*.

Molecular Mechanism of Transcription in Prokaryotic Cells

The molecular mechanism of transcription involves following steps :

1. **Recognition of Promoter Region on DNA**

   The sigma factor of ***RNA polymerase*** recognises the – 35 region and – 10 region of promoter. The core enzyme of ***RNA polymerase*** then binds lightly to the promoter region.

2. Binding of RNA Polymerase to DNA Duplex

The core enzyme ***RNA polymerase*** binds to a specific **promoter** in the DNA double helix. This site is located on the 5′ end upstream of the initiation site of the gene to be transcribed. It initiates separation of DNA strands and thus determines where to start RNA synthesis. The promoter also determines which DNA strand is to be transcribed.

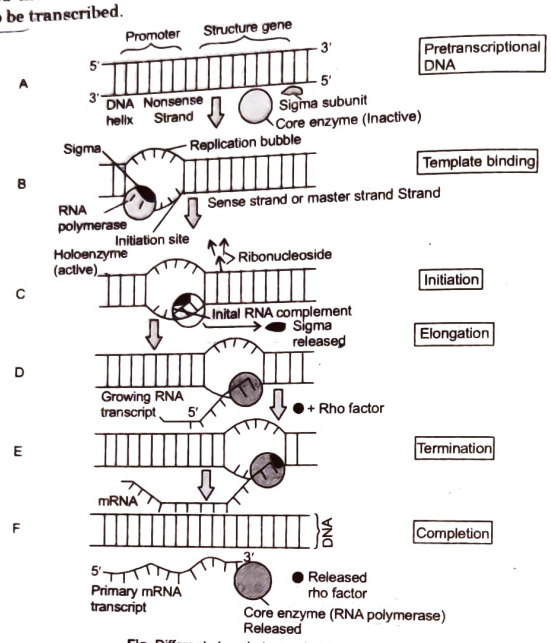


Fig. Different steps in transcription

3. Exposure of DNA Bases and Initiation of Polynucleotide Chain

RNA polymerase initiates local unwinding and separation of two strands of DNA. The separation begins from about the middle of Pribnow box which is 6-10 base long segment of DNA prior to the initiation point of RNA synthesis, approximately 10 – 15 base pairs are separated, leading to the formation of **transcription bubble**.

The separation of DNA strands exposes the bases of the DNA for the initiation of synthesis of polynucleotide chain of mRNA. Out of the two DNA strands only one, i.e., 3′ → 5′ strand functions as template. It is called **master strand** or **nonsense strand** or template strand. The complementary strand corresponding to master strand is called **coding strand** or **sense strand** or **non-template strand**. It does not participate in transcription of RNA.

4. Base Pairing : The ribonucleoside triphosphate molecules, i.e., adenosine triphosphate (ATP), guanosine triphosphate (GTP), cytidine triphosphate (CTP) and uridine triphosphate (UTP) start binding to the nucleotides of antisense strand or template strand of DNA from the initiation point. The ribonucleotides are found floating in the nucleus and serve as raw material for RNA synthesis. These are formed by the activation (phosphorylation) of ribonucleoside monophosphates, viz., adenosine monophosphate (AMP), guanosine monophosphate (GMP), cytidine monophosphate (CMP) and uridine monophosphate (UMP) through union with ATP. The enzyme ***phosphorylase*** catalyses the activation process. The ribonucleoside triphosphates get joined to the base of the DNA template chain one by one by hydrogen bonding according to the base pairing rule, i.e., A–U, U–A, C–G, G–C. This is brought about by ***RNA polymerase***. Thus, enzyme ***RNA polymerase*** not only initiates but also catalyses polymerisation, i.e., elongation of ribonucleotide chain in 5′ → 3′ direction.

5. Conversion of Ribonucleoside Triphosphates to Ribonucleoside Monophosphates

The various ribonucleoside triphosphates on linking to the DNA template chain break off their high-energy bonds. This changes them to ribonucleoside monophosphates that are the normal components of RNA, and sets free pyrophosphate group (P~P). Pyrophosphate contains a high-energy bond (~). It undergoes hydrolysis by the enzyme ***pyrophosphatase***. This releases energy and sets free inorganic phosphate P_i. The first ribonucleotide triphosphate retains all the three phosphates and is, thus, chemically distinct from the other nucleotides added after it.

$$P \sim P + H_2O \xrightarrow{\text{Pyrophosphatase}} 2P_i + \text{Energy}$$

6. Elongation of RNA Polynucleotide Chain

With the energy so released, each ribonucleoside monophosphate joined to DNA template chain then joins the ribonucleotide arrived earlier, making the RNA chain longer. The enzyme ***RNA polymerase*** catalyses the formation of **phosphodiester bonds** between successive nucleotides and requires a divalent ion Mg^{++} and Mn^{++}.

When a few nucleotides (about 10) are added, the sigma factor is released from the enzyme ***RNA polymerase***. About 10 – 12 nucleotides of the growing RNA chain are bonded to template DNA at any one time.

mRNA synthesis on DNA template takes place in 5′ → 3′ direction so that the successive nucleotides are attached at 3′ end of growing RNA strand. Core enzyme moves along DNA, unwinds the two strands exposing the nucleotides of template DNA. As a result complementary copy of RNA is transcribed on 3′ → 5′ strand of DNA. The transcribed RNA molecule is called **RNA transcript** or **primary RNA transcript**.

7. Termination of Transcription and Separation of RNA Chain

As transcription proceeds, the hybrid DNA-RNA molecule dissociates, partly freeing the RNA molecule under synthesis. When the polymerase core enzyme reaches a **terminator sequence** on the DNA, it leaves the DNA. The terminator sequence lies in the template strand and consists of two inverted repeat separated by about six bases and followd by a sequence of A's. The string of A's in template gives rise to a sequence of U's at the 3′ end of mRNA. It is the terminator sequence in mRNA.

1. Without termination factor, and
2. With terminator factor (rho)

(i) **Rho-protein Dependent Termination:** The chain terminating protein ***rho (ρ)*** acts as an accessory termination factor. It acts by disrupting the RNA-DNA-hybrid. It dissociates the RNA transcript from template and terminates transcription.

(ii) **Rho-protein Independent Termination :** This type of RNA transcript termination depends on a specific DNA sequence that ends RNA transcript with a stable hairpin structure. Such termination sites have two or more stretches of G–C pairs arranged in the form of inverted repeat followed by a string of adenine (poly A) in the master strand. The released mRNA is called **primary transcript**. The self complementary residues in the transcript fold into a hairpin loop, ending in poly U sequence at 3′ end. It undergoes processing to become functional. From nucleus, it comes out into the cytoplasm through nuclear pores.

8. Reformation of DNA Helix : As the RNA chain grows, the transcribed region of the DNA molecule gets hydrogen bonded to the sense strand and the two are spirally coiled to assume the original double helical form. When the last ribonucleotide is added, the RNA polymerase and RNA chain are fully released from the DNA, which also completes its winding into a double helix. The protective protein coat is added again to the DNA duplex.

9. Processing of RNAs: The originally transcribed RNA molecules are called **primary transcripts**. In both prokaryotes and eukaryotes, they undergo extensive changes to become functional. This is called **processing of RNAs** or **post-transcriptional modifications of RNAs**.

Molecular Mechanism of Transcription in Eukaryotic cells

Transcription in eukaryotic cells is more complex and differs from that in prokayotes in following features:

1. Three different ***RNA polymerases*** are required in the transcription of three different types of RNAs.
2. **Eukaryotic promoters** are more varied than prokaryotic promoters. There are three different types of promoters for three different polymerases. Some eukaryotic promoters are located downstream from the transcription start point while some are present upstream.
3. **Transcription factors:** Many transcription factors are involved in binding of eukaryotic *RNA polymerase* to DNA.

Table : Differences between prokaryotic and eukaryotic transcription

Prokaryotic Transcription	Eukaryotic Transcription
1. Occurs in the cytoplasm.	1. Occures in the nucleus.
2. There is no definite phase for its occurrence.	2. Takes place in the G1 and G2 phases of cell cycle.
3. A single RNA polymerase synthesises all the	3. Three RNA polymerases I, II and III synthesise three types of RNA (mRNA, tRNA, rRNA) rRNA, mRNA, and tRNA respectively.
4. Coupled transcription-translation is the rule.	4. Coupled transcription-translation in not possible.
5. Initiation of transcription does not need any proteins or initiation factors.	5. Initiation of transcription requires proteins called transcription factors. These are TFIIA, TFIIB, TFIID, TFIIE, TFIF and TFIIH. These recognise TATA box.
6. Pre-RNA molecules are released and processed cytoplasm.	6. Pre-RNAs are released and processed in the in the nucleus.
7. RNA polymerase is a complex of five polypeptides.	7. RNA polymerases are complexes of 10 to 15 polypeptides.
8. Transcriptional unit has more than one genes (Polycistronic)	8. Transcriptional unit has only one gene. **(Monocistronic).**
9. The mRNA primary transcript was fewer surplus nucleotides.	9. The mRNA primary transcript has a large number of surplus nucleotides.
10. The 23S, 16S, 5S rRNAs are formed from a single primary transcript.	10. The 28S, 18S, 5.8S and 5S rRNAs are formed from two primary transcripts.

Eukaryotic RNA Polymerases

The three nuclear RNA polymerases are:

1. RNA Polymerase I

It is found in the nucleolus and is responsible for the synthesis of ribosomal RNA. The RNA molecule synthesised in the nucleolus is precursor to three ribosomal RNAs(28S, 18S, 5.8S rRNA).

2. RNA Polymerase II

It is found in the nucleoplasm and transcribes protein-encoding gene. It synthesise precursors of mRNA. It also synthesises most small nuclear RNAs (snRNAs) which are needed in post-transcriptional processing of primer RNA.

3. RNA Polymerase III

It is also found in nucleoplasm. It is responsible for the synthesis of a variety of small RNAs including tRNA and 5S rRNA.

Table : List of different types of RNA polymerases and RNA transcribed by them

Types of RNA Polymerases	Types of RNA synthesised
Type I *polymerase*	28S rRNA, 18S rRNA, 5.8 RNA
Type II *polymerase*	mRNA, SnRNA, ScRNA
Type III *polymerase*	tRNA, 5S rRNA and other small RNAs
Mitochondrial *RNA polymerase*	Mitochondrial RNAs inside mitochondria
Chloroplast *RNA polymerase*	Chloroplast RNAs inside chloroplast

Transcription Factors

These are specific proteins present in the nucleoplasm and are required for the correct functioning of RNA polymerases during initiation of RNA transcription. They bind to TATA box and are also called **TATA binding proteins (TBP)**. These are classified into two categories :

1. General Transcription Factors or Basal Transcription Factors

These are needed for the transcription of all the genes transcribed by a particular *RNA polymerase*. They are designated as TFI, TFII and TFIII and correspond to the three *RNA polymerases*.

2. Additional Transcription Factors or Specific Transcription Factors

These are needed for the transcription of particular specific genes under specific circumstances. These are responsible for regulating gene expression and, therefore, control the expression of individual genes.

Transcription Factors needed for RNA Polymerase II

Transcription factors TFIIB, TFIID, TFIIE, TFIIF and TFIIH are general transcription factors, required for ***RNA polymerase II*** to start RNA transcription. TFIID has a TATA binding protein (TBP) which binds specifically to the TATAA consensus sequence. TFIID also binds to the TBFIIB forming TATAA-TFIID-TBIIB complex at the promoter. With the help of TFIIF factor ***RNA polymerase*** binds to this complex.

The release of *RNA polymerase* II from promoter and its movement along template DNA require other three TFII complexes, TFIIE, TFIIH and TFIIJ. TFIIH unwinds DNA around the transcription site (**helicase activity**) and phosphorylates the tail of largest subunit of RNA polymerase at the C-terminal. This end is also called **carboxy-terminal domain (CTD).**

Table : Transcription factors associated with RNA polymerase II

TBP	binds to TATA box, a part of TFIID
TFIID	includes TBP; recognises Polymerase II-specific promoter; first to bind to TATA box
TFIIA	binds upstream of TATA box; required for binding of RNA Pol II to promoter
TFIIB	binds downstream of TATA box; required for binding of RNA Pol II to promoter
TFIIF	accompanies RNA Pol II as it binds to promoter
TFIIE	required for promoter clearance and elongation of ribonucleotide chain
TFIIH	phosphorylates the tail of RNA Pol II, retained by polymerase during elongation
TFIIJ	required for promoter clearance and elongation.

Functional or Consensus Sequences in Prmoter Region in Eukaryotes

Sequence of nitrogenous bases similar to that found in prokaryotes are present in the promoter region of eukaryotes also. There are different promoters for all the *RNA polymerases*.

(A) Promoters for Polymerase II

The promoter region for *RNA polymerase* II is formed of three regions: the **initiation box**, **TATA box** and a variety of **upstream elements**.

TATA Box or Goldburg Hogness Box

The consensus sequence in TATA box is a hexanucleotide sequence, consisting of **5'...TATAAA...3'**. It is usually representd by **TATXAX**, where X may be T or A. It is comparable to Pribnow sequence of prokaryotic promoter. It is the site of assembly of a pre-initiation complex and determines the site of initiation of transcription. It lies about 25bp upstream of initiation box. It is rich in AT sequences.

The assembly of pre-initiation complex (transcription factors and polymerase) is essential for initiation o transcription.

In addition, eukaryotic genome has proximal control elements, named **GC box** with base sequence GGGCG at – 100 site and a **CAAT box** with base sequence **GCCCAATCT**.

(B)Promoter for RNA Polymerase I

The promoter for RNA polymerase I has an **upstream control element** and a **core promoter**. The co promoter site is rich in GC sequences. The protein (**UBFI upstream binding factor for pol I**) recogni and binds to both the upstream control element and the core promoter. Later, **SLI (selectivity factor** **pol I**) binds to DNA in association with UBFI. Finally, **RNA polymerase I** binds and transcribes rRNA ge

The promoter for *RNA polymerase III* for both 5S RNA and tRNA are located within the gene itself. The recognition sites for them are downstream of the start site. Two factors TFIIIC and TFIIIB are needed for the binding. Firstly, TFIIIC (or TFIIIA) binds to both sites and induces TFIIIB to bind to the promoter near the start sites and TFIIIB binds *RNA polymerase III* to promoter in DNA.

RNA-Directed RNA Synthesis

In RNA viruses whose genome is RNA, RNA is transcribed from RNA. These viruses induce the formation of an enzyme ***RNA-directed RNA polymerase*** or ***RNA replicase*** in the host cell. ***RNA replicase*** acts like ***DNA-directed RNA polymerase***.

Post-Transcriptional Modification of RNA or RNA Processing

The immediate product of transcription is called **primary transcript**. It is biologically inactive. The process of chemical modifications in the primary transcript to generate the final functional RNA product is called **RNA processing**. In RNA processing.

- Larger RNA precursors are cut into smaller RNAs by ***ribonuclease-P enzymes*** (cleaving),
- Unwanted nucleotides are removed by enzymes named ***nucleases*** (**splicing**),
- Useful regions are rejoined by ***ligase*** enzyme (union),
- Certain nucleotides are added at the end enzymatically (**terminal addition**),
- Molecule may fold on itself to assume proper shape (**folding**), and
- Some nucleotides may be modified (**nucleotide modification**)

Processing of Pre-tRNA Transcript

Eukaryotic pre-tRNA molecules are biochemically altered. The process has been studied in detail in years, where following four processes are needed for functional tRNA molecules to be formed:

- Trimming of ends by the cleavage of phospho-ester bonds.
- Splicing for removing an intron.
- Addition of some terminal sequences.
- Heterocyclic base modification, usually by **methylation**.

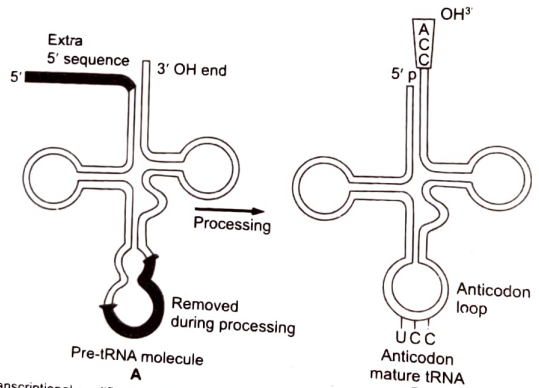


Fig. Post-transcriptional modification or processing of a pre-tRNA molecule in E. coli. **A.** Pre-tRNA molecule; **B.** Mature tRNA: Processing involves removal of extra segment at 5′ end and of intron sequence from anticodon loop and the addition of CCA sequence at the 3′ OH end

In prokaryotes pre-tRNA molecule undergoes only a few of the above modification steps. In some prokaryotic tRNA, as some introns are removed, in some, terminal sequences are added and in some others, a few heterocyclic and ribose rings are modified. The endonuclease enzyme, ***ribonuclease P***, catalyses removal of a segment from 5′ end of pre-tRNA by hydrolytic cleavage.

Processing of rRNA-Transcript

The primary transcript of rRNA contains precursors for all the three rRNAs (23S, 16S and 5S). The processing involves its trimming to smaller segments of appropriate size with the help of ribonucleases. It has already been discussed in biogenesis of rRNA.

Processing of Pre-mRNA Transcript

Most prokaryotic mRNA transcripts require very little or no post-transcriptional modification to be changed into functional mRNA. In prokaryotes, often translation of mRNA begins while it is being transcribed. This is called coupled **transcription-translation**. But in eukaryotes, the primary-mRNA transcript molecules have copy of both exons and introns. Therefore, they are spliced and chemically modified in the nucleus before being transported to the ribosomes for translation. Following three biochemical processes occur during mRNA processing in eukaryotes :

- RNA capping at 5′ end
- Addition of polyadenylate tail at 3′ end, and
- Splicing or removing the introns.

1. RNA Capping

The **5′ end of primary mRNA transcript** is modified by the removal of a phosphate from

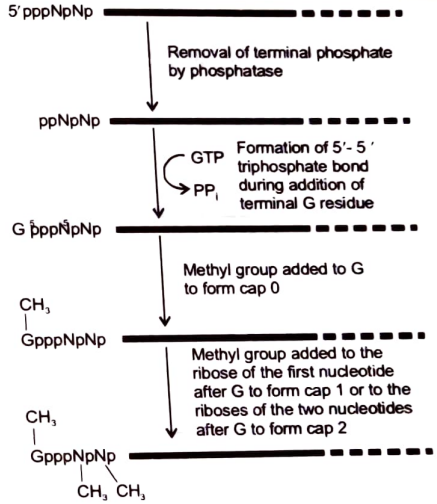


Fig. Steps involved in the formation of 5G cap

the triphosphate functional group. A guanosine monophosphate (GMP) gets attached to this diphosphate at 5′ end, resulting in the formation of an unusual 5′ – 5′ triphosphate covalent linkage. The guanine residue at the end of this linkage is methylated at N_7. The **methylated guanosine** forms a cap at 5′ end of mRNA (Cap 0). Additional capping may occur by another methylation of hydroxyl groups of ribose of first one or two nucleotides in some higher eukaryotes (Cap 1 or Cap 2). Only cap 0 is formed in lower eukaryotes.

Capping of RNA is probably necessary for making mature mRNA and to protect it against cleavage by exonucleases.

2. Poly A Addition

The **3′ end of primary mRNA transcript** is modified by the addition of a polyadenylate (**poly A**) tail. It is catalysed by ***polyadenylate polymerase (poly A polymerase)*** enzyme. The poly A tail is formed of 20 to 250 nucleotide residues. Before the addition of poly A tail, an endonuclease removes a few nucleotides from the 3′ end of the transcript. The cleavage site has tail recognition sequence – AAUAAA and GU rich sites and is recognised by a polyadenylation complex.

The ***polyadenylation complex*** is formed by the joining of poly A adenylate and poly A binding protein (PABP) with cleavage-polyadenylation specificity factor (CPSF) and cleavage stimulation factor (CST). This complex binds to the tail recognition sequence – AAUAAA – and GU-rich sites and stimulates the process of cleavage of nucleotides and addition of poly A tail.

Most eukaryotic mRNA molecules have a 3′ poly A tail. It is thought to stabilise mRNA molecule by increasing resistance to cellular nucleases and helps it to cross the nuclear membrane.

3. RNA Splicing

The process of cutting out introns from primary RNA transcript and rejoining the exons to form functional mRNA is called **RNA splicing**. It occurs only in eukaryotic cells which have split genes.

Mechanism of RNA Splicing

The mechanism of RNA splicing depends on specific base sequences at the junctions between introns and exons. The sequence at the 5′ end of mRNA intron, called **5′ splice site**, starts with **GU** and **3′ splice site** ends with **AG**.

Spliceosome

In the middle of the intron is present a special adenine residue. It is used as a **branch site** during splicing. The concensus recognition sequences for these sites are:

1. 5′ splice site: 5′ ...AG↓**GU**AAGU...3′
2. 3′ splice site: 5′ ...YYYYYYNCAG↓...3′
3. Branch site: 5′ ...UACU**A**AC...3′

The process of RNA splicing is catalysed by one RNA-protein complex, called **spliceosome**. It is formed of four small RNA-protein complexes called **small nuclear ribonucleoproteins** (**snRNPs** or, **snurp**) and some **additional proteins**. Each **snRNP** contains one or two small molecules of RNA, called **snRNA** (small nuclear RNA). However, the total number of snRNPs in a spliceosome is five, namely U1, U2, U4, U5 and U6. Of these U4 and U6 are present together in one snRNP.

Genetic Code

Genetic Code : The theory which is widely accepted till now was proposed by **F.H.C. Crick**. The theory holds the existence of a genetic code and its smallest unit which codes for one amino acid is known as **codon**.

A codon (code word) is the nucleotide or nucleotide sequence in mRNA which codes for a particular amino acid, whereas the genetic code is the sequence of nitrogenous bases in mRNA molecule which enclose information for the synthesis of protein molecules.

Genetic Code is a Triplet Code : The main problem of genetic code was to determine the exact number of nucleotides in a codon which codes for one amino acid. Since there are only four nitrogenous bases in mRNA for 20 amino acids, combination of only one or two nitrogenous bases can not provide sufficient code words for 20 amino acids. A **singlet code** consisting of only one nucleotide provides just four codons A, C, G and U. These are insufficient to code for 20 amino acids. Similarly a combination of two nitrogenous bases (**doublet code**) provides $4 \times 4 = 16$ codons still insufficient for 20 amino acids.

Triplet tcodons of mRNA for amino acids

	U	C	A	G	
U	UUU, UUC } Phe UUA, UUG } Leu	UCU, UCC, UCA, UCG } Ser	UAU, UAC } Tyr UAA Ochre (Terminator) UAG Amber (Terminator)	UGU, UGC } Cys UGA } Terminator UGG } Try	U C A G
C	CUU, CUC, CUA, CUG } Leu	CCU, CCC, CCA, CCG } Pro	CAU, CAC } His CAA, CAG } Gln	CGU, CGC, CGA, CGG } Arg	U C A G
A	AUU, AUC, AUA } IIu AUG } Met	ACU, ACC, ACA, ACG } Thr	AAU, AAC } Ası. AAA, AAG } Lys	AGU, AGC } Ser AGA, AGG } Arg	U C A G
G	GUU, GUC, GUA, GUG } Val	GCU, GCC, GCA, GCG } Ala	GAU, GAC } Asp GAA, GAG } Glu	GGU, GGC, GGA, GGG } Gly	U C A G

Gamow (1954) pointed out the possibiiity of three letteer code, i.e. each codon consisting of three introgenous bases.

Nature and characteristics of Genetic code

The characteristics of genetic code are as follows:

1. **Genetic Code is a Triplet Code :** A codon of the present day genetic code that specifies one amino acid in a polypeptide chain comprises of a sequence of three nitrogenous bases on mRNA is a specific sequence.
2. **Genetic Code is Commaless :** Genetic code on mRNA is read continuously without internal punctuation along the reading frame.
3. **Genetic Code is Non-overlapping :** The codons in mRNA are read in successive groups of three nucleotides in 5′→3′ direction. Translation of genetic code in an mRNA begins at the correct point and there is **non-overlapping** of codons i.e., genetic code is **non-overlapping**.
4. **Genetic Code is Unambiguous :** The genetic code inside the cell medium (*in vivo*) is said to be nonambiguous because a particular codon always codes for the same amino acid. No doubt the same amino acid may be coded by more than one codons (**degeneracy**), but one codon never codes for two different amino acids.
5. **Genetic Code is almost Universal :** The same genetic code is said to be present in all kinds of living organisms including viruses, bacteria, unicellular and multicellular organisms.
6. **Genetic Code is Degenerate :** Since there are $4^3 = 64$ possible codons for only 20 amino acids that are found in proteins, it was concluded that in most cases a single amino acid is coded for by two to several different codons. This multiple system of coding is known as **degenerate system** or **degenerate genetic code**.

Reasons for Degeneracy : Wobble Hypothesis (Wobble Occurs in the Anticodon)

Wobble or **Fluctuating Base : Crick** (1966) proposed **Wobble hypothesis**. According to this hypothesis, the base in first position of anticodon on tRNA is usually an abnormal base, like inosine, pseudouridine, tyrosine, etc. These abnormal bases are able to pair with more than one type of nitrogenous bases in the third position of the codon on mRNA. For example, inosine (I) can pair with A, C or U. This base is called a **Wobble base** or **fluctuating base**

The last two nucleotides of an anticodon on tRNA are strictly complementary to first two nucleotides of codon in mRNA as per classical rule of pairing. But the pairing of third base on mRNA with first base of anticodon of tRNA is less stringent and may occur in a nonclassical fashion, so that in some cases a single tRNA may base pair with more than one codon. This is called **Pairing Wobble** or **Wobble Base Pairing**.

Biological Significance of Degeneracy of Genetic Code

Degeneracy of genetic code is a boon for the survival of living beings, because of the following reasons :

1. It is protective device against mutations. The synonym codons usually differ only in the base occupying third position in the codon. The first two nitrogenous bases are important in coding. Therefore, mutations that change the third base of the codons often go unnoticed and amino acid in the polypeptide chain remains unchanged.
2. Chain termination mutations usually lead to the synthesis of incomplete and nonfunctional proteins. This may lead to the appearance of recessive character. Thus occurrence of three termination codons helps in reducing the probable harmful effect of certain mutations.

7. **Genetic Code has Chain Initiation (start) and Chain Termination (stop) Codons :** A reading frame of the genetic message always has a specific start and stop signals for protein synthesis. The codon present in the beginning of the cistron known as **initiation codon** marks the beginning of the message for a polypeptide chain. The initiation codon is AUG and it codes for the amino acid, methionine or formylmethionine.

Similarly, the last codon of a cistron helps in reading the termination of polypeptide chain. This is known as **termination codon**. There are three termination codons: UAA, UGA and UAG. Earlier when the function of these codons was not known, these were called **nonsense codons**. No tRNA molecule has anticodon for these codons. Therefore, synthesis of polypeptide chain stops here, that is why these codons are called **stop codons** or **termination codons**.

8. **Genetic Code has Collinearity :** The codons in DNA and mRNA and the corresponding amino acid residues in the polypeptide chain have a linear arrangement which has been demonstrated by the studies of T_4 mutants. These produce incomplete head molecules. These mutants can be shown to map in linear sequence by the recombination technique. This suggests that the code is **collinear**.

Protein Synthesis

Proteins are giant molecules (macromolecules) of utmost biological importance because these not only form the building material of a cell but act as enzymes controlling all the biochemical reactions going on in the living system and also control the expression of phenotypic traits.

Chemically, proteins are **polymers of amino acids**. The amino acid molecules are linked together in linear fashion forming a long **polypeptide chain**. The adjacent amino acids are joined together by **covalent peptide bond** between the carboxyl group of one amino acid and the amino group of the other.

Concepts About Flow of Genetic Information

1. **One Way Flow of Information : Crick** (1958) suggested that DNA transmits the genetic information for protein synthesis by transcribing mRNA (transcription) and other types of RNAs. This information encoded in the ribonucleotide sequences in RNA is then translated into the sequence of amino acids (**translation**). This is known as **one way flow of information** or **central dogma**.
2. **Circular Flow of Information: Barry Commoner** (1968) suggested about circular flow of information. The DNA transcribes RNA (mRNA, tRNA, etc). RNA translates the information into protein, protein then directs the synthesis of RNA and RNA dictates the synthesis of DNA.
3. **Inverse Flow of Information: Temin** (1970) described the presence of an enzyme, ***RNA dependent DNA polymerase***, which controls the synthesis of DNA from a single strand of RNA. **D. Baltimore** (1970) described the presence of this enzyme in certain RNA tumour viruses. In tumour viruses, RNA is genetic RNA but in certain cases the nongenetic RNA synthesised under the control of DNA may also control the synthesis of DNA by inverse flow of information. This is known as **central dogma reverse** or **reverse transcription** (**H. Temin, Baltimore** along with **R. Dulbecco** were awarded Nobel Prize in Medicine in 1976).

Raw Material for Protein Synthesis

Following machinery and substances are required for protein synthesis:

1. Ribosomes 2. Amino acids 3. mRNA 4. tRNA
5. Enzymes :
   (*a*) Amino-acid activating enzymes : ***amino acyl-tRNA synthetase***, and
   (*b*) ***Peptide polymerase*** system
6. Adenosine triphosphate (ATP) as an energy source
7. Guanosine triphosphate (GTP) for synthesis of peptide bond
8. Soluble protein and transfer factors
9. Various inorganic cations such as Mg^{2+}, K^+ and Na^+.

Mechanism of Protein Synthesis

The central dogma of protein synthesis comprises of following two steps:

I. Transcription II. Translation

Translation (Assembly of Polypeptide Chain)

Translation is the second step of central dogma. It involves a change in the language from nucleotide order in an mRNA molecule to amino acid sequence in a polypeptide. In translation, the order of nucleotides of mRNA is read as triplet codons and determines the order of amino acids in the growing polypeptide chain.

The process of translation involves follwing steps :

- Activation of amino acids.
- Attachment of activated amino acids with tRNA (Amino-acetylation of tRNA).
- Stages of translation: Initiation, elongation and termination of polypeptide chain.
- Modification of released polypeptide chain.

1. Activation of Amino Acids

Amino acids in the cytoplasm occur in inactive form and they can not take part in protein synthesis. Hence these are activated by giving them energy. The activation is facilitated through **ATP**, whose molecules unite with the amino acids forming highly reactive **amino acid phosphate-adenyl complexes** which are known as **aminoacyl adenylates**. The process of activation is governed by specific enzyme, ***aminoacyl tRNA synthetase***. Usually each amino acid has its own specific ***aminoacyl tRNA synthetase enzyme***. Hence, there are as many enzymes as the number of amino acids (i.e. 20). Each enzyme has double specificity. The specificity is provided by **recognition region**. It recognises its own amino acid and finds out its own tRNA. The mechanism of activation can be represented as below :

$$AA + ATP \xrightarrow{\text{AA activating enzyme}} AMP \sim AA + PP$$

Activated amino acid — Pyrophosphate

Where, AA represents amino acid
ATP represents adenosine triphosphate
AMP represents adenonsine monophosphate
PP represents pyrophosphate (inorganic)

2. Attachment of Activated Amino Acids with tRNA or Formation of Aminoacyl-tRNA (Changing of tRNA)

The enzyme bound activated amino acids, **aminoacyl adenylates**, become attached to the 3' end of their respective tRNA molecules. The attachment is catalysed by the same enzymes, ***aminoacyl transfer RNA synthetases***, that catalyse activation of their amino acid. The product thus formed is known as **aminoacyl transfer RNA complex** (aminoacyl tRNA). The reaction is as under:

$$AA \sim AMP + tRNA \xrightarrow[\text{enzyme}]{\text{AA tRNA synthetase}} AA \sim tRNA + AMP$$

This is important to note that particular varieties of amino acids join specific tRNA molecules. This means that for 20 amino acids at least, there are 20 different tRNA molecules and 20 different enzymes.

3. Stages during Translation

Process of translation can be separated into the following steps: (1) Initiation, (2) Elongation, and (3) Termination. Though the process of translation includes the same three steps in both prokaryotes and eukaryotes, these are more complex in eukaryotic cells.

Translation in Prokaryotes

1. Initiation of Polypeptide Chain in Prokaryotes (Formation of Initiation Complex)

In prokaryotes initiation involves following four steps :

1. Three **initiation factors (IFs)**, called **IF1**, **IF2** and **IF3** bind to the small sub-unit of ribosome with GTP attached to IF2.
2. **Initiator tRNA** with its amino acid and mRNA binds to the ribosomal subunit. In *E. coli* and other bacteria the initiator tRNA is called **formylatable tRNA**, and is represented as $tRNA^{f\text{-}Met}$. It carries **N-formyl-methionine (f-Met)**, which is a modified methionine with a formyl group on its nitrogen atom. In prokaryotes, each polypeptide chain starts with N-formyl-methionine amino acid. IF_2 with its GTP helps to identify the location of N-formylmethionine in $tRNA^{f\text{-}Met}$ in the P-site of 30S subunit of ribosome. It is the only aminoacyl tRNA that can bind to the P-site of small ribosomal subunit.
   The initiation factors, GTP, mRNA and 30S subunit of ribosome all collectively constitute **30S preinitiation complex**.
3. Anticodon of $tRNA^{f\text{-}Met}$ base pairs with the initiation codon, **AUG**, the first codon on the 5′ end of mRNA. AUG is brought into correct position when mRNA binds to 30S ribosomal subunit by its ribosome binding site. This completes the formation of **30S preinitiation complex**.
   The ribosomal binding site on mRNA is also called **leader sequence** or Shine-Dalgarno **Sequence** (after its discover).

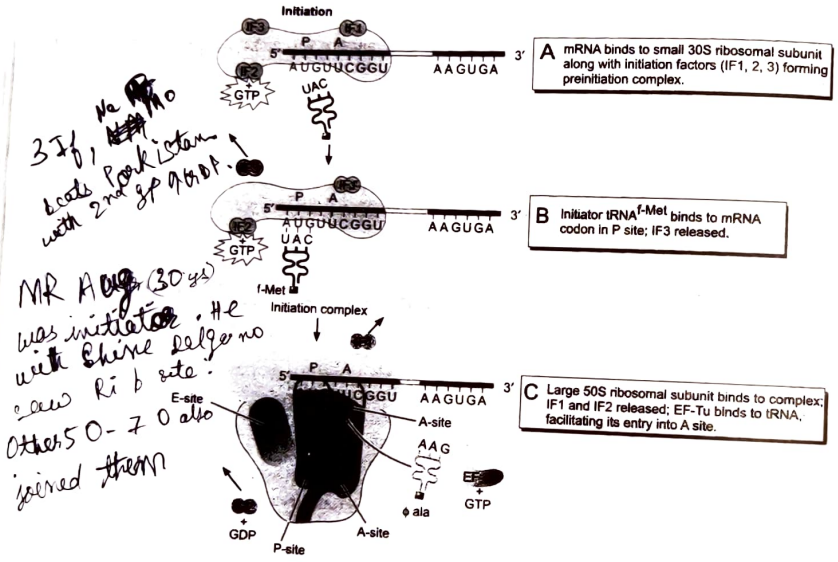


Fig. Steps in the formation of initiation complex for the initiation of polypeptide chain in *E. coli*

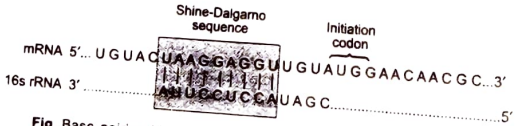


Fig. Base pairing between Shine-Dalgarno sequence (leader sequence) in mRNA and complementary region near 3′ terminus of 16S tRNA

4. **30S preinitiation complex** formed this way joins with 50S ribosomal subunit, generating **70S initiation complex**. The energy needed for the binding of 50S subunit is provided by the hydrolysis of GTP Mg^{2+} ions are required during this process.
   When 50S subunit of ribosome joins with the 30S preinitiation complex, the f-Met-$tRNA^{f\text{-}Met}$ occupies its P site. This enables the pairing of f-Met-tRNA anticodon with AUG initiation codon in mRNA. Thus, the reading frame is defined.

2. Elongation of Polypeptide Chain

After the formation of 70S mRNA-f-Met-$tRNA^{f\text{-}Met}$ complex, the elongation of polypeptide chain begins by the regular addition of amino acids in following steps:

1. Binding of AA-tRNA at Site-A of Larger Subunit of Ribosome

The large subunit of each ribosome has two slots for the attachment of two molecules of tRNA. These are called **P-site** (**peptidyl** or **donor site**) and **A-site** (**aminoacyl** or **acceptor site**). The incoming aminoacyl-tRNA complex (AA-tRNA) attaches to the acceptor site and base pairs with mRNA codon present in the ribosome's A site. In the first round of elongation, the codon in a site is immediate next to the start codon. It receives the second aminoacyl tRNA. The tRNA carrying peptide chain shifts to the peptidyl or donor site, i.e. f-Met-$tRNA^{f\text{-}Met}$ attaches to the P-site.

The binding of new aminoacyl tRNA needs two **elongation factors** (**EFs**), namely **EF-Tu** and **EF-Ts**. The required energy is provided by the hydrolysis of 2GTP. **EF-Tu** is a dimer of two proteins, each of which binds to one GTP. **EF-Tu** conveys aminoacyl tRNA to A site and is released after the transfer of aminoacyl tRNA. **EF-Ts** regenerates EF-Tu-2GTP for the next aminoacyl tRNA.

2. Formation of Peptide Bond

With f-Met-$tRNA^{f\text{-}Met}$ now at the **P**-site and second aminoacyl tRNA at **A**-site, a peptide bond is formed between carboxyl group of f-Met amino acid from peptidyl tRNA and amino group (NH_2^+) of amino acid from aminoacyl-tRNA of site A. As a result, two amino acids are now attached to the second tRNA, present at A-site.

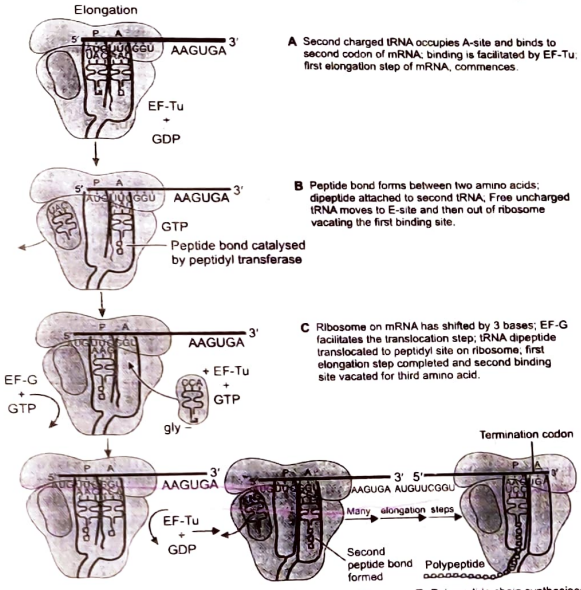


Fig. Formation of peptide bond and elogation of growing polypeptide chain in prokaryotes

Peptide bond formation is catalysed by ***peptidyl transferase***. It is an enzymatic activity inherent in 23S rRNA of large ribosomal subunit (50S). Thus, 23S rRNA is a ***ribozyme***. The energy for peptide bond formation is provided by the hydrolysis of 'high energy' ester bond by which f-Met was attached to its tRNA.

3. Translocation (Movement of peptidyl-tRNA from A-site to P-site)

After the formation of peptide bond, $tRNA^{f\,Met}$ on **P**-site is without amino acid and tRNA at **A**-site has a dipeptide. Three movements occur at this point :

- The uncharged tRNA moves to leave site on ribosome and is finally released in the cytoplasm to start a new polypeptide chain.
- The **dipeptidyl tRNA** (with two amino acids) moves from **A**-site to **P**-site leaving the A-site vacant, this is called **translocation**. This requires enzyme ***translocase***, and an elongation factor **EF-G** with a bounded **GTP** which attaches to ribsome. The GTP provides energy for translocation as EF-G leaves the ribosome.
- As the peptidyl tRNA translocates, it remains hydrogen-bonded to mRNA and pulls mRNA along with it. As a result, ribosome moves along the length of mRNA in 5′ → 3′ direction so that the next codon (i.e. third codon) on mRNA is available at A-site to receive next aminoacyl tRNA having correct anticodon.

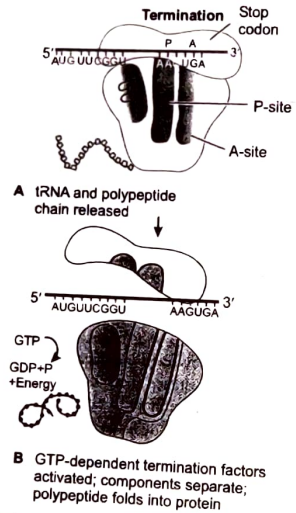


Fig. Diagram showing termination of translation and release of polypeptide chain

3.Termination of Polypeptide Chain

Termination of polypeptide chain occurs when one of three termination codons (UAA, UAG or UGA) present on mRNA at the 3′ end of each cistron arrives at the ribosome's **A**-site. There are no tRNA molecules that recognise these termination codons. Releasing factors RF_1 and RF_2 are needed for chain termination because these recognise termination codons and terminate translation. $\mathbf{RF_1}$ is specific for UAG and $\mathbf{RF_2}$ is specific for UGA. The releasing factor with the termination codons forms a complex which induces the enzyme ***peptidyl transferase*** to catalyse the termination and release of polypeptide chain (**Lipmann**, 1973). Factor $\mathbf{RF_3}$ stimulates RF_1 and RF_2. The free ribsome now dissociates into two subunits whith the help of dissociation factor IF_3.

Table : Various factors involved during translation

Process	Factor	Role
Initiation	IF_1	Stabilises 30S subunit
	IF_2	Binds **f-Met-tRNA** to 30S mRNA complex; stimulates GTP hydrolysis.
	IF_3	Binds 30S subunit to mRNA, dissociates ribosomes (70S) into subunits (30S + 50S) following termination.
Elongation	EF-Tu	Bring aminoacyl-tRNA to the A-site.
	EF-Ts	Generates active EF-Tu
	EF-G	Stimulates translocation of dipeptidyl tRNA (GTP dependent).
Termination	RF_1	Catalyses release of the polypeptide chain from tRNA and dissociation of the translation complexes; specific for UAA and UAG termination codons.
	RF_2	Behaves like RF_1, specific for UGA and UAA codons.
	RF_3	Stimulates RF_1 and RF_2.

Polysomes or Polyribosomes

During protein synthesis, when polypeptide chain has grown to 25 amino acid residues, the **AUG** initiation codon of encoding mRNA is completely free of the ribosome to start a new polypeptide chain. Formation of a new polypeptide chain is then initiated. Now two ribosomes are attached to one mRNA molecule. When second ribosome has also moved along mRNA and has a 25 amino acids long polypeptide chain, a third riboisome also attaches to initiation codon of mRNA. This process of movement and reinitiation continues until mRNA is covered with a series of ribosomes aligned at a distance of one ribosome per 80 nucleotides. Such a large translation unit with several ribosomes attached to one mRNA is called **polyribosome** or simply a **polysome**.

Translation in Eukaryotes

The process of polypeptide synthesis in eukaryotes follows the same general pattern as in the case of prokaryotes. It can be discussed under the following headings:

1.Initiation of Polypeptide Chain in Eukaryotes

The process of initiation of polypeptide chain in eukaryotes is more complex and needs at least twelve proteins, marked as eIFs (**eukaryotic initiation factors**). The initiation involves following steps :

- The initiation factors eIFIA and eIF3 bind to the 40S ribosomal subunit.
- GTP binds to eIF2 and this complex binds to initiator aminoacyl tRNA (methionyl $tRNA^{Met}$), forming **Met-$tRNA^{Met}$ eIF2-GTP**. In mammals initiation factor IF2 has three subunits α, β and γ. The α-subunit of factor eIF2 binds to eIF2. γ-binds to Met-$tRNA^{Met}$ and eIF2; β is supposed to be a recycling factor.
- Met-$tRNA^{Met}$ -eIF2-GTP associates with 40S subunit to form **40S preinitiation complex**.
- The 40S preinitiation complex (40S-Met $tRNA^{Met}$-eIF2-GTP) binds to 5′ end of mRNA.
- mRNA is brought to the 40S subunit of ribosome by eIF4E.
- The eIF4E recognises and binds to the 5′ cap of mRNA.
- The eIF4G binds to both eIF4E at the 5′ cap and to PABP (poly A binding protein) at the 3′ poly A tail and also to eIF4A and eIF4B.
- The small subunit (40S) of ribosome with initiatioin Met-$tRNA^{Met}$ + eIF2α + GTP slides along mRNA to reach initiation codon AUG and gets associated with 60S subunit of ribosome. Factor eIF5 releases eIF2 and eIF3 factors and brings about the association of 60S subunit with the initiation complex GTP hydrolyses to provide energy for this binding :

Table : List of initiation factors from reticulocytes of eukaryotes and their functions

Sl. No	Factor	Structure and Mass (in daltons)	Function
1.	eIF1	Monomer, 15,000	Assists mRNA binding to ribosomal subunit.
2.	eIF2	Trimer: Contains following three chains	Binding Met tRNA to initiation codon
		(*i*) α-chain; 35,000	Binds to GTP
		(*ii*) β-chain, 38,000	Recycling factor
		(*iii*) γ-chain; 55,000	Binds to Met-tRNA
3.	eIF3	Multimer; 75,00,000	Binding mRNA
4.	eIF4A	Multimer; 15,000	Assists mRNA binding (also binds ATP).
5.	eIF4B	Monomer; 15,000	Assists mRNA binding and unbinding.
6.	eIF4E eIF4G	Monomeer; 15,000	Binding 60S subunit to 40S subunit mRNA.
7.	eIF4H	Monomer	Unknown
8.	eIF4F	Multimer; 2,00,000	Binding mRNA5′ end; unwinding of DNA.
9.	eIF5 eIF5B	Monomer; 1,50,000	Releasing eIF2 and eIF3.
10.	eIF6	Monomer; 23,000	Prevents joining of 40S and 60S subunits of ribosomes.

2.Elongation of Polypeptide Chain in Eukaryotes

The second or next codon in mRNA base pairs with the appropriate AA-tRNA at the **A**-site. This requires the presence of elongation factor I (**EFI**) and energy from GTP. A peptide bond is formed between amino group of amino acid on tRNA at A-site with carboxyl group of amino acid on tRNA at **P**-site. This is catalysed by ***peptidyl transferase*** enzyme present in 60S subunit of ribosome. The discharged tRNA in the **P**-site is immediately released. Enzymes ***translocase*** or ***G-factor*** also called **EF2** causes translocation of newly formed p-tRNA from **A** to **P**-site. The energy is obtained from GTP. The process is repeated again and again adding amino acids to the polypeptide chain.

3.Termination of Polypeptide Chain

When termination codon on mRNA reaches the **'A'** site of ribosome, the growth or elongation of polypeptide chain stops, because there is no tRNA for termination codons. The releasing factors **RF1** or **RF2** with the help of enzyme ***peptidyl transferase*** release polypeptide from 'P' site and also tRNA carrying the polypeptide. Finally ribosome dissociates into 60S and 40S subunit. **RF1** recognises the termination codons UAG and UAA while **RF2** recognises UGA and UAA.

In eukaryotes only one releasing factor **eRF1** is present. GTP is required for the binding of this factor to ribosome. On hydrolysis, GTP provides energy for the release of **eRF1** from the ribosome.

4.Modification of Released Polypeptide Chain or Polypeptide Folding

The formyl group of the first amino acid (methionine) of the released polypeptide chain is removed by the enzymes deformylases. Some other enzymes like exopeptidases remove some of the amino acids either from N-terminal end or the C-terminal end or from both ends of the polypeptide chain. Finally, this polypeptide chain alone or along with other chains gets folded to assume the tertiary or quarternary structure and is modified inot a functional enzyme.

Inhibition of Protein Synthesis

Some antibiotics inhibit protein synthesis in bacteria. The list below shows the effect of various antibiotics:

1. **Tetracycline :** Inhibits binding of aminoacyl tRNA to ribosome.
2. **Streptomycin :** Inhibits initiation of translation.
3. **Neomycin :** Inhibits interaction between tRNA and mRNA.
4. **Chloramphenicol :** Inhibits action of enzyme ***peptidyl transferase*** and formation of peptide bonds.
5. **Erythromycin :** Inhibits translocation of ribosome along mRNA.

Post-Translational Processing of Proteins

The products of translation are **polypeptides**. They are biologically inactive. Several steps of folding and biochemical modifications are needed, then only they are able to perform their specific function. Also most proteins are synthesised in the cytoplasm (except those formed in mitochondria and chloroplasts) Therefore, they are to be transported to cellular regions where they have to carry out specific biological function or catalyse specific reaction.

1.Protein Folding

The polypeptide chains represent the **primary structure**. They undergo folding immediately after the chains begin to emerge from the ribosomes in the endoplasmic reticulum. As a result of folding, the polypeptide chains assume stable three-dimensional secondary and tertiary structures. The secondary structure is achieved by the formation of hydrogen bonds, electrostatic interactions and disulphide bonds. The overall folding process is described as cooperative, because each folding step facilitates and initiates the formation of another favourable interaction.

Chaperones bind to the amino (N) terminal portion of the nascent polypeptide chain and stabilise it in an unfolded configuration during the synthesis of the polypeptide. The completed polypeptide is then released from the ribosome and is able to fold into its correct three-dimensional conformation.

Molecular Chaperones and Protein Folding: For smaller proteins, the protein folding is a spontaneous, self assembly process. But larger proteins assume their native structure with the help of special proteins, called **molecular chaperones**, which help in correct folding and protect and stabilise transitory intermediates. They are also called **accessory proteins**. These were first discovered in *Drosophila* and were originally described as **heat-shock proteins**. Now they have been isolated from all prokaryotes and eukaryotes. **Molecular chaperones** mainly belong to two families, **Hsp 70** and **Hsp 60**. They act as catalysts and carry out following functions.

- Folding by binding to unfolded or partially folded polypeptide chains.
- Preventing formation of unwanted interaction that do not provide biologically active structure.
- Stabilising interaction that may be otherwise unstable.
- Avoiding improper or unfruitful hydrophobic interactions.
- Avoid aggregation and precipitation of proteins during their synthesis.
- Some chaperones couple ATP hydrolysis to protein-folding process.
- Bind to folding proteins to hide their exposed hydrophobic amino acid residues.

2.Biochemical Modifications

The biological changes in proteins to achieve their native functional form are:

1. **Proteolytic Cleavage :** The amino acid **'N-formyl-methionine'** in prokaryotes and methionine in eukaryotes attached to the N-terminus of polypeptides chain are removed by hydrolysis. This is called **proteolyitc cleavage** and is an essential step. In approximately 50 per cent of eukaryotic proteins an acetyl group is added to their amino end.
2. **Amino Acid Modification :** The activity of enzymes and some other proteins is altered by **phosphorylation**, **methylation** or **hydroxylation** of some amino acids. For example, hydroxyl groups (OH) of serine, threonine and tyrosine are modified by phosphoryl group from ATP. The hydroxylation of proline and lysine changes the protein to collagen (a structural protein).
3. **Attachment of Carbohydrates :** In glycoproteins, carbohydrate molecules are covalently bonded to the proteins. These provide immunological protection, cell-cell recognition and blood clotting.
4. **Addition of Prosthetic Group :** Many enzymes can function in association with a covalently bound **cofactor** or **prosthetic group**. The prosthetic groups are heme, FAD, biotin and pantothenic acid.
5. **Trimming of Polypeptide Chain :** In some cases long polypeptide chains are trimmed by enzyme **exopeptidase**. The amino acid residues are removed one by one either from N- or C-terminus. For example, long polypeptide chain of preinsulin is cut to a small functional insulin of only 51 amino acids.

6. **Signal Molecules are added or removed** from the N-terminal end of some proteins. The proteins are synthesised in cytoplasm and are transported to ER, Golgi complex, mitochondria or chloroplasts or they are excreted out of the cell.

   For transportation to their specific destination proteins have a sequence of up to 30 amino acids at their N-terminal. This is called **signal sequence**. Once a protein has reached its destination the signal sequence is removed before it assumes the functional status.

7. **Polypeptide Chains are also complexed with metal ions or prosthetic group.** The tertiary and quarternary level of protein structure is attained only when some metal atom is complexed with it. Iron in case of haemoglobin and Mg in case of chlorophyll is essential for their quarternary structure.

Regulation of Gene Expressions

Gene Regulation in Prokaryotes

In prokaryotes, the gene regulatory systems are adapted to provide maximum growth rate in a particular environment and gene expression is controlled primarily at transcription level with some control at the translational level. Transcriptional control is accoplished by using **on off** mechanism. It means a particular gene is switched on and synthesis of its mRNA is induced only when the gene product is needed. When the gene product is not needed, the gene is switched off and synthesis of its mRNA stops. The post-transcriptional control (i.e. control at translation level) is achieved by rapid degradation of mRNA molecules involved in this synthesis and modification of nascent polypeptide chain or chains into functional proteins.

Strategies of Gene Regulation

In bacterial cell protein synthesis or gene expression is mostly controlled at transcription level. This regulation may be **negative** or **positive**.

1. **Negative Regulation :** In negative regulation an **inhibitor** or **repressor** is present in the cell. The binding of inhibitor prevents or suppresses transcription of a relevant gene. A signal molecule or **inducer** is needed to react with and remove the inhibitor so as to allow the initiation of transcription. The negative regulation of gene expression is also called **repression**.
2. **Positive Regulation :** In this sytem, an **effector** or **inducer** molecule activates the promoter which promotes transcription of mRNA. There is no inhibitor inducer interaction. The positive regulation of gene action is also called **induction** and the substance that induces gene action is called **inducer**. The inducer may be a protein, a small molecule or a molecular complex.

   Positive regulation can be achieved in any one of the two ways :

   - Transcription will continue until a molecular signal binds to the activator molecule and causes it to dissociate from DNA.
   - The activator-molecular signal complex binds to the promoter. If the signal molecule dissociates, the activator falls off DNA, stopping the transcription.

1. **Mechanisms of Gene Regulation At transcription Level**

Induction and Repression

A set of genes will be switched on when there is necessity to metabolise a new substrate. When these genes are switched on, enzymes are produced. This phenomenon is called **induction**. Similarly, when a metabolite needed by bacterium is provided in excess in the medium, bacterium stops its synthesis and the gene associated with its metabolism is turned off. This is called **feedback repression**.

1. **Induction and Inducible System (Inducible Operon)**

   In genetic induction the gene or genes are induced or 'switched on' to produce mRNA, needed for the synthesis of required enzyme or enzymes. The substance which induces the gene for protein synthesis or enzyme production is known as **inducer**. The phenomenon of induction can be demonstrated by growing *E. coli* in different nutritional media as follows :

   - When *Escherichia coli* are grown in a culture medium containing glycerol, they produce all those enzymes, which are needed for the breakdown of glycerol. The synthesis of all other enzymes is maintained at the minimum level.

In the above experiments substrate glycerol act as inducers for the synthesis of required enzymes. The enzymes whose synthesis can be induced by adding substrate are called **inducible enzymes**. The gene complex responsible for the synthesis of inducible enzymes is called '**inducible system**'.

2. **Repression or Repressible System**

In repression the activity of gene or genes is suppressed and the synthesis of specific protein is stopped or reduced. The substance which stops or suppresses protein synthesis is known as **repressor**.

For example, in minimal culture medium, *E. coli* cells can synthesise all the enzymes needed for the synthesis of different amino acids. But when a particular amino acid is added in the culture medium, the enzymes needed for its synthesis are not produced. The enzymes, whose synthesis is checked by the addition of end product, are known as **repressible enzymes**. The genes associated with these enzymes form **repressible system** and the substance that represses the synthesis of repressible enzymes is called a repessor.

As a matter of fact, some enzymes are normally present in the cell but cease to be synthesised when high concentrations of their end products are present. These end products are called **corepressors**. The regulator gene produces a substance called **aporepressor**. The aporepressor unites with corepressor to form the functional **repressor molecule**. This repressor molecule inhibits transcription of mRNA from all these structural genes which are responsible for the synthesis of enzymes of synthetic pathway. Most repressible enzyme systems occur in anabolic pathway.

1. The Lactose System in E. coli

Jacob and **Monod** (1961), while studying catabolism of lactose in E. coli, suggested that the action of most genes is regulated at the transcription level by **induction** and **repression** phenomena. Three enzymes, namely, ***β-galactosidase, galactose permease*** and ***thiogalactoside transacetylase*** are needed for lactose catabolism. The genes or cistrons (DNA segments) for these three enzymes are represented by ***lac Z***, ***lac Y*** and lac A. These are called **structure genes** and are located in a linear sequence in the bacterial chromosome. The action of these genes is controlled in a coordinated fashion by **regulatory genes**.

***Lac* Operon System**

Jacob and **Monod** explained regulation of the activity of these *lac* genes by **operon model.** The operon model for lactose catabolism is called ***lac* operon**. It consists of two kinds of genes:

1. **Structure Genes :** These genes are segments of DNA that are associated with the synthesis of those enzymes that are needed for catabolism of lactose. In ***lac* operon** there are three structure genes:
   - *lac Z* gene for ***β-galactosidase*** enzyme
   - *lac Y* gene for ***galactose permease*** enzyme and
   - *lac A* gene for ***thiogalactoside transacetylase***

   These genes produce a single polycistronic mRNA molecule

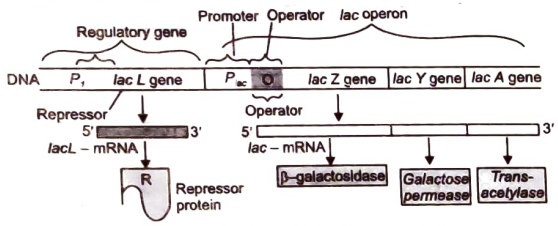


Fig. Genetic map of *lac* operon in *E.coli*.

2. **Control Genes or Regulatory Genes :** These genes control the activity of structural genes and lie immediately adjacent to the structural genes. There are three control geyes, namely:
   - **Operator gene (O) :** It lies just before, i.e. upstream of first structural gene and overlaps the promoter sequence. It controls transcription of mRNA from structure genes. It is under the control of repressor molecule, produced by the regulator on repressor gene.

- **Promoter gene (P) :** It lies immediately adjacent to operator gene. RNA polymerase binds at this site. Transcription of an operon begins at promoter, the site where ***RNA polymerase*** binds for the transcription of mRNA from structural genes. It controls rate of mRNA synthesis.
- **Repressor or regulator gene (L) :** It lies outside the operon and produces a **repressor substance.** This repressor binds to operator gene and suppresses the transcription of structural genes.

Functioning of Lac Operon

Beckwith (1967), **Epstein** and **Beckwith** (1968) and **Martin** (1969) have described the operation of *lac* operon in *E. coli* in the following ways:

1. In the absence of an inducer-lactose, the regulator gene L produces a repressor protein (R) which binds strongly to the operator site and prevents its transcription. As a result, the structure genes do not synthesise mRNA and proteins are not formed.
2. When an inducer-lactose is introduced in to the medium, it enters the cell and gets modified in such a way that it binds to the repressor. The repressor now fails to bind to the operator. The operator is, therefore, free and induces the ***RNA polymerase*** to bind to the initiation site on promoter and to transcribe the cistrons into a polycistronic *lac* mRNA. This codes for the three enzymes necessary for lactose catabolism.

Definition of Operon

An operon is a group of closely linked structural genes (or cistrons) and the associated control genes (operator and promoter genes) which regulate the genetically controlled metabolic acitivity.

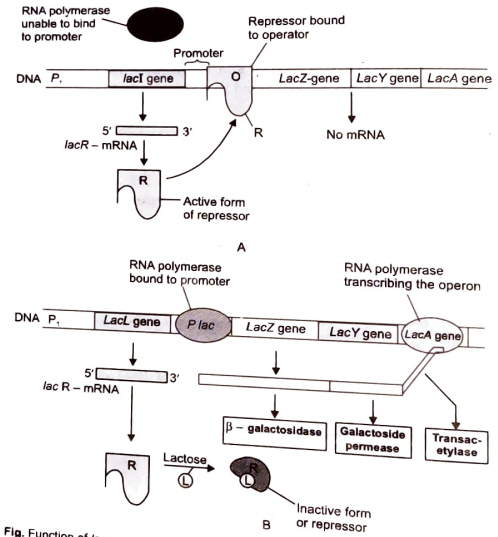


Fig. Function of *lac* operon in Escherichia coli. **A.** Lactose absent, repressor active and bound to operator, operon repressed; transcription blocked and no mRNA synthesised; **B.** Lactose present, repressor bound to lactose and inactivated, repressor on longer binds the operator, operon derepressed, transcription begins and mRNA synthesised

2. Allosteric Regulation of *Lac* Repressor

Repressor proteins are **allosteric**, i.e. these exist in two conformational forms. In one form the protein is active and in the other form it is inactive. When effector molecule binds to the protein, the protein molecule undergoes conformational change and becomes inactive. But change is reversible. As soon as effector moves away, protein becomes active again by returning to the original form.

Role of Cyclic AMP in Catabolite Repression (Positive Control)

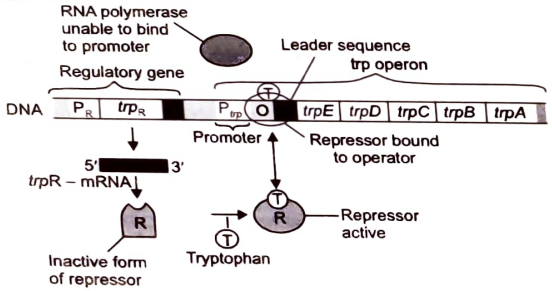
Small molecules of cyclic AMP (cAMP) are distributed in bacteria and multicellular eukaryotes. Cyclic AMP is synthesised enzymatically by adenyl cyclase. Its concentration is regulated indirectly by glucose metabolism.

1. When bacteria are growing in culture medium containing glucose, the cAMP concentration in the cells is quite low.
2. In a medium containing glycerol or any other carbon source that cannot enter the biochemical pathway used to metabolise glucose, the concentration of cAMP becomes high. It regulates the activity of *lac* operon as well as of some other operons.
3. *E. coli* contains a protein called **catabolite activator protein (CAP).** It is encoded by a gene **Crp**.
4. Mutants of either Crp or adenyl cyclase gene are unable to synthesise *lac* mRNA. It means both CAP and cAMP are needed for lac mRNA synthesis.
5. CAP and cAMP bind together forming **cAMP-CAP** unit. The complex unit regulates lac system. It is a **positive regulator**. It functions independent of *lac* repression system.
6. In the absence of cAMP-CAP complex, the ***RNA polymerase*** binds weakly to promoter. The weak binding fails to initiate transcription.
7. cAMP-Crp complex binds to a base sequence in DNA in the promoter region to initiate transcription.

Tryptophan Operon: A Repressible Operon System

Tryptophan operon (*trp* operon) in *E. coli* is responsible for controlling synthesis of amino acid **tryptophan**. Its synthesis is carried out by five enzymes. The genes encoding these enzymes are located adjacent to one another in succession and are named *trpE, trpD, trpC, trpB* and *trpA*. Gene *trpE* is the first one to be translated. Gene **promoter**, **operator,** two regions called leader and **attenuator** are located close to gene ***trpE***. The repressor gene ***trpR*** is present far from this gene cluster.

1. Tryptophan operon is switched on for normal transcription when tryptophan amino acid is not accumulated in the medium, i.e. in short supply.
2. This operon is switched off only when tryptophan accumulates in the medium.



A Tryptophan present, binds to repressor and activated repressor binds to the operator, operon repressed

The regulatory protein of *trp* operon is produced from the *trpR* gene. Mutations either in this gene or in the operator gene causes constitutive initiation of transcription of *trp* mRNA. This protein synthesised from mutant gene is called **aporepressor**. It does not bind to the operator unless tryptophan is present. The aporepressor and tryptophan molecule join to form active ***trp repressor***, which binds to the operator to switch it off.

3. Attenuation : Regulation after the Initiation of Transcription

Charles Yanofsky and his colleagues found that *trp* operon has a regulatory site between promoter, operator and the first structural gene *trp* E. This site of DNA is called leader sequence (L). It is non-coding DNA segment which is transcribed into leader mRNA at 5′ end of mRNA. This *trp* leader mRNA is 162 nucleotides long. It is sensitive to tryptophan level and exercises an additional control on the transcription of polycistronic *trp* mRNA. The effect of this control element, i.e. of leader sequence is called **attenuation**, because it reduces or attenuates synthesis of mRNA.

Structure of Leader mRNA

The 162 bases long leader mRNA has following segments:

1. It has an initiation codon **AUG** at 5′ end and a stop codon **UGA** in the same reading frame. This region encodes a polypeptide, consisting of 14 amino acids. It is called **leader polypeptide**.
2. The leader polypeptide mRNA contains two adjacent **tryptophan codons** at position 10 and 11. It means the 10th and 11th amino acids in leader polypeptide are tryptophans.
3. Leader mRNA has four distinct regions denoted as 1, 2, 3 and 4. These regions can base pair with each other in different ways, forming several hairpin structures. For example:
   - In a stable form the free leader mRNA shows base pairing between 1 and 2 and between 3 and 4 regions.
   - In low concentration of tryptophan, ribosome is stalled in region 1 so that regions 2 and 3 base pair together. Transcription of region 4 and of subsequent structure genes is completed.
   - In high concentration of tryptophan in the medium, the 3 and 4 regions get base paired and synthesis of protein is interrupted on reaching the 2nd region.
4. After region 4 is present a string of eight U lying just prior to structure gene *trp E*.
5. The part of leader mRNA comprising of regions 3, 4 and a string of eight U is called **attenuator**. When base pairing between 3 and 4 regions creates a hairpin, the attenuator acts as a transcription termination signal and *RNA polymerase* and growing RNA chain detaches from DNA.

Mechanism of Attenuation

During translation, a ribosome attaches to the first binding site (the initiation codon) on *trp* mRNA. Transcription of mRNA and its translation follow closely.

1. When tryptophan level in the medium is low, the concentration of tryptophenyl-tRNA (i.e. tRNA molecule carrying tryptophan) is also low. When ribosome arrives at the tryptophan codons in leader sequence, it waits for the arrival of tryptophenyl—tRNA and blocks region I. As a result, a hairpin structure is formed by pairing between regions 2 and 3.

   Thus, region 3 is unable to pair and form hairpin with region 4. The termination hairpin loop is not formed and transcription of structural genes of tryptophan operon continues to produce polycistronic *trp* mRNA.
2. When tryptophan level in the medium is high, the tryptophenyl tRNA cencentration is also high. The ribosome moving from the initiation codon, passes over tryptophan codons and stops at stop codon. Thus, it blocks region 2 of *trp* mRNA. This permits pairing of segments 3 and 4 and a 3-4 hairpin loop is formed. This along with poly U causes the termination of mRNA and structural genes are not transcribed, stopping the tryptophan synthesis.

   A regulatory mechanism of this type can not exist in eukaryotes because transcription and translation can not be coupled. In eukaryotes, transcription occurs in the nucleus and translation in the cytoplasm.

4. Feedback Inhibition or End Production Inhibition

In case the end product of a biosynthetic pathway is not utilised, it accumulates inside the cell and stops further synthesis of that substance. This is known as **feedback inhibition** of the gene action. The inhibition is caused by the binding of the accumulated end product to the enzyme. For example, **Umbarger** (1961) demonstrated that **isoleucine synthesis** in *E. coli* can be inhibited if some isoleucine is added to the medium.

By the binding of isoleucine to the first enzyme, the enzyme is inactivated. This phenomenon of inhibition at the enzyme level is known as **allosteric interaction**.

Thus, in feedback or end product inhibition the accumulated end product of a biosynthetic pathway combines loosely with the first enzyme in the pathway. This complex does not bind to the operator or promoter site but the tertiary structure of enzyme is modified which makes the enzyme inactive. This **allosteric transition** of enzyme molecule blocks its enzyme activity and prevents overproduction of end product and also the intermediate metabolites.

In end product inhibition the expression of gene is regulated after proteins have been synthesised. Thus, it is **post-translational control of gene action**.

Gene Regulation at Translation Level

Translational regulation of protein synthesis refers to the control of number of copies of a particular protein translated from a polycistronic mRNA. It varies from gene to gene and there is a gradient of translation decreasing from 5′ terminus to 3′ terminus of mRNA.

This depends on :

- Varying efficiency of initiation of translation.
- Different spacing between chain termination codon and next initiation codon.
- Different sensitivity of various regions of mRNA to degradation.
- Inhibition of a particular gene by its gene product (**feedback inhibition**).
- Synthesis of ribosomes.

Gene Regulation in Eukaryotes

Though there are certain basic similarities between prokaryotic and eukaryotic regulation of gene expression. The transcriptional regulation is more complex in eukaryotes. There are several dramatic differences between the two because :

1. Eukaryotes do not have typical operons as found in bacteria.
2. Eukaryotes have many more genes than prokaryotes. Their expression is regulated differently in different tissues and at different stages of development and differentiation.
3. Transcriptional regulation in eukaryotes involves many more regulatory proteins and DNA control elements such as **promoters, enhancers, activators, insulators, mediators** and other upstream or downstream nucleotide sequences.
4. Some eukaryotic regulatory proteins interact directly with DNA as in prokaryotes many regulatory proteins act on DNA indirectly, via interactions with other proteins.
5. The regulatory proteins are synthesised by ribosomes in the cytoplasm but transcription of RNA occurs in the nucleus. Therefore, the regulatory proteins are to be transported into the nucleus.
6. In eukaryotes DNA is associated with histone proteins and form **nucleosomes**. Nucleosomes do not allow DNA transcription. Removal of histones is essential for transcription to start.
7. In eukaryotes long sections of DNA are frequently folded tightly into heterochromatin. In heterochromatic stage, they become inaccessible to RNA polymerase and transcription factors.
8. Introns are present in most eukaryotic genes but only few in prokaryotic genes.
9. In eukaryotes RNA is synthesised in the nucleus and is transported through nuclear pores to the cytoplasm where it is translated.
10. Eukaryotes possess mechanism for rearrangement of DNA segments in a controlled manner by increasing number of copies of genes (gene amplification).
11. Regulatory regions are much larger than those of prokaryotes and these may be away from promoters.
12. In eukaryotes most genes are subjected to positive control. Repressors are rare and usually behave differently to those in bacteria.

Levels of Gene Regulation in Eukaryotes

In eukaryotes gene expression is regulated at the following levels:

1. Genome level.
2. Transcription of RNA.
3. Processing of RNA transcript.
4. Transport of RNA out of nucleus.
5. Degradation of mature RNA.
6. Translation of mRNA.
7. Modification of processing of proteins.
8. Degradation of protein products.

II. Rregulation of Gene Expression at Transcription Level

In eukaryotes DNA is associated with histones (the basic proteins) and forms nucleosomes. Therefore, in eukaryotes transcriptional control on genes includes :

- Acetylation of histones
- Chromatin remodelling complexes
- Methylation of nucleotides
- Control elements
- Transcription factors
- Mediators
- Insulator
- Regulatory proteins

1. Transcriptional Control by Acetylation of Histones

Acetylation of histone tails promotes disaggregation of neighbouring nucleosomes. All the four types of histones (H2A, H2B, H3 and H4) can be acetylated, but H4 is acetylated the most. Acetylated histones form less condensed chromatin and provide more chances for *RNA polymerase* and other proteins to approach DNA helix. Acetylation is achieved by enzymes ***histone acetyl transferases (HATs)*** and deacetylation is carried out by ***histone deacetylases (HDACs)***.

HATs activate the genes whereas HDACs suppress them. Several proteins which were previously called coactivators belong to HAT category. Similarly, several so called corepressors are HDACs proteins.

2. Transcriptional Control by Chromatin Remodelling Complexes

Chromatin remodelling complexes are protein polymers that rearrange histones in the nucleosome so that promoter region is exposed and RNA polymerase can get access to it. These complexes are formed of 2 to 12 proteins. They bind strongly to nucleosomer or DNA and loosen the winding of DNA. These are of two types: Small complexes formed of 2 to 6 polypeptides and are called **initiation switches** and larger complexes have 8 to 12 polypeptides and are designated as **Swi/Snf complexes**. The former binds to histones and the latter to DNA. The remodelling of nucleosomes is achieved in two ways:

1. By sliding the nucleosomes along a DNA molecule and exposing nucleotide sequences for transcription.
2. By rearranging, histones loosen the ribosomes and expose nucleotide sequences.

3. Transcriptional Control by Methylation of DNA (Silencing of Gene)

Methylation of nitrogenous bases in DNA silences genes and is often used to control gene expression during development in higher organisms. Silencing involves methylation of cytosine in 5'.....CG....3' sequences.

The methyl groups project into major groove of DNA and interfere with the binding of transcription factors. Also methylated cytosine-guanine sequences are recognised by **methyl-cytosine binding proteins** (MeCPs). When bound to methylated-CG-sequences, MeCPs help in removal of acetyl groups from histones.

This leads to the condensation of DNA to form heterochromatin, which is inaccessible for transcription. In animals about half of the functional genes are located close to – **CG – islands** or clusters of –CG–sequences. The housekeeping genes that are expressed in all cells and tissues, possess nonmethylated–CG–islands, and are always acetylated and active. In contrast, the –CG–islands of **tissue specific genes** get nonmethylated only in those tissues where these genes are expressed.

4. Transcriptional Control by Control Elements

The protein-coding genes(structural genes) of eukaryotes have two types of controlling elements or sequences. These are **promoter elements** and **enhancer elements**.

(i) Promoter Elements

They are located upstream of initiation site of coding genes. They serve two important functions. Some promoters identify the point of initiation of transcription. For example, TATA element is present at the site where transcription is to begin. Other promoters are regulatory. These control whether transcription is to occur or not. The transcription factors bind to these promoter sites. These promoter elements may produce **positive regulatory proteins** or **negative regulatory proteins**. Accordingly, the promoter elements are called **positive regulatory elements** and **negative regulatory elements**. The former activates transcription of structure genes, while latter turns off transcription.

(ii) Enhancer or Activator Elements

The enhancers are nucleotide sequences that determine the maximal transcription of a gene by binding specific transcription factors, especially during development or in different cell types. They are located several kilobases away from the promoter either upstream or downstream. They are formed of a cluster of recognition sites and therefore bind several regulatory proteins. Binding of transcription factors to them influences polymerase activity and consequently the gene expression. Enhancers need not lie close to its own promoter. It can increase transcription from any promoter within its neighbourhood. When an enhancer switches a gene on, the DNA between it and the promoter loops out.

5. Transcriptional Control by Transcription Factor

In eukaryotes, initiation of transcription needs either the formation of **transcription complex** between initiation sequences of DNA, ***RNA polymerase*** and several transcription factors or regulation of binding of RNA polymerase to a pre-existing transcription complex by a transcription factor.

Transcription factors are protein complexes that respond to specific signals and in accordance regulate expression of protein coding genes by binding to the control region of DNA. They are of following two types:

1. **General Transcription Factors:** They bind initially to the TATA box and are needed for the initiation of all the transcriptions. They are TFIIB, TFIID and TFIIH. They are needed for the recognition and binding to the promoter.
2. **Specific Transcription Factors:** They bind to general transcription factors in response to specific signals to initiate transcription of specific genes. They are TFIIA, TFIIC, TFIIF. Typical specific transcription factors have following common features:
   - Transcription factors respond to some specific stimulus that signals to switch on one or more genes.
   - Transcription factors can enter the nucleus through nuclear pores.
   - They recognise and bind to a specific promoter sequence on DNA.
   - They make contact with the transcription apparatus.
   - Transcription factors have at least two domains. One domain binds to promoter DNA and another interacts with the transcription apparatus.

6. Transcriptional Control by Mediator

Mediator is also a protein complex consisting of about 20 protein subunits. it sits on top of ***RNA polymerase II***, receives signals from activators or enhancers and passes them to *RNA polymerase* of the transcription apparatus. Some subunits of mediator act in a positive manner in controlling transcription of genes while others act negatively.

7. Transcriptional Control by Insulator

Insulators are regions of DNA rich in GC and CG sequences that are quickly methylated. Also insulators have clusters of nucleotide sequences that bind copies of special zinc finger proteins which are also called **insulator binding proteins (IBPs)**. On binding to insulation IBPs block enhancer but on being methylated the insulator fails to bind to IBPs and end bind and switch on the gene. For example, in vertebrates the Igf2 gene (encoding insulin-like growth factor-II) and gene H19 are located close together and have common insulator and enhancer. When insulator is not methylated, the protein CTCF binds to the insulator and enhancer switches on the H19 gene. But when insulator and H19 gene are methylated, the CTCF protein does not bind to them and H19 gene is not switched on. It allows the enhancer to activate the Igf2 gene.

8. Transcriptional Control by Regulatory Proteins

The key participant in transcription is ***RNA polymerase***. Its activity is controlled or mediated by two types of regulatory proteins, the activators and repressors and also the enzymes that cut or modify DNA, recognise palindromes or inverted repeats in DNA. Both types of these proteins recognise and bind to specific base sequences in DNA and turn the polymerase **'on'** or **'off'**. The regulatory proteins have descrete binding domains. They can read specific base sequences and bind to them. They also have **protein binding domains**, by which they bind to ***RNA polymerase*** or other regulatory proteins.

The action of regulatory proteins is also influenced by other molecular signals like small metabolites and hormones. Their binding can increase or decrease the affinity of protein DNA interaction.

Classes of Regulatory Proteins

Based on the presence of common structural motifs, the proteins can be placed in the following three classes:

1. Helix-turn-helix motif and helix-loop-motif (HLH)
2. Zinc finger motif
3. Leucine zipper motif

1.Helix-Turn-Helix Motif (HTH)

It is the most common DNA binding domain found in prokaryotic regulatory proteins. It is a dimer formed of about 20 amino acid residues arranged in two short α-helices of 7 to 9 amino acids. The two helices are labelled as α2 and α3 and are connected by a β turn. One of these helices fits into the major groove of DNA and makes contact with the bases. Proteins with HTH and HLH motifs identify and bind as dimers to the inverted repeats in DNA. Therefore, this helix is called **recognition helix**.

In helix-loop-helix the turn of loop is longer. In HLH motif, the first α-helix binds to DNA, whereas in HTH motif the second α-helix binds to DNA. HTH motif is used both in prokaryotes and eukaryotes, whereas HLH motif is used in eukaryotes only.

2.Zinc Finger Motif

This motif consists of central zinc atom surrounded with 25-30 amino acid residues. Zinc is usually bound to two cysteine molecules which lie in a very short piece of β-sheet having two histidines and a β-hairpin.

The regulatory proteins with zinc-finger motif bind to DNA in the major groove and wrap around the axis of double helix. The transcription factor TFIIIA from *Xenopus laevis* and glucocorticoid receptor protein are examples of having zinc-finger motif.

3.Leucine Zipper Motif

Leucine zipper motifs are found in many eukaryotic transcription factors. A leucine zipper motif is formed of two α-helices. Each α-helix has hydrophobic zones and basis ends. The two α-helices are joined by their hydrophobic zones but their basic ends remain wide apart and fit into the major groove of DNA. The regulatory protein with leucine zipper assumes zipper motif and has two functional domains.

- **Protein binding domain** or **dimerisation domain** of α-helix is formed of about 30 amino acid residues, rich in leucine. The leucine residues are four of five in number and are spaced at an interval of seven amino acids. These form the hydrophobic side chains. The leucine hydrophobic side chains of two α-helices are held together by hydrophobic interactions.

- **DNA binding domain** of a-helix is rich in positively charged amino acids (lysine and arginine). It lies next to leucine zipper and binds to DNA.

The binding of metabolites or hormones to regulatory proteins can increase or decrease the affinity of protein DNA interaction.

III. Gene Regulation at Post-Transcription Level or Gene Regulation at Processing of RNA Primary Transcript

1. Post-transcriptional Control by Choice of Splice Site

In eukaryotes all three types of RNAs are synthesised as precursor molecules. Gene expression can be regulated at the level of processing of these precursor RNA molecules into functional RNAs. The post transcriptional control is exercised by two regulatory actions:

- Choice of splice site, and
- Choice of poly (A) site

2. Regulation of Gene Expression by RNA

Most biologists believe that only proteins regulate gene expression. The regulatory proteins sense and bind a small molecule, such as a hormone or metabolite and then bind to DNA or RNA to induce or suppress synthesis of certain proteins. It has now been discovered that some forms of RNA also control and initiate changes in gene expression. Following two types of RNAs are found to control gene expression.

(i) **Ribozymes or Riboswitches :** Some ribozymes both in prokaryotes and eukaryotes are found to identify the presence of some metabolities or hormones in the cell and act to regulate the concentration of proteins that act on these small molecules. These unique ribozymes are described as metabolite-responsive **genetic switches** or **riboswitches**. These were discovered during vitamin synthesis in bacteria. For example, which vitamins like B_1 and B_{12} reach certain concentration inside the cell, they interact with special RNA to relay the message for slowing down the translation. The RNA segment that acts as riboswitch precedes the coding part of mRNA molecule. It makes the mRNA inactive by cleaving the biologically active mRNA or the metabolite attached to the riboswitch and switches off the translation process.

(ii) **RNA Interference :** Some RNA molecules can suppress or stop the expression of certain genes. This phenomenon is called **post-transcriptional gene silencing (PTGS)** or **RNA interference (RNAi)**.

This phenomenon of RNA interference was discovered by **Richard Jorgensen** and colleagues while working to darken the purple colour of petunia flowers by introducing additional copies of the concerned gene. He found that the expression of introduced genes as well as of the similar endogenous gene for pigment was suppressed instead of being enhanced.

The biochemical mechanism of RNAi has been studied in *Drosophila*. The incorporated double-stranded RNA (dsRNA) is attached by an ATP dependent endonuclease enzyme, named **DICER**. This enzyme catalyses cleavage of the long dsRNA into small fragments of 21-23 base pairs. These dsRNA fragments are called **small interfering RNAs (siRNAs)**. They assemble into endonuclease-containing complexes, called **RNA-induced silencing complexes (RISCs)**. Their double-stranded RNA unwound into single strands. Finally, the complexes join with single-stranded regions of the target mRNA and cleave it to prevent translation.

3. Control on Transport of RNA

The transport of mRNAs from nucleus to the cytoplasm is another important control point in eukaryotic gene regulation. A spliceosome retention model has been proposed for this regulation. The processing of pre-mRNA involves snRNPs (small ribonucleoprotein particles). While undergoing processing pre-mRNA is retained in the nucleus by snRNAs forming spliceosomes. When all introns are removed from mRNA and all exons are joined, spliceosomes also dissociate and free mature mRNAs are released to move out of the nucleus through nuclear pores.

Gene Mutation

A gene mutation is abrupt inheritable qualitative or quantitative change in the genetic material of an organism The first scientific study of mutation was started in 1910, when **Morgan** started his work on fruitfly, *Drosophila melanogaster* and reported white eyed male individuals among red eyed male individuals. The discovery of white eyed mutants in *Drosophila* was followed by an extensive search of other mutants of *Drosophila* by **Morgan** and his co-workers and other geneticists.

Mutations occur frequently in the nature and have been reported in many organisms, e.g., *Drosophila*, mice and other rodents, rats, rabbits, guinea pigs and man.

Any change in sequence of nucleotides in the DNA will result in the corresponding change in the nucleotide sequence of mRNA. This may result in alignment of different tRNA molecules on mRNA (during protein synthesis). Thus, the amino acid sequence and hence, the structure and properties of the enzyme formed will be changed. This defective enzyme or structural protein may adversely affect the trait controlled by the protein. In consequence, a mutant phenotype makes its expression.

Kinds of Mutations

1.Classification of Mutation According to Type of Cells

According to their occurrence in somatic and germinal cells following types of mutations have been classified:

A. Somatic mutations. The mutations occuring in non-reproductive body cells are known as **somatic mutations**. Examples of somatic mutation have been reported in *Oenothera lamarckiana* (Hugo de Vries) and several other cases including man. In man somatic mutation causes several fatal diseases such as *paraoxysmal noctural hemoglobinura, circumscribed neurofibroma, unilateral retinoblastoma* and *heterochromia of the iris.*

B. Gametic mutations. The mutations occuring in gamete cells (e.g., sperms and ova) are called **gametic mutations**. Such mutations are heritable and of immense genetical significance. The gametic mutations only form the raw material for the natural selection.

2.Classification of Mutations According to the size DNA Quality

According to size following two types of mutations have been recognised:

A. Point mutation: When heritable alterations occur in a very small segment of DNA molecule, i.e., a single nucleotide or nucleotide pair, then this type of **mutations** are called "**point mutations**". The point mutations may occur due to following types of subnucleotide change in the DNA and RNA.

1. **Deletion mutations :** The point mutation which is caused due to loss or deletion of some portion (single nucleotide pair) in a triplet codon of a cistron or gene is called **deletion mutation**. Deletion mutations have been frequently reported in some bacteriophages (Phage T_4).

2. **Insertion or addition mutation :** The point mutations which occur due to addition of one or more extra nucleotides to a gene or cistron are called **insertion mutations**. The insertion mutations can be artificially induced by certain chemical substances called **mutagens** such as arcidine dye and proflavin. A proflavin molecule, it is believed insert between two successive bases of a DNA strand, thereby stretching the strand lengthwise.

   The mutations which arise from the insertion or deletion of individual nucleotides and cause the rest of the message downstream of the mutation to be read out of phase, are called **frameshift mutations**. They result in the production of an incorrect, hence, inactive protein, due to which the death of the cell may occur.

3. **Substitution mutation:** A point mutation in which a nucleotide of a triplet is replaced by another nucleotide, is called **substitution mutation**. The substitution mutation affect only a particular triplet codon. Such an altered code word (triplet codon) may designate a different amino acid and may result in the production of a protein with a single amino acid substitution. The substitution mutations alter the phenotype of an organism variously and are of great genetical significance. They may be of following types :

   (*i*) Transition : When a purine (e.g., adenine) base of a triplet codon of a cistron is substituted by another purine base (e.g., guanine) or a pyrimidine (e.g., thymine) is substituted by another pyrimidine base, (e.g., cytosine) then such kind of substitution is called **transition**. The transitional substitution mutations occur due to tautomerization.

Such uncommon forms of DNA bases are generated by single proton shifts and are called **rare states** or **tautomers**. A tautomeric shift is believed to occur when the amino (NH_2) form of adenine is changed to an imino (NH) form. Similarly, a tautomeric shift may occur in thymine changing it form the keto (C = O) form to the rare enol (COH) form. When a base occurs in its rare or tatuomeric state, it cannot be linked to its normal partner. However, a purine, such as adenine can in its rare state forms a bond with cytosine (besides thymine), provided the cytosine is in its normal state.

Effect of Chemical Mutagens on Nucleotide Sequence

(*a*) Alteration in Resting Nucleic Acid

1. **Deamination:** Some chemical substances such as nitrous acid causes transitional mutation due to oxidative deamination of DNA bases. In the process of oxidative deamination, the amino group (NH_2) of a DNA base is replaced by hydroxyl (OH) group by the chemical mutagen. Thus, adenine is deaminated into **hypoxanthine** by nitrous acid.

   By tautomeric shift the hypoxanthine (HX) is converted into more common or keto-tautomer which pairs with cytosine. The A : T pair, thus, can be converted to a G : C pair.

2. **Hydroxylamine (HA = NH_2. OH) and hydrazine (HZ = NH_2 NH_2).**

   When DNA is treated with hydroxylamine (HA), its cytosine base is the strongest reacting base. Hydroxylamine probably cause hydroxylation of cytosine at amino group giving rise to **hydroxylcytosine**, which then subsequently pair with adenine. Thus, hydroxylamine (HA) induces in DNA a GC → AT base pair transiton.

   The hydrazine affects DNA by breaking of rings of uracil and cytosine giving rise to **pyrazolone** and **3-aminopyrasole**, respectively. The treatment of RNA or DNA with **anhydrous hydrazine** results in the destruction of their pyrimidines.

3. **Alkylating agents :** Some alkylating agents carry one, two, or more alkyl groups in a reactive form and act as strong mutagens. Examples of some moist extensively studied alkylating agents include **diethyl sulphate (DES), dimethylsulphate (DMS), methylmethane sulphonate (MMS), ethylethance sulphonate (EES)** and **ethyl methance sulphonate (EMS)**.

(*b*) Alteration during Replication of Nucleic Acid

1. **Base analogues :** Certain chemical substances have molecular structure similar to the usual DNA bases that, if they are available, such **analogues** may be incorporated into a replicating DNA strand. For example, **5-bromouracil** (5BU) or its nucleoside **5-bromodeoxyuridine** (5-BUdR) in its usual (keto) form is a structural analogue of thymine (5-methyl uracil) and it will substitute for thymine.

2. **Inhibiton of precursors of nucleic acids :** There are some mutagens which interfere with the synthesis of nitrogen bases of nucleic acids such as purines or pyrimidines. Often lack of one base either causes breaks or causes pairing mistakes. For example, **azaserine** (a potent alkylating agent) inhibits purine synthesis and **urethance** (a mild alkylating agent) is an inhibitor of pyrimidine synthesis.

3. **Transversion :** The substitution mutation when involves the substitution or replacement of a purine with a pyrimidine or vice versa then that type of substitution mutation is called **transversion mutation**.

   Effects of Physical Conditions on Nucleotide Sequence

   High temperature and low pH value are known to affect depurination or loss of purine bases. The removal of a purine from a strand of DNA leaves a gap at that point. At the time of replication, it would be possible for any of the four bases to insert in the complementary newly formed strand. If the inserted nucleotide contained a purine, the complementary strand would contain a transversion.

B. Multiple mutations or gross mutations : When changes involving more than one nucleotide pair, or entire gene, then such mutations are called **gross mutations**. The gross mutations occur due to rearrangements of genes within the genome and may be of the following types :

1. The rearrangement of genes may occur within a gene. Two mutations within the same functional gene can produce different effects depending on gene whether they occur in the cis or trans position.
2. The rearrangement of gene may occur in number of genes per chromosome. If the numbers of gene replicas are non-equivalent on the homologus chromosomes, they may cause different types of phenotypic effects over the organisms.
3. Due to movement of a gene locus new type of phenotypes may be created, especially when the gene is relocated near heterochromatin. The movement of gene loci may take place due to following method:

   (*i*) **Translocation :** Movement of a gene may take place to a non-homologous chromosome and this is known as **translocation**.

   (*ii*) **Inversion :** The movement of a gene within the same chromosome is called **inversion**.

3.Classification of Mutation According to the Origin

According to the mode of origin, following two kinds of mutations have been recognised :

1. **Spontaneous mutations:** The spontaneous mutations occur suddenly in the nature and their origin is unknown. They are also called "**background mutation**" and have been reported in many organisms such as *Oenothera*, *maize*, bread molds, microorganisms (bacteria and viruses), *Drosophila*, mice, man, etc.
2. **Induced mutations:** Besides naturally occurring spontaneous mutations, the mutations can be induced artificially in the living organisms by exposing them to abnormal environment such as radiation, certain physical conditions (i.e., temperature) and chemicals. The substances or agents which induce artificial mutations are called **mutagens** or **mutagenic agents**.

Mutagenic Agents

The Mutagenic agents are of the following kinds :

A. Radiations : The radiations which are important in mutagenesis are of two categories: one type is **ionizing radiations** such as X-rays and gamma rays; alpha and beta rays; electrons, neutrons, protons and other fast moving particles. The second type is **non-ionizing radiations** such as ultraviolet and visible light. Both types of radiations induce mutations by following methods:

Dimerization : The ultraviolet radiation produces several effects on DNA, one being the formation of chemical bonds between two adjacent pyrimidine molecules in a polynucleotide and particularly, between adjacent thymine residues. As the two thymine residues associate, or **dimerize** to form a **dimer**, their position in the DNA helix becomes so displaced that they can no longer form hydrogen bonds with the opposing purines and thus regularity of the helix becomes distorted. Thus, dimerization interferes with the proper base pairing of thymine with adenine, and may result in thymine's pairing with guanine. This will produce a T-A to C-G transition.

B. Temperature as mutagen : The rate of all chemical reactions are influenced by temperature. It is reported that the rate of mutation is increased due to increase in temperature. For example, an increase of 10°C temperature increases the mutation rate two or three fold. Temperature probably affects both thermal stability of DNA and the rate of reaction of other substances with DNA.

A chemical mutation can cause mutation only when it enters in the nucleus of the cell. It can affect the chromosomal DNA by following two ways :

(1) Direct gene change : Certain chemical mutagens affect DNA directly. They affect the constituents of DNA only when DNA is not replicating. For example, nitrous acid converts adenine into hypoxanthine and cytosine to uracil by deamination. Like the nitrous acid, nitrogen mustard, formaldehyde, epoxides, dimethyl and diethyl sulphonate, methyl and ethyl methanesulphonate (MMS and EMS) and nitrosoguanadine (NG) also have direct mutagenic effect on the DNA molecule.

(2) Copy error : Certain chemical compounds, called **base analogues** (e.g., 5-bromouracil, 2-aminopurine, etc.) closely, resemble with certain DNA bases and are, therefore, act as mutagens.1 During DNA replication, they are incorporated by DNA in place of the normal DNA bases. Certain

other base analogues such as urethane triazine, caffiene (in coffee, tea and soft drinks), phenol and carcinogens, acridines (proflavin, etc.), have mutagenic effects. Certain inorganic substances such as manganese chloride is mutagenic for many organisms, as, they are the compounds which bind calcium and, thus, interfere with the integrity of the chromosome structure.

4.Classification of Mutation According to the Direction

According to their mode of direction following types of mutations have been recognised :

(A) Forward mutations : In an organism when mutations create a change from wild type to abnormal phenotype, then that type of mutations are known as **forward mutations**. Most mutations are forward type.

(B) Reverse or back mutations : The forward mutations are often corrected by error correcting mechanism, so that an abnormal phenotype changes into willd type phenotype. The may be of the following types :

5.Classification of Mutation According to Magnitude of Phenotypic Effect

According to their phenotypic effects following kinds of mutations may occur :

1. **Dominant mutations:** The mutations which have dominant phenotypic expression are called dominant mutations. For example, in man the mutation disease **aniridia** (absence of iris of eyes) occurs due to a dominant mutant gene.
2. **Recessive mutations:** Most types of mutations are recessive in nature and so they are not expressed phenotypically immediately. The phenotypic effects of mutations of a recessive gene is seen only after one or more generations, when the mutant gene is able to recombine with another similar recessive gene.
3. **Isoalleles:** Some mutations alter the phenotype of an organism so slightly that they can be detected only by special techniques. Mutant genes that give slightly modified phenotypes are called isoalleles. They produce identical phenotypes in homozygous or heterozygous combinations.
4. **Lethal mutations:** According to their effects on the phenotype mutations may be classified as lethals, subvitals and supervitals. **Lethal mutations** result in the death of the cells or organisms in which they occur. **Subvital mutations** reduces the chances of survival of the organism in which they occur. **Supervital mutations**, in contrast, cause the improvement of biological fitness under certain conditions.

6.Classification of Mutation According to Consequent Change in Amino Acid Sequence

1. **Missence mutations :** They change the meaning of a codon, changing one amino acid into another.
2. **Temperature sensitive mutations or Ts mutations :** If the substitution produces a protein that is active at one temperature (typically 30°C) and inactive at a higher temperature (usually 40 – 42°C).
3. **Nonsense or chain termination mutations :** They arise when a codon for an amino acid is mutated into a termination codon (U AG, UAA or UG A), resulting in the production of a shorter protein.

   Since, temperature-sensitive and chain termination mutations exhibit the mutant phenotype only under certain conditions, they are called **conditional mutations**; they are the most versatile and useful mutations.
4. **Silent mutations :** They change a nucleotide but not the amino acid sequence because they affect the *third position* of the codon, which is usually less important in coding. This is a silent mutation because it leaves the protein sequence unchanged.

7.Classification of Mutation According to the Types of Chromosomes

According to the types of chromosomes, the mutations may be of following two kinds :

1. **Autosomal mutations :** This type of mutation occurs in autosomal chromosomes.
2. **Sex chromosomal mutations :** This type of mutation occurs in sex chromosomes.

Significance of Mutation

The vast majority of mutations are deleterious to the organism and are kept at low frequency in the population by the action of natural selection. Mutant types are generally unable to complete equally with wild type individuals. Even under optimal environmental conditions many mutants appear less frequently than expected.

EXERCISE – I

MCQ TYPE QUESTIONS

1. The process by which DNA of nucleus passes genetic information to m RNA
   (a) Translocation (b) Transcription
   (c) Translation (d) Transportation
2. The genes are responsible for growth and differentiation in an oraganism through regulation of
   (a) Translocation
   (b) Transformation
   (c) Transduction and translation
   (d) Translation and transcription
3. One strand of DNA non template has base sequence CAG, TCG, GAT what will be the sequences of base in the m RNA
   (a) AGC, CTA, CAT (b) GTC, AGC, CTC
   (c) CAG, UCG, CTC (d) GAC, TAG, CTA
4. The Okazaki fragment in DNA chain growth
   (a) Result in transcription
   (b) Polymerize in the 3' to5' direction and forms replication fork
   (c) Prove semi conservative nature of DNA replication
   (d) Polymerize in the 5' to 3'; direction and explain the 3' to 5' direction DNA replication
5. Which of the following step of translation does not consume high energy phosphate bond
   (a) Translocation
   (b) Amino acid activities
   (c) Peptidyl transferase reaction
   (d) Aminoacyl t-RNA to a site
6. The process by which m rna is made by the DNA and protein by mRNA are respectively called as
   (a) Transcription and translation
   (b) Translation and transcription
   (c) Synthesis of m RNA and protein
   (d) Replication of mRNA and protein
7. Which amino acids are present in histones
   (a) Lysine and histidine (b) Valine and histidine
   (c) Arginine and lysine (d) Arginine and histidine
8. After mutation at genetic locus the character of an organism changes due to the change in
   (a) Protein synthesis pattern
   (b) RNA transcription pattern
   (c) Protein structure
   (d) DNA replication
9. In eukaryotic cell transcription, RNA splicing and RNA capping takes place inside the
   (a) Ribosome (b) Nucleus
   (c) Dictyosomes (d) ER
10. Balbiani rings are sites of
   (a) DNA replication
   (b) RNA and protein synthesis
   (c) Synthesis of lipids
   (d) Synthesis of polysaccharides
11. Which of these is a base analogue
   (a) Nitrous acid (b) Colchicine
   (c) 5-Bromouracil (d) Caffeine
12. If an isolated strain of DNA is kept at 82-90°C, then
   (a) It changes into RNA
   (b) It divide into multiple pieces
   (c) No effect
   (d) It uncoils into helixes
13. Which site of t-RNA molecule hydrogen bonds to an m-RNA molecule
   (a) Codon
   (b) Anticodon
   (c) 5' end of the t-RNA molecule
   (d) 3' end of the t-RNA molecule
14. Uridine monophosphate is found in
   (a) Centromere (b) Lysosome
   (c) Cell Wall (d) RNA
15. Removal of introns and joining the exons in a defined order in a transcription unit is called
   (a) Splicing (b) Tailing
   (c) Transformation (d) Capping
16. The enzyme which helps to cut strand of DNA duplex to release tension of coiling of two strands is
   (a) DNA ligase (b) DNA polymerase I
   (c) Topoisomerase (d) Helicase
17. A DNA molecule in which both strands have radioactive thymidine is allowed to duplicate in an environment containing non-radioactive thymidine. What will be the correct number of DNA molecules that contains some radioactive thymidine after three duplications
   (a) There will be four such molecules
   (b) There will be eight such molecules
   (c) There will be only one such molecule
   (d) There will be two such molecules

18. DNA consists of two complementary nucleotide chains. If the sequence of nucleotide in one of the chains is AGCTTCGA, then the nucleotide sequence in the other chain shall be
   (a) TAGCATAT (b) GATCCTAG
   (c) TCGAAGCT (d) GCTAAGCT
19. The enzyme that breaks H_2 bonds in DNA is
   (a) Helicase (b) Topoisomerase
   (c) Ligase (d) Polymerase
20. Exon part of mRNAs has code for
   (a) Protein (b) Lipid
   (c) Phospholipid (d) Carbohydrate
21. Which of the following is not correct?
   (a) A/T = 1 (b) A + T = G + C
   (c) A + G = C + T (d) None of these
22. The function of tRNA is
   (a) Production of mRNA
   (b) Production of ribosomes
   (c) Production of microsomes
   (d) Selection of amino acids
23. The base sequence for a nucleic acid segment is given CAG, AGC, CGA, CCA. From this it can definitely be concluded that it is a segment of
   (a) DNA strand (b) mRNA strand
   (c) tRNA strand (d) data not sufficient
24. The 3'-5' phosphodiester linkage inside a polynucleotide chain serve to join
   (a) One DNA strand with the other DNA starnd
   (b) One nucleoside with another nucleoside
   (c) One nucleotide with another nucleoside
   (d) One nitrogenous base with pentose sugar
25. DNA or RNA segment tagged with a radioactive molecule is called
   (a) Plasmid (b) Vector
   (c) Probe (d) Clone
26. Which one of the following also acts as a catalyst in a bacterial cell?
   (a) 23 sr RNA (b) 5 sr RNA
   (c) sn RNA (d) hn RNA
27. hnRNA undergoes two additional processing. Out of which, in one of them an unusual nucleotide is added to the 5' end of hnRNA. This is known as
   (a) Capping (b) Tailing
   (c) Splicing (d) Termination
28. Removal of RNA polymerase III from nucleoplasm will affect the synthesis of
   (a) tRNA (b) hnRNA
   (c) mRNA (d) rRNA
29. Centromere is a part of chromosome which helps in the
   (a) Division of centrosomes
   (b) Formation of spindle fibers
   (c) Movement of chromosomes
   (d) Formation of nuclear spindle
30. The distance between two genes in a chromosome is measured in cross-over units which represent
   (a) Ratio of crossing over between them
   (b) Percentage of crossing over between them
   (c) Number of crossing over between them
   (d) None of these
31. In polytene chromosomes dark bands are visible. These bands are formed by the apposition of
   (a) Protein particles
   (b) Chromomeres on chromonemata
   (c) Nucleosomes
   (d) None
32. Heterochromatin remains condensed in which part of chromosome
   (a) Secondary constriction- I
   (b) Secondary constriction- II
   (c) Telomeres
   (d) Both (a) and (b)
33. The genes, which are confined to differential region of Y chromosome only, are called
   (a) Mutant
   (b) Autosomal
   (c) Holandric
   (d) Completely sex-linked
34. In sex linkage, the speciality is
   (a) Atavism
   (b) Criss-cross inheritance
   (c) Reversion
   (d) Gene flow
35. "Cytochimeras" means
   (a) Cells having haploid number of chromosomes
   (b) Cells having two nuclei
   (c) Cells having different chromosomes than other vegetative cells
   (d) None of the above
36. What are allosomes?
   (a) Granular structures on chromosomes
   (b) Node like structures on chromosomes
   (c) Sex chromosomes
   (d) None of the above

37. In mRNA, AUG is the initiation codon and UAA, UAG and UGA are termination codons, Therefore Polypeptide cannot be synthesized beyond any of these triplets to the end of mRNA. Then which one of the following mRNA can be translated completely

(a) AUG UUC UCC UGG UAA UAU
(b) AUG UUC UCC UGA UGG UAU
(c) AUC ACGUAU UUC UGA CUC
(d) AUG UAU UUC UGC CUC UAG

38. Wild type E. coli cells are growing in normal medium with glucose. They are transferred to a medium containing only lactose as the sugar. Which one of the following changes takes place

(a) The Lac-Operon is repressed
(b) All Operons are induced
(c) E. Coli cells stop dividing
(d) The Lac-Operon is induced

39. When more than one codon code for the same amino acid, this is called as

(a) Universal nature of genetic code
(b) Redundancy of genetic code
(c) Punctuation in genetic code
(d) Continuous nature of genetic code

40. Genes that are involved in turning on or off the transcription of a set of structural genes are Called

(a) Polymorphic genes
(b) Operator genes
(c) Redundant gene
(d) Regulatory gene

41. A naturally occurring coding strand composed of alternating C and U residues would result in the formation of

(a) A polypeptide containing alternating leu and ser residues
(b) A polypeptide containing either leu or ser residues
(c) A polypeptide containing only leu residues
(d) A polypeptide containing only ser residues

42. In E. coli a an operator gene combine with

(a) Inducer gene to switch on structural gene action
(b) Inducer gene to switch off structural gene action
(c) Regulator protein (repressor) to switch off structural gene action
(d) Regulator protein to switch on gene action

43. Differentiation of organs and tissues in a developing organism, is associated with

(a) Developmental mutations
(b) Differential expression of genes
(c) Lethal mutations
(d) Deletion of genes

44. Given below is a sample of a portion of DNA strand giving the base sequence on the Opposite strands. What is so special shown in it?

5′ GAATTC 3′
3′ CTTAAG 5′

(a) Palindromic sequence of base pairs
(b) Replication completed
(c) Deletion mutation
(d) Start codon at the 5′ end

45. One of the parents of a cross has a mutation in its mitochondria. In that cross the parent is taken as male during segregation of F2 progenies that mutation in found in

(a) All the progenies
(b) fifty percent of the progenies
(c) one-third of the progenies
(d) none of the progenies

46. A sequential expression of a Set of human genes occurs when a steroid molecule bind to the

(a) Transfer RNA (b) messenger RNA
(c) DNA sequence (d) Ribosome

47. E. coli cell with a mutated z gene of the lac operon cannot grow in medium containing only lactose as the source energy because

(a) In the presence of glucose, *E. coli* cells do not utilise lactose
(b) They cannot transport lactose from the medium into the cell
(c) Lac operon is constitutively active in these cells
(d) They cannot synthesize functional beta galactosidase

48. The cause of Cat-cry syndrome is due to

(a) Loss of segment of X chromosomes
(b) Loss of the segment of 5th chromosome
(c) Loss of segment of Y chromosome
(d) None of the above

49. When two gene are situated very close to each other in a chromosome

(a) The percentage of crossing over between is very high
(b) Hardly any cross over is detected
(c) No crossing over is can take place between them
(d) Only double cross over can take place between them

50. During transcription holoenzyme RNA polymerase binds to a DNA sequence and the DNA assumes a saddle like structure at the point. What is that sequence called

(a) CAAT box (b) GGTT box
(c) AAAT box (d) TATA box

51. When a cluster of genes show linkage behaviour they

(a) induce cell division
(b) Do not show a chromosome map
(c) Show recombination during meiosis
(d) Do not show independent assortment

52. Marriages between close relatives and cousins is not advisable because

(a) More mutation can occur
(b) More recessive defects are likely to appear
(c) More chances are there of Rh blood group anomalies
(d) More chances are there for multiple births

53. Haploids are more suitable for mutation studies than the diploids. This is because

(a) All mutations, whether dominant or recessive are expressed in haploids
(b) Haploids are reproductively more stable than diploids
(c) Mutagens penetrate in haploids more effectively than diploids
(d) Haploids are more abundant in nature than diploids

54. Select the two correct statements out of the four (A-D) given below about lac operon.

A. Glucose or galactose may bind with the repressor and inactivate it
B. In the absence of lactose the repressor binds with the operator region
C. The z-gene codes for permease
D. This was elucidated by Francois Jacob and Jacque Monod

The correct statements are

(a) A and B (b) B and C
(c) A and C (d) B and D

55. Removal of RNA polymerase III from nucleoplasm will affect the synthesis of

(a) rRNA (b) tRNA
(c) hnRNA (d) mRNA

56. A single RNA transcript that encodes more than one polypeptide is

(a) Operon (b) Transcription factor
(c) Polycistronic mRNA (d) Regulatory protein

57. Recombination hot spots in the genome of sexually reproducing organism exhibit

(a) Highly elevated rates of meiotic recombination
(b) Highly elevated rates of mitotic recombination
(c) Highly elevated rates of homologous recombination
(d) Highly elevated rates of non-homologous recombination

58. Germline aneuploidy is typically detected through

(a) Real-time PCR (b) QF-PCR
(c) Karyptyping (d) All of these

59. (*i*) Transposase is very unstable, so it is not commonly accumulated to high concentrations
(*ii*) The efficiency of transposition decreases as the distance between the transposase gene and its binding site decreases.
(*iii*) Translation of the Tn10 transposase gene is repressed by an antisense RNA.

Which of the above statements are true about the transposable elements?

(a) i, ii (b) ii, iii
(c) i,iii (d) i,ii,iii

60. Ribosomes are very old structures found in all organisms (except viruses) from bacteria to humans. As a result, molecular taxonomists are looking at relationships based upon which rRNA sequences?

(a) 16S (b) 18S
(c) 30S (d) all of these

61. An RNA molecule folds back upon itself to form a "hairpin" structure held together by a region of base pairing. One segment of the molecule in the paired region has the base sequence 5′-GCAUUGA-3′. What is the base sequence with which this segment is paired?

(a) 5′-UCAAUGC-3′ (b) 5′-GCAUUGA-3′
(c) 5′-TCAATGC-3′ (d) 5′-GCATTGA-3′

62. Which DNA polymerase is located in mitochondria and chloroplasts?

(a) DNAP-α (b) DNAP-β
(c) DNAP-γ (d) DNAP-δ

63. What is the function of DNAP-α?

(a) It is believed to catalyze discontinuous synthesis of the lagging strand
(b) It is believed to catalyze continuous synthesis of the lagging strand
(c) Unknown
(d) DNA repair pathways

64. Each eukaryotic chromosome apparently contains a
(a) 1 giant molecule of DNA
(b) 2 giant molecule of DNA
(c) 3 giant molecule of DNA
(d) 4 giant molecule of DNA

65. Nucleosome conserved core consists of a
(a) 16 nucleotide pair DNA
(b) 72 nucleotide pair DNA
(c) 146 nucleotide pair DNA
(d) All of these

66. In which phase the coiled 300° chromatin fibers are inturn coiled into highly condensed structures?
(a) Interphase (b) Anaphase
(c) Telophase (d) Metaphase

67. DNA containing sequences that are repeated more than 10^5 times is called
(a) Repetitive DNA
(b) Highly repetitive DNA
(c) Middle repetitive DNA
(d) None of these

68. Euchromatin consists of
(a) Unique sequences
(b) Single copy sequences
(c) Middle repetitive sequences
(d) All of these

69. Which studies on isolated chromatin have provided a solid framework for a rapidly emerging picture of chromosome structure in eukaryotes?
(a) Chemical analysis (b) Electron microscopy
(c) X-ray diffraction (d) All of these

70. Where are the histones replaced by another class of small basic proteins called protamines?
(a) Brain (b) Nerve
(c) Sperm (d) Muscle

71. Histones are basic because they contain
(a) 20-30% Arginine and Leucine
(b) 20-30% Tryptophan and Lysine
(c) 5-10% Arginine and Lysine
(d) 20-30% Arginine and Lysine

72. The function of highly repetitive DNA sequences is
(a) Structural role in chromosomes
(b) Involvement in chromosome pairing during meiosis
(c) Involvement in crossing-over
(d) All of these

73. The nascent DNA is always synthesized in which direction?
(a) 3′ → 5′ direction (b) 5′ → 3′ direction
(c) Both (a) & (b) (d) None of these

74. Heat shock proteins in co-operation with SSB, leads to
(a) DNA unwinding and active replication
(b) DNA overwinding and active replication
(c) DNA underwinding and no replication
(d) None of these

75. In the formation of replication fork, the role of the DNA polymerase is to
(a) Initiate nascent, daughter strand synthesis
(b) Elongation of template DNA
(c) Initiates RNA molecules
(d) Synthesis of RNA molecule

76. Polymerase accessory factors of the polymerase III holoenzyme are
(a) β, γ, τ (b) γ, δ and τ
(c) β, γ, δ, δ and τ (d) None of these

77. Deoxynucleotide polymerase is caused by
(a) DNA ligase (b) SSB proteins
(c) DNA polymerases (d) Helicases

78. Function of DNA primase is to :
(a) Processive unwinding of DNA
(b) Initiate synthesis of RNA primers
(c) Initiate synthesis of DNA primers
(d) Both (b) and (c)

79. Which of the following proteins prevent premature reannealing of dsDNA?
(a) DNA polymerases (b) Helicase
(c) SSB proteins (d) DNA primase

80. What is the function of topoisomerases?
(a) Processive unwinding of DNA
(b) Prevent premature re-annealing of dsDNA
(c) Relieve torsional strain that results from helicase-induced unwinding
(d) Both (b) and (c)

81. The DNA is synthesized continuously on the
(a) Leading strand (b) Forward strand
(c) Both (a) and (b) (d) Lagging strand

82. The DNA which is synthesized in short fragments are called
(a) Okazaki fragments (b) Primosomes
(c) Leading fragments (d) None of these

83. The function of pol-III (E.coli) is
(a) DNA repair
(b) DNA proof reading
(c) Both (a) and (b)
(d) Processive, leading strand synthesis

84. DNA proof reading and repair is done by
(a) DNA pol-I (b) DNA pol-III
(c) DNA pol-I & III (d) DNA pol-II

85. What initiates synthesis of RNA primers?
(a) DNA polymerases (b) DNA primase
(c) SSB proteins (d) Both (a) & (c)

86. The function of DNA polymerase γ(mammalian) is
(a) DNA repair
(b) DNA proof reading
(c) Mitochondrial DNA Synthesis
(d) None of these

87. The nicking-resealing enzymes are called
(a) DNA helicases
(b) DNA topoisomerases
(c) cyclin-dependent protein kinases
(d) All of these

88. DNA synthesis occurs during which phase of the cell cycle?
(a) S-phage (b) G1-phase
(c) G2-phase (d) all of these

89. Cell prepares for DNA synthesis in which phase and for mitosis in which phase
(a) G1 and S (b) G2 and S
(c) G1 and G2 (d) S-phase only

90. Which CDK's levels rise in late G1 phase?
(a) Cyclin D (b) Cyclin C
(c) Cyclin A (d) None of these

91. The CDK's which are activated by D-cyclins are
(a) CDK4 and CDK6 (b) CDK2 and CDK4
(c) CDK2 and CDK6 (d) None of these

92. Cyclin E and CDK2 form a complex in
(a) Early G1 phase (b) Late G1 phase
(c) S-phase (d) All of these

93. What is the function of cylin D
(a) Transition from G2 to M
(b) Initiation of DNA synthesis in early S-phase
(c) Progression past restriction point at G1/S boundary
(d) All of these

94. CDK2 is involved in
(a) Progression past restriction point at G1/S boundary
(b) Initiation of DNA synthesis in early S-phase
(c) Both (a) & (b)
(d) Transition from G2 to M

95. Match the following

PROTEINS	FUNCTIONS
1. DNA polymerase III	P. Catalyzes the formation of negative super coils
2. DNA pol-I	Q. Catalyzes covalent joining of okazaki fragments
3. DNA ligase	R. Removes RNA primers and replaces them with new DNA strands
4. DNA gyrase	S. Catalyzes polymerization of new chains

(a) 1-Q; 2-R; 3-S; 4-P (b) 1-S; 2-R; 3-Q; 4-P
(c) 1-R; 2-Q; 3-S; 4-P (d) 1-Q; 2-S; 3-P; 4-R

96. The nucleotide in the promoter adjacent to the transcript initiation site is designated as
(a) +1 (b) –1
(c) +15 (d) None of these

97. The necessity of 5′ caps is
(a) Subsequent processing of the primary transcript mRNA
(b) For the translation of the mRNA
(c) For protection of mRNA against exonucleolytic attack
(d) All of these

98. The metalloenzyme molecule present in RNA polymerase is
(a) 2 Zn^{+2} molecules (b) 3 cobalt molecules
(c) 2 Mg^{+2} molecules (d) 2 Fe^{+3} molecules

99. How many distinct nuclear DNA dependent RNA polymerases do the Mammalian cells possess?
(a) 2 (b) 5
(c) 3 (d) None of these

100. Which RNA polymerase has 14 smaller subunits?
(a) RNA-pol-I (b) RNA-pol-II
(c) RNA-pol-III (d) All of these

101. Mammalian RNA polymerase II is
(a) Insensitive to α-amanitin
(b) Highly sensitive to α-amanitin
(c) Intermediately sensitive to α-amanitin
(d) None of these

102. What are the major product(s) of RNA polymerase-III?
(a) rRNA, tRNA (b) tRNA/5sRNA
(c) tRNA/mRNA (d) mRNA/5sRNA

103. In eukaryotes, the process of PIC formation is facilitated by
(a) Multiple General Transcription Factors (GTFs) TFIA, B
(b) GTFs, TFIA, F
(c) GTFs, TFIIA, B, D, E, F & H
(d) TFII A, B, C, D & D

104. Which of the following can block the translocation of RNA polymerase during transcription?
(a) α-amanitin (b) β-amanitin
(c) BAL (d) Biotin

105. Elongation of RNA molecule is from
(a) 5′ to 3′ end continues cyclically
(b) 3′ to 5′ end continues cyclically
(c) 3′ to 5′ end discontinuous
(d) parallel to its template

106. Which enzyme precedes and follows the progressing RNA to prevent the formation of super-helical complexes?
(a) Helicases (b) Topoisomerases
(c) RNA polymerases (d) Both (b) & (c)

107. Termination of the synthesis of the RNA molecule in bacteria signal is recoginized by
(a) Termination protein (b) Rho factor
(c) Both (a) & (b) (d) α and β subunits

108. The Rho (ρ) factor is
(a) ATP depeadent RNA stimulated helicase
(b) ATP independent RNA stimaluted helicase
(c) ATP dependent DNA stimulated ligase
(d) It is stimulated by RNAP

109. Rho is an ATP-dependent RNA stimulated helicase that disrupts
(a) Superhelical complexes
(b) Nascent DNA
(c) Nascent RNA
(d) Nascent RNA-DNA complex

110. What is the approximate length of bacterial promoters?
(a) 40 nucleotides in length
(b) 400 nucleotides in length
(c) 3 nucleotides in length
(d) 8 nucleotides in length

111. More proximal to the transcription start site about 10 base pair upstream is a
(a) Closed complex (b) TATA box
(c) Open complex (d) None of these

112. Approximately 35bp upstream of the transcription start site, there is a consensus sequence to which RNAP binds is
(a) 5′-TGTTGACA 3′ (b) 3′-TGTTGACA 5′
(c) 5′-TTTGGACA 3′ (d) 5′-UUUGGAAG3′

113. What is TF11D?
(a) Complex of TBR
(b) Complex of TAF's
(c) Both (a) & (b)
(d) Complex of TAF's with DPE

114. When does the formation of basal transcription complex begin?
(a) When TF11D binds to the TATA box
(b) Interactions between ADs of these proteins and the rest of the transcription machinery.
(c) When TF11D binds to the DPE
(d) All of these

115. The structure of TF11D complexes have been determined by
(a) Spectrometry of 70° resolution
(b) Electron microscopy at 30° resolution
(c) Southern blot technique
(d) Both (b) & (c)

116. The promoter and promoter-proximal cis-active upstream elements confer
(a) Fidelity
(b) Frequency of initiation upon a gene
(c) Both (a) & (b)
(d) Increased binding affinity of the cognate trans factor

117. A 3rd class of sequence elements can either increase or decrease the rate of transcription Initiation of eukaryotic gener as
(a) Enhancers (b) Repressors
(c) Silencers (d) All of these

118. Match the following

Element	Factor
P. TATE BOX	1. oct 1, 2, 4, 6*
Q. CAAT BOX	2. HSF
R. GC BOX	3. TBP
S. Heat shock	4. Myo D
T. Ig Octamer	5. ATF*
	6. NF-Y*

	P	Q	R	S	T
(a)	3	6	4	2	1
(b)	6	3	5	2	1
(c)	5	6	4	2	1
(d)	3	6	5	2	1

119. Match the following

General mechanisms	Specific Components
P. Basal componetns	1. SP, ATF
Q. Co-activators	2. TAFs
R. Activators	3. TBP, TF11A
	4. TAFs, Srbs
	5. ATF, TAFs

	P	Q	R
(a)	3	4	1
(b)	4	1	3
(c)	1	2	5
(d)	2	3	4

120. The factor of AP1 element is
(a) SP1 (b) FOS
(c) SRF (d) NF1

121. The larger hnRNA contains exons that will form the mature.
(a) mRNA (b) tRNA
(c) rRNA (d) None of these

122. Number of unique proteins present in eukaryotic transcription complex apparatus are
(a) About 50 (b) About 4-8
(c) About 20 (d) None of these

123. Eukaryotic RNA-polymerase-11 consists of
(a) 3 subunits (b) 6 subunits
(c) 9 subunits (d) 12 subunits

124. Which of the following facilitates the binding the mRNA to 43S pre-initiation complex?
(a) 5′-methyl GTP (b) 3′-methyl GTP
(c) Poly (a)- tail (d) None of these

125. The regulation of elF-4E controls
(a) The rate of initiation
(b) The rate of elongation
(c) The rate of termination
(d) None of these

126. What is the role of 4E in protein synthesis?
(a) It is responsible for recognition of the mRNA cap structure
(b) It is responsible for recognition of the mRNA poly (a) tail structure
(c) Formation of the 80S initiation complex
(d) All of these

127. How is the peptide sequence determined in elongation process of protein synthesis?
(a) By the order of the codons in the mRNA
(b) By the activation of elF-4E
(c) By the order of the codons in the tRNA
(d) All of these

128. Which of the following factor forms a ternary complex with GTP and enters amino-acyl t-RNA?
(a) Initiation factors 2B (b) Elongation Factor-1A
(c) Elongation factor-2 (d) Both (b) & (c)

129. At initiation polypeptide site is occupied by amine-acyl-tRNA methionine. This reaction is catalyzed by :
(a) Peptidyl transferase (b) Peptidyl translocase
(c) Polypeptidase (d) All of these

130. Which of the following factor binds and displaces the peptidyl tRNA from the A site to the P site?
(a) Elonation factor EF-1 (b) Elongation factor EF-2
(c) Both (a) & (b) (d) None of these

131. Which of the following is the stop/terminating codon/of mRNA
(a) UAA (b) UGA
(c) UAG (d) All of these

132. Which factors recognizes that a stop codon resides in the A site?
(a) Elongation factors EF-2
(b) EF-1
(c) RF
(d) RF3

133. The releasing factors are protein that hydrolyze
(a) The peptidyl tRNA bond when a stop codon occupies the A site
(b) The peptidyl tRNA bond when a stop codon occupies the P site
(c) The peptidyl tRNA bond when initiation codon occupies the A site
(d) Both (b) & (c)

134. Multiple ribosomes on the same mRNA molecule forms
(a) Polyribosome (b) Polysome
(c) Both (a) & (b) (d) Ribozyme

135. Which of the following antibiotic prevents the binding of aminoacyl tRNA's to the A site?
(a) Puromycin (b) Tetracycline
(c) Diphtheria toxin (d) Both (b) & (c)

136. Which antibiotic inhibit protein synthesis on all ribosomes?
(a) Puromycin (b) Cycloheximide
(c) Ricin (d) Tetracycline

137. Which is the structural analogue of tyrosinyl-tRNA
(a) Puromycin (b) Cycloheximide
(c) Tetracycline (d) None of these

138. The function of rich in protein synthesis of eukaryote is
(a) Inactivates eukaryotic 28S ribosomal RNA by providing the N-glycolytic cleavage
(b) Revoval of single adenine
(c) Both (a) & (b)
(d) Prevents the binding of aminoacyl-tRNAs to the A site

139. Lac I repressor protein molecule has a high affinity for
(a) Promoter (b) Operator locus
(c) Regulator locus (d) None of these

140. The role of positive regulator
(a) Increases gene operation
(b) Decreases gene expression
(c) Both (a) & (b)
(d) Maintain equilibrium

141. The Epson is the corrdinated unit of genetic expression in
(a) Bacteria (b) Virus
(c) Mammalian (d) All of these

142. Thiogalactoside transacetylase is also known as
(a) Galactoside O-acetyltransferase
(b) β-galactoside permease
(c) β-galactosidase
(d) None of these

143. The dual control mechanism causes the sequential utillzation of glucose and lactose in two distinct growth phases, known as
(a) Diauxie (b) Inducer
(c) CAP (d) Both (b) & (c)

144. Which genes are not subjected to regulation?
(a) Inducible genes (b) Constitutive genes
(c) Regulatory genes (d) Both (b) & (c)

145. Tryptophan pyrrolase of liver is induced by
(a) Tryptophan (b) Constitutive genes
(c) Tyrosine (d) None of these

146. Match the following

Gene/Regulatory element	Function
P. Regulatory gene	1. Encodes β-galactosidase
Q. Operator	2. Codes for repressor
R. Promoter	3. Encodes β-galactoside permease
S. Structural gene	4. Binding site of repressor
	5. Binding site of RNAP and CAP-cAMP complex

	P	Q	R	S
(a)	2	3	4	5
(b)	2	4	5	3
(c)	3	4	1	2
(d)	3	1	2	4

147. What would be the result of inactivation by mutation of the regulator genes in the E.coli lactose operon?
(a) Constitutive synthesis of the lac enzymes
(b) Uninducibility of the lac enzymes
(c) No β-galactosidase activity
(d) All of these

148. Write the partial diploid genotype for a strain that will produce b-galactosidase constitutively and permeability inducible
(a) $\frac{i^+o^cz^+y^-}{i^-o^-z^-y^+}$ (b) $\frac{i^+o^cz^+y^-}{i^+o^+z^-y^+}$
(c) $\frac{i^+o^+z^+y^+}{i^+o^cz^+y^+}$ (d) None of these

NUMERICAL TYPE QUESTIONS

1. In a given DNA segment the number of nucleotides of guanine is 75 and those of thymine is 75. The total number of nucleotides in the segment will be

2. If a length of DNA has 45,000 base pairs, how many complete turns will the DNA molecule take?

3. Depending upon size and centromere position, the 46 chromosomes have been divided into a number of groups

4. What would be the number of chromosomes of the aleurone cells of a plant with 42 chromosomes in its root tip cells?

5. In his experiment on the chemistry of DNA, Chargaff estimated the base composition of human sperm and found that adenine constituted 31%, guanine 19%. The quantity of cytosine in the DNA of human somatic cells is likely to be

6. How many nucleosomes are found in helical coil of 30 nm chromatin fibre

7. In the hexaploid wheat, the haploid (n) and basic (x) numbers of chromosomes are:

8. If a haploid human genome contains 3×10^9 nucleotide-pairs and the average molecular weight of a nucleotide-pair is 600, how many copies of the human genome are present, on average, in 1μg of human DNA?

9. If the haploid genome of the small plant Arabidopsis thaliana contains 7.7×10^7 nucleotide pairs, how many copies of the A. thaliana genome are present, on average, in 1μg of A. thaliana DNA?

10. What is the mass copy of the human genome?

11. What is the mass of one copy of the A. thaliana genome?

12. The length of the Lamp-brush chromosomes is

13. How much amount of DNA is being polymerized by E.coli pol-III during one cycle on the leading strand?

14. In mammalian cells, the polymerase is capable of polymerizing about how many nucleotides per second?

15. The replication process is completed in about 30 mins, at replication rate of

16. How many peptide bonds are synthesized by a single mammalian ribosome for every minute?

17. How many identical subunits are assembled into a lac repressor molecule?

EXERCISE – II
(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. What will be the binding status of regulatory proteins in *lac* operon when concentrations of both lactose and glucose are very low in the culture medium?
(a) Only the repressor remains bound to the operator
(b) Only the cyclic AMP-Catabolic Activator Protein (cAMP-CAP) complex remains bound to the CAP binding site
(c) Neither the repressor nor cAMP-CAP complex remain bound to their respective binding sites
(d) Both the repressor and cAMP-CAP complex remain bound to their respective binding sites

2. Based on their function, find the **ODD** one out.
(a) miRNA (b) siRNA
(c) shRNA (d) snRNA

3. Which set of the following events occurs during the elongation step of translation?
P. Attachment of mRNA with the smaller subunit of ribosome
Q. Loading of correct aminoacyl-tRNA into the A site
R. Formation of a peptide bond between the aminoacyl-tRNA in the A site and the peptide chain that is attached to the peptidyl-tRNA in the P site
S. Dissociation of the ribosomal subunits
T. Translocation of peptidyl-tRNA from the A site to the P site of the ribosome
(a) P, Q and R (b) P, Q and T
(c) Q, R and T (d) R, S and T

4. A DNA sequence, 5′-ATGGACGTGCTTCC CAAAGCATCGGGC–3′, is mutated to obtain
P. 5′-ATGGACGTGCTTC**a**
CAAAGCATCGGGC–3′
Q. 5′-ATGGACGTGCTTCCC**g**
AAAGCATCGGGC–3′
R. 5′-ATGGACGTGCTTCC–
AAAGCATCGGGC–3′
S. 5′-ATGGA CGTGCTTCCCAA**t**
GCATCGGGC–3′
T. 5′-ATGGACG**a**GCTTCCC
AAAGCATCGGGC–3′
[Point mutations are shown in the **lower case** or '–' within the sequences]
Which of the above mutant sequences **DO NOT** have frame-shift?
(a) P, Q and S (b) P, S and T
(c) Q, R and S (d) Q, S and T

2015

5. How many 3-tuples are possible for the following amino acid sequence ?
MADCMWDISEASE
(a) 4 (b) 5
(c) 11 (d) 12

6. Match the drugs in Group I with their mechanism of action in Group II.

Group I
P. Paclitaxel
Q. Colchicine
R. Etoposide
S. Methotrexate

Group II
1. Inhibits protein translation
2. Inhibits microtubule depolymerization
3. Inhibits DNA replication
4. Alkylates DNA
5. Inhibits dihydrofolate reductase
6. Inhibits microtubule polymerization

(a) P-1, Q-6, R-3, S-4
(b) P-2, Q-6, R-3, S-5
(c) P-1, Q-3, R-6, S-5
(d) P-2, Q-3, R-6, S-4

7. An *in vitor* translation system can synthesize peptides in all three reading frames of the RNA template. When 5′-UCUCUCUC---$(UC)_n$---UCUCUCUC - 3′ was used as the template in this *in vitro* translation system, the synthesized peptides contained 50% each of serine and leucine. When 5′-CCUCCUCCU---$(CCU)_n$---CCUCCU-3′ was used as the template, the synthesized peptides contained 33.3% each of serine, leucine, and proline. Deduce the condon for proline.
(a) UCU
(b) CUC
(c) CCU
(d) UCC

8. Cytoplasmic extract from the wild type strain of a bacterium has the ability to convert a colorless substrate (S) to a colored product (P) via three colorless intermediates X, Y and Z, in that order. Each step of the pathway involves a specific enzyme coded by a distinct gene. Four mutant strains (a^-, b^-, c^-, d^-) were isolated, whose extracts are incapable of producing the colored product in the presence of S. In a series of experiments, extracts from the individual mutants were incubated with X, Y or Z and secored for color development. The data are summarized in the table below. (Yes : color developed, No : no color developed)

Mutants	Compounds		
	X	Y	Z
a^-	No	No	No
b^-	No	Yes	Yes
c^-	Yes	Yes	Yes
d^-	No	No	Yes

Based on the data, which one of the following is the correct order of enzymes involved in the pathway ?

(a) $S \xrightarrow{d} X \xrightarrow{c} Y \xrightarrow{b} Z \xrightarrow{a} P$

(b) $S \xrightarrow{a} X \xrightarrow{d} Y \xrightarrow{b} Z \xrightarrow{c} P$

(c) $S \xrightarrow{b} X \xrightarrow{a} Y \xrightarrow{c} Z \xrightarrow{d} P$

(d) $S \xrightarrow{c} X \xrightarrow{b} Y \xrightarrow{d} Z \xrightarrow{a} P$

2014

9. The 5' ends of the mature forms of the prokaryotic mRNAs and tRNAs are

(a) a triphosphate group in mRNAs and a monophosphate group in tRNAs

(b) triphosphate groups in both mRNAs and tRNAs

(c) monophosphate groups in both mRNAs and tRNAs

(d) a monophosphate group in mRNAs and a triphosphate group in tRNAs

10. Topological winding number of a 2.0 kb covalently closed circular DNA was found to be 191 with a writhing number of –4. Hence, its LINKING NUMBER and the NUMBER OF BASE PAIR PER TURN when the molecule is laid flat on the surface is _____ and_____, respectively.

(a) 187, 10.69 (b) 195, 10.25

(c) 200, 10.00 (d) 187, 10.50

11. The percentage SIMILARITIES and IDENTITIES, respectively, between the two peptide sequences given below will be _____% and _____%.

Peptide I : Ala-Ala-Arg-Arg-Gln-Trp-Leu-Thr-Phe-Thr-Lys-Ile-Met-Ser-Glu

Peptide II: Ala-Ala-Arg-Glu-Gln-Tyr-Ile-Ser-Phe-Thr-Lys-Ile-Met-Arg-Asp

(a) 80, 80 (b) 80, 60

(c) 60, 60 (d) 90, 60

12. For their efficient translation, eubacterial mRNAs possess a Shine-Dalgarno sequence for its recognition by an anti-Shine-Dalgarno sequence (ASD) in the ribosomes. The correct statement is

(a) ASD is present in 5S rRNA

(b) ASD is present in 23S rRNA

(c) ASD is present in 16S rRNA

(d) ASD is formed by the interaction of the 16S rRNA with the 23S rRNA upon docking of the 50S subunit on the 30S subunit of the ribosomes

13. The active site in the alpha/beta barrel structures is usually located

(a) inside the barrel

(b) at the amino side of the strands

(c) at the carboxy side of the strands

(d) at any arbitrary site

2013

14. Under alkaline conditions, DNA is more stable than RNA because

(a) RNA forms secondary structures

(b) RNA is a single stranded molecule

(c) RNA has uracil in place of thymidine

(d) RNA is susceptible to hydrolysis

15. Which one of the following modifications is common toboth protein and DNA?

(a) SUMOylation (b) Nitrosylation

(c) Methylation (d) Ubiquitination

16. The nucleotide analogue used in DNA sequencing by chain termination method is

(a) 1',3'-dideoxy nucleoside triphosphate

(b) 2',3'-dideoxy nucleoside triphosphate

(c) 2',4'-dideoxy nucleoside triphosphate

(d) 2',5'-dideoxy nucleoside triphosphate

17. The RNA primer synthesized during the replication process in bacteria is removed by

(a) DNA gyrase (b) primase

(c) DNA polymerase I (d) DNA polymerase II

18. Match the entries in **Group I**with the entries in **Group II.**

Group I	Group II
P. RNAse P	1. Polyadenylation
Q. RNase H	2. Splicing
R. snRNAs	3. Ribozymes
S. CstF	4. DNA-RNAhybrids

(a) P-3, Q-4, R-2, S-1 (b) P-4, Q-3, R-2, S-1

(c) P-3, Q-2, R-1, S-4 (d) P-2, Q-4, R-1, S-3

2012

19. In mismatch correction repair, the parental DNA strand is distinguished from the daughter strand by _____

(a) acetylation (b) phosphorylation

(c) methylation (d) glycosylation

20. A protein is phosphorylated at a serine residue. A phosphomimic mutant of the protein can be generated by substituting that serine with

(a) glycine (b) alanine

(c) aspartate (d) threonine

21. A truncated polypeptide is synthesized due to a nonsense mutation. Where would you introduce another mutation to obtain a full-length polypeptide?

(a) Ribosomal protein gene

(b) Transfer RNA gene

(c) DNA repair gene

(d) Ribosomal RNA gene

22. The direction of shell coiling in the snail Limnaea peregra is a classic example of

(a) chromosomal inheritance

(b) extra-chromosomal inheritance

(c) chromosomal translocation

(d) homologous recombination

23. Which one of the following DNA sequences carries an invert repeat?

(a) ATGAGCCCCGAGTA
TACTCGGGGCTCAT

(b) ATGAGCCGGCTCTA
TACTCGGCCGAGAT

(c) ATGAGCCGAGCCTA
ACTCGGCTCGGAT

(d) ATGAGCCTATGGTA:
TACTCGGATACCAT

24. In zinc finger proteins, the amino acid residues that coordinate zinc are

(a) Cys and His (b) Asp and Glu

(c) Arg and Lys (d) Asp and Arg

2011

25. Molecular chaperones are class of proteins that facilitate

(a) the proper folding of newly synthesized proteins

(b) unfolding of newly synthesized proteins

(c) degradation of newly synthesized proteins

(d) targeting of newly synthesized proteins

26. Diphtheria toxin, tetracycline and streptomycin inhibit

(a) DNA repair (b) DNA replication

(c) transcription (d) translation

27. Determine the correctness or otherwise of the following **Assertion (A)** and the **Reason (R).**

Assertion : N-methyl-N'-nitro-N-nitrosoguanidine (NTG) is an effective chemical mutagen.

Reason : Mutations induced by NTG mainly are the GC → AT transitions.

(a) both (A) and (R) are true and (R) is the correct reason for (A)

(b) both (A) and (R) are true but (R) is not the correct reason for (A)

(c) (A) is true but (R) is false

(d) (A) is false but (R) is true

28. Determine the correctness of the following statements

I. Enhancer sequences are those DNA sequences that are involved in increasing the rate of DNA replication.

II. Enhancer sequences work by binding with eukaryotic gene activator factors.

(a) only I is true (b) only II is true

(c) both I and II are true (d) both I and II are false

29. Match the terms in **Group I** with their associated functions in **Group II.**

Group I

P. Shine-Dalgarno sequences

Q. Leucine zipper

R. Aminoacyl tRNA synthetase

S. RNA interference (RNAi)

Group II

1. Aminoacylation of tRNA
2. Gene silencing
3. Ribosome binding and facilitation of translation initiation
4. Transcription factors

	P	Q	R	S
(a)	3	4	1	2
(b)	4	3	2	1
(c)	2	3	1	4
(d)	3	2	4	1

30. Determine the correctness or otherwise of the following **Assertion (A)** and the **Reason (R).**

Assertion : Isopropylthiogalactoside (IPTG) is a gratuitous inducer of lactose operon.

Reason : Gratuitous inducers are chemical analogs which behave like natural inducer but they do not serve as substrate for the enzymes that are subsequently synthesized.

(a) both (A) and (R) are true and (R) is the correct reason for (A)

(b) both (A) and (R) are true but (R) is not the correct reason for (A)

(c) (A) is true but (R) is false

(d) (A) is false but (R) is true

31. Which of the following characteristics with respect to bacterial DNA polymerase III are TRUE?

P. Initiation of chain synthesis

Q. 5′ – 3′ polymerization

R. 3′ – 5′ exonuclease activity

S. 5′ – 3′ exonuclease activity

(a) P and Q only (b) Q and R only

(c) Q and S only (d) P and S only

2010

32. Peptidyl transferase activity resides in

(a) 16S rRNA (b) 23 rRNA

(c) 5S rRNA (d) 28 rRNA

33. During transcription

(a) DNA Gyrase introduces negative supercoils and DNA Tbpoisomerase I removes negative supercoils

(b) DNA Topoisomerase I introduces negative supercoils and DNA Gyrase removes negative supercoils

(c) both DNA Gyrase and DNA Tbpoisomearse I introduce negative supercoils

(d) both DNA Gyrase and DNA Tbpoisomearse I remove negative supercoils

34. Nuclease-hypersensitive sites in the chromosomes are sites that appear to be

(a) H2 and H4 histone free

(b) H1 and H2 histone free

(c) H3 and H4 histone free

(d) Nucleosome free

35. Determine the correctness or otherwise of the following **Asertion (a)** and the **Reason (r)**

Assertion : MTT assay is used to determine cell viability based on the principle of colour formation by DNA fragmentation.

Reason : MTT assay is used to determine cell viability based on the colour development by converting tetrazolium soluble salt to insoluble salt.

(a) both (a) and (r) are true and (r) is the correct reason for (a)

(b) both (a) and (r) are true and (r) is not the correct reason for (a)

(c) (a) is true but (r) is false

(d) (a) is false but (r) is true

36. Match the following antibiotics in Group I with their mode of action in Group II

Group I	Group II
P. Chloramphenicol	1. Binds to DNA gyrase
Q. Norfloxacin	2. Binds to RNA Polymerase
R. Puromycin	3. Inhibits peptidyl transferase
S. Rifampicin	4. Mimics aminoacyl tRNA

(a) P-1, Q-3, R-2, S-4

(b) P-3, Q-1, R-2, S-4

(c) P-3, Q-1, R-4, S-2

(d) P-4, Q-2, R-3, S-1

2009

37. Which of the following are commonly used as reporter genes?

P. NPTgene Q. Luciferase gene

R. CFTR gene S. GFPgene

(a) Q, S (b) R,S

(c) P, R (d) P, Q

38. Determine the correctness or otherwise of the following Assertion (a) and Reason (r):

Assertion (a): IPTG (Isopropylthiogalactoside) is a gratuitous inducer of lac operon.

Reason (r) : IPTG is an efficient inducer, but not a substrate of *lac* operon

(a) Both (a) and (r) are true and (r) is the correct reason for (a)

(b) Both (a) and (r) are true but (r) is not the correct reason for (a)

(c) (a) is true but (r) is false

(d) (a) is false but (r) is true

2008

39. Zinc fingers are characteristics of

(a) blood clotting proteins

(b) RNA chaperones

(c) DNA binding proteins

(d) lysosomal hydrolases

2007

40. Si RNA(s) interfere at

(a) transcriptional level

(b) post-transcriptional level

(c) DNA replication level

(d) translational level

41. Presence of CX_{2-4} $CX\phi X_8HX_3H$ sequence in a protein suggest that it is

(a) a protein kinase (b) GTP binding protein

(c) zinc finger protein (d) lipase

42. Mobile genetic elements present in human genome are

P. long interspersed elements (LINEs)

Q. short interspersed elements (SINEs)

R. P elements

S. IS elements

(a) Q, R

(b) P, Q

(c) P, R

(d) Q, S

2006

43. The enzymes that can be used in 5′ end labeling of DNA are

(P) alkaline phosphatase

(Q) DNA ligase

(R) terminal transferase

(S) polynucleotide kinase

(a) P, S (b) R, Q

(c) P, R (d) R, S

NUMERICAL TYPE QUESTIONS

2013

1. The total number of fragments generated by the complete and sequential cleavage of the polypeptide given below by Trypsin followed by CNBr is _____

Phe-Trp-Met-Gly-Ala-Lys-Leu-Pro-Met-Asp-Gly-Arg-Cys-Ala-Gln

2. In a genetic study, 80 people were found to have alleles for polydactyly. Only 36 of them were polydactylous. What is the extent of penetrance percentage? _____

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (b)	2. (d)	3. (c)	4. (d)	5. (a)	6. (a)	7. (c)	8. (c)	9. (b)	10. (b)
11. (c)	12. (d)	13. (b)	14. (d)	15. (a)	16. (c)	17. (d)	18. (c)	19. (a)	20. (a)
21. (b)	22. (d)	23. (d)	24. (c)	25. (c)	26. (a)	27. (a)	28. (a)	29. (c)	30. (b)
31. (b)	32. (a)	33. (c)	34. (b)	35. (d)	36. (c)	37. (d)	38. (d)	39. (b)	40. (b)
41. (a)	42. (c)	43. (b)	44. (a)	45. (b)	46. (c)	47. (d)	48. (b)	49. (b)	50. (d)
51. (d)	52. (b)	53. (a)	54. (d)	55. (b)	56. (c)	57. (a)	58. (d)	59. (d)	60. (a)
61. (a)	62. (c)	63. (a)	64. (a)	65. (c)	66. (d)	67. (b)	68. (d)	69. (d)	70. (c)
71. (d)	72. (d)	73. (b)	74. (a)	75. (a)	76. (c)	77. (c)	78. (b)	79. (c)	80. (c)
81. (c)	82. (a)	83. (d)	84. (d)	85. (b)	86. (c)	87. (b)	88. (a)	89. (c)	90. (d)
91. (a)	92. (b)	93. (c)	94. (b)	95. (b)	96. (b)	97. (d)	98. (a)	99. (c)	100. (c)
101. (b)	102. (b)	103. (c)	104. (a)	105. (a)	106. (b)	107. (c)	108. (a)	109. (d)	110. (a)
111. (b)	112. (a)	113. (c)	114. (a)	115. (b)	116. (c)	117. (d)	118. (a)	119. (a)	120. (b)
121. (a)	122. (a)	123. (d)	124. (a)	125. (a)	126. (a)	127. (a)	128. (b)	129. (a)	130. (b)
131. (d)	132. (c)	133. (a)	134. (c)	135. (b)	136. (a)	137. (a)	138. (c)	139. (b)	140. (a)
141. (a)	142. (a)	143. (a)	144. (b)	145. (a)	146. (b)	147. (a)	148. (b)		

Numerical Type Questions

1. 300 2. 4,500 3. 7 4. 63 5. 19% 6. 06

7. $n = 21$ and $x = 7$

8. 3.04×10^5 copies of human genome

9. 1.18×10^7 copies of genome 10. 3.3×10^{-6} µg 11. Approximately 8.4×10^{-8} µg 12. 800 µm

13. 0.5MB 14. 100 15. 3×10^5 bp/min 16. 400 17. 4

EXERCISE – II

MCQ Type questions

1. (d)	2. (d)	3. (c)	4. (b)	5. (c)	6. (b)	7. (c)	8. (d)	9. (a)	10. (a)
11. (b)	12. (c)	13. (c)	14. (d)	15. (c)	16. (b)	17. (c)	18. (a)	19. (c)	20. (c)
21. (b)	22. (b)	23. (b)	24. (a)	25. (a)	26. (d)	27. (a)	28. (b)	29. (a)	30. (a)
31. (b)	32. (b)	33. (a)	34. (d)	35. (d)	36. (c)	37. (a)	38. (b)	39. (c)	40. (b)
41. (c)	42. (b)	43. (a)							

Numerical Type Questions

1. (5) 2. (45)

EXPLANATIONS

EXERCISE - I

MCQ TYPE QUESTIONS

1. **Transcription:** transcription is the formation of RNA over the template of DNA.
2. By the process of transcription and translation, these proteins are formed which are responsible for growth and differentiation
3. The non template strand has same sequence as in mRNA but thymine is replaced by Uracil.
4. The Okazaki fragment in DNA chain growth polymerizes in the 5' to 3' direction and explains the 3' to 5' direction DNA replication.
5. Translocation of mRNA by DNA in nucleus is called transcription and formation of polypeptide chain on ribosome with the help of mRNA is called translation
6. Formation of mRNA by DNA in nucleus is called transcription and formation of polypeptide chain on the ribosome with the help of mRNA is called translation
7. Histones are low molecular weight protein with high molecular proportion of free charged basic amino acid arginine and lysine
8. Changed proteins are formed in the mutation due to the reason that mRNA transcribed from the mutated gene and abnormal m ran introduces different amino acids in the polypeptide chain formed with changed message.
9. Mature m RNA comes out from nucleus into cytoplasm only after completion of splicing, capping and tailing.
10. These rings contain active DNA or RNA and protein are synthesized here.
11. 5-bromo uracil(Bu) is a base analogue of thymine. Transitions are caused by base analogues.
12. The phenomenon of separating of two strand of DNA molecule by breaking of hydrogen bond at the temperature 90°C is the uncoiling of DNA helix
13. Anticodon arm is responsible for recognizing and binding codons in the m-RNA.
14. Uracil base is found in RNA, in this way, uridine monophosphate is the nucleotide of RNA.
15. Spliceosomes cut introns from hn-RNA and exons are joined by RNA ligase. It is called splicing.
16. The enzyme helicase unwind the helix while topo-isomerase break and release tension of strands of DNA. Topo-isomerase also takes part in recombination.
17. Because of semi conservative method of DNA replication only 2 molecules of DNA will have some radioactive thymidine.
18. The nucleotide sequence in the other chain is TCGAAGCT.
19. Enzymes helicase unwinds the DNA helix and unzips the two strands of DNA.
20. Exon is the active part of mRNA and intron is the inactive part of mRNA, which codes the formation of specific protein.
21. According to Chargaff rules A + T = G + C.
22. Because single amino acid is attached on tRNA.
23. Data not sufficient for a nucleic acid because T is absent in the base sequence
24. 3'-5' phosphodiester bond is formed between carbon 3 of one nucleotide and carbon 5 of the other nucleotide.
25. Probe - 15-30 long radioactive/ non radioactive segments DNA/RNA that is used in hybridization with DNA segment.
26. 23 sr RNA is catalytic RNA.
27. In capping an unusual nucleotide is added to the 5' end of hnRNA. In tailing, an adinilate residues are added at 3'end in template independent manner.
28. RNA polymerase III forms tRNA in Eukaryotes.
29. Centromere joins the microtubules of chromosomal spindle fibers and helps in the movement of chromosome during cell division. Centromere also plays an important role in the polymerization of tubulin protein used in the formation of microtubules. Hence it is helpful in attaching the chromosome with spindle fibers.
30. The cross over percentage differs for different pairs of genes because the amount of crossing over depends on the frequency with which breaks in the chromatids occur between the two genes. This frequency further depends specially on the distance of these genes which are apart on the chromatids.
31. Polytene chromosomes contain several dark stained regions called bands separated by lighter or less stained area called interbands. These dark bands are heterochromatin area.
32. Each chromosome has a centromere but in some cases secondary construction is also present, more specifically called as "secondary constriction-II". Its position is constant for particular chromosome heterochromatin is condensed in this region. In man "secondary constriction-II" is present in chromosome number 1, 10, 13, 16 and Y. nuclear organizer is called "secondary constriction- I".
33. The genes present on the differential part of Y chromosome are passed directly from father to son and are called as holandric genes.
34. Linkage is the pattern of assortment of genes that are located on the same chromosome. It is important because if the genes are located relatively far apart crossing over is more likely to occur between them than in they are located close together. In sex linkage, the speciality is criss-cross linkage.
35. Chimeras are the individual having the different genotypes in the different parts.
36. Sex chromosomes are also called idiochromosomes/ allosomes.
37. Because in this mRNA, AUG is the initiation codon and UAG is the termination codon.
38. Lac operon: cluster of genes that regulates production of enzymes needed to metabolize lactose in bacterial cells (E. coli). It is an inducible operon.
39. The genetic code is degenerate, i.e., it lacks specificity and one amino acid often has more than one code triplet. Only methionine and tryptophan have single triplet codon.
40. Operator gene controls the activity of structural gene which synthesizes m-RNA (transcription).
41. Because CUC and UCC code for respectively Leucine and Serine.
42. Operator gene is a part of DNA between promoter and structural gene, where repressor binds to stop activity of RNA polymerase.
43. Differential expression of genes is responsible of different type of protein by which different cells of organs are formed.
44. 5' GAATTC 3' / 3' CTTAAG 5' } They are palindromic sequences
45. Because mitochondrial genes also show maternal inheritance because all the mitochondria a zygote has come from the cytoplasm of the ovum.
46. ex. Oval albumin synthesis in oviduct of chick is influenced by estrogen (steroids) (transcriptional control)
47. The enzyme like lactase or β-galactosidase which is formed in response to the presence of its called inducible enzymes.
48. Cat-cry or cri-due chat syndrome is caused by the deletion of short arm of chromosome no. 5 (5p-) so, the genotype of effected individual in 46, XX, 5p- in females and 46, XX, 5p- in males.
49. Frequency of crossing over is proportional to the distance of gene.
50. TATA box is present in eukaryotic promoter region. It has a resemblance with pribnow box of prokaryotes. TATA box was identified by Dr. Hogness and so it is also called hogness box. During the process of transcription the RNA polymerase binds to the TATA box due to which DNA assumes a saddle like structure in this place.
51. When a cluster of genes show linkage behaviour they do not show independent assortment because they are located very close to each other.
52. Marriages between close relatives and cousins is not advisable because recessive lethal defects are carried in heterozygous conditions and express themselves only when mating takes place between two carrier individuals or close relatives.
53. Haploid is a ploidy (number of sets of chromosomes in the nucleus) condition of having a single set of each chromosome in a cell or cell nucleus. In most animals, only the gametes (reproductive cells) are haploid. They are used in plant breeding to obtain pure lines that are homozygous by using colchicines to double the chromosome number or relying on spontaneous mutations. Haploids are more suitable for mutation studies because haploids generally and mostly represent gametes or germ cells. As such, the mutations occurring in those germ cells will be transferred and subsequently expressed in the future generations.
54. The most direct way to control the expression of a gene is to regulate its rate of transcription; that is, the rate at which RNA polymerase transcribes the gene into molecules of messenger RNA (mRNA). In the absence of lactose, the repressor protein encoded by the I gene binds to the lac operator and prevents transcription. Binding of allolactose to the repressor causes it to leave the operator. This enables RNA polymerase to transcribe the three genes of the operon. The single mRNA molecule that results is then translated into the three proteins. The gene encoding the lac repressor is called the lac I gene. It happens to be located just upstream of the lac promoter.
55. Transcription initiation by eukaryotic RNA polymerase (Pol) III relies on the TFIIE-related sub-complex. RNA polymerase (Pol) III is highly specialized for the production of short non-coding RNAs. RNA polymerase (pol) III transcription is responsible for the transcription of small, untranslated RNAs involved in fundamental metabolic processes such mRNA processing (U6 snRNA) and translation (tRNAs).

RNA pol III transcription contributes to the regulation of the biosynthetic capacity of a cell and a direct link exists between cancer cell proliferation and deregulation of RNA pol III transcription. Mostly, it results in the synthesis of t-RNA.

56. An mRNA that codes for more than one protein is a polycistronic mRNA. Polycistronic mRNA carries several open reading frames (ORFs), each of which is translated into a polypeptide. These polypeptides usually have a related function (they often are the subunits composing a final complex protein) and their coding sequence is grouped and regulated together in a regulatory region, containing a promoter and an operator. Most of the mRNA found in bacteria and archaea is polycistronic.

57. Recombination between homologous DNA sequences occurs in all organisms, and the resultant exchange of information is critical for the survival of species. Recombination is an essential cellular process catalyzed by proteins explicitly expressed for this purpose. It provides an effective means of generating genetic diversity that is important for evolution. Recombination, together with mutation, generates the raw material of evolution, is essential for reproduction and lies at the heart of all genetic analysis. Recombination is known to occur at discrete regions of the genome called hotspots. Genetic rearrangements in the cells that form sperm and eggs ensure that the set of genes passed on to every sperm and egg cell is unique. Thus, recombination hotspots exhibit elevated levels of meiotic recombination.

58. Aneuploidy is defined as an abnormal number of chromosomes. Syndromes caused by an extra or missing chromosome are among the most widely recognized genetic disorders in humans. Different organisms have widely varying chromosome complements. Germline aneuploidy is typically detected through karyotyping, a process in which a sample of cells is fixed and stained to create the typical light and dark chromosomal banding pattern and a picture of the chromosomes is analyzed. Other techniques include Fluorescence In Situ Hybridization (FISH), Quantitative Polymerase Chain Reaction (PCR) of Short Tandem Repeats, Quantitative Fluorescence PCR (QF-PCR), Quantitative Real-time PCR (RT-PCR) dosage analysis, Quantitative Mass Spectrometry of Single Nucleotide Polymorphisms, and Comparative Genomic Hybridization (CGH).

59. Transposases use many different mechanisms for cutting and pasting DNA. Transposase is an enzyme that binds to the ends of a transposon and catalyzes the movement of the transposon to another part of the genome. Transposase Tn5 is a member of the RNase superfamily of proteins which includes retroviral integrases. Because DNA transposition events are inherently mutagenic, the low activity of transposases is necessary to reduce the risk of causing a fatal mutation in the host, and thus eliminating the transposable element. Transposable elements (TEs), also known as "jumping genes," are DNA sequences that move from one location on the genome to another. Some silenced TEs are inactive because they have mutations that affect their ability to move from one chromosomal location to another; others are perfectly intact and capable of moving but are kept inactive by epigenetic defence mechanisms such as DNA methylation, chromatin remodeling, and miRNAs.

60. Ribosomal RNA (rRNA) is a molecule in cells that forms part of the protein-synthesizing organelle known as a ribosome and that is exported to the cytoplasm to help translate the information in messenger RNA (mRNA) into protein. The three major types of RNA that occur in cells include rRNA, mRNA, and transfer RNA (tRNA). In bacteria the gene that has proved to be the most informative for investigating evolutionary relatedness is 16S rRNA, a sequence of DNA that encodes the RNA component of the smaller subunit of the bacterial ribosome (16S refers to the rate of sedimentation, in Svedberg units, of the RNA molecule in a centrifugal field). The 16S rRNA gene is present in all bacteria, and a related form occurs in all cells, including those of eukaryotes. Analysis of the 16S rRNA sequences from many organisms has revealed that some portions of the molecule undergo rapid genetic changes, thereby distinguishing between different species within the same genus.

61. A hairpin loop is an unpaired loop of messenger RNA (mRNA) that is created when an mRNA strand folds and forms base pairs with another section of the same strand. The resulting structure looks like a loop or a U-shape. Hairpins are a common type of secondary structure in RNA molecules. In RNA, the secondary structure is the basic shape that the sequence of A, C, U, and G nucleotides form after they are linked in series, such a folding or curling of the nucleic acid strand. mRNA hairpins can be formed when two complementary sequences in a single mRNA molecule meet and bind together, after a folding or wrinkling of the molecule. Hairpin loops can also form in DNA molecules, but are most commonly observed in mRNA. The sequence of the paired segment '5'-UCAAUGC-3'' is in antiparallel complementary.

62. Polymess-γ is located in organelles mitochondria and chloroplast. Thus DNAP-g is probably responsible for replicattrion of the DNA present in these organelles. Recent evidence has shown that DNAP-α and δ are both required for DNA replication in nuclei.

64. Each eukaryotic chromosome apparently contains a single giant molecule of DNA that extends from one end of the chromosome through the centromere to the other end of the chromosome.

65. Nucleosome contains highly conserved core consisting of a 146 nucleotide pair long segment of DNA wrapped around an octamer of histones.

67. DNA containing sequences that are repeated more than 10^5 times is called highly repetitive DNA, Highly repetitive sequences are usually located in heterochromatin.

68. Euchromain consists of both unique/ single copy sequences and base sequences repeated to 10 to 10^5 times (middle repetitive sequences)

69. Recently, however, chemical analysis, electron microscopy and X-ray diffraction studies on isolated chromatin (the complex of the DNA chromosomal proteins and other chromosome constituents isolated from nuclei).

70. 5 major histones, called H1, H2a, H2b, H3 and H4 are present in almost all cell types (a few exception exist, most notably some sperm, where the histones are replaced by another class of small basic proteins called protamines)

71. Histones are basic because they contain 20-30% arginine and tysine, 2 positively charged aminoacids. The exposed $-NH_3^+$ groups of arginine and lysine allow histones to act as polycations.

72. Functions of highly repetitive DNA sequences include

1. Structural/organizational roles in chromosomes
2. Involvement in chromosome pairing during meiosis
3. Involvement in crossing over/recombination
4. Protection of important structural genes, like histone, rRNA or ribosomal protein genes
5. A repository of unessential DNA sequences for use in the future evolution of the species.

76. The polymerase III holoenzyme (the dnaC gene product in E.coli) binds to template DNA as part of a multi-protein complex that consists of several polymerase accessory factors (β, γ, δ, and τ).

81-82.

On the leading (forward) strand, the DNA is synthesized continuously on the lagging strand (retrograde), the DNA is synthesized in short (1 - 5kb) fragments, the so called okazak fragments.

83-84, 86 :

E.coli	Mammalian	Function
I	α	Gap filling and synthesis of lagging strand
II	ε	DNA proof reading and repair
	β	DNA repair
	γ	Mitrochondrial DNA synthesis
III	δ	processive, leading strand synthesis

(90-93) :

The cyclins are a family of proteins whose concentration increases and decreases throughout the cell cycle-tnus their name. The cyclins turn on, at the appropriate time, different cyclin dependent protein kinases (CDKs) that phosphorylated substrates essential for progression through the cell cycle. E.g. Cyclin D levels rise in late G1 phase 4 d or progression beyond the start (yeast) or restriction point (mammals); The point beyond which cells irrevocably proceed into the S/DNA synthesis phase.

The cyclin D activate CDK4 and CDK6. These 2 kinases are also synthesized during G1 in cells undergoing active division. The D cyclins and CDK4 and CDK6 are nuclear proteins that assume as a complex in the G1 phase.

94. Cyclin E and CDK2 form a complex in late G1. Cyclin E is rapidly degraded, and the released CDK2 then forms a complex with cyclin A. This sequence is necessary for the initiation of DNA synthesis in S-phase.

96. The starting point of transcription corresponds to the 5′ nucleotide of the mRNA. This is designated position +1. The numbers increase as the sequence proceeds downstream. This convention makes it easy to locate particular regions, such as intron and exon boundaries. The nucleotide in the promoter adjacent to the transcription initiation site is designated –1 and

these negative numbers increase at the sequence proceeds upstream away from the initiation site.

99. 3, each of these DNa-dependent RNA polymerases is responsible for transcription of different sets of genes.

(101, 102 & 104) :

Form of RNAP	Sensitivity to α-Amantin	Major products
I(A)	Insensitive	rRNA
II(B)	High sensitivity	mRNA
III(C)	Intermediate sensitivity	tRNA/5srRNA

One peptide toxin from the mushroom Amantia phalloicles, α-amanitin, is specific differential inhibitor of the eukeryotic nuclear DNA-dependent RNAPs and as such has proved to be a powerful research tool. α-amanitin blocks the translocation of RNAP during transcription.

103. RNA polymerases interact with unique cis-active regions of genes, termed promoters, in order to form pre-initiation complexes (PICs) capable of initiation. In eukaryotes, the process of PIC formation is facilitated by multiple general transcription factors (GTSs) TF11A, B, D, E and H.

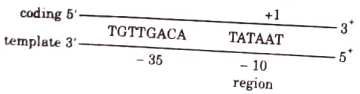
105. RNA synthesis is a cylical process and involves initial elongation and termination. Initiation of formation of the RNA molecule at its 5′ end then follows, while-elongation of the RNA molecule, from 5′ to its 3′ end continuous cyclically anti-parallel to its template.

(107-109).

Terminntion of the synthesis of the RNA molecule in bacteria is signaled by sequence in the template strand of the DNA molecule- a signal that is recognized by a termination protein, the rho (ρ) factor. Rho is an F-dependent RNA-stimulated helicase that disrupts the nascent RNA-DNA complex.

110. Bacterial promoters are relatively simple and are approximately 40 nucleotides (49 bp/ four turns of DNA double helix) in length, a region small enough to be covered by an E.coli RNA holo polymerase.

111. More proximal to transcription start site – about 10 nucleotides upstream - is a 6 - nucleotide pair A + T rich sequence (5′ TATAAT-3′). These conserved sequence elements comprising the promoter are shown in figure (TATA BOX)



(113-114):

The TATA box is bound by 34kDa TATA binding protein (TBP), which in turn binds several other proteins called TBP-associated factors (TAFs). This complex of TBP and TAFs is referred to as TF 11D. Binding of TF11D to the TATA box sequence is though to represent the first step in the formation of the transcription complex on the promoter.

117. A 3rd class of sequence elements can either increase/ decrease the rate of transcritpion initiation of eukaryotic genes. These elements are called either enhancers - Increase or repressors (silencers) decrease, depending on which effect they have. Enhancers and silencers can function in an orientation-dependent fashion.

(118 & 120):

Some of the transcriptional control elements, their consensus sequences, and the factors that bind to them which are found in mammalian genes transcribed by RNA polymerase II

Element	Consensus Sequence	Factor
TATA Box	TATAAT	TBP
CAAT Box	CCAATC	C/EBP*, NF-Y*
GC box	GGGCGG	SP1*
	CAACTGAC	MyoD
	T/CGGA/CN$_S$GCCAA	NF1*
Lg octamer	ATGCAAAT	OCT1, 2, 4, 6*
AP1	TGAG/CTC/AA	Jun, FOS, ATF
Serum response	GATTCCCATA	SRF
Heat shock	(NGAAN)$_3$	HSF

General mechanisms	Specific components
Casal components	TBP, TF11A, B, E, F & H
Co-activators	TAFs(TBP + TAFs) = TF11Ds
Activators	SP1, ATF, CTF, AP1 etc.

124. The 5′ terminals of most mRNA molecules in eukaryotic cells are capped. This methyl-GTP can facilitates the binding of mRNA to the 43S pre-initiation complex.

(125-126).

The 4F complex is particalarly important in controlling rate of protein translation. 4F is a complex consisting of 4E which binds to the m7G cap structure at the 5′ end of the mRNA and 4G, which serves as a scaffolding protein.

4E is responsible for recognition of the mRNA cap structure and is a rate limiting step in translation. This process is regulated at 2 levels. Insulin and mitogenic growth factors result in phosphorylation of 4E on Ser 209.

127. Elongation is a multistep process/ cyclic process on the ribosome in which 1 amino acid at a time is added to the nascent peptide chain. The peptide sequence is determined by the order of the codons in the mRNA.

128. Binding of Aminoacyl-tRNA to the A-site.

In the complete 80s ribonsome formed during the process of initiation, the A site is free. The binding of the proper aminoacyl tRNA in the A site requires proper codon recognition. Elongation factor EF1-A forms a ternary complex with GTP and the entering aminoacy-tRNA. This complex then allows the aminoacyl tRNA to enter the A site with the release of EF1-A, GDP and phosphate. GTP hydrolysis is catalyzed by an active site on the ribosome. EF1A-GDP then recycles to EF1A-GTP with the aid of other soluble factors and GTP.

129. The α-amino group of the new aminoacyl-tRNA in the A site carries out a nucleophillc, attack on the esterified carboxy group of peptidyl-tRNA occupying the P site. At initiation, this site is occupied by aminoacyl-tRNA methionine. This reaction is catalyzed by a peptidyl transferase, a component of the 28SRNA of 60S ribosomal subunit.

130. EF2 binds and displaces the peptidyl tRNA from A site to the P site. In turn, the deacylated tRNA is in the E site from which it leaves the ribosome. The EF-2 compound is hydrolyzed to EF2-GDP, effectively moving the mRNA forward by one codon and leaving the A site for occupancy by another ternary complex of amino acid tRNA-EF1A-GTP and another cycle of elongation.

131. After multiple cycles of elongation culminating in polymerization of the specific aqs into a protein molecules, the stop or terminating codon of mRNA (UAA, UAG, UGA) appears in the A site.

(132-133).

RF1 (Releasing factor) recognizes that a stop codon resides in the A site. RF1 is bound by a complex consisting of RF3 with bound GTP. This complex, upon hydrolysis and release, the 80s ribosome dissociates into it's 40s and 60s subunits, which are then recycled. Therefore, the releasing factors are proteins that hydrolyzes the peptide-tRNA bond when a stop codon occupies the A site. The mRNA is then released from the ribosome, which dissociates into its component 40s and 60s subunits, and another cycle can be repeated.

135. The most useful members of this class of antibiotics (e.g. tetracycline's, linocomycin, erythromycin and chloramphenicol) do not interact with components of eukaryotic ribosomal particles and thus are not toxic to eukaryotes.

Tetracyline prevents the binding of aminoacyl-tRNA to the A site.

(136-137).

Puromycin is a structural analog of tyrosinyl-t-RNA. Puromycin is incorporated via the A site on the ribosome into the carboxyl terminal position of a peptide but causes the premature release of the polypeptide. Puromycin, as a tyrosinyl-tRNA analog, effectively inhibits protein synthesis in both prokaryotes and eukaryotes.

138. Ricin, an extremely toxic molecule isolated from the castor castor bean, inactivates eukaryotic 28S ribosomal RNA by providing the. N-glycolytic cleavage/ removal of a single adenine.

139. Four identical subunits with molecular weights of 38,000 assemble into lac repressor molecule. The lac I repressor protein molecule, the product of lacI, has a high affinity (K_a about 10^{-13}) for the operator locus.

140.

	Rate of gene expression	
	Negative regulation	Positive regulation
Regulator present	Decreased	Increased
Regulator absent	Increased	Decreased

(142-143).

The *lac* operon uses a two-part control mechanism to ensure that the cell expends energy producing β-galactosidase, β-galactoside permease and thiogalactoside transacetylase (also known as galactoside O-acetyltransferase) only when necessary. It achieves this with the *lac* repressor, which halts production in the absence of lactose, and the Catabolite activator protein (CAP), which assists in production in the absence of glucose. This dual control mechanism causes the sequential utilization of glucose and lactose in two distinct growth phsases, known as diauxide.

NUMERICAL TYPE QUESTIONS

1. Number of guanine is equal to cytosine and that of thymine is equal to adenine, hence the total number of nucleotides is sum of all the four bases.
2. In each turn of the double helix structure of DNA, there are 10 base pairs each placed at a distance of 3.4A. Hence, if the length of DNA has 45,000 bp, the complete turns of DNA molecule is 4,500.
3. Depending upon the position of centromere and relative length of two arms, human chromosomes are of three types: metacentric, sub-metacentric, and acrocentric. The photograph of chromosomes is artificially arranged in the order of descending length in seven group (A to G).
4. Aleurone is triploid and root tip is diploid.
5. 19%; because G is always attached with C (GC). Thus, percentage of C will always be equal to G.

6. According to radial loop model each chromosome has 1 or 2 interconnected scaffolds made of non-histone chromosomal proteins. The scaffold bears a large member of the lateral loop all over it. Each lateral loop is 30 nm thick fibres similar to chromatin fibre.

7. Wheat has a haploid number of seven chromosomes. That means for wheat, x = 7. Now, in case of hexaploidy of the wheat, we have the hexaploid chromosome number = 6x. The diploid number of chromosome for a normal wheat plant would be given by 2n. That means we have, 2n = 6x. Now, we know x = 7. Therefore, 6x = 6 × 7 = 42 and 2n = 6x = 42 which implies that n = 21.

8. Using an average molecular weight/ nucleotide-pair of 660, the molecular weight of the entire human genome is 1.98×10^{12} g (1 mole = no. of grams equivalent to the molecular weight) to human DNA will contain, on average, 6.02×10^{23} molecules (Avogadro's number = number of molecules [here copies of genome] present in one mole of substance). One gm will contain, on average 3.04×10^{11} ($6.02 \times 10^{23}/1.98 \times 10^{12}$) copies genome. i.e. 3.04×10^{5} copies of human genome per μg

9. By analogous calculation 1 mg of Arabidopsis thaliana DNA contains, on average, 1.18×10^{7} copies of the genome.

10. One copy of human genome weights approximately 3.3×10^{-12} g (1.98×10^{12} g/ mole/ 6.02×10^{23} molecules/mole) or 3.3×10^{-6} μg

13. In E.coli, polymerase III functions at the replication fork, of all polymerases; it catalyzes the highest rate of chain elongation and is the most processive. It is capable of polymerizing 0.5 Mb of DNA during one cell cycle on the leading strand.

14. In mammalian cells, the polymerase is capable of polymerizing about 100 nucleotides/second, a rata atleast tenfold slower than the rate of polymerization of deoxynucleotides by the bacterial DNA polymerase complex.

17. Four identical subunits with molecular weights of 38,000 assemble into lac repressor molecule. The lac I repressor protein molecule, the product of lacI, has a high affinity (K_c about 10^{-13}) for the operator locus.

EXERCISE – II

MCQ TYPE QUESTIONS

1. The binding status of regulatory proteins, in *lac* operon when concentrations of both lactose and glucose are very low in the culture medium will be both the repressor and cAMP-CAP complex remain bound to their respective binding sites.

2. miRNA is associated with RNA interference, nuclear export. siRNA and shRNA also functions in RNA interference. snRNA is involved in spliceosomal reaction.

3. On the ribosome, the anticodon of the incoming aminoacyl-tRNA is matched against the mRNA codon positioned in the A-site. The translational machinery select the aminoacyl-tRNA carrying the matching anticodon out of 64 different combinations. During this proof-reading, aminoacyl-tRNAs with non-cognate anticodons are thrown out of the ribosome and replaced by new amino acyl-tRNAs that are to be checked. When the right aminoacyl-tRNA enters the A-site the growing polypeptide in the P-site is almost immediately linked to the new amino acid in the A-site via a peptide bond. Formation of the peptide bond is catalysed by the ribosome itself. The reaction leaves an empty tRNA in the ribosomal P-site and the new peptidyl-tRNA in the A-site.

4. A frameshift mutation is a genetic mutation caused by insertions or deletions of a number of nucleotides in a DNA sequence that is not divisible by three. Due to the triplet nature of gene expression by codons, the insertion or deletion can change the reading frame resulting in a completely different translation from the original. Therefore P, S and T do not have frameshift.

5. Number of 3 tuples = L – k + 1

   L → Total length of sequence = 13

   k → tuples = 3

   Then 3 tuples = 13 – 3 + 1 = 11

6.

Group I	Group II
P. Paclitaxel	2. Inhibits microtubule depolymerization
Q. Colchicine	6. Inhibits microtubule polymerization
R. Etoposide	3. Inhibits DNA replication
S. Methotrexate	5. Inhibits dihydrofolate reductase

7. Only two codons are possible in 1st case : UCU, CUC

   3 codons are possible in 2nd case : CCU, CUC, UCC

   Due to wobble hypothesis the unique amino acid codon in 2nd case is only CCU hence a new amino acid is synthesized.

8. $S \xrightarrow{c} X \xrightarrow{b} Y \xrightarrow{d} Z \xrightarrow{a} P$

9. In the more matured forms of RNAs in case of prokaryotic organisms, the 5' end of mRNA consists of a triphosphate group while there is a single monophosphate group present at the 5' terminal of tRNA molecules.

10. Given topological winding no. of a closed circular DNA = 191

    Writhing number = - 4

    So, linking number = Winding No. + Writhing No. = 191 – 4 = 187 when the molecule is laid flat on the surface

    Number of base pair per turn = Bases/ winding no. = 2kb/ 191 = 2048/ 191 = 10.722

11. Peptide I has 15 sequences out of which 9 are identical to the 15 sequences of Peptide II.

    So, the percentage identity between the two sequences will be = 9/15 × 100 % = 60 %

    Now, arginine and glutamine, leucine and isoleucine and glutamate and aspartate are similar amino acids = 3 similar + 9 identical amino acids between the two peptides

    So percentage similarity = 12/15 × 100% = 80%

12. Shine-Dalgarno (SD) sequence is a ribonucleotide sequence for the formation of the preinitiation complex between a 30S ribosomal subunit and an mRNA molecule. It is a seven-nucleotide, pyrimidine-rich sequence near the 3? terminus of 16S (prokaryotic) or 18S eukaryote ribosomal RNA, found in many organisms, that forms base pairs with a complementary three- to seven-nucleotide, purine-rich sequence (the Shine-Dalgarno sequence) preceding the initiation codon in many mRNA molecules, thereby allowing the initiating ribosomal subunit to bind the mRNA so as to discriminate between AUG or GUG (i.e. initiation) triplets and to select the one at the beginning of a cistron. Anti Shine Dalgarno sequences are present in 16s rRNA molecules for the recognition of SD sequences in eubacterial mRNA.

13. The most frequent and most regular of the domain structures are the alpha / beta domains. These domains consist of a central parallel or mixed beta-sheet surrounded by alpha-helix. All the glycolytic enzymes are alpha / beta structure proteins. There are three maim classes of alpha / beta structure proteins, TIM barrel, Rossman fold, and horseshoes fold (leucine-rich motif). The alpha/beta barrel fold is adopted by most enzymes performing a variety of catalytic reactions, but with very low sequence similarity. The cluster centers are found to occur in the middle of the strands or in the C-terminal of the strands. In most cases, the residues forming the clusters are part of the active site or are located close to the active site. The active site is majorly present at the carboxy-terminal side of the α sheets or the β strands.

14. DNA is the repository of genetic information gathered over incalculable millions of years and it is stored in a stable form inside the cell. However, chemical molecules have some degree of instability and often breakdown with time. But DNA molecules have to remain stable. The presence of the –OH group on the second carbon chain of the ribose sugar makes a ribopolynucleotide less stable than a deoxyribose molecule. The presence of 2'-OH group on the ribose sugar makes it susceptible to nucleophilic attack in the presence of OH (ions) on the 5'-phosphorous atom, thus causing breakage of the phosphodiester link and forming a 2',3' cyclic phosphate. The –OH ion facilitates the reaction because it can generate a 2'–O ion from the 2'-OH group, which attacks the phosphorous atom and converts the phosphodiester group into a 2', 3' –cyclic nucleotide, thus breaking the polynucleotide chain. Hydrolysis of the cyclic nucleotide produces a mixture of 2' and 3' nucleotides at the breakpoint. Moreover, the double stranded DNA has relatively small grooves as opposed to the larger grooves on RNA molecules. This provides ample docking space for damaging enzymes.

15. The addition of a methyl group (-CH_3) to a molecule is known as methylation. Extensive methylation of cytosine in DNA is correlated with reduced transcription. DNA methylation is one of several epigenetic mechanisms that cells use to control gene expression. Proteins can be methylated on the side-chain nitrogen of arginine and lysine residues or on carboxy-termini. Protein methylation is a way of subtly changing the primary sequence of a peptide so that it can encode more information. This common posttranslational modification is implicated in the regulation of a variety of processes including protein trafficking, transcription and protein-protein interactions.

16. DNA sequencing is the determination of the precise sequence of nucleotides in a sample of DNA. The most popular method for doing this is called the dideoxy method or Sanger method. DNA is synthesized from four deoxynucleotide triphosphates. The dideoxy method gets its name from the critical role played by synthetic nucleotides that lack the -OH at the 3' carbon atom. A dideoxynucleotide can be added to the growing DNA strand but when it is, chain elongation stops because there is no 3' -OH for

the next nucleotide to be attached to. For this reason, the dideoxy method is also called the chain termination method. The DNA sample is divided into four separate sequencing reactions, containing the four standard deoxynucleotides (dATP, dGTP, dCTP and dTTP) and the DNA polymerase. To each reaction is added only one of the four dideoxynucleotides (ddATP, ddGTP, ddCTP, or ddTTP). These dideoxynucleotides are the chain-terminating nucleotides, lacking a 3′ -OH group required for the formation of a phosphodiester bond between two nucleotides during DNA strand elongation. Incorporation of a dideoxynucleotide into the nascent (elongating) DNA strand therefore terminates DNA strand extension, resulting in various DNA fragments of varying length. The dideoxynucleotides are added at lower concentration than the standard deoxynucleotides to allow strand elongation sufficient for sequence analysis.

17. DNA Polymerase I is a template-dependent DNA polymerase which catalyzes 5′→3′ synthesis of DNA. DNA Polymerase I catalyzes the template-directed polymerization of nucleotides into duplex DNA in a 5′→3′ direction. In addition to polymerase activity, this DNA polymerase exhibits 3′ to 5′ and 5′ to 3′ exonuclease activity. It is able to utilize nicked circular duplex DNA as a template and can unwind the parental DNA strand from its template. Because of its exonuclease activity in the 3′ to 5′ direction, the polymerase can remove or excise the RNA primers synthesized during replication.

18. Ribonuclease P (RNase P) is the endoribonuclease that generates the mature 5′-ends of tRNA by removal of the 5′-leader elements of precursor-tRNAs. Although it carries out a biochemically simple reaction, RNase P is a complex ribonucleoprotein particle composed of a single large RNA and at least one protein component. RNase H (Ribonuclease H) is an endoribonuclease that specifically hydrolyzes the phosphodiester bonds of RNA which is hybridized to DNA. The snRNAs, along with their associated proteins, form ribonucleoprotein complexes (snRNPs), which bind to specific sequences on the pre-mRNA substrate. This intricate process results in two sequential transesterification reactions. These reactions will produce a free lariat intron and ligate two exons to form a mature mRNA. This results in the splicing. CstF is recruited by cleavage and polyadenylation specificity factor (CPSF) and assembles into a protein complex on the 3′ end to promote the synthesis of a functional polyadenine tail, which results in a mature mRNA molecule ready to be exported from the cell nucleus to the cytosol for translation.

19. Mismatch repair deals with correcting mismatch of the normal bases using BER or NER enzyme systems. The system assumes that parental strands are methylated and freshly synthesized daughter strands are non methylated.

20. To investigate the effect of serine 78 phosphorylation on p21 activity, replacement of serine 78 with aspartic acid is done, creating the phosphomimic $p21^{*78D}$.

21. Non sense mutations causes premature stop to be introduced that would lead to truncated or incomplete protein synthesis. If the protein synthesis has to be continued, another mutation in t-RNA gene would continue be advised to get full length polypeptide

22. The direction of shell coiling in snail is a dassical example for extra chromosomal inheritance, as in this it is determined by maternal gene effects and not that of the offspring.

23. An inverted repeat (or IR) is a sequence of nucleotides followed downstream by its reverse complement. It is a sequence found in identical (but inverted) form, for example, at the opposite ends of a transposon.

    Here, the DNA sequence which carries an inverted repeat is:

    ATGAGCCCCGAGTA

    TACTCGGGGCTCAT

24. Zinc fingers coordinate zinc ions with a combination of cysteine and histidine residues. They can be classified by the type and order of these zinc coordinating residues (e.g., Cys_2His_2, Cys_4, and Cys_6).

25. The folding of many newly synthesized proteins in the cell depends on a set of conserved proteins known as molecular chaperones. These prevent the formation of misfolded protein structures, both under normal conditions and when cells are exposed to stresses such as high temperature. Significant progress has been made in the understanding of the ATP-dependent mechanisms used by the Hsp70 and chaperonin families of molecular chaperones, which can cooperate to assist in folding new polypeptide chains.

26. Diphtheria toxin consists of a single polypeptide. Proteolysis yields two fragments (A and B) which are held together by a disulfide bond. The toxin binds to EGF-like domain of Heparin-binding EGF-like growth factor (HB-EGF) through fragment B and is internalized with HB-EGF by receptor-mediated endocytosis. Diphtheria toxin catalyzes the ADP-ribosylation of, and inactivates, the elongation factor eEF-2. In this way, it acts to inhibit translation during eukaryotic protein synthesis. Tetracyclines are antibiotics which inhibit the bacterial growth by stopping protein synthesis. Three different specific mechanisms of tetracycline resistance have been identified so far: tetracycline efflux, ribosome protection and tetracycline modification. Streptomycin is a water-soluble aminoglycoside derived from Streptomyces griseus. Streptomycin is an aminoglycoside antibiotic that is currently used largely in the therapy of active tuberculosis. Like other aminoglycosides, streptomycin is thought to act by binding to bacterial ribosomes and inhibiting protein synthesis. Nevertheless, streptomycin is considered bacteriocidal as well as bacteriostatic.

27. N-methyl-N'-nitro-N-nitrosoguanidine is a highly reactive chemical that introduces alkyl radicals into biologically active molecules and thereby prevents their proper functioning. It could be used as an antineoplastic agent, but it might be very toxic, with carcinogenic, mutagenic, teratogenic, and immunosuppressant actions. It could also be used as a component of poison gases. This substance is reasonably anticipated to be a human carcinogen. The mutations induced by this mutagen are mostly purine-pyrimidine transitions of GC → AT.

28. Enhancer sequences are regulatory DNA sequences that, when bound by specific proteins called transcription factors, enhance the transcription of an associated gene. Regulation of transcription is the most common form of gene control, and the activity of transcription factors allows genes to be specifically regulated during development and in different types of cells. Transcription factors can bind to enhancer sequences located upstream or downstream from an associated gene, resulting in stimulation or enhancement of transcription of the related gene. Enhancer sequences act upon genes on the same DNA molecule; however, enhancer sequences can be located thousands of base pairs away from the transcription start site of the gene being regulated. Because DNA is folded and coiled in the nucleus, the enhancer may actually be located near the transcription start site in the folded state.

29. The Shine-Dalgarno sequence is a ribosomal binding site in the mRNA. The leucine zipper is a three dimensional structural motif in proteins. These motifs are usually found as part of a DNA-binding domain in various transcription factors, and are involved in regular gene expression. Aminoacyl tRNA synthetase is an enzyme that catalyzes the esterification of a specific amino acid or its precursor to one of all its compatible cognate tRNAs to form an aminoacyl tRNA. RNAi works for the gene silencing (Small interfering RNA is mainly used in gene silencing and Antisense technology).

30. Gratuitous inducer is an analogue of a natural inducer that is capable of inducing an operon while not serving as a substrate for the enzyme being induced. IPTG is a gratuitous inducer. It induces gene expression but itself is not metabolized by the enzyme.

32. Peptidyl transferase is an amino acyltransferase and forms peptide links between adjacent amino acids using tRNA during translation process. This peptidyl transferase activity is mediated by 23 ribosomal RNA. This ribozyme 23 rRNA is the transferase component in bacteria. Also in eukaryotic cells, the same 23 rRNA holds the subunit containing peptidyl transferase component acting as the ribozyme.

33. DNA gyrase catalyse ATP dependant –ve supercoiling as these are topoisomerases which influence the topological state of DNA. It is type II topoisomerase which induce –ve supercoils (or relax +ve supercoils) by looping the template and cutting one of the double helices and then 2 ends are twisted for –ve supercoils. Type I topoisomerase unwinds –ve supercoils to control protein synthesis and facilitate DNA replication, and ease out problems due to unwinding of double helical forms.

34. Region of eukaryotic chromosome DNA that is specifically vulnerable to nuclease attack perhaps because it is not wrapped in histone as nucleosomes are called the nuclease hypersensitive sites in the chromosomes. Thus they are H_2 and H_4 free, H_1 and H_2 free, H_3 and H_4 histone free.

35. The MTT assay is a colorimetric assay which is used for measuring the activity of cellular enzymes that reduce the tetrazolium dye, MTT, to its insoluble formazan dye, giving a purple color. These assays measure cellular metabolic activity via NAD(P)H-dependent cellular oxidoreductase enzymes and may, under defined conditions, reflect the number of viable cells present. MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide, a yellow tetrazole), is reduced to purple formazan in living cells. A solubilization solution (usually either dimethyl sulfoxide, an acidified ethanol solution, or a solution of the detergent sodium dodecyl sulfate in diluted hydrochloric acid) is added to dissolve the insoluble purple formazan product into a

colored solution. Reduction of MTT and other tetrazolium dyes increases with cellular metabolic activity due to elevated NAD(P)H flux.

36. Chloramphenicol binds to the peptidyl transferase enzyme to inhibit transfer of the growing polypeptide to the next amino acid occupying the "Acceptor" site. Norfloxacin is a nalidixic acid analogue and one of the most potent DNA gyrase inhibitors. Puromycin is a secondary metabolite of Streptomyces alboniger that blocks protein biosynthesis. Puromycin is a structural analogue of the 32 end of aminoacyl-transfer RNA. Rifampicin inhibits bacterial RNA polymerase by binding to the beta subunit of RNA polymerase.

37. Commonly used reporter genes that induce visually identifiable characteristics usually involve fluorescent and luminescent proteins. Examples include the gene that encodes jellyfish green fluorescent protein (GFP), which causes cells that express it to glow green under blue light, the enzyme luciferase, which catalyzes a reaction with luciferin to produce light, and the red fluorescent protein from the gene dsRed.

38. IPTG (I-8000, I-8050) is an artificial inducer of the lac operon. IPTG binds to the lac repressor and inactivates it, preventing its binding to the lac operator and thus inducing the expression of beta-galactosidase. IPTG cannot be hydrolyzed and broken down by the cell and is therefore called a "gratuitous" inducer.

39. Zinc fingers function as interaction modules that bind DNA, RNA, proteins, or other small, useful molecules.

40. RNA interference (RNAi) also called post transcriptional gene silencing (PTGS), is a biological process in which RNA molecules inhibit gene expression, typically by causing the destruction of specific mRNA molecules. Two types of small ribonucleic acid (RNA) molecules: microRNA (miRNA) and small interfering RNA (siRNA) - are central to RNA interference.

41. A zinc finger protein is a DNA-binding protein domain consisting of zinc fingers ranging from two in the *Drosophila* regulator ADR1, the more common three in mammalian Sp1 up to nine in $TF_{111}A$. They occur in nature as the part of transcription factors conferring DNA sequence specificity as the DNA-binding domain.

42. Mobile genetic elements are the nucleotide sequences that can change their position in the genome. There are two categories of mobile genetic elements: DNA mobile genetic elements or DNA transposons and retroelements.

There are 3 kinds of retroelements:

- LINE(Long Interspersed Nuclear Elements)
- SINE(Short Interspersed Nuclear Elements)
- LTR(retrovirus like elements with Long Terminal Repeats)

43. Alkaline phosphatase is a hydrolase enzyme responsible for removing phosphate groups from many types of molecules, including nucleotides, proteins, and alkaloids. The process of removing the phosphate group is called dephosphorylation. Polynucleotide kinase (or PNK) is a T7 bacteriophage (or T4 bacteriophage) enzyme that catalyzes the transfer of a gamma-phosphate from ATP to the free hydroxyl end of the 5' DNA or RNA.The resulting product could be used to end-label DNA or RNA, or in a ligation reaction.

NUMERICAL TYPE QUESTIONS

1. Given sequence: Phe-Trp-Met-Gly-Ala-Lys-Leu-Pro-Met-Asp-Gly-Arg-Cys-Ala-Gln

First it is digested with Trypsin and than with Cyanogen Bromide in a process of sequential cleavage. Cyanogen bromide hydrolyzes peptide bonds at the C-terminus of methionine residues. Trypsin cleaves peptide chains mainly at the carboxyl side of the amino acids lysine or arginine, except when either is followed by proline.

After trypsin digestion, fragments generated are:

Phe-Trp-Met-Gly-Ala-Lys; Cys-Ala-Gln; Leu-Pro-Met-Asp-Gly-Arg

After CNBr digestion, the fragments generated are:

Phe-Trp-Met; Gly-Ala-Lys; Cys-Ala-Gln; Leu-Pro-Met; Asp-Gly-Arg

2. Given, No. of polydactylous persons = 36

No. of people having the allele for polydactyly = 80

Therefore, the extent of penetrance in the population = 36/80 x 100 % = 45 %